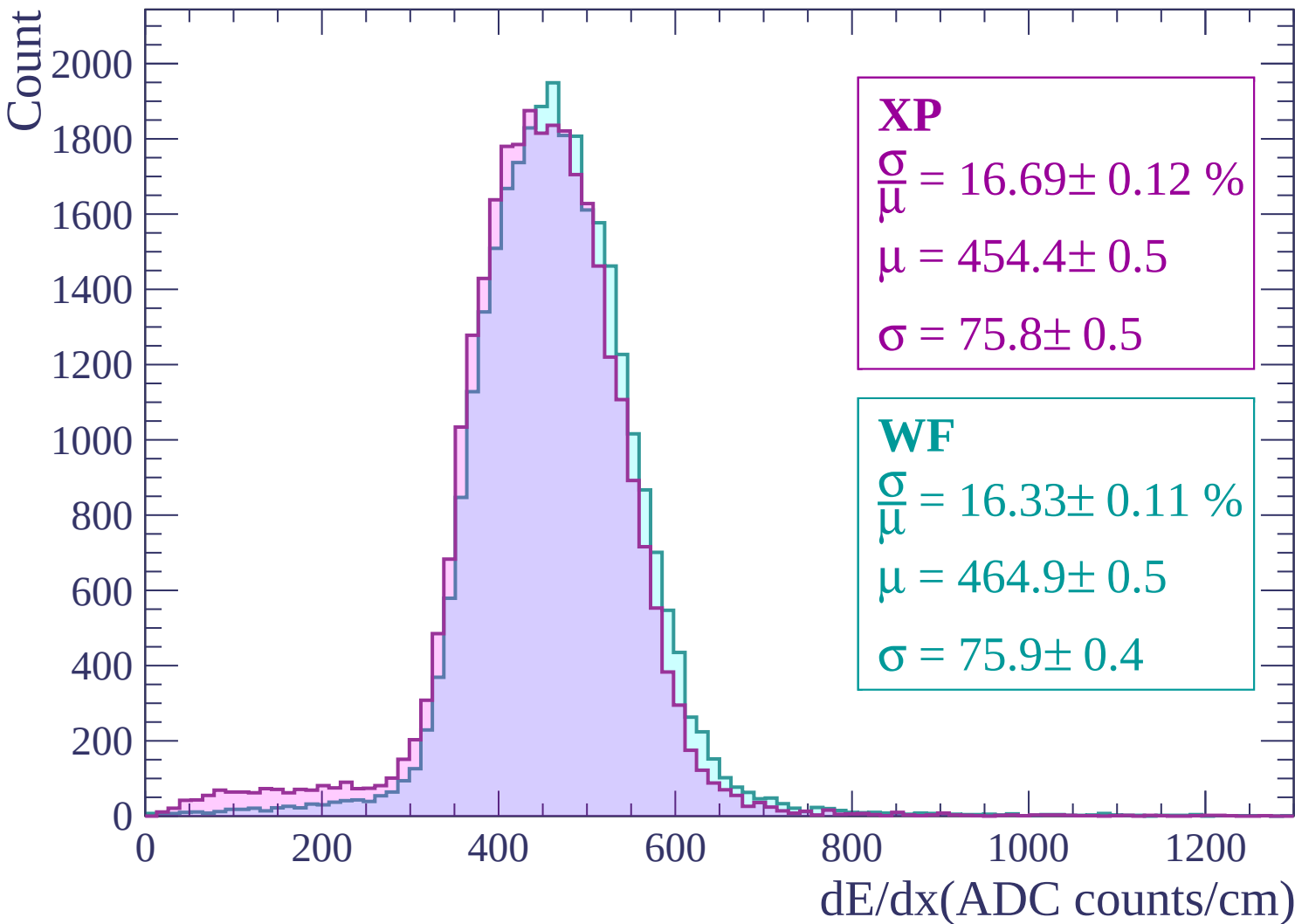
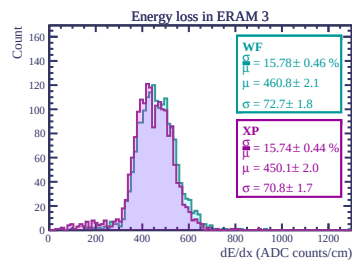
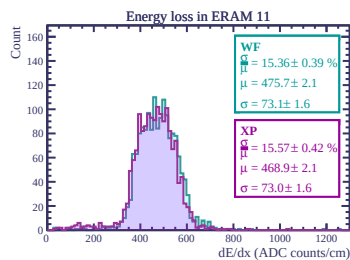
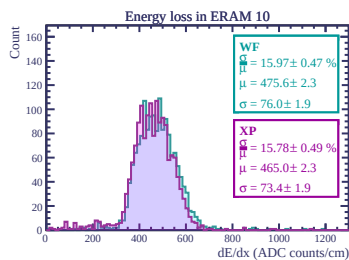
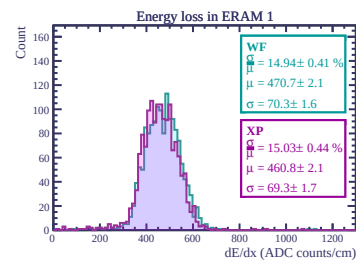
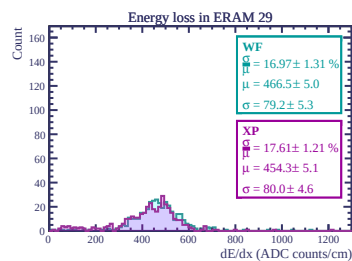
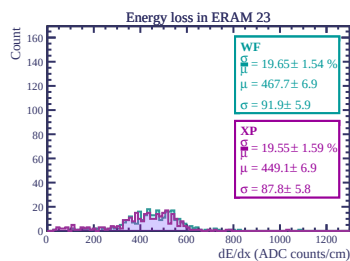
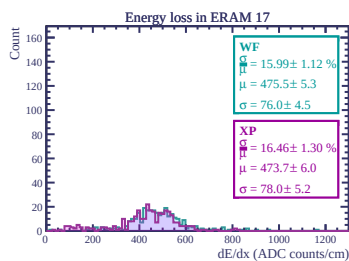
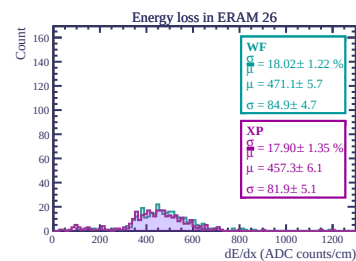
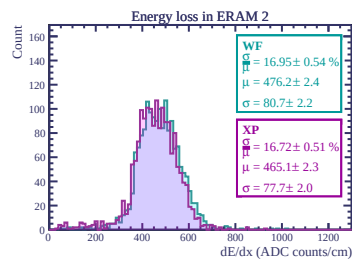
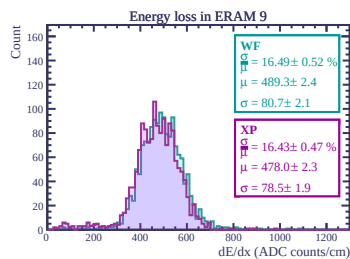
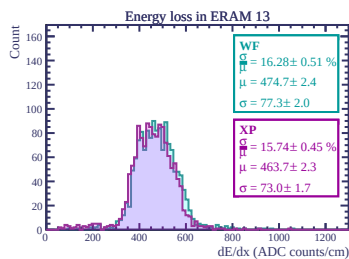
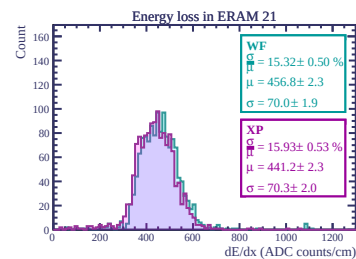
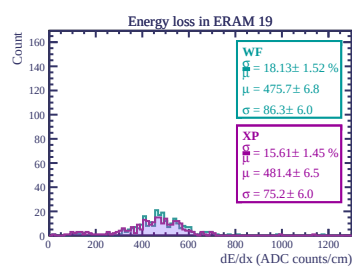
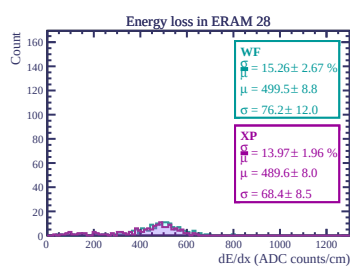
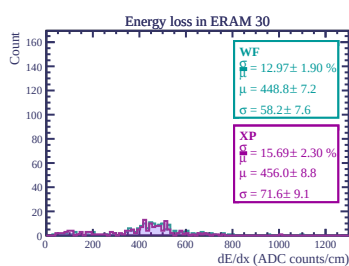
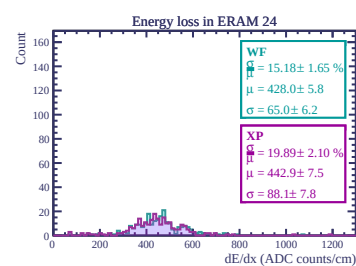
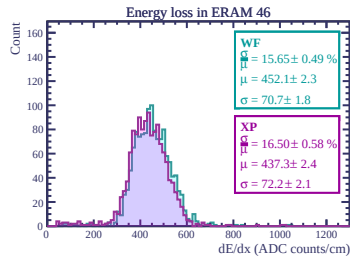
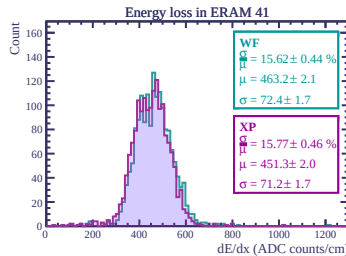
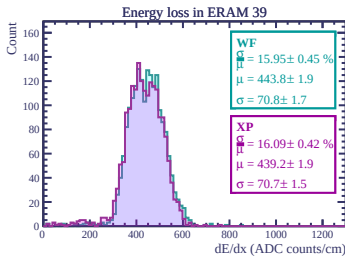
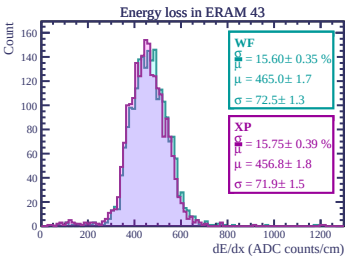
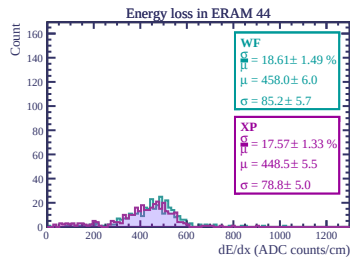
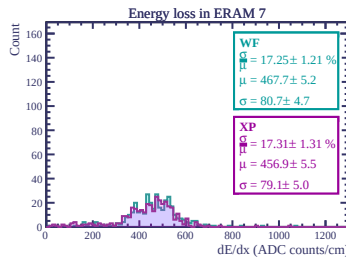
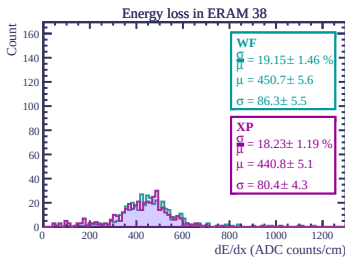
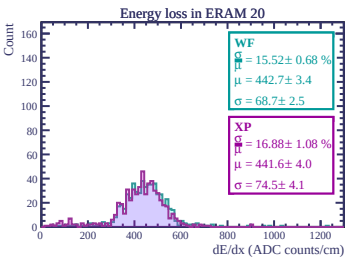
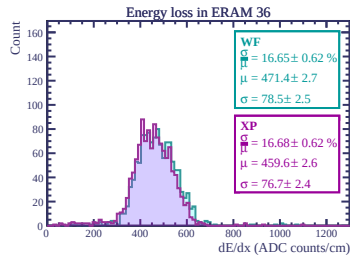
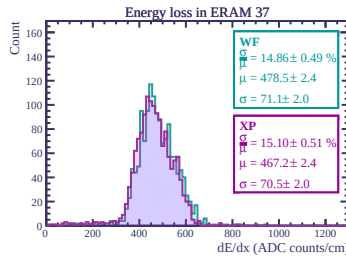
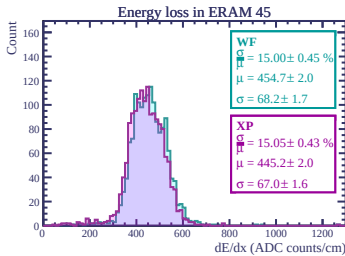
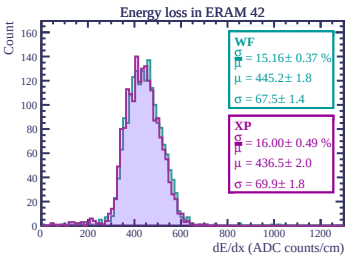
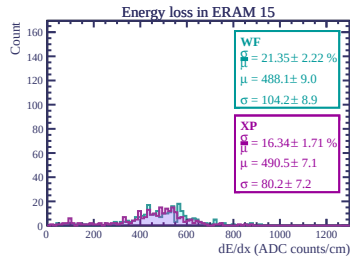
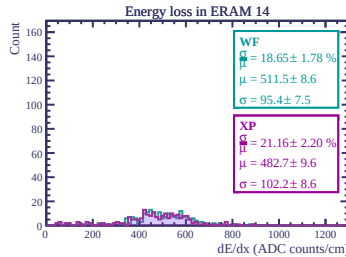
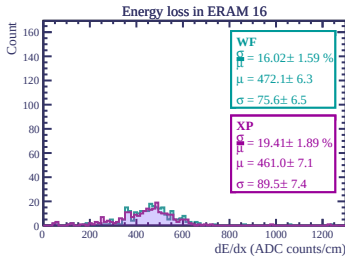
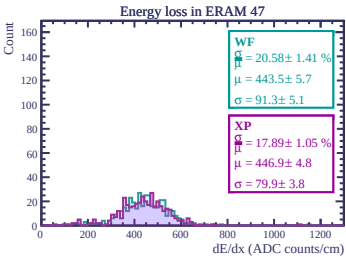
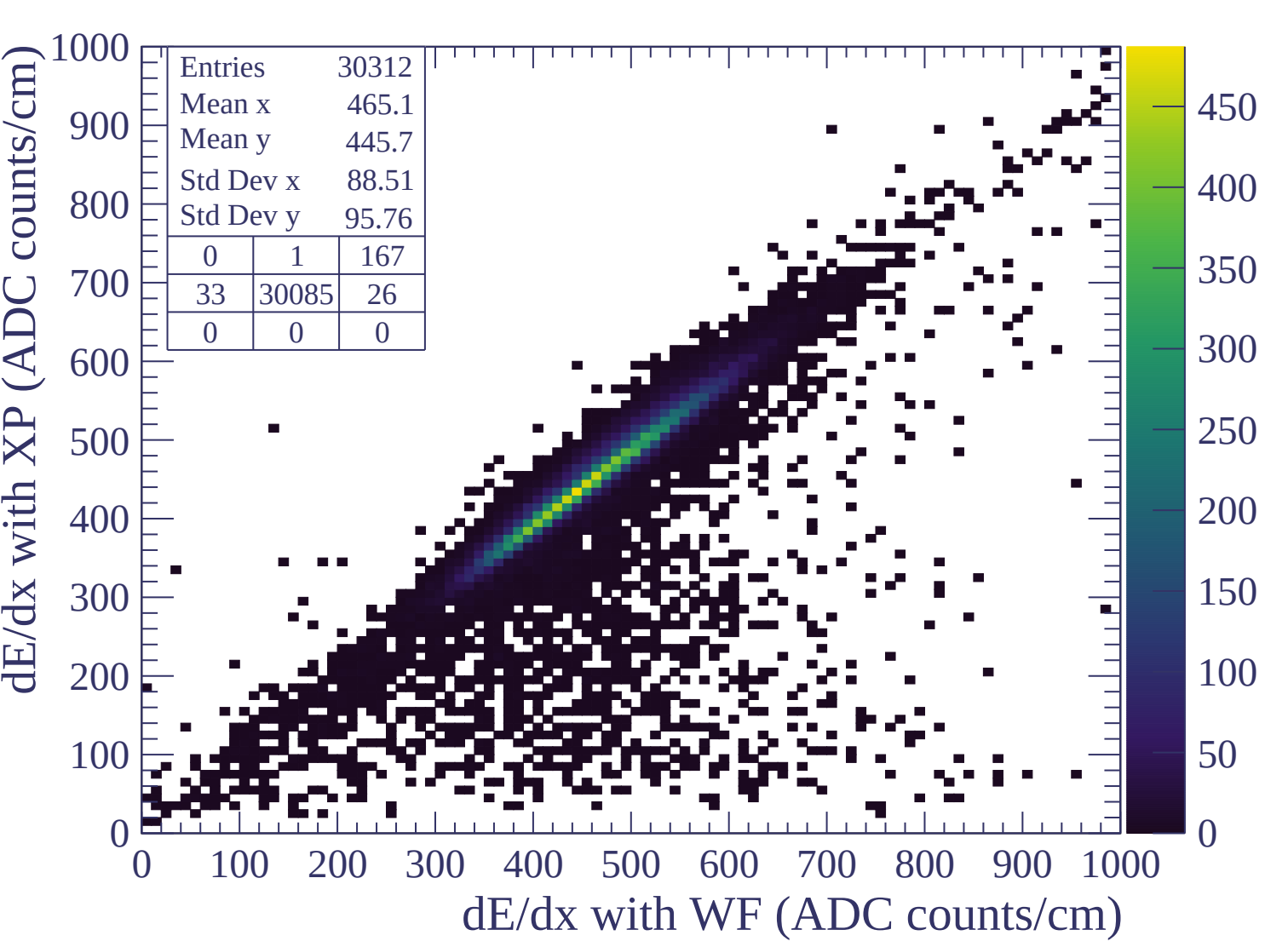
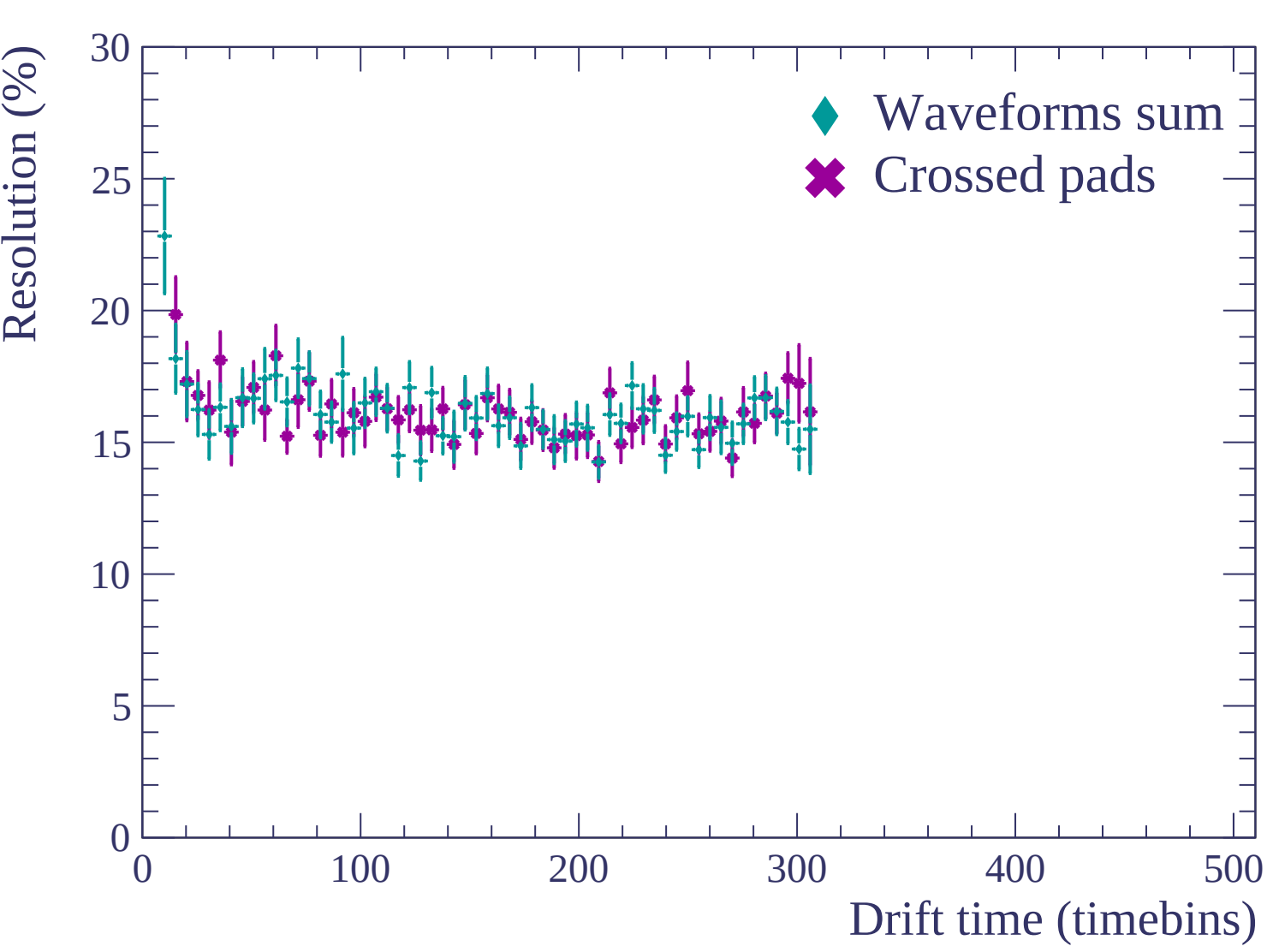


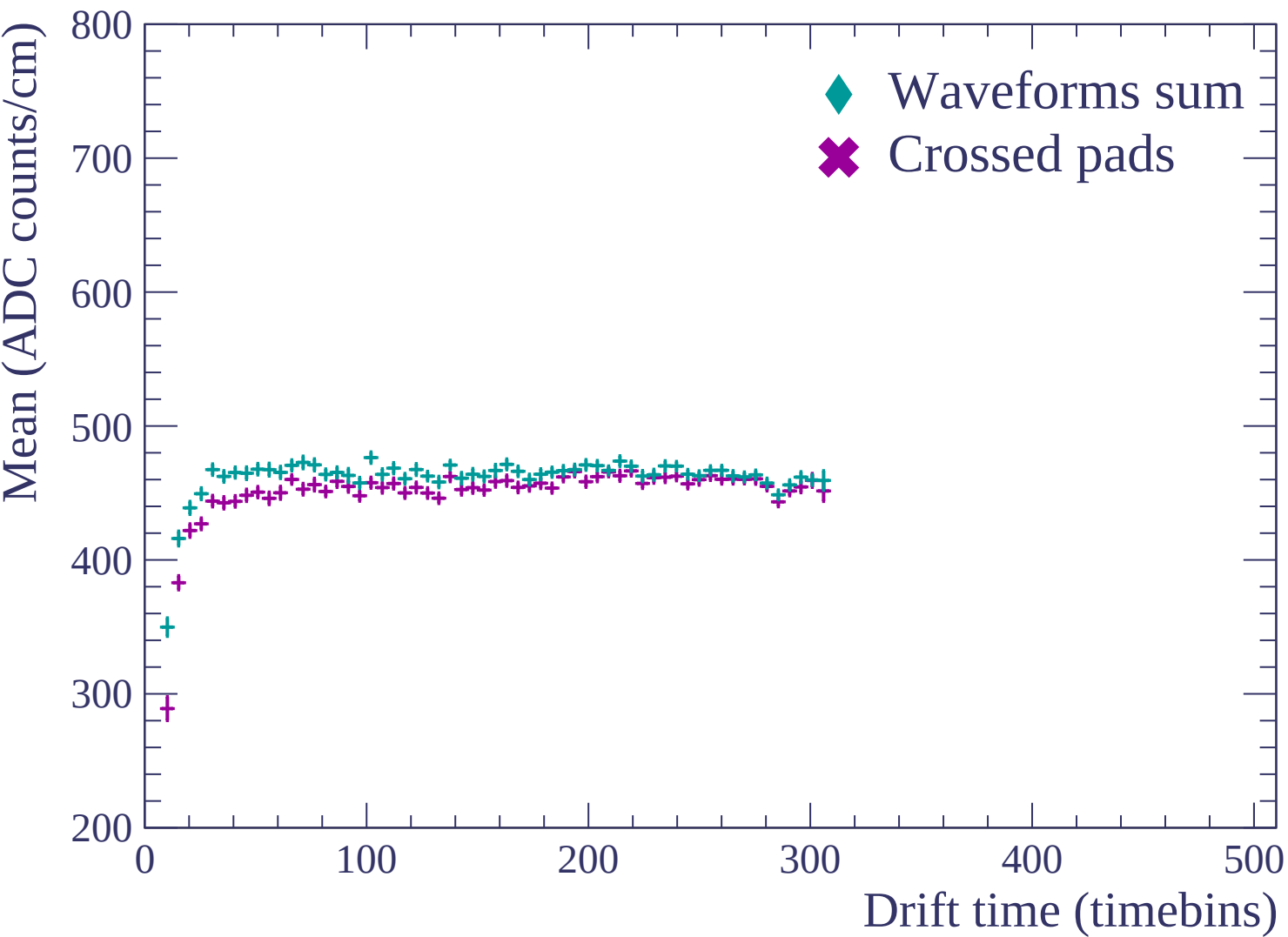


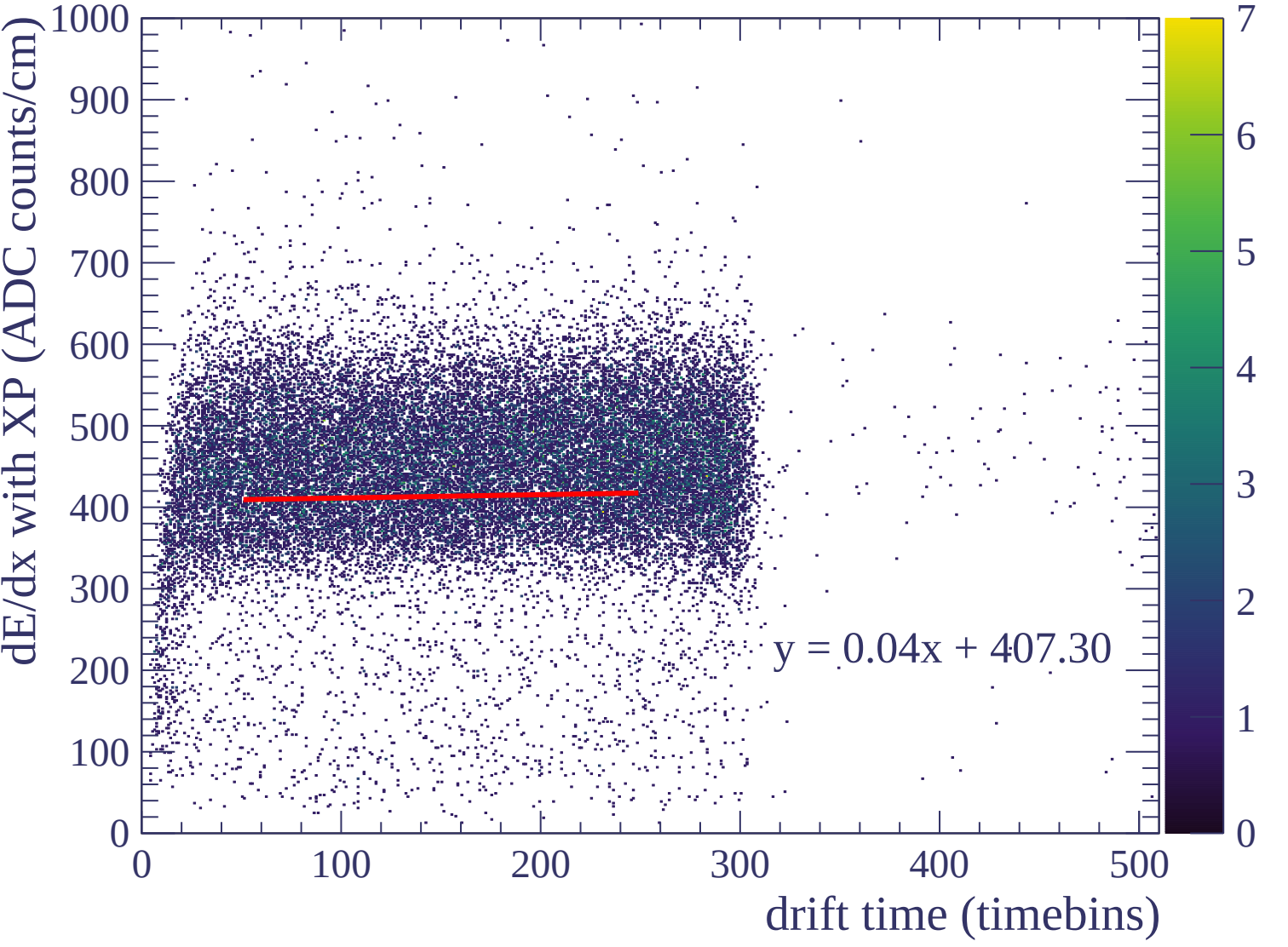


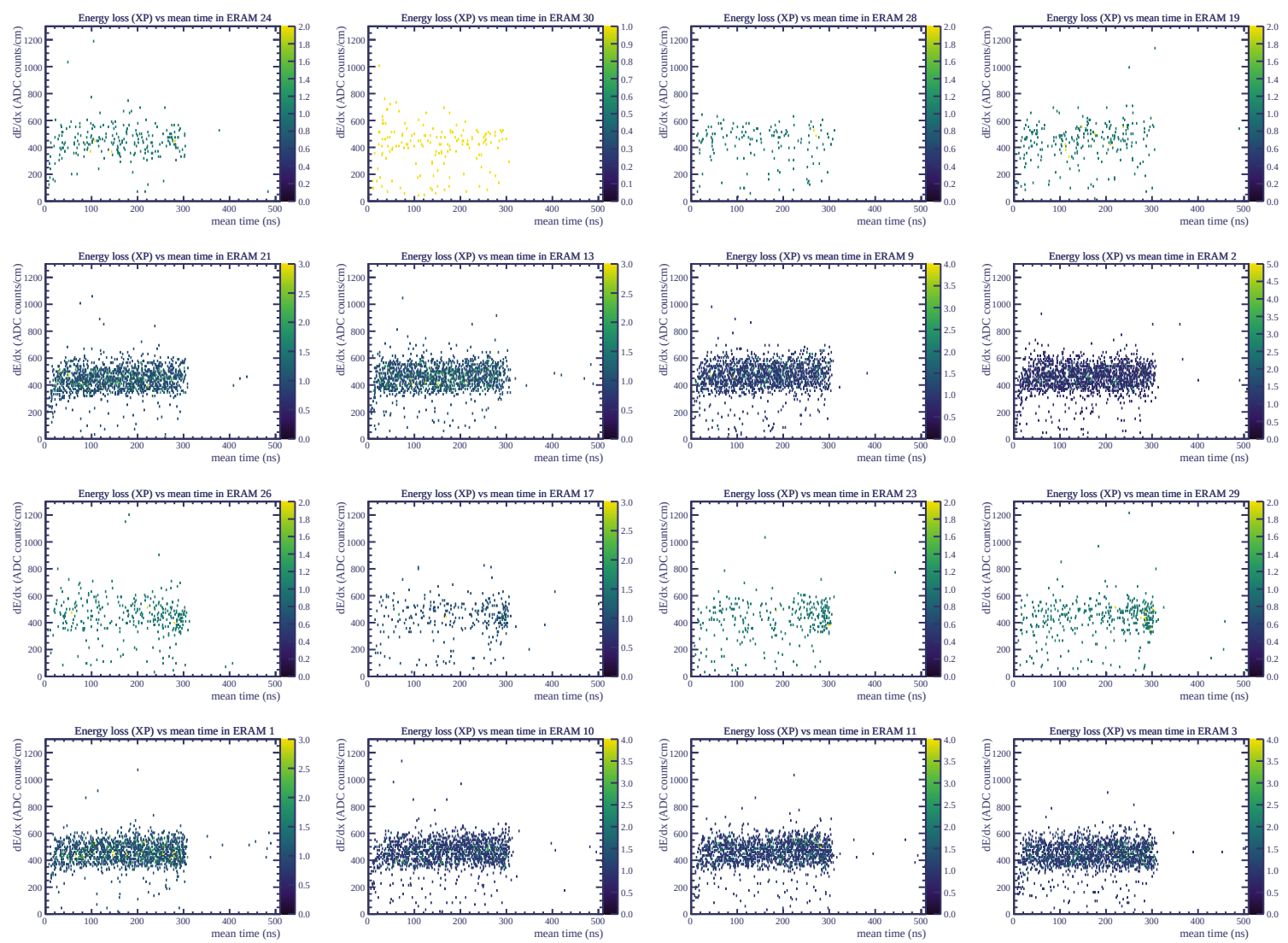


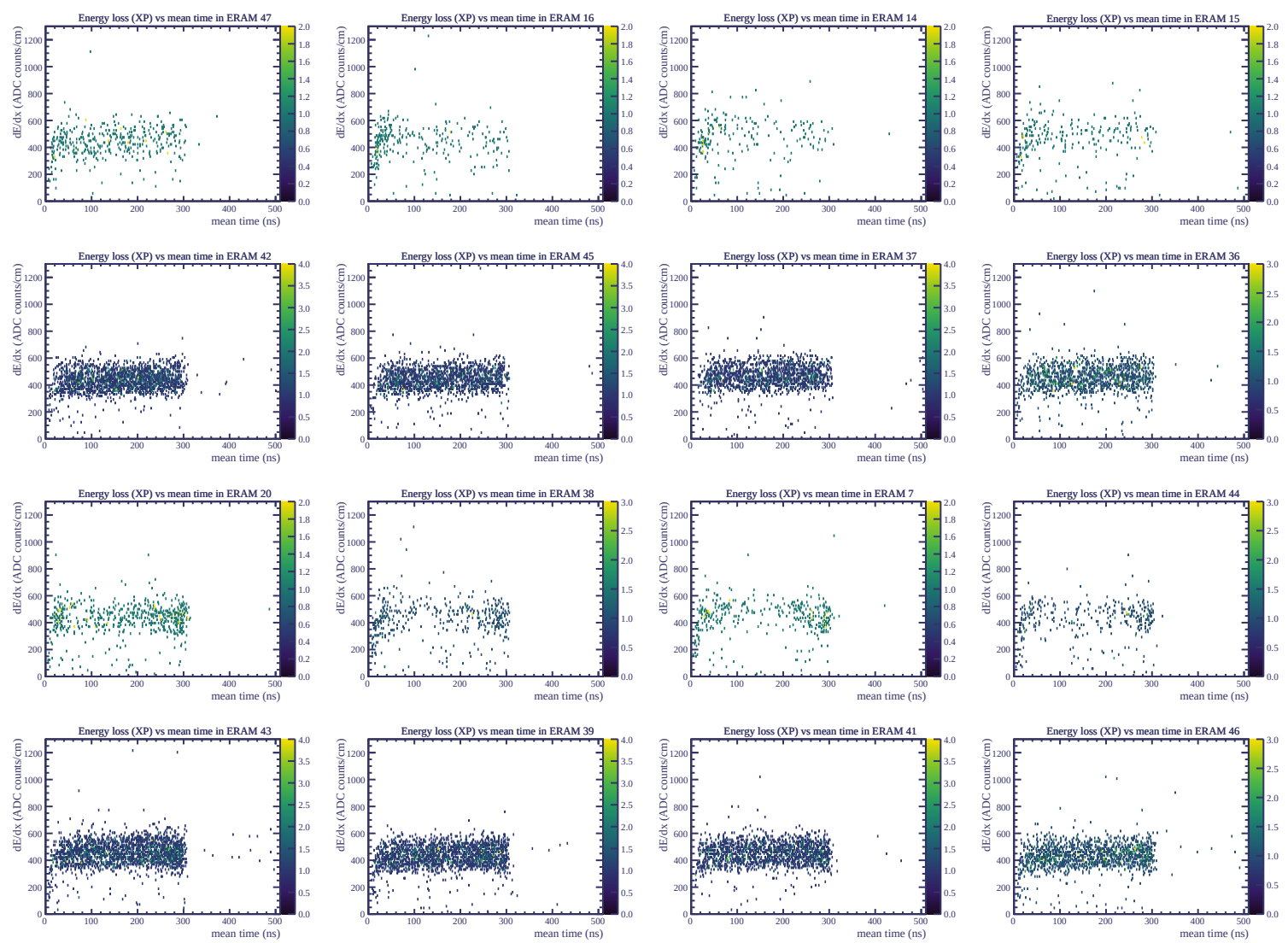


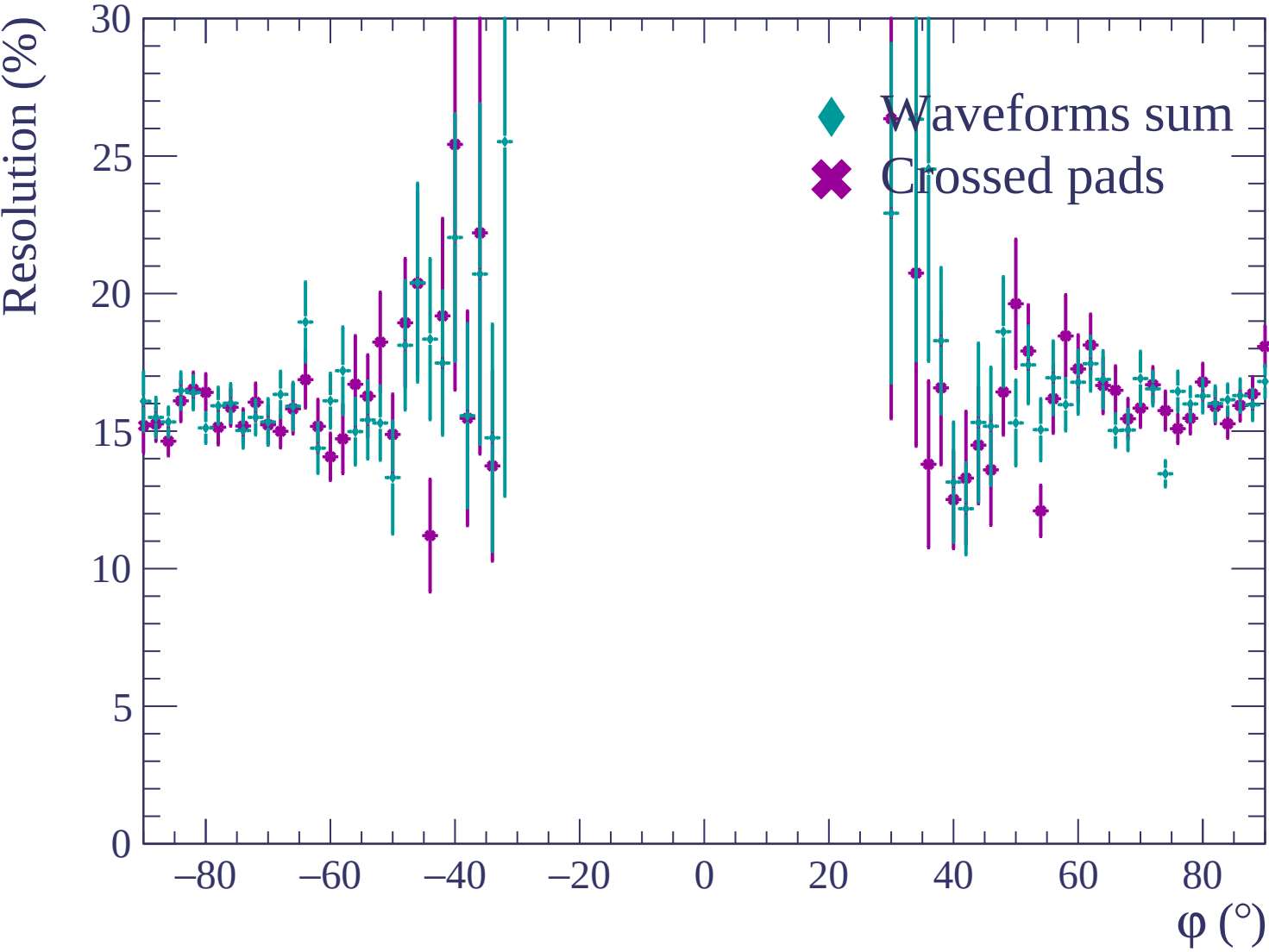


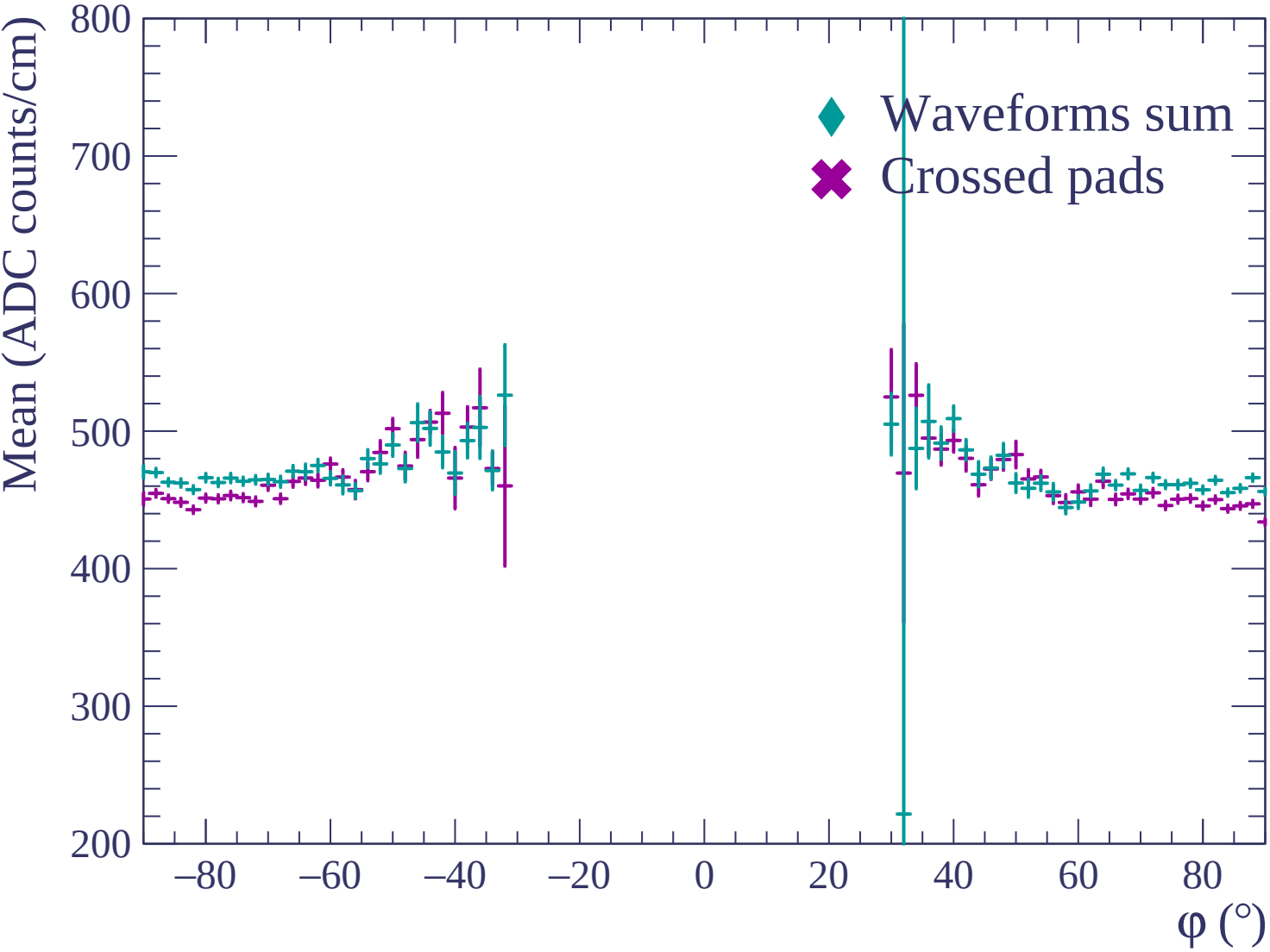


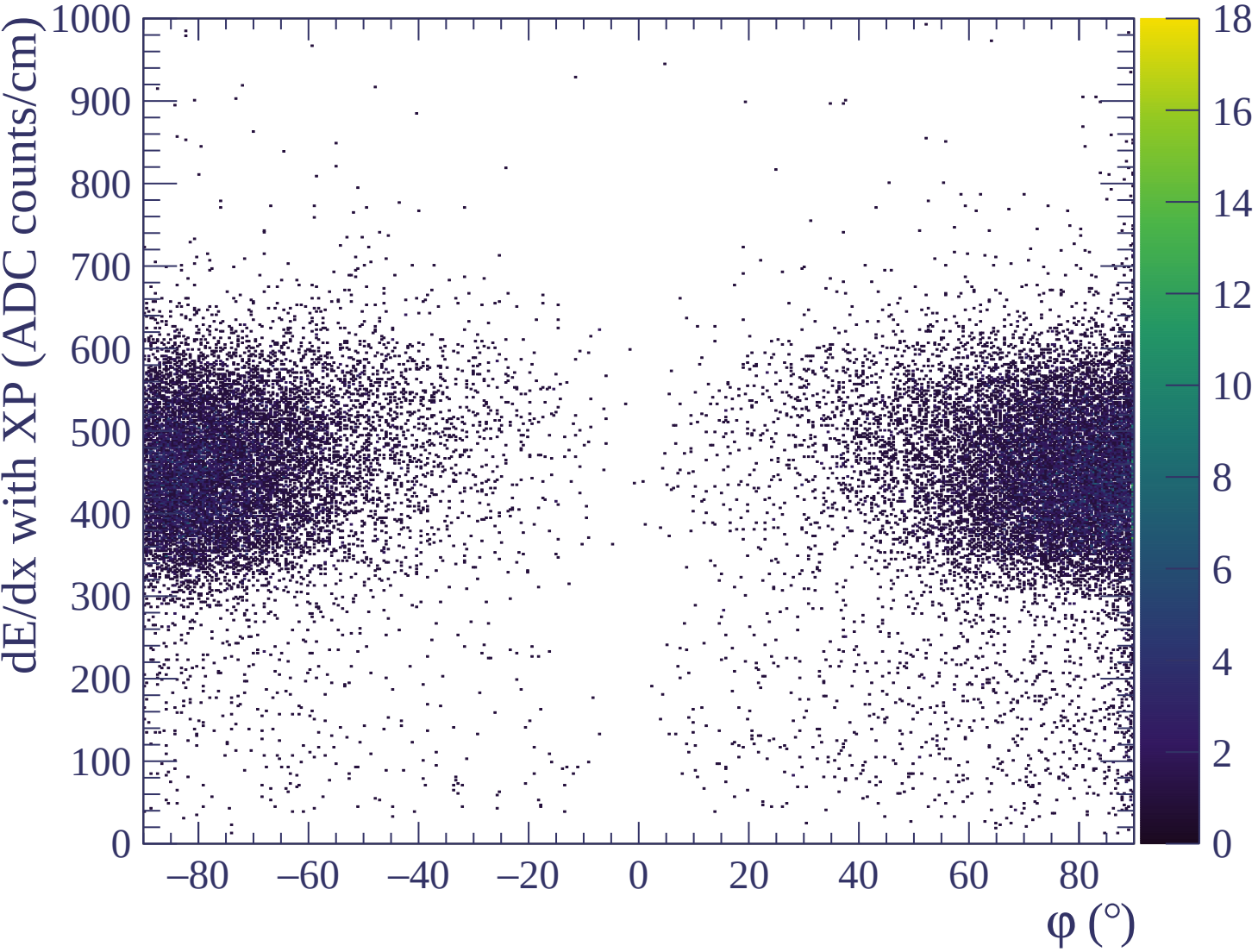


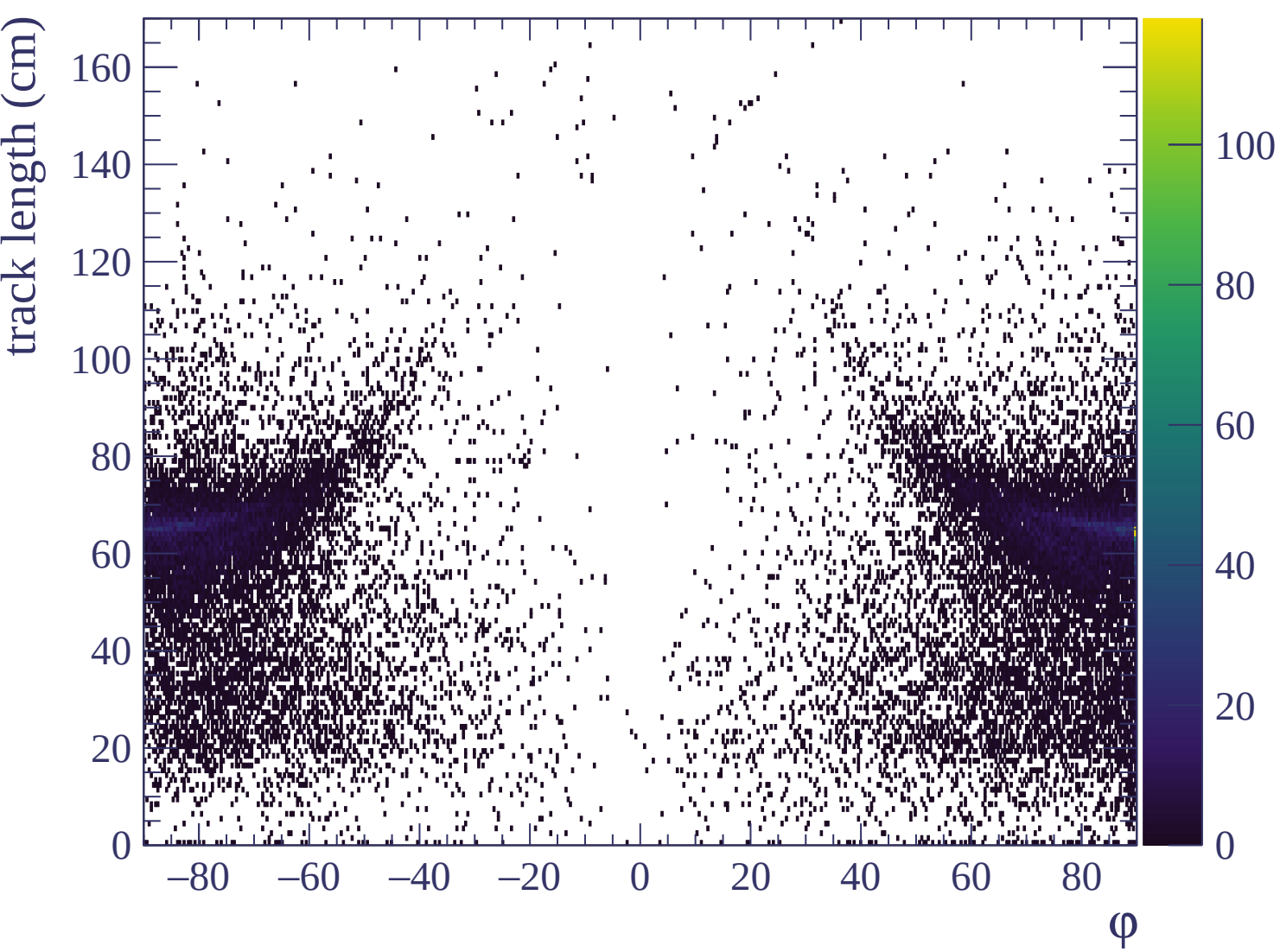


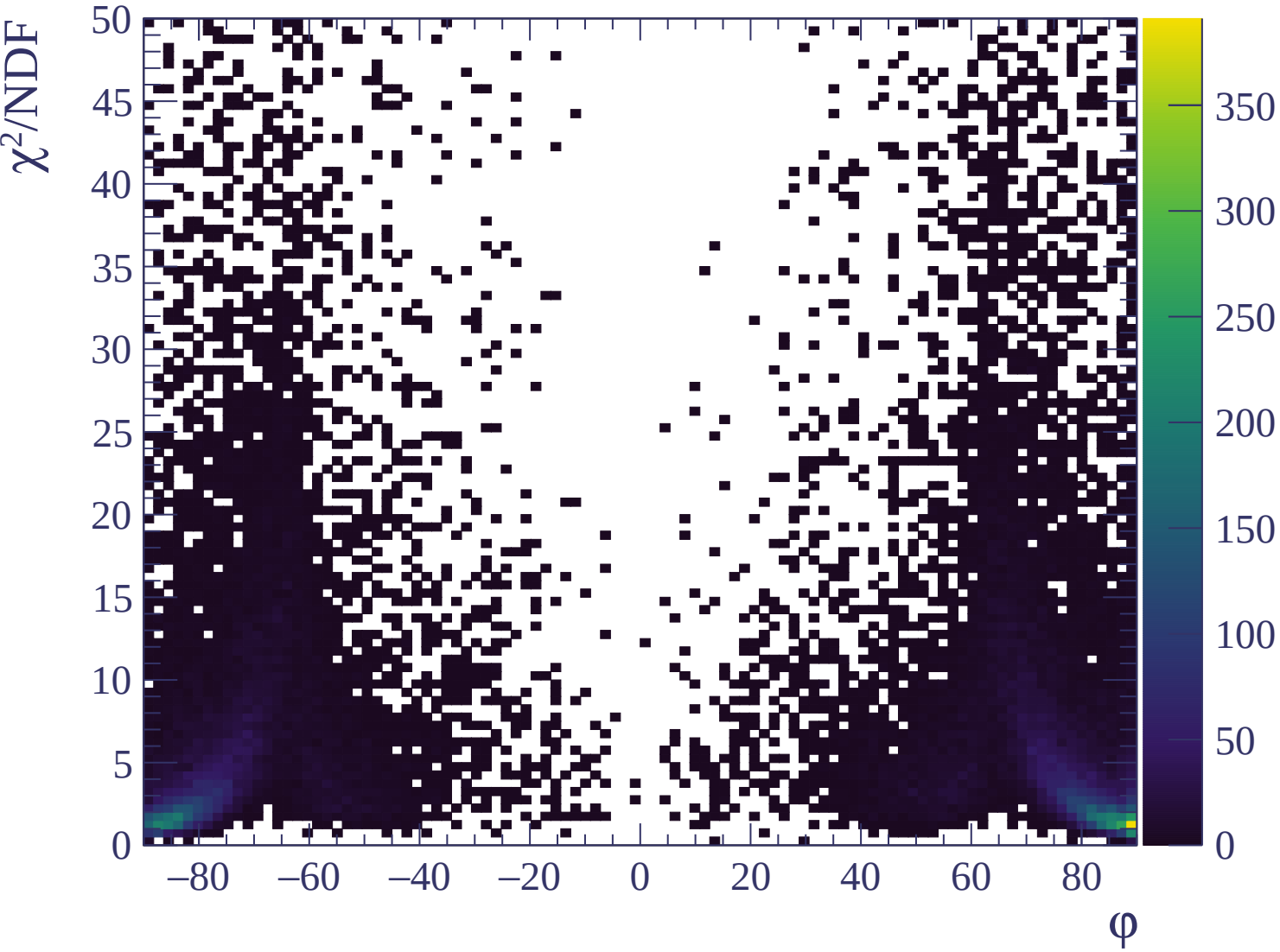


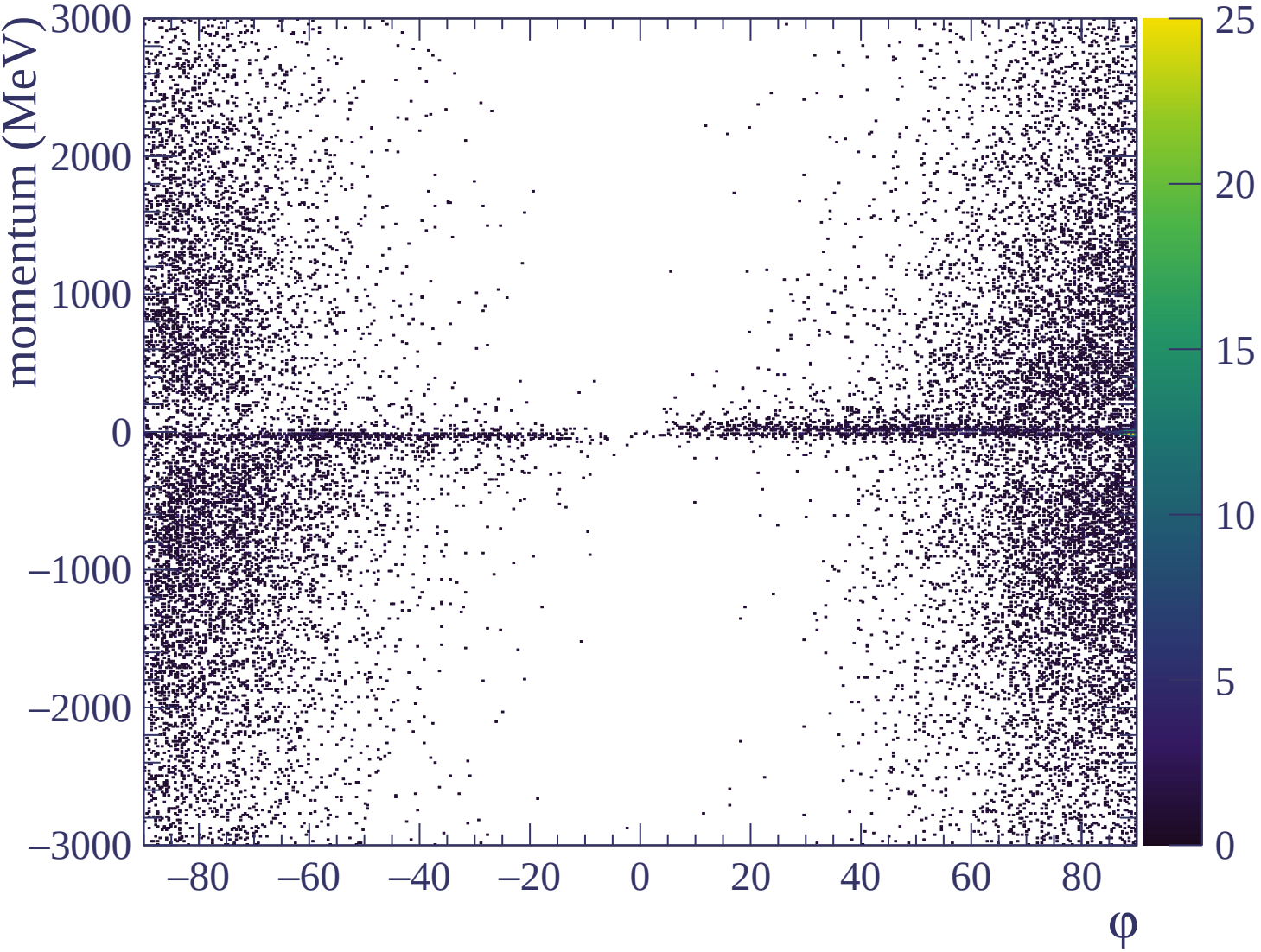


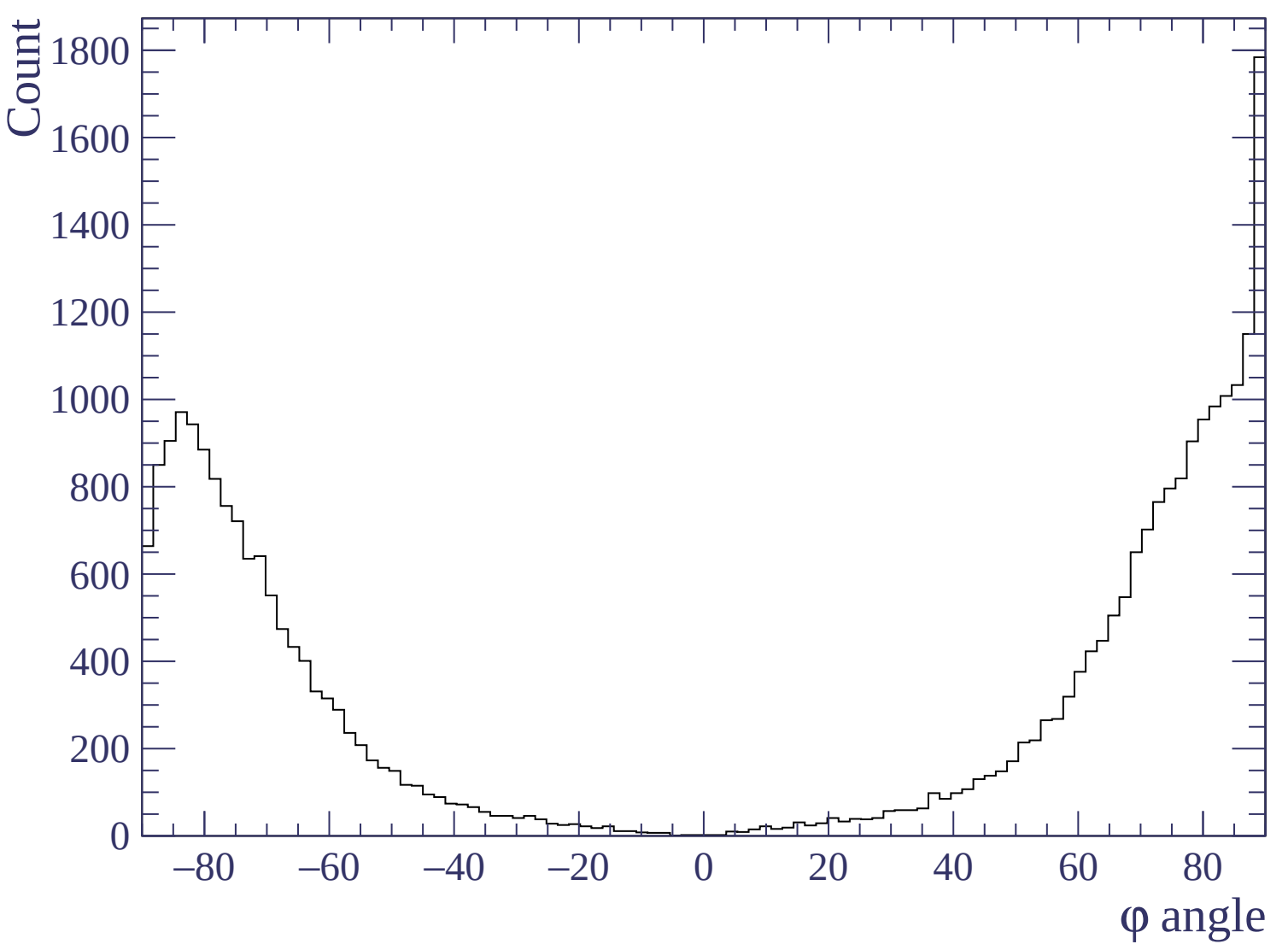


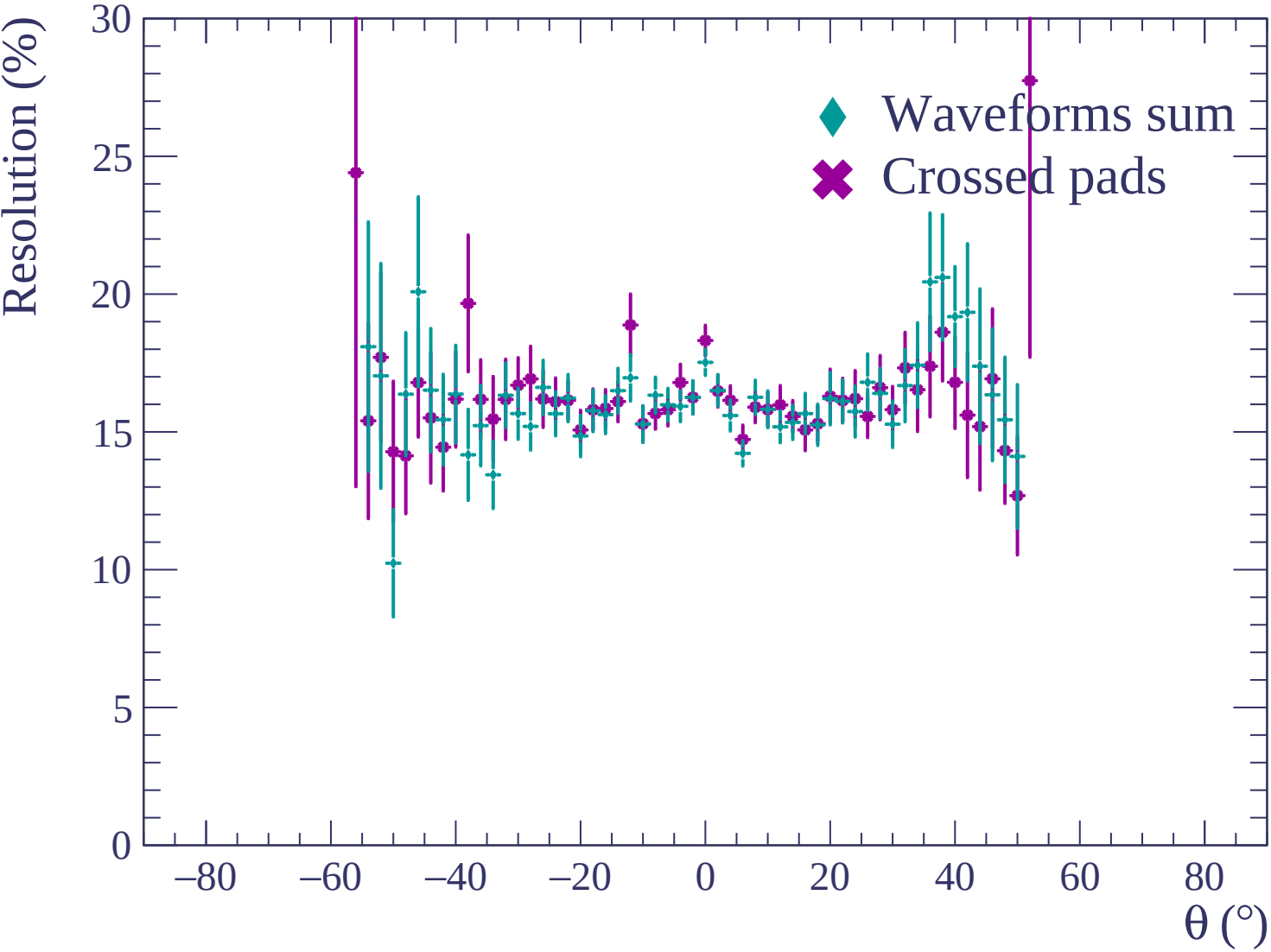


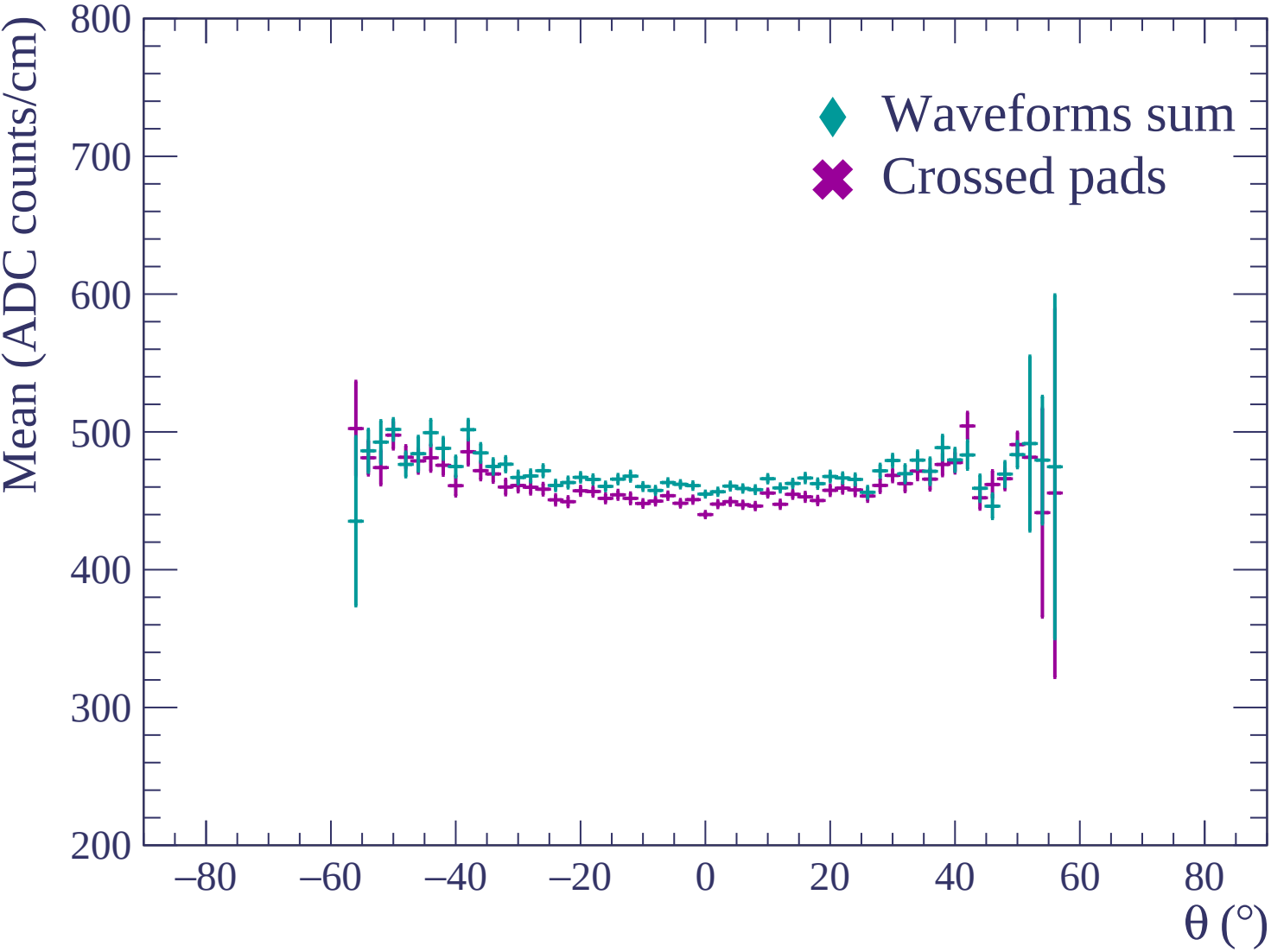


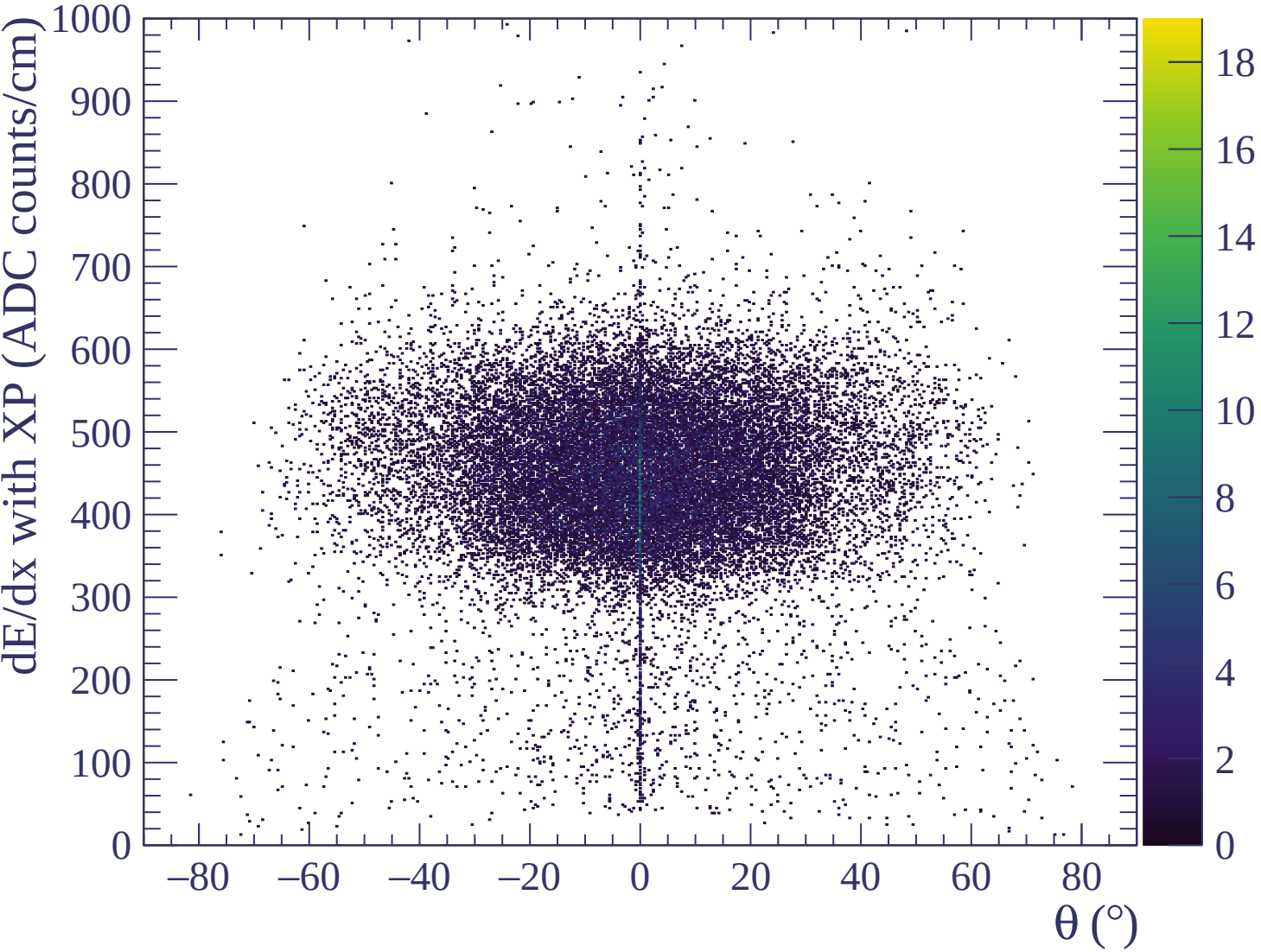


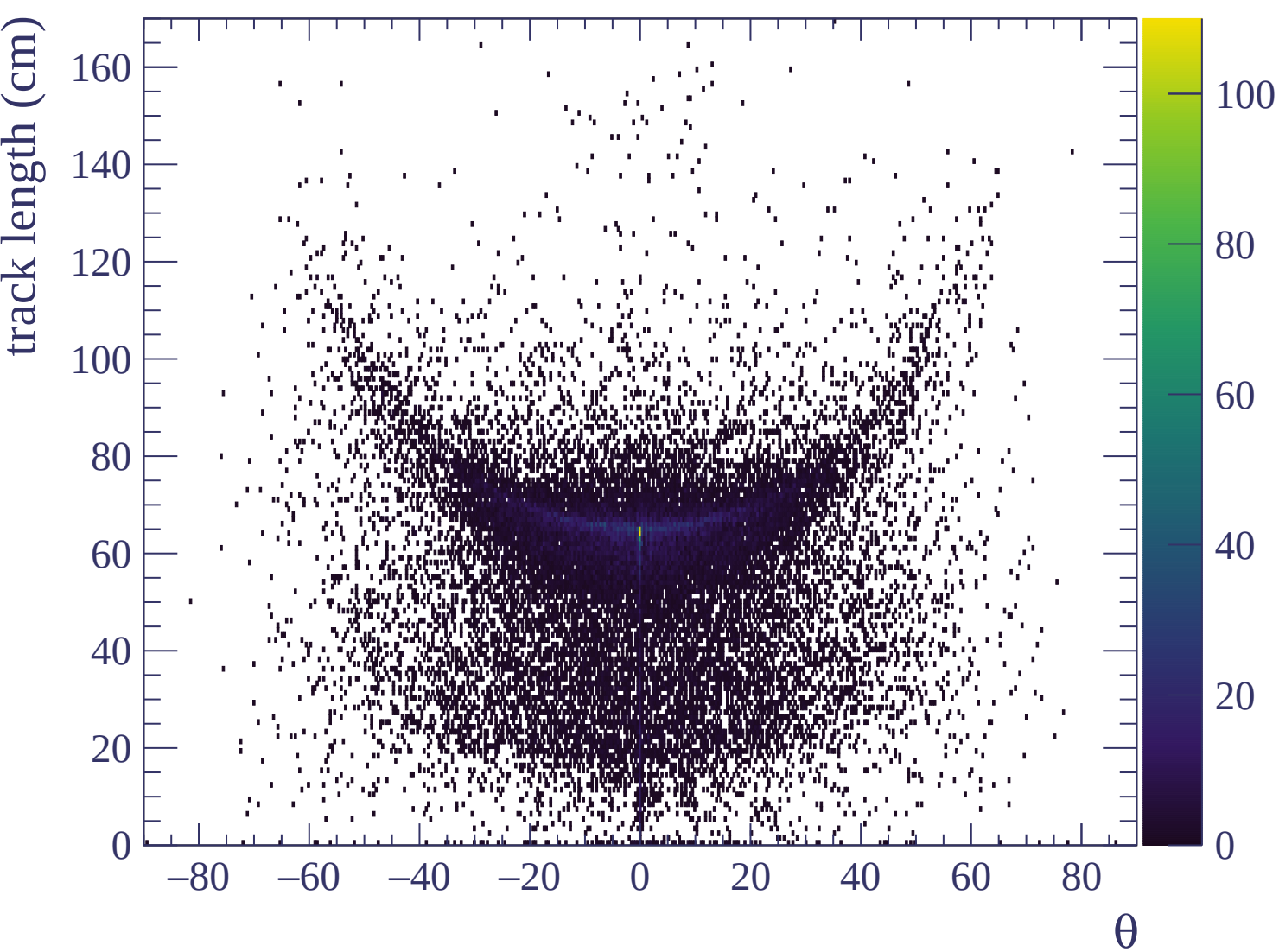


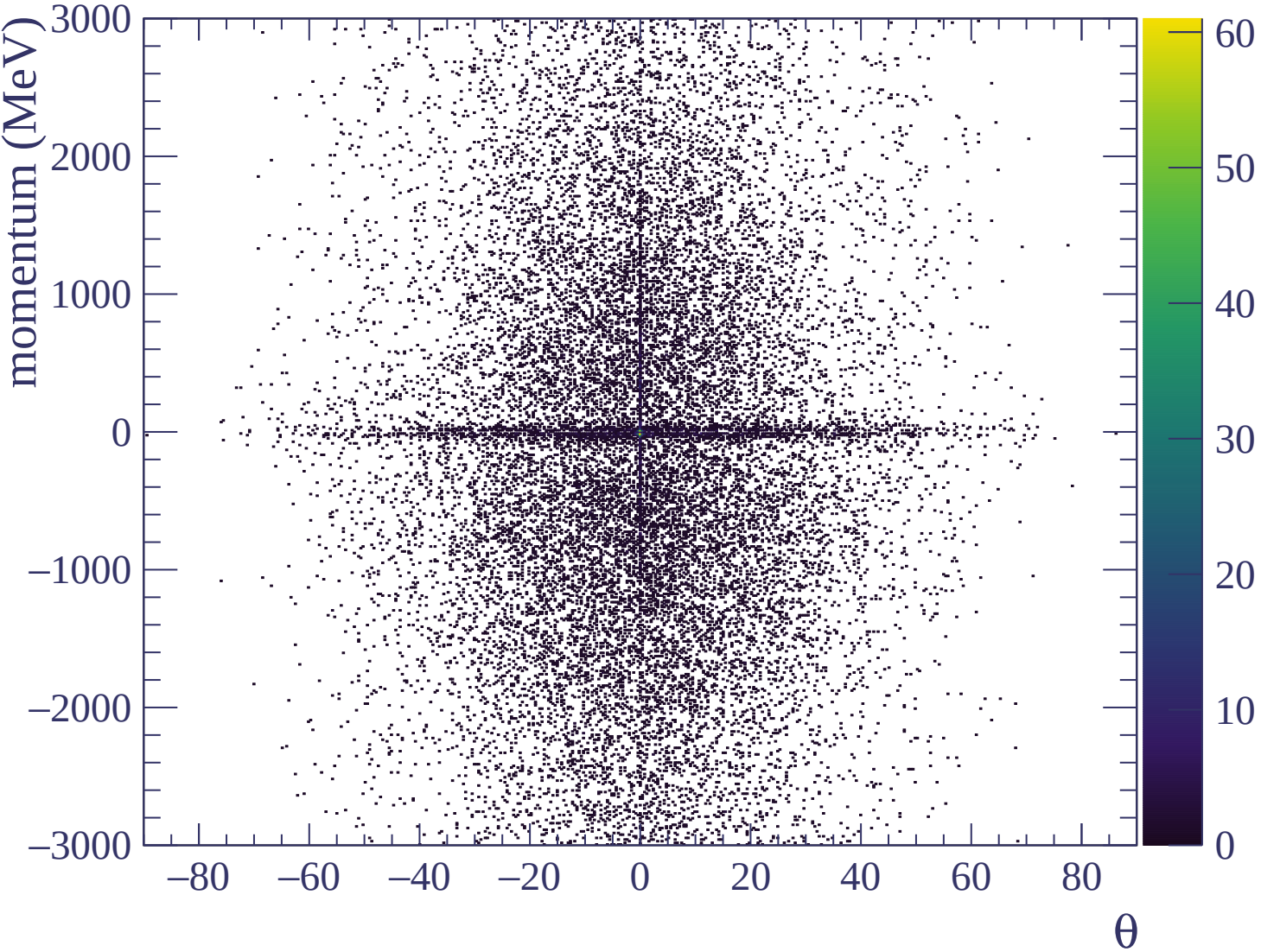


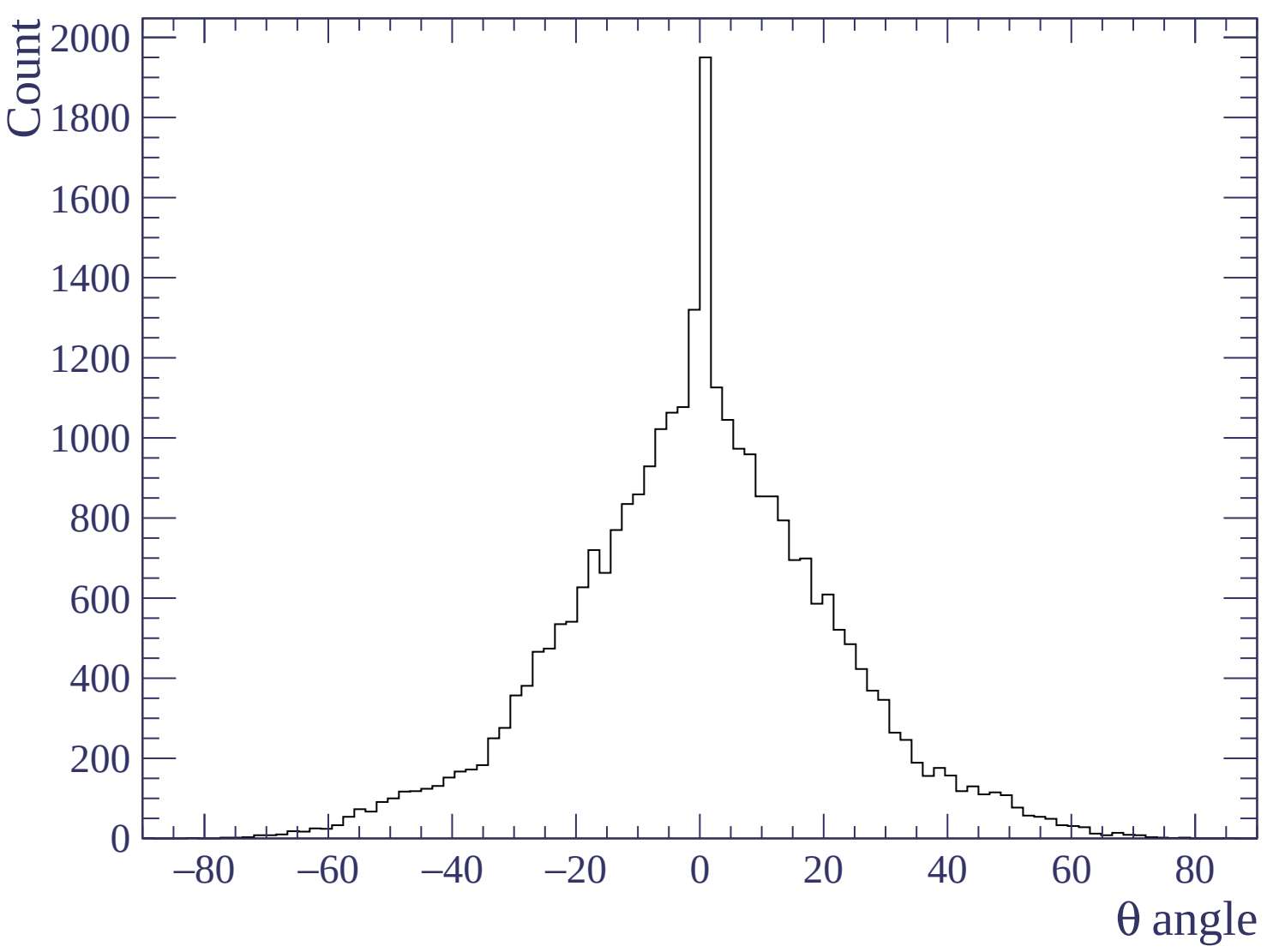


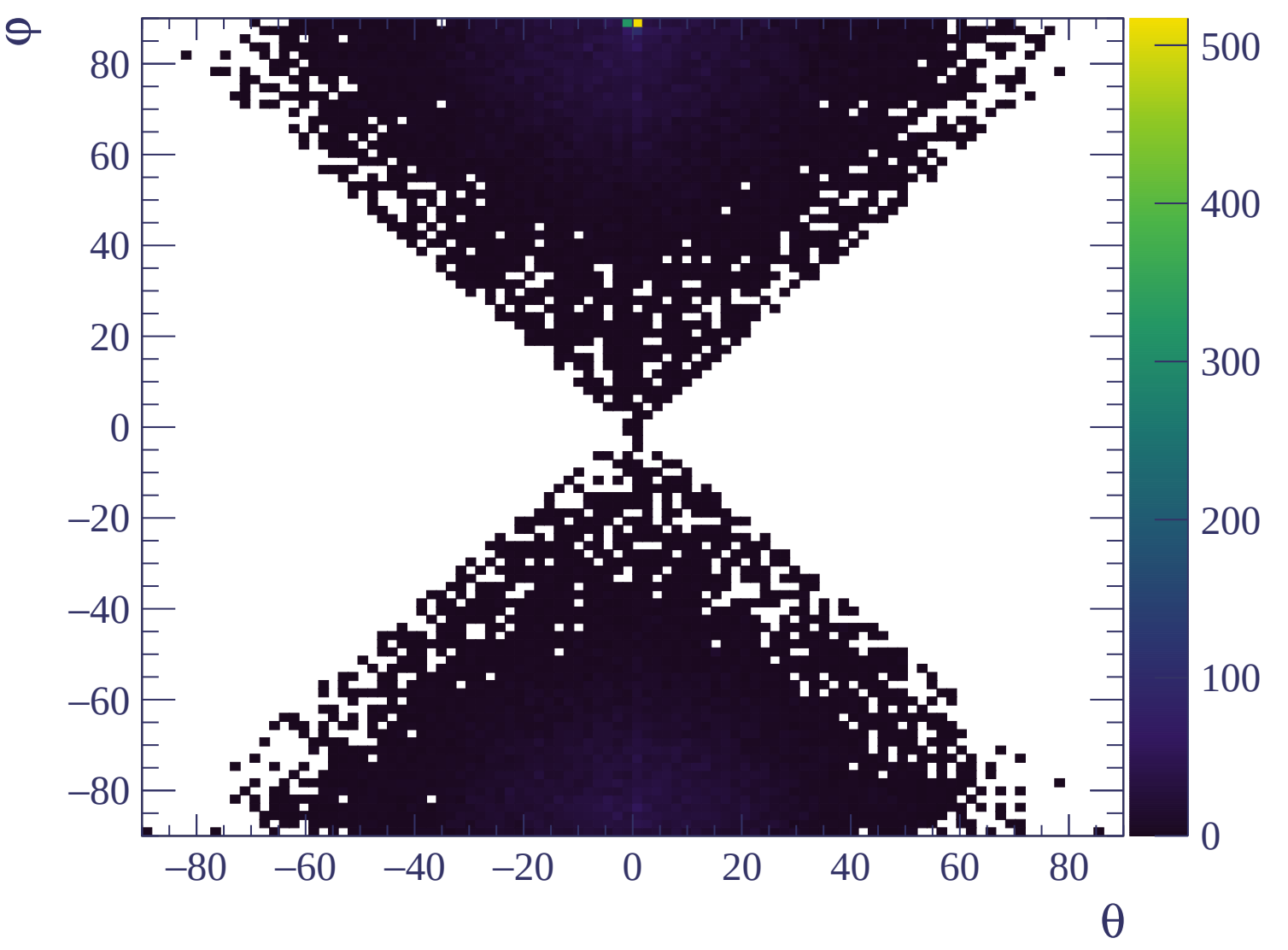


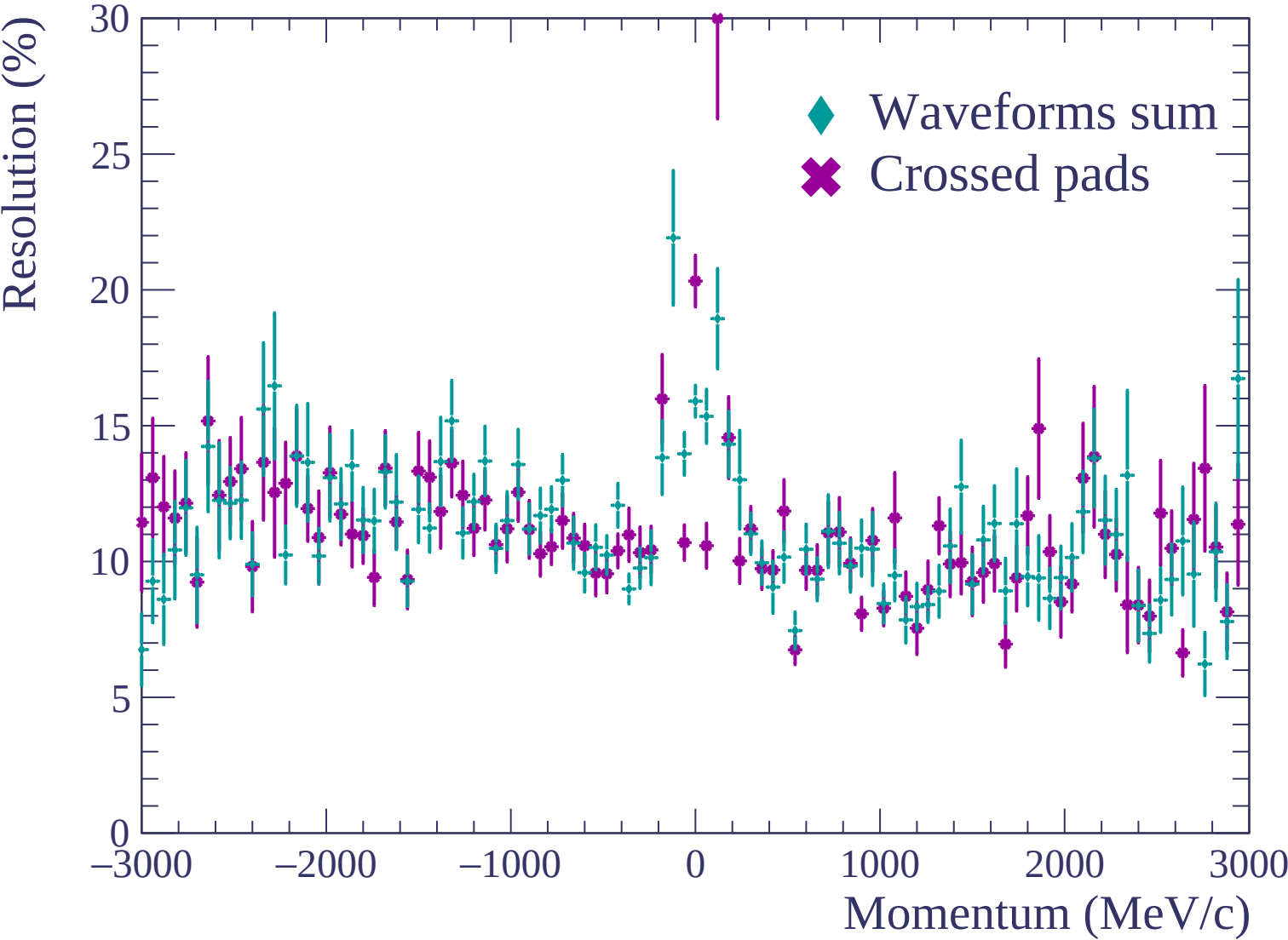


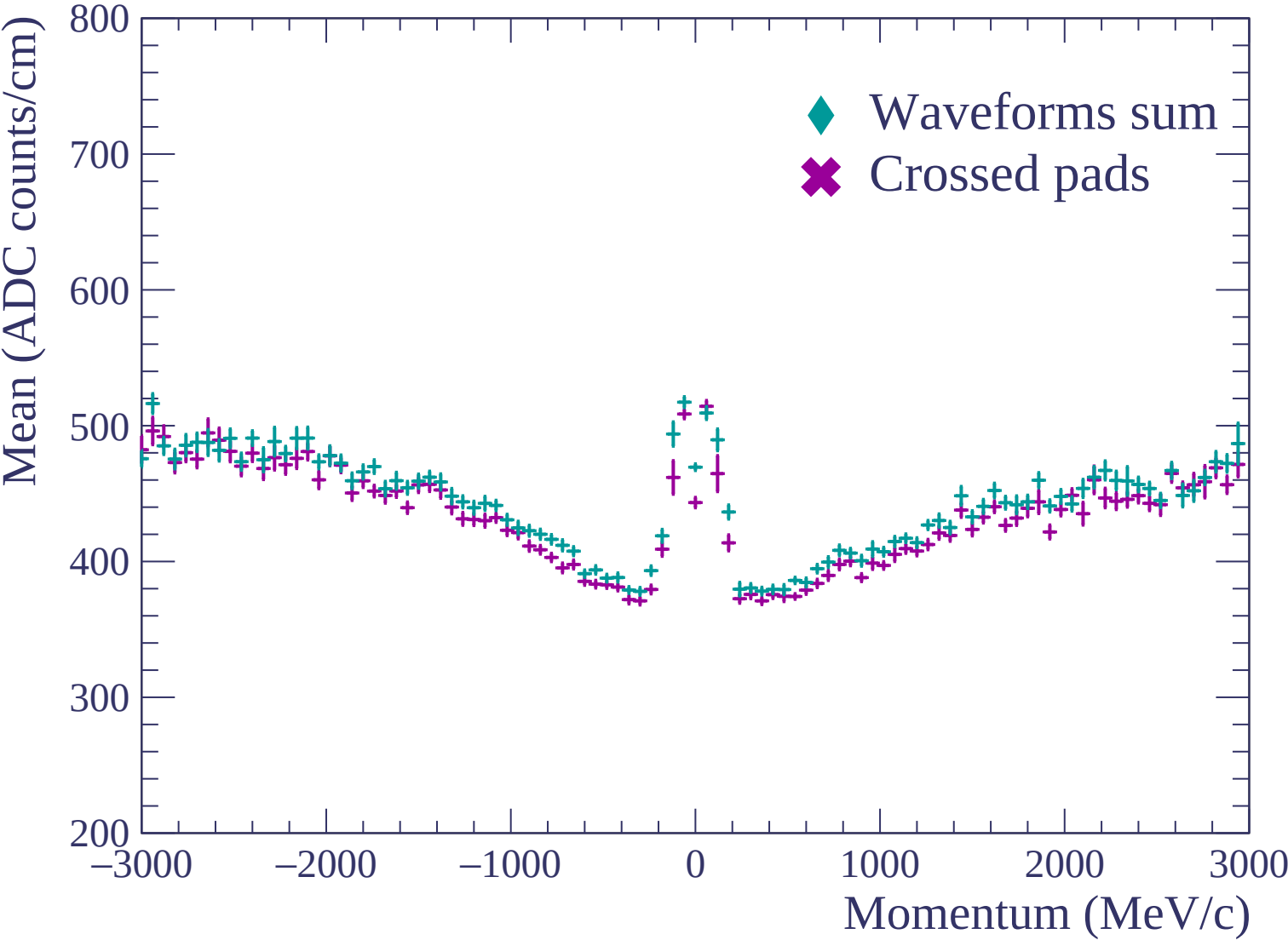


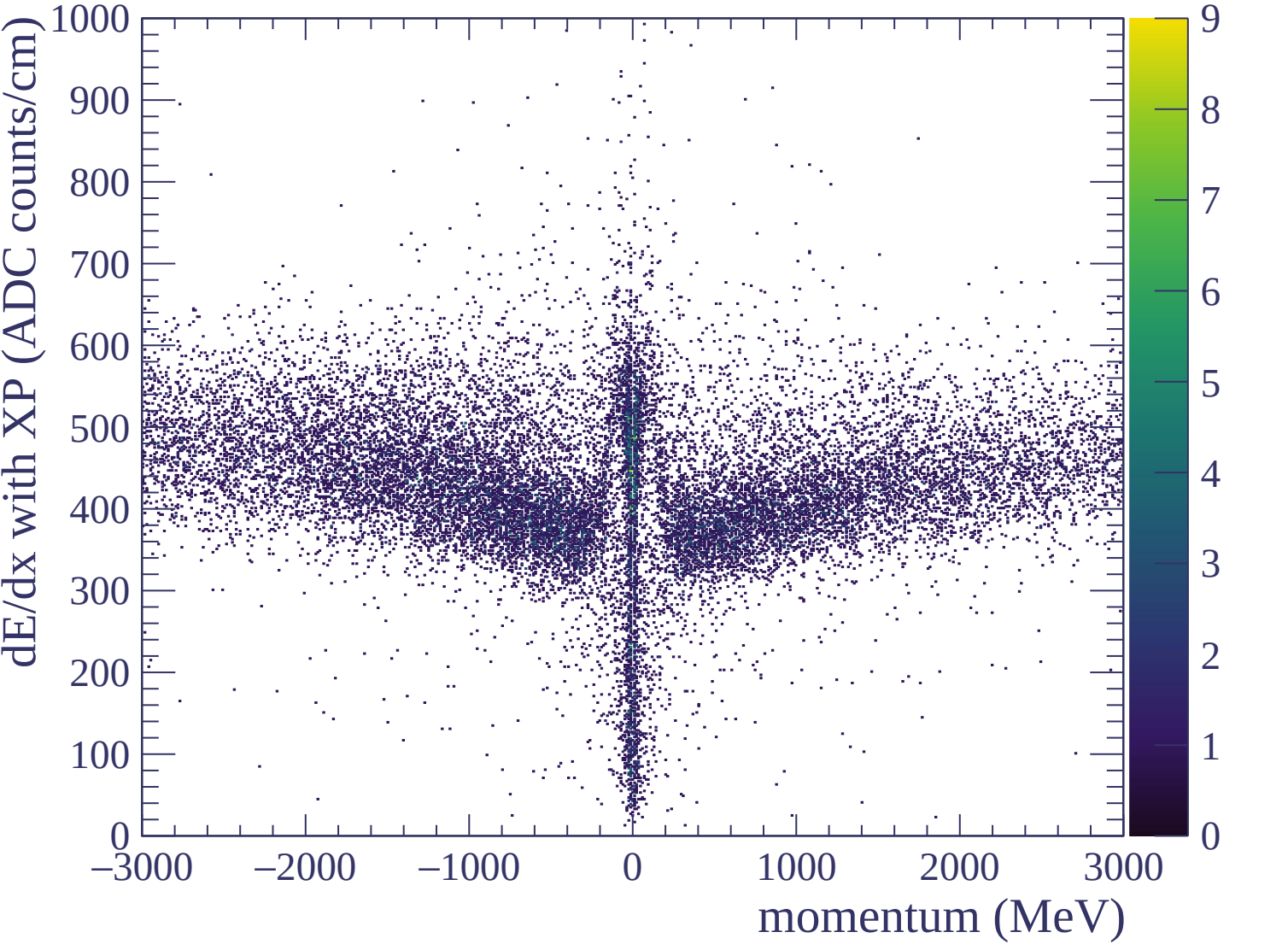


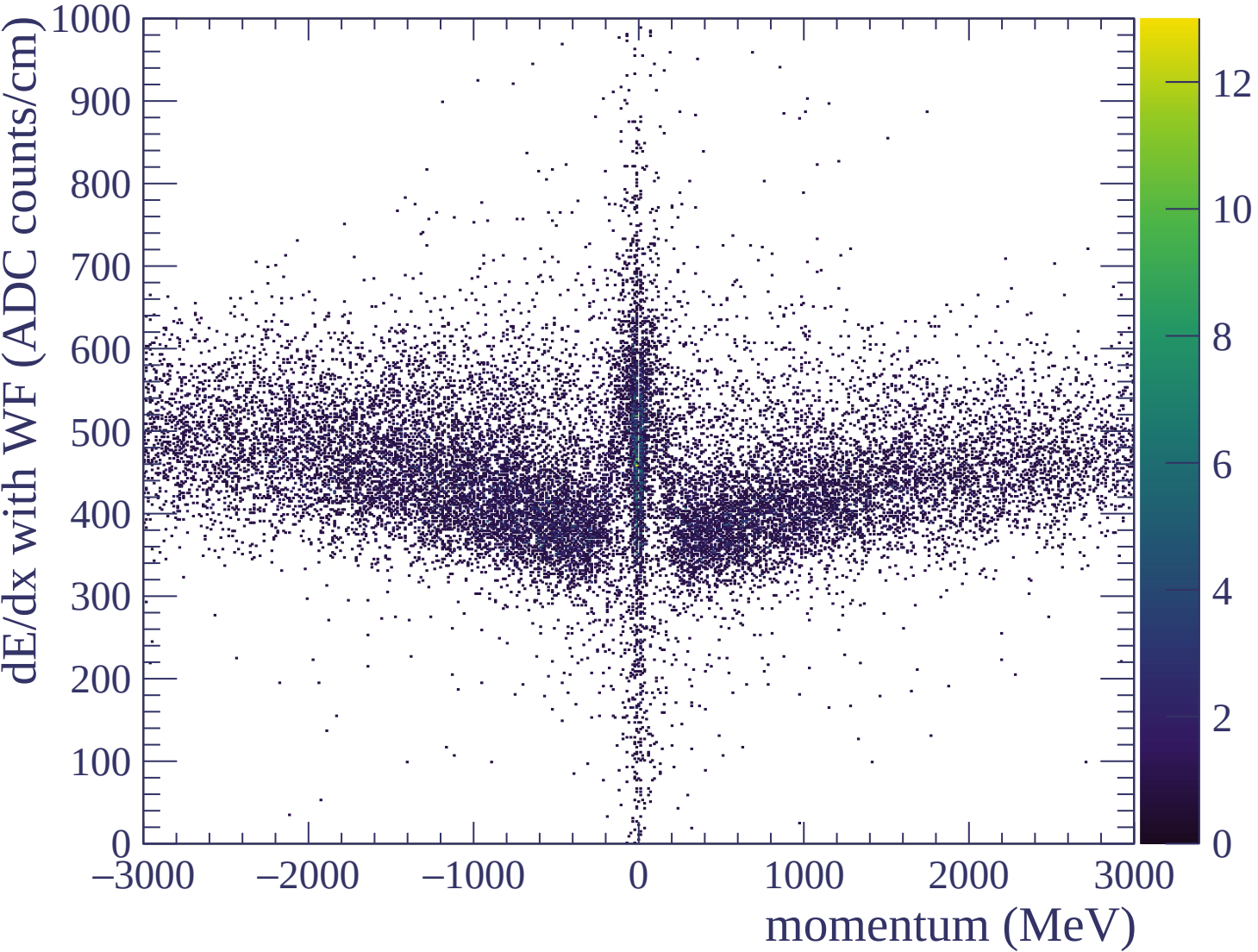


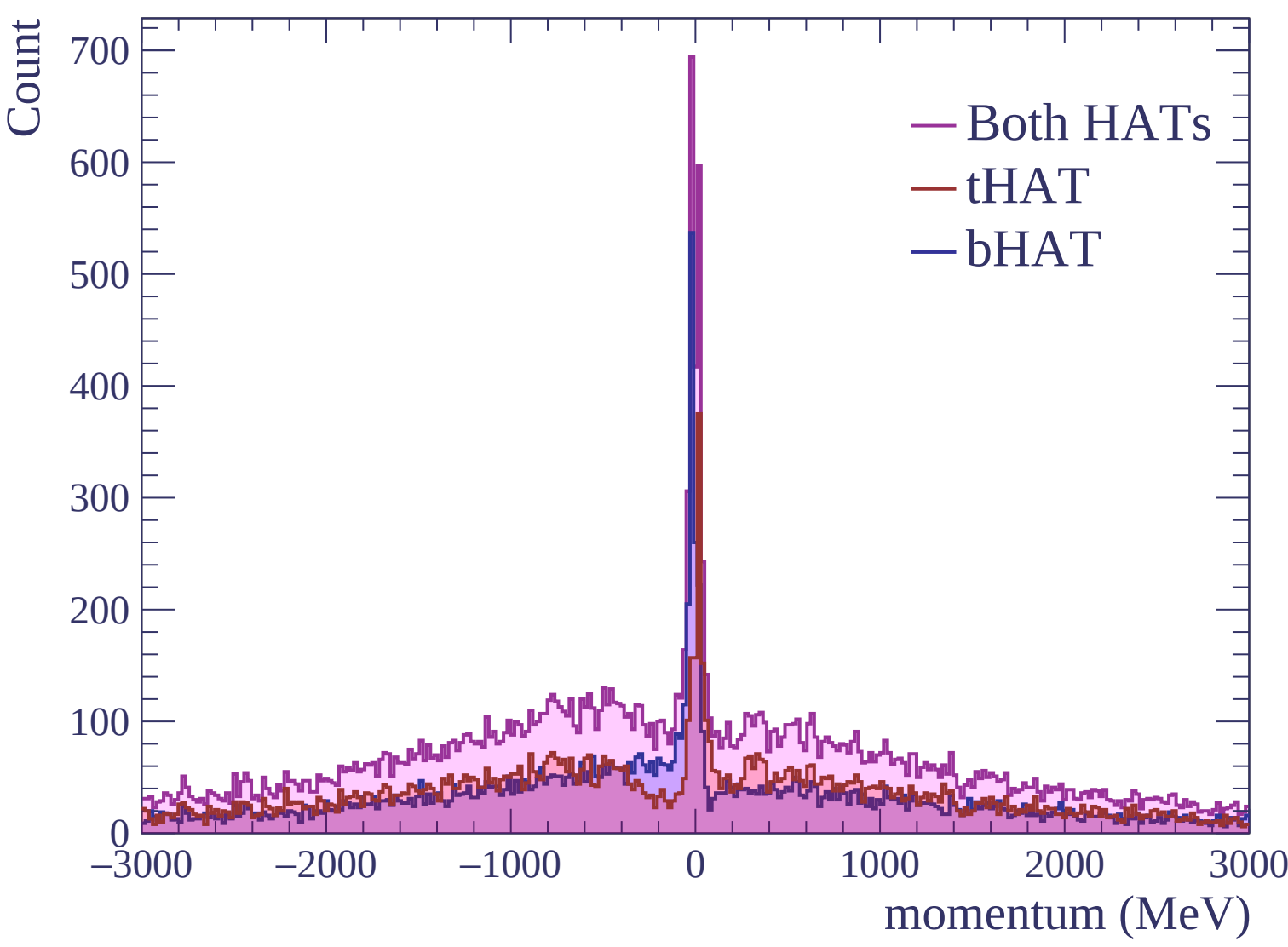


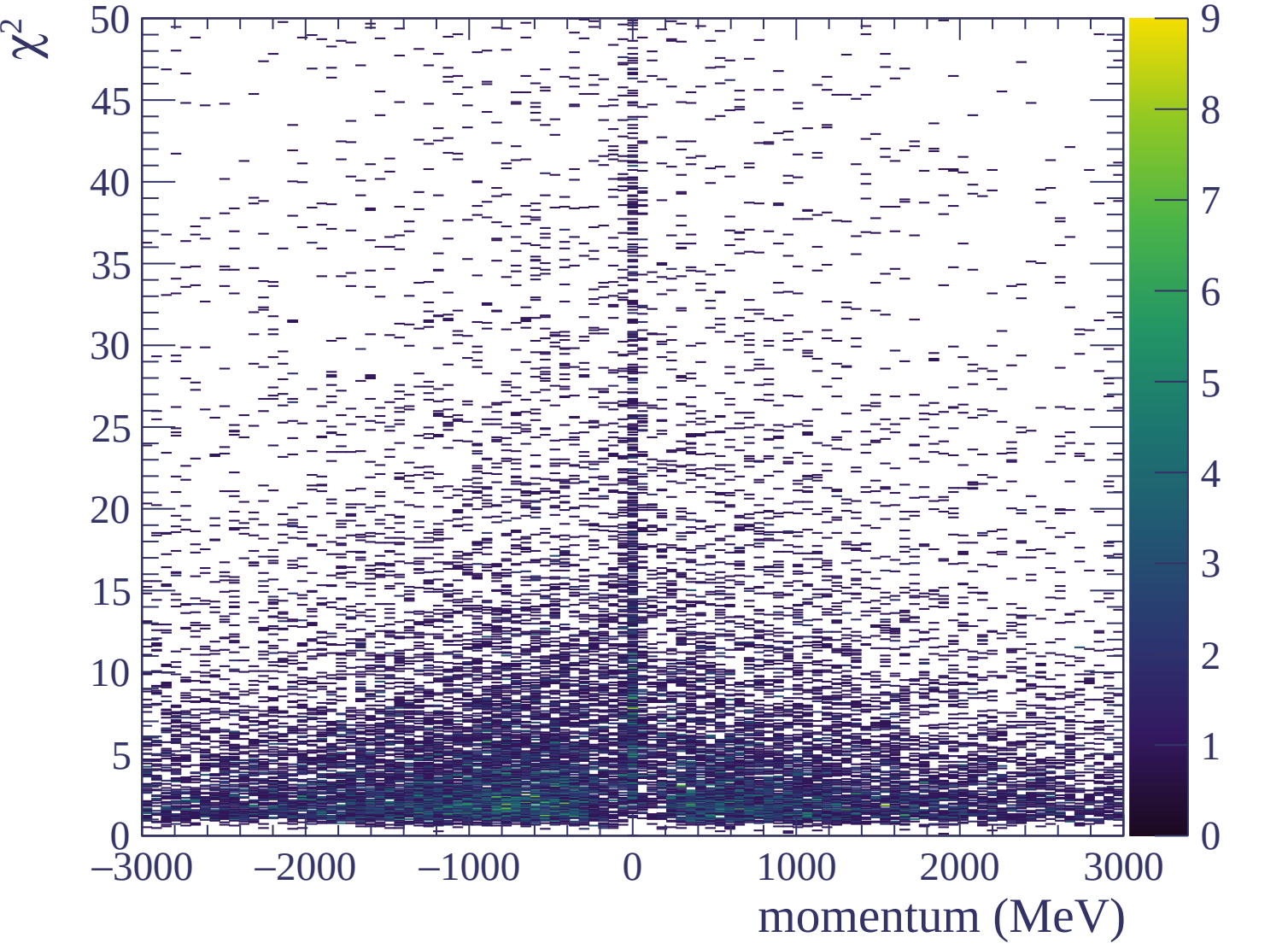


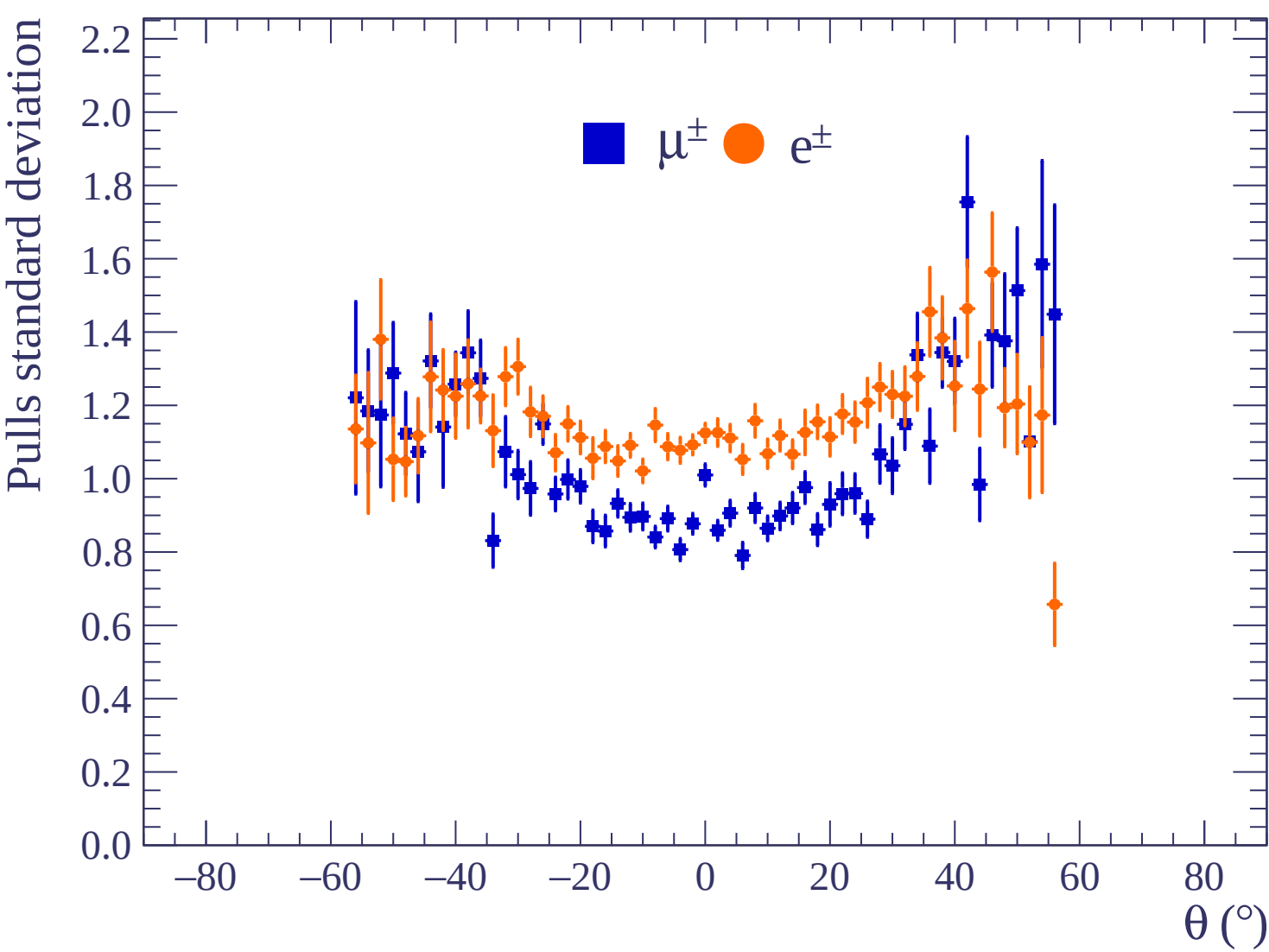


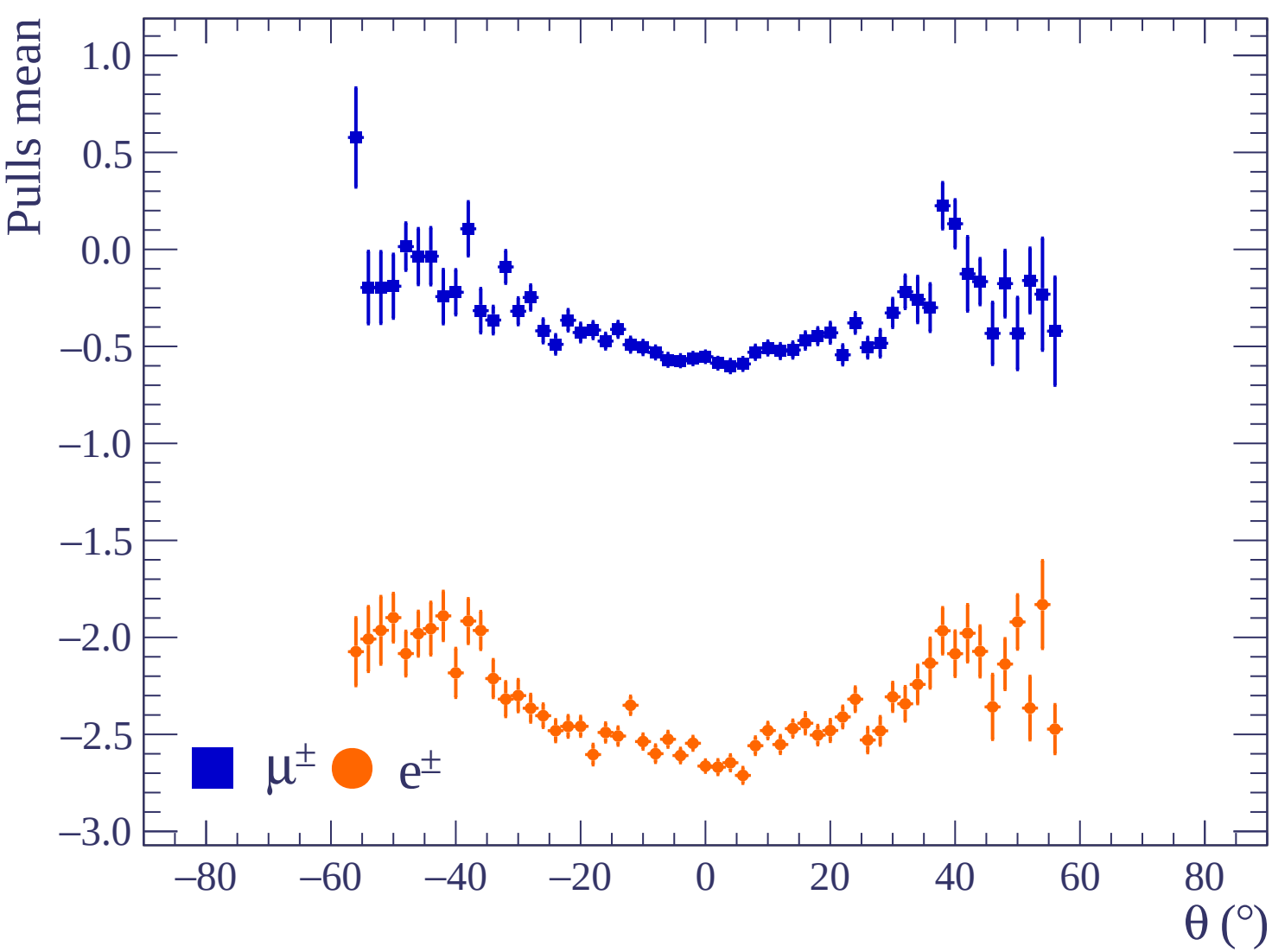


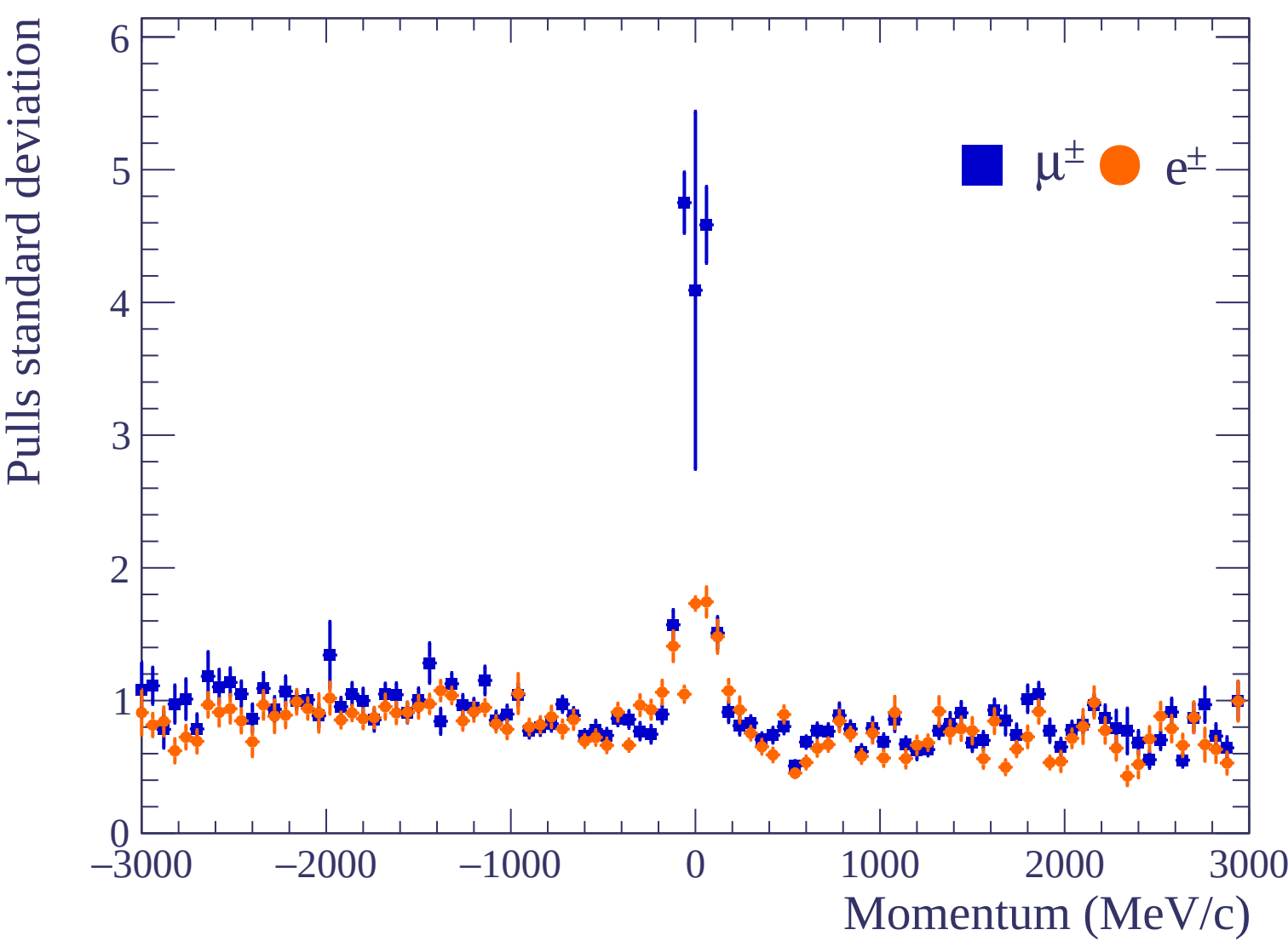


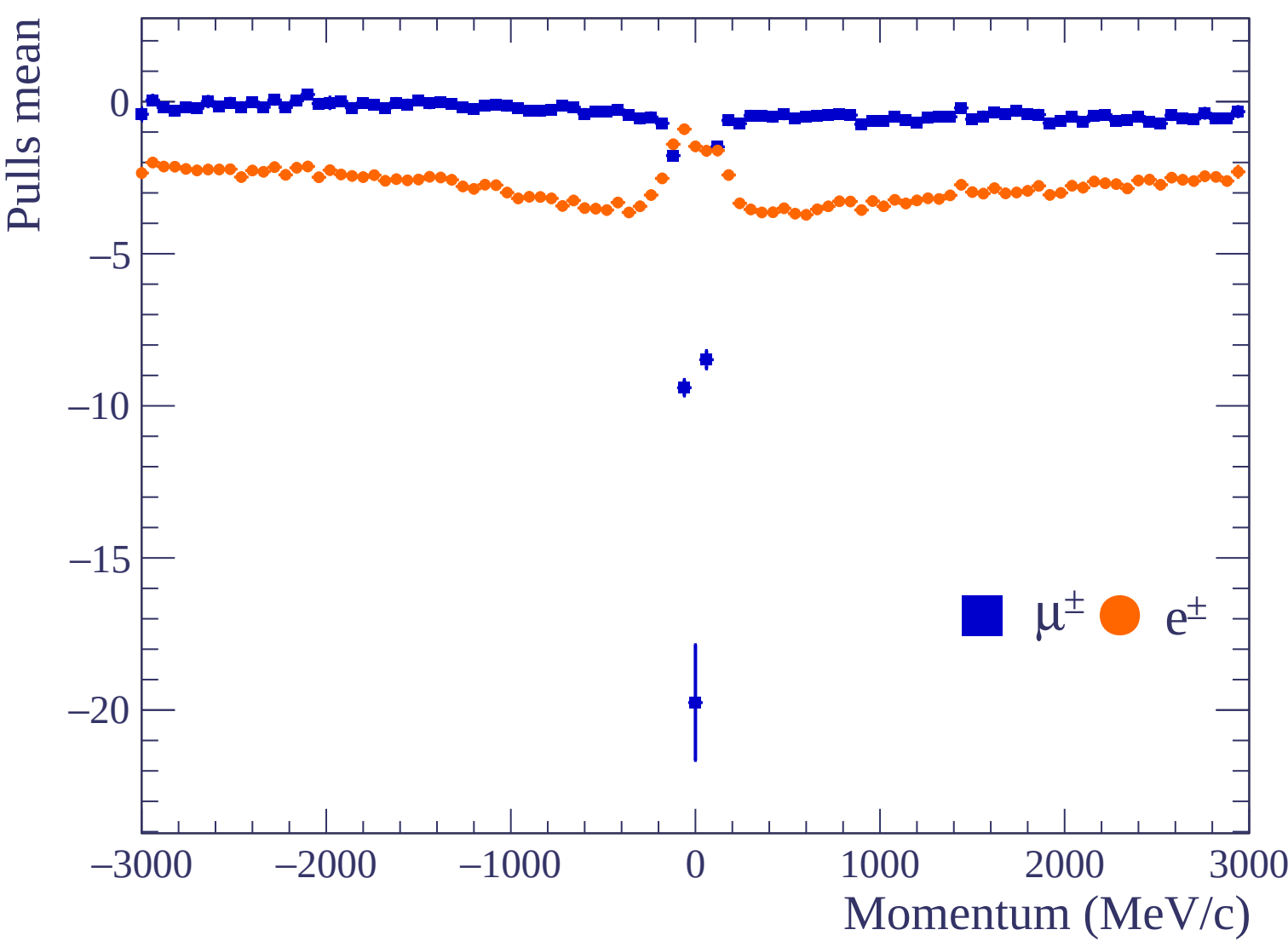


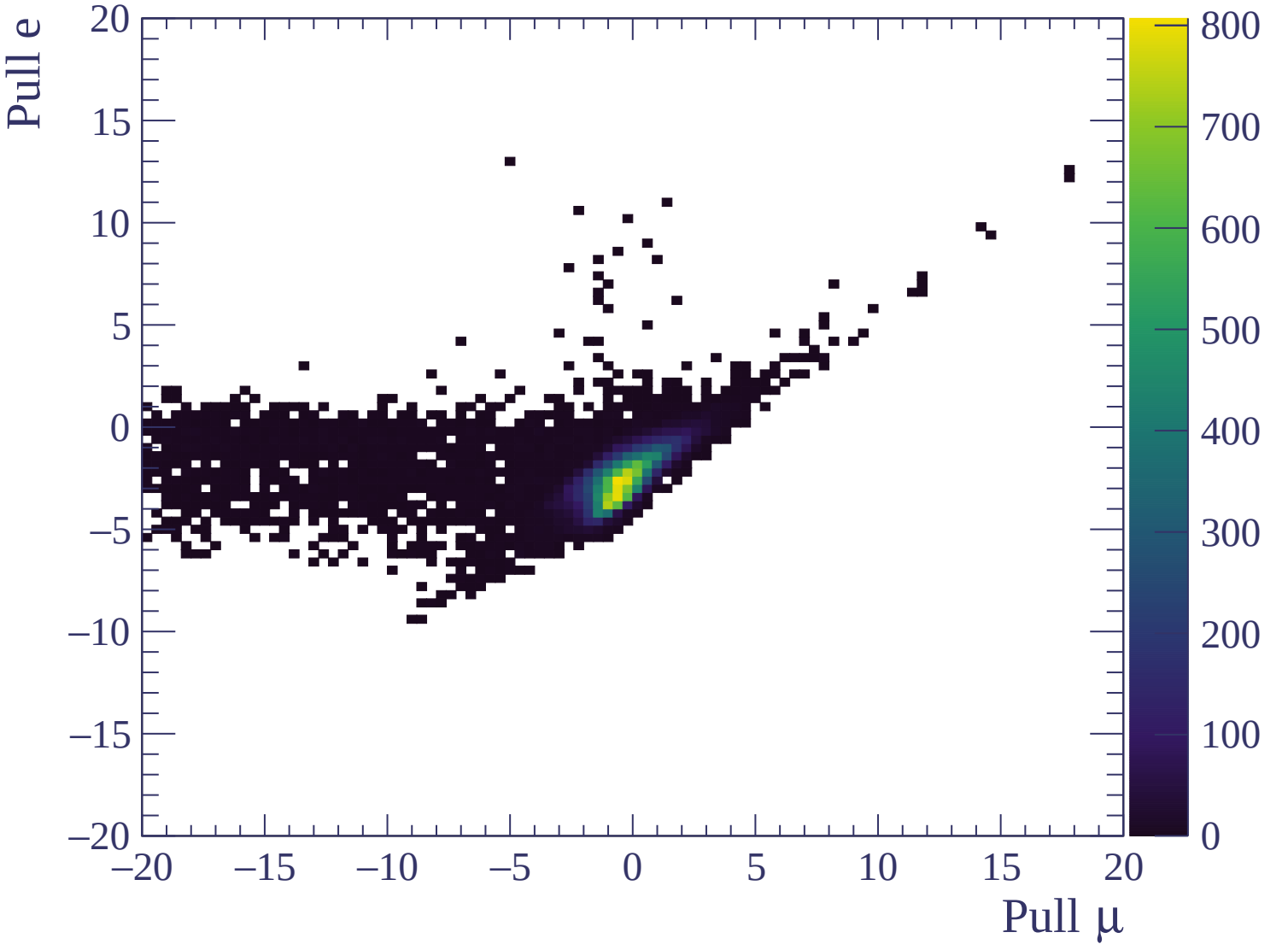


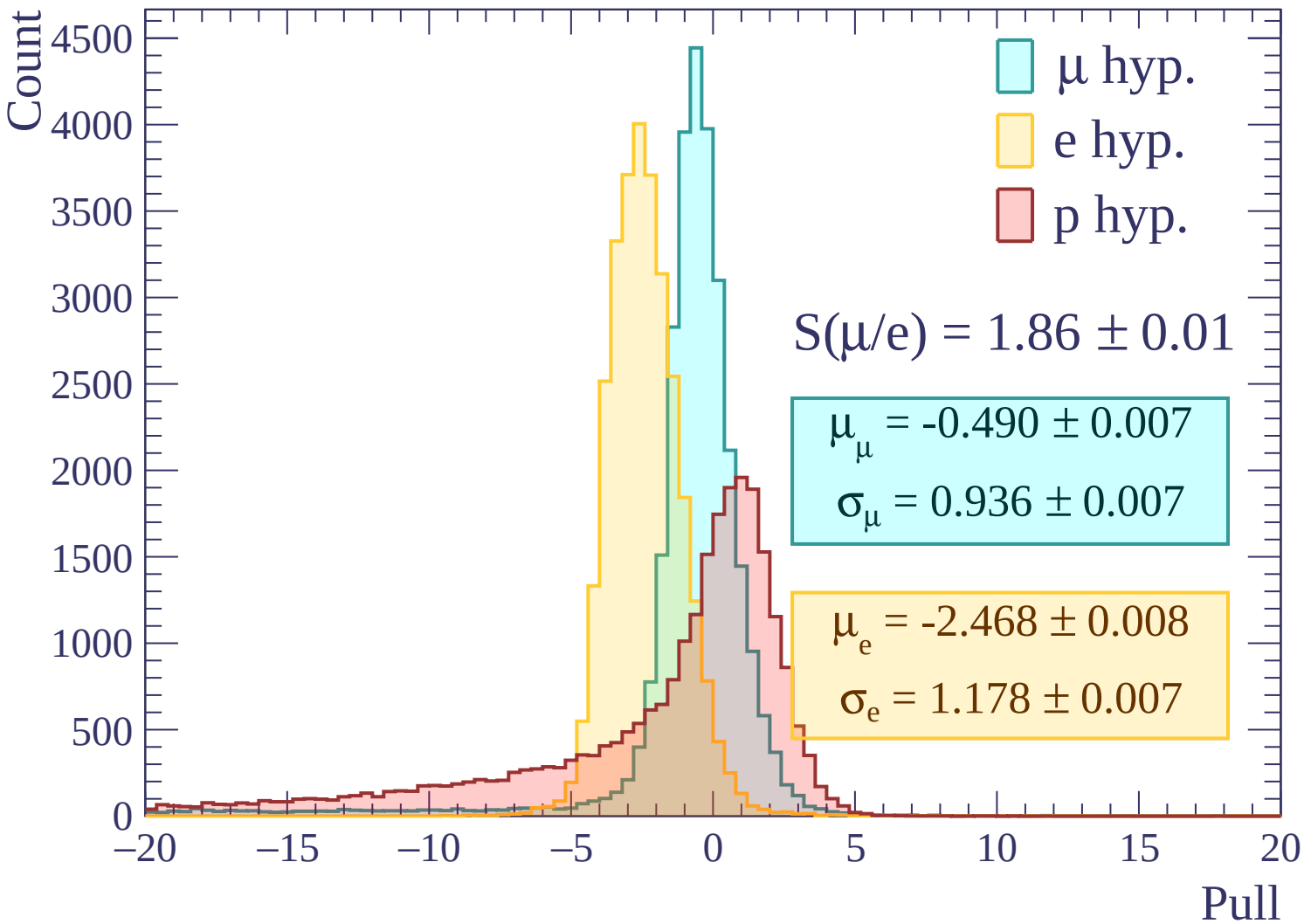


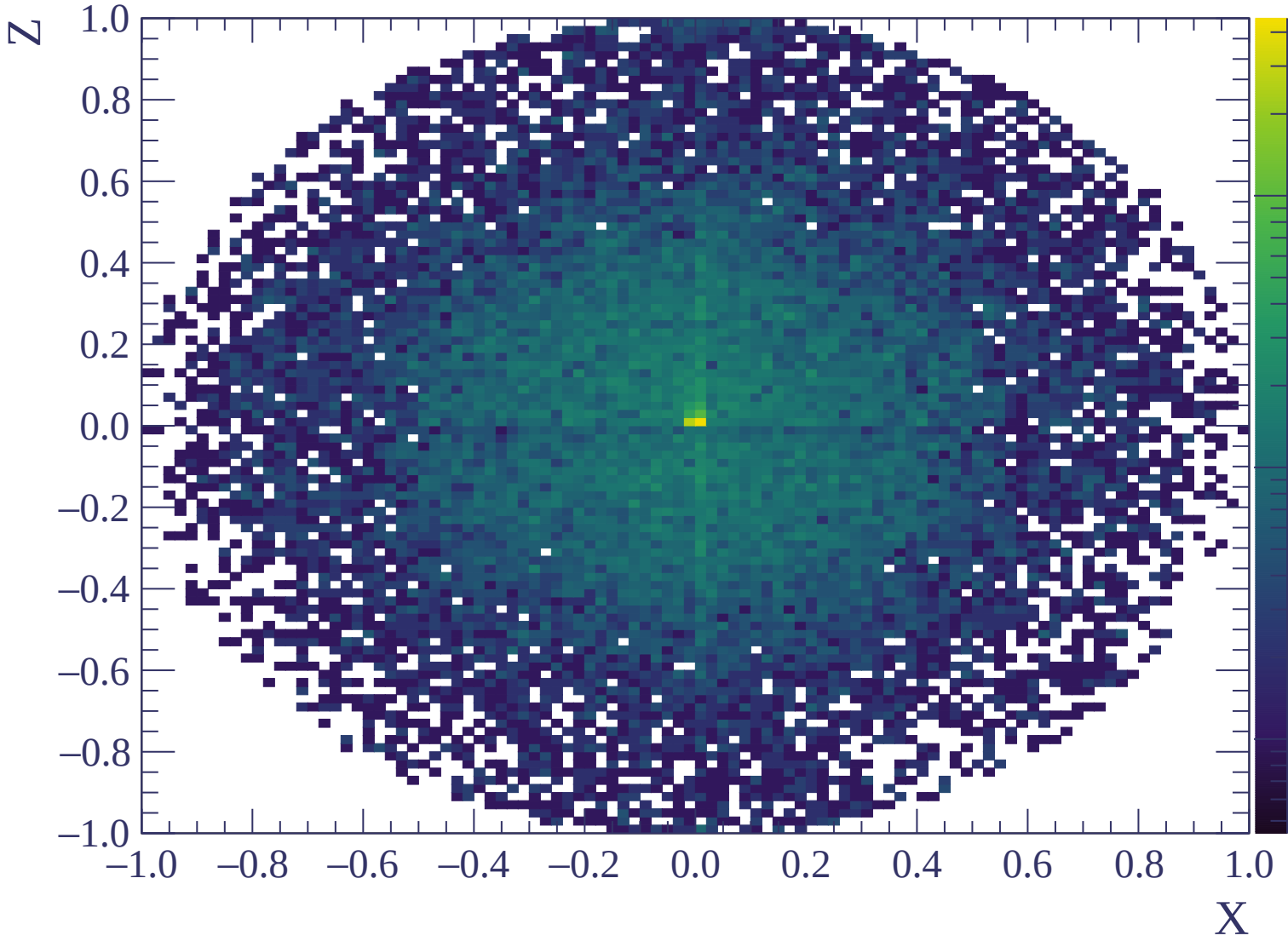


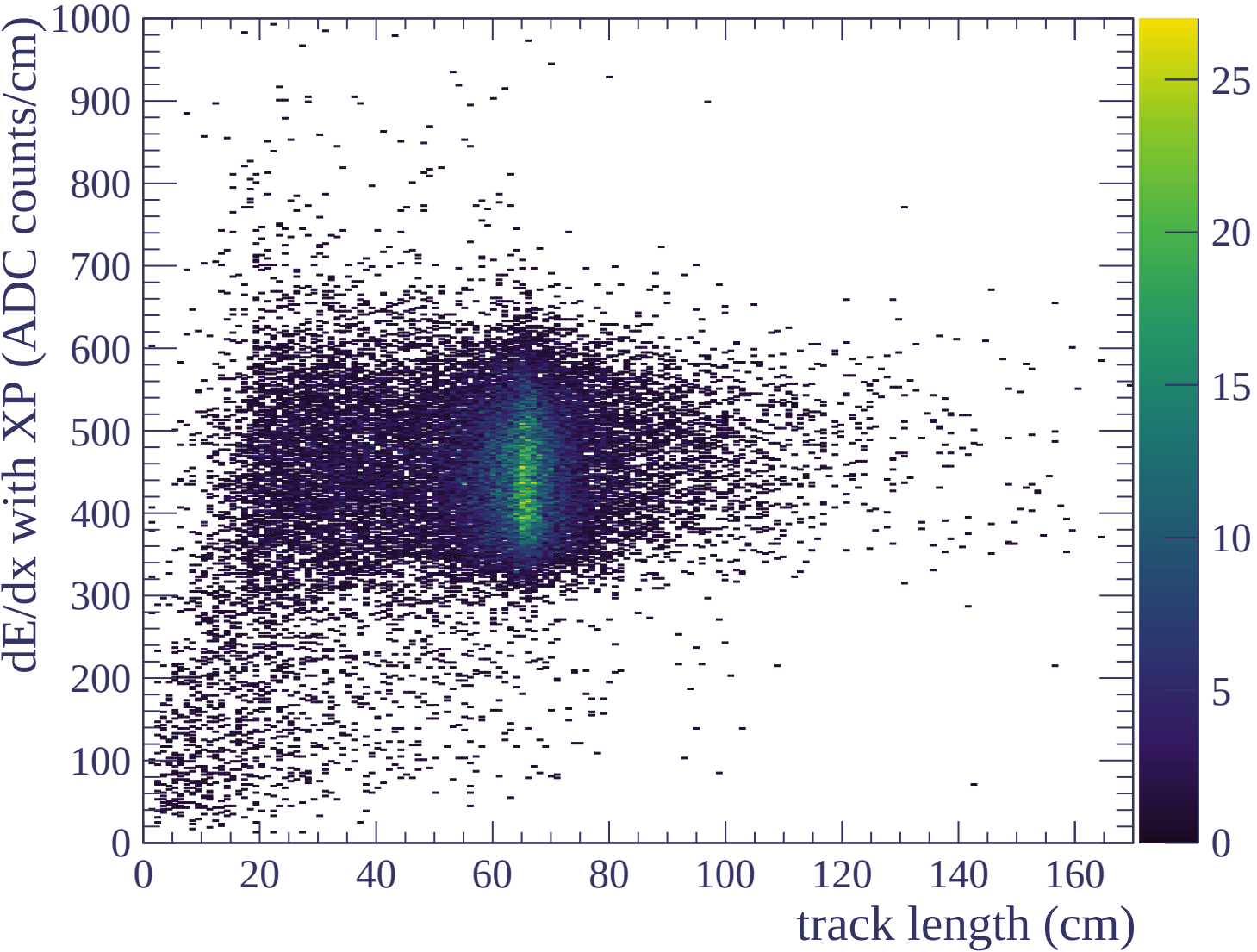


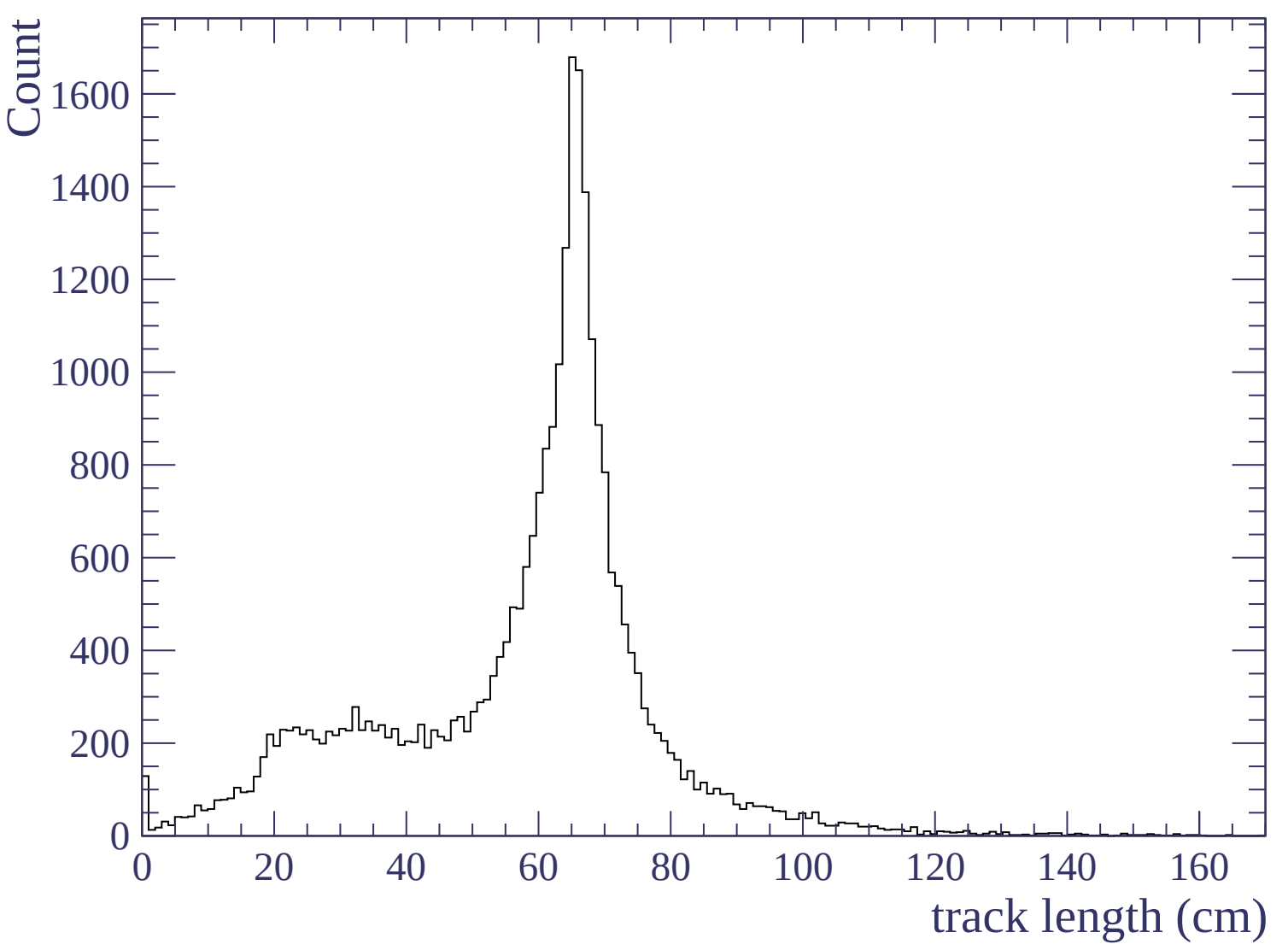


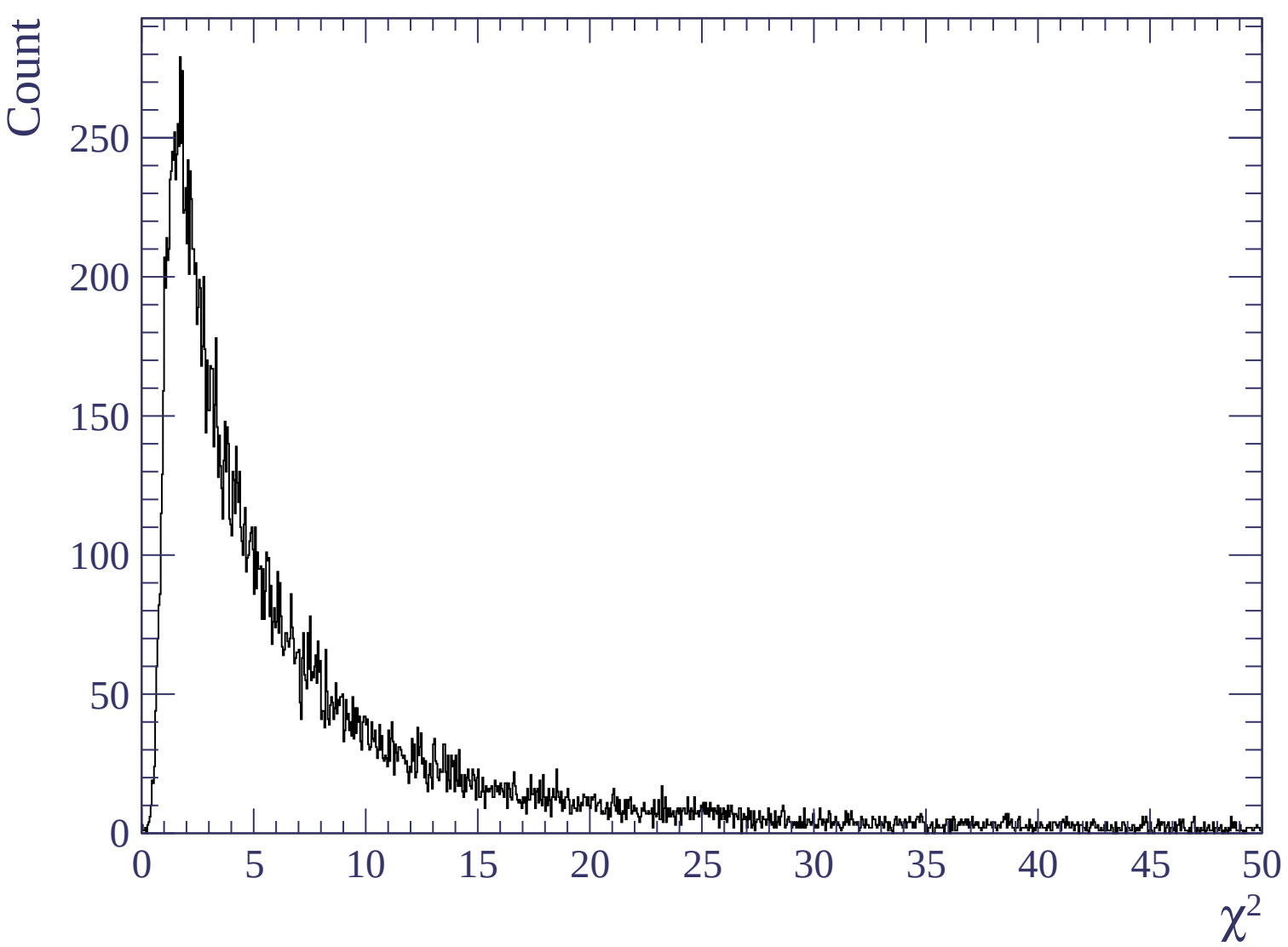


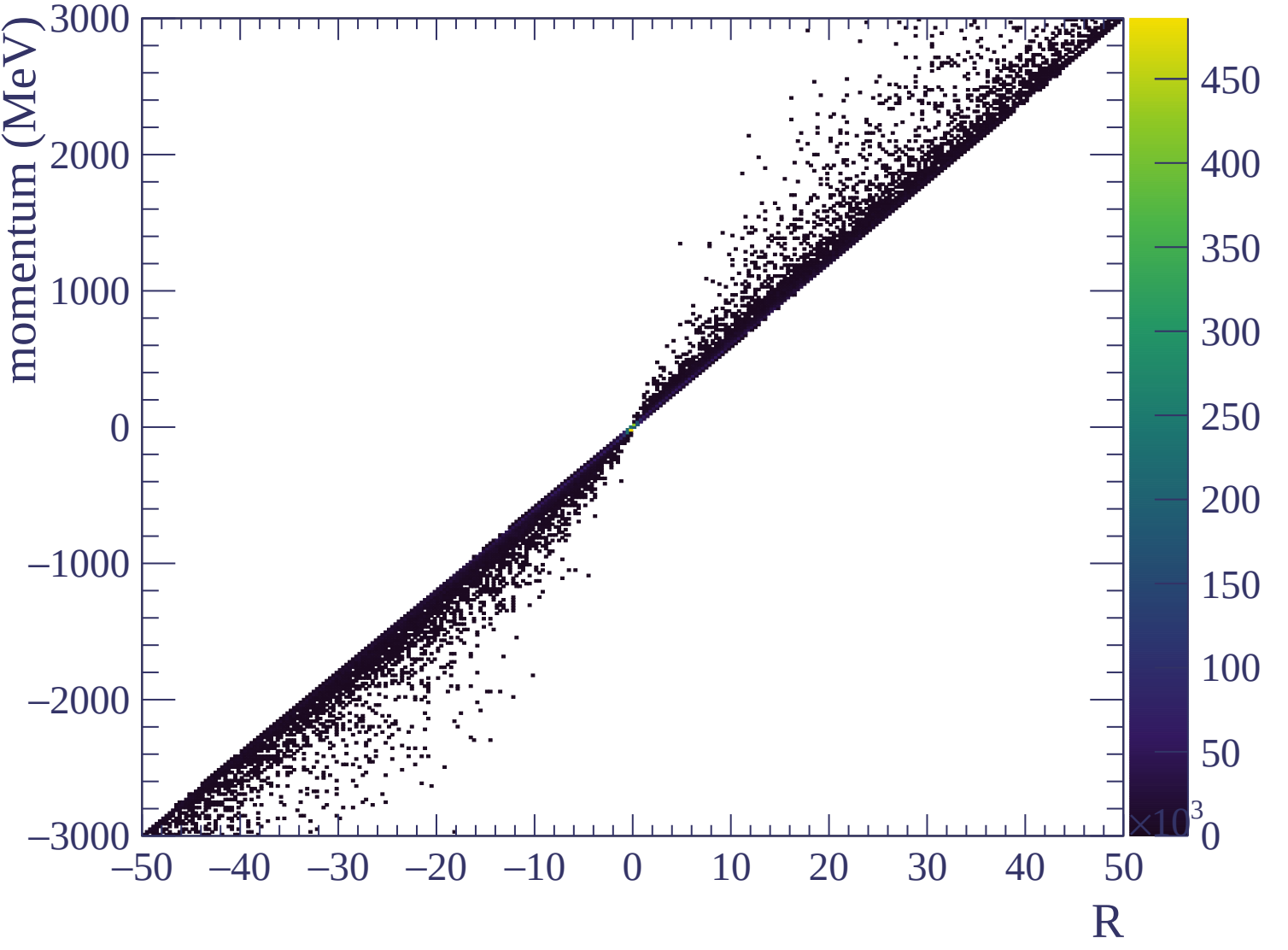


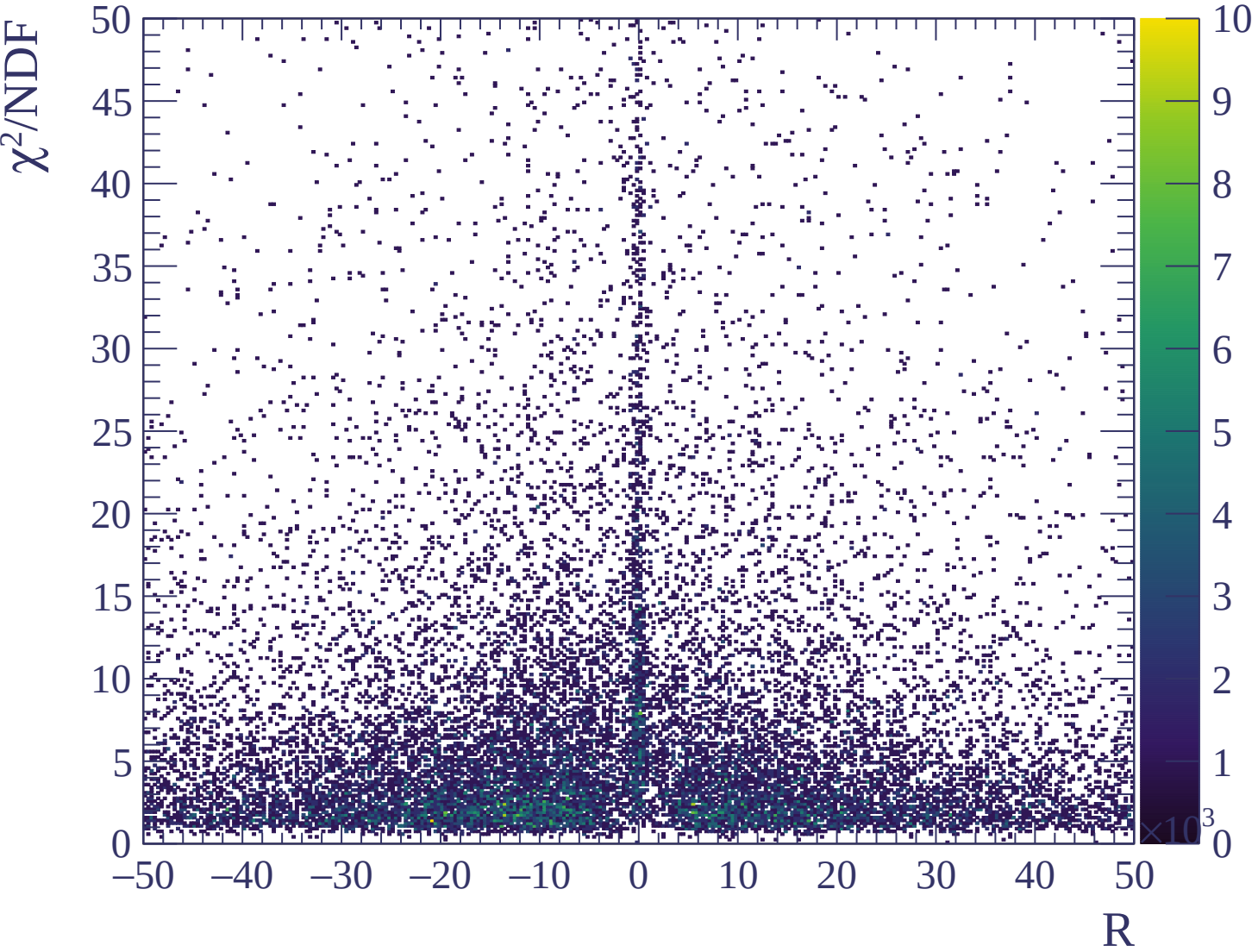






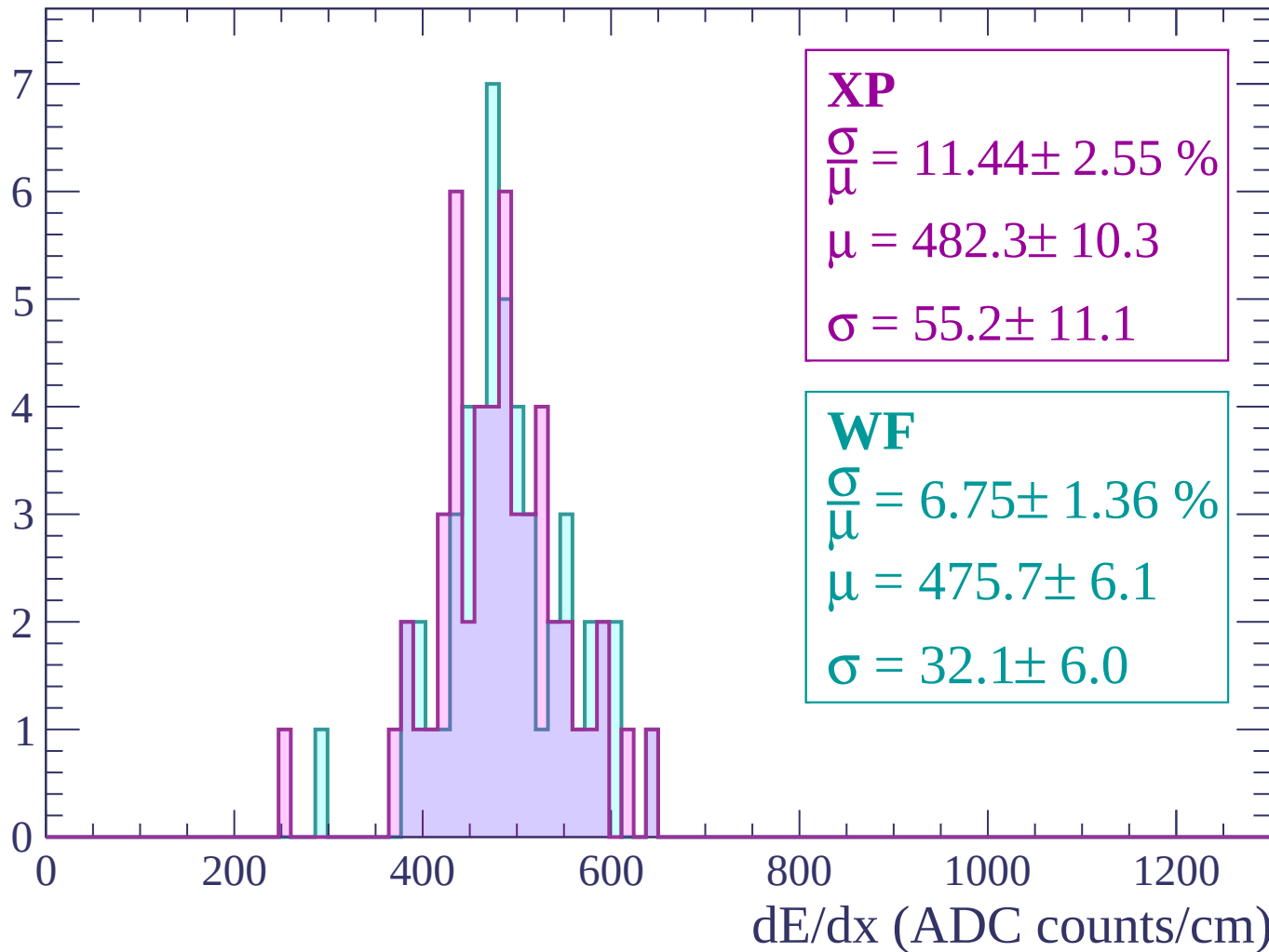






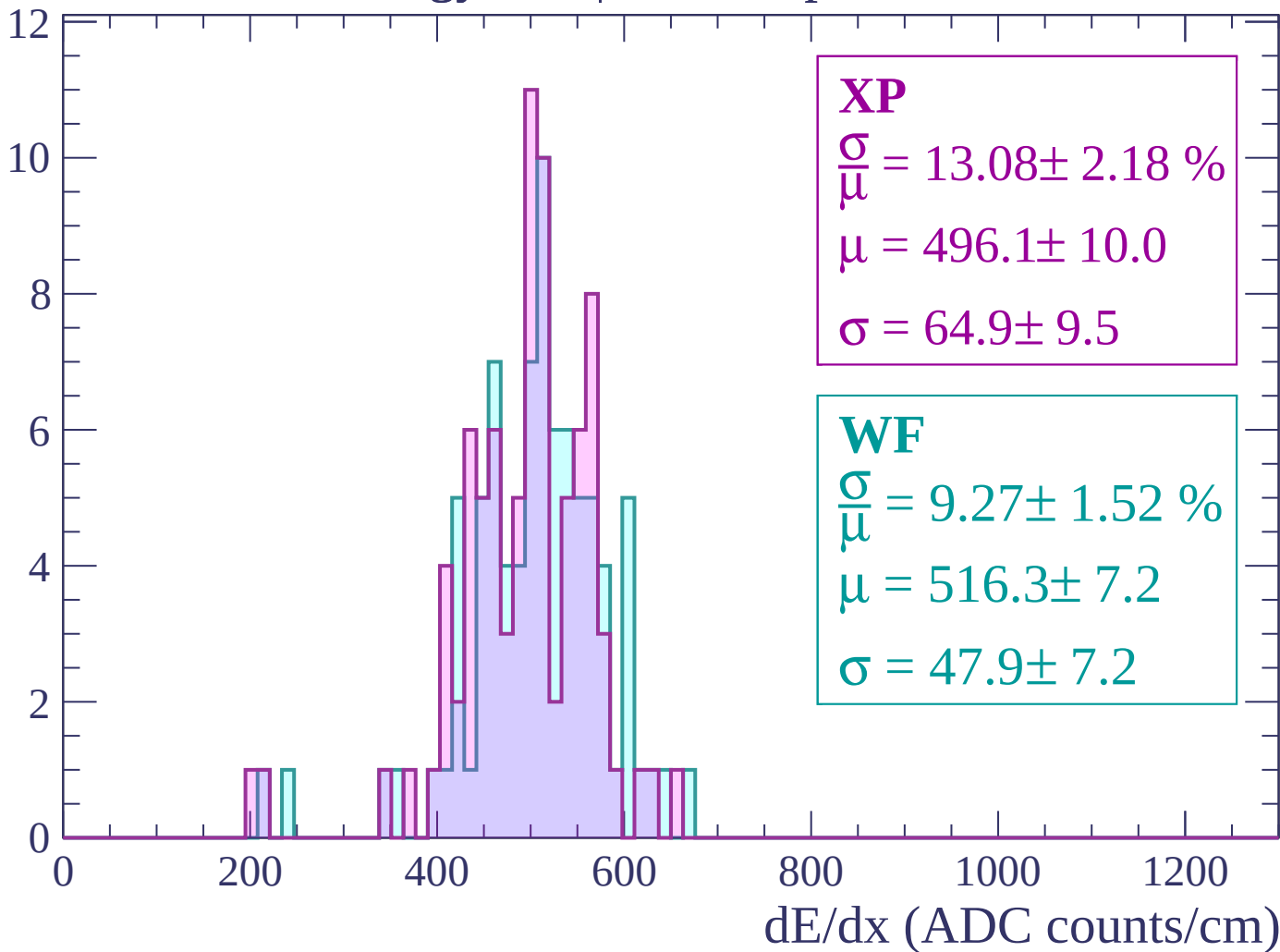
Energy loss | -3000 < p < -2940

Count



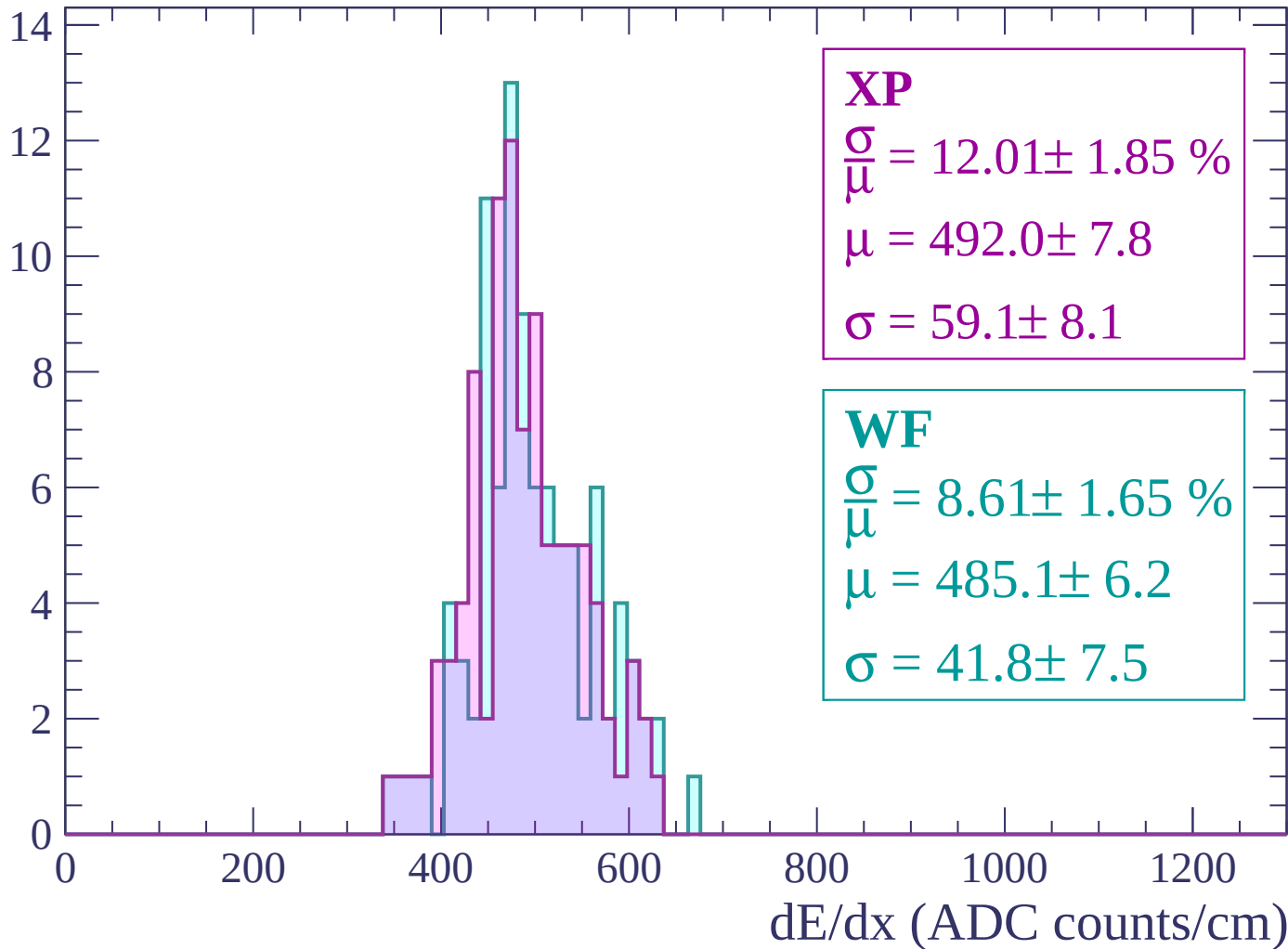
Energy loss | -2940 < p < -2880

Count



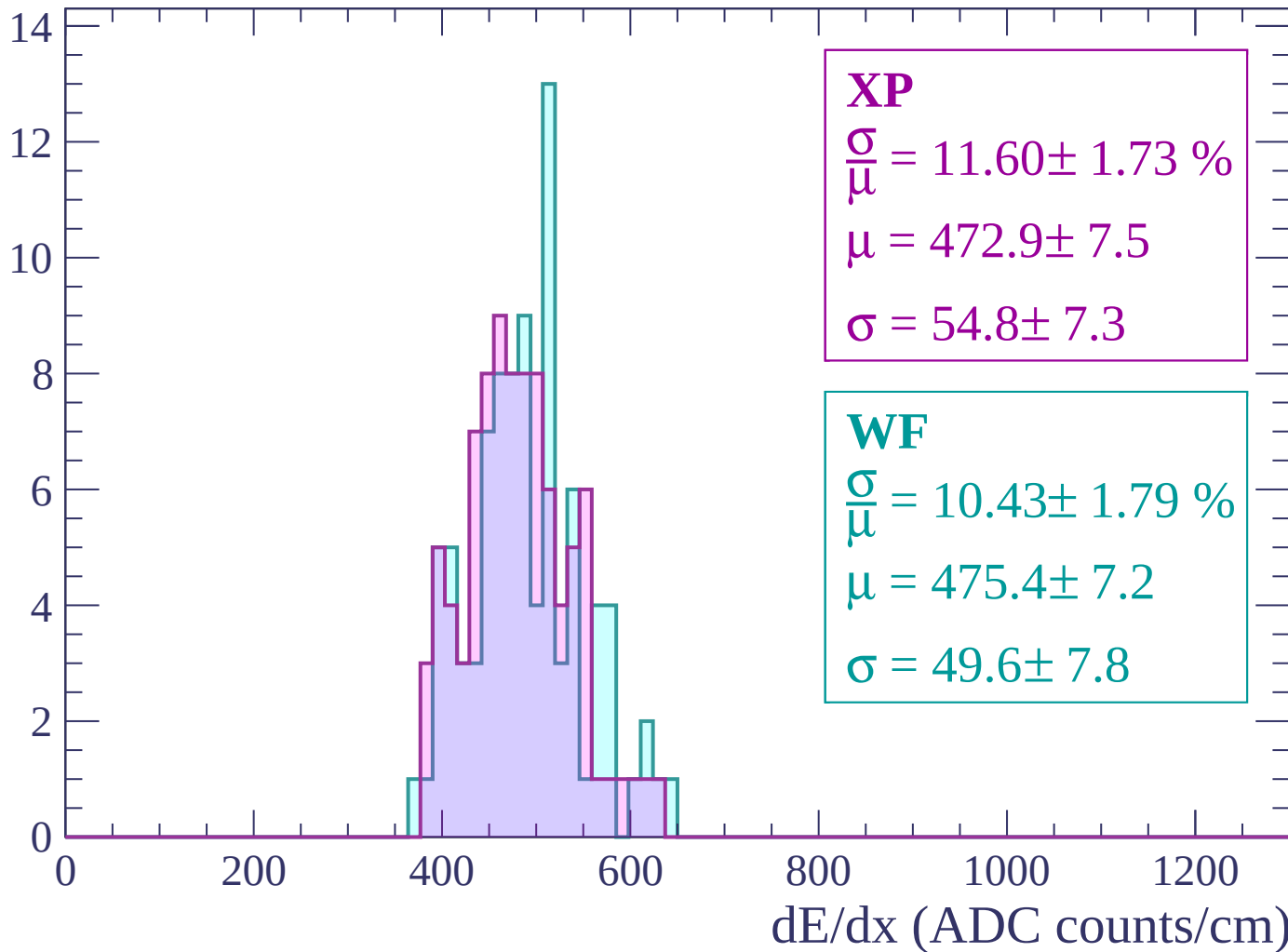
Energy loss | -2880 < p < -2820

Count



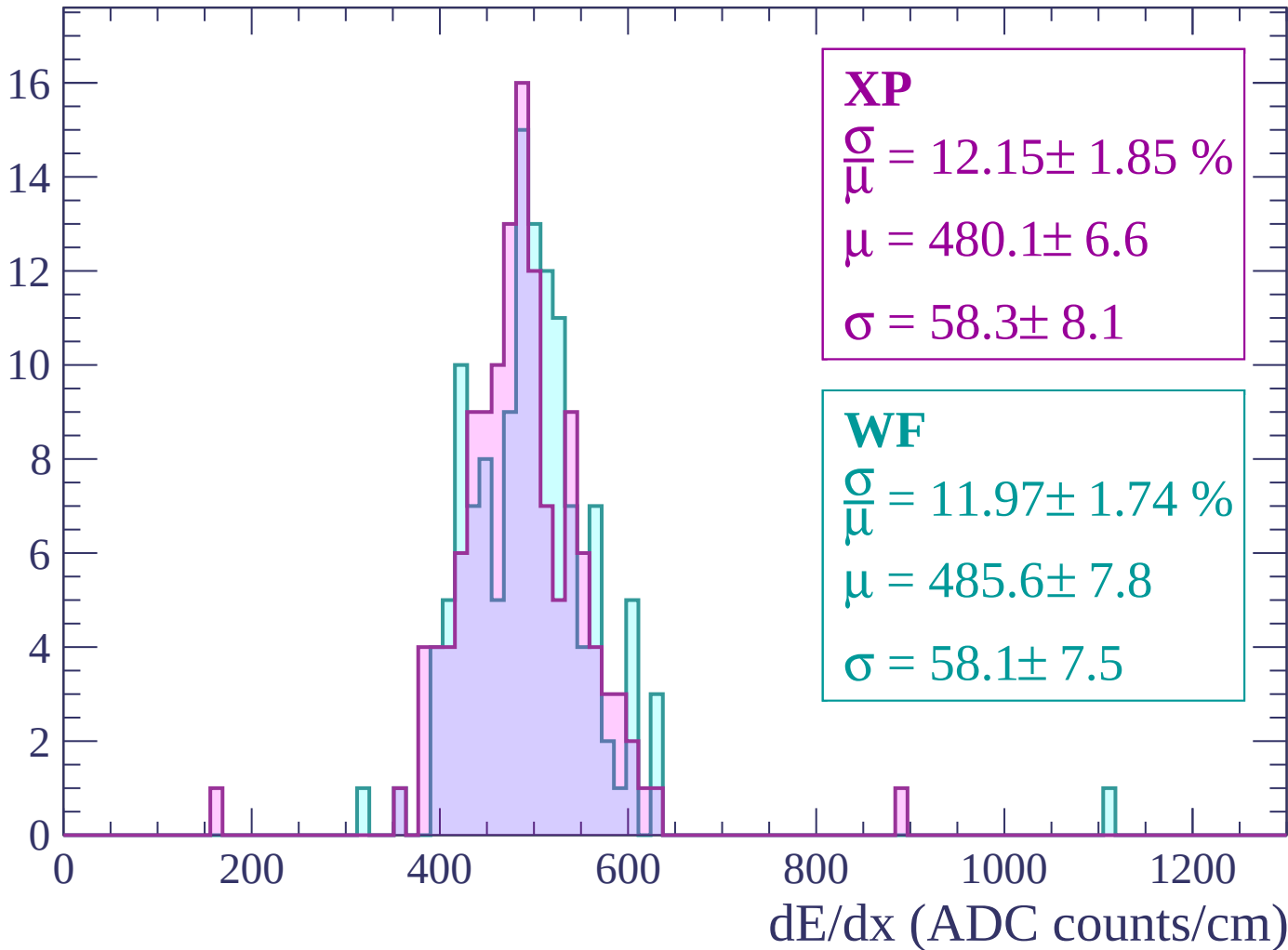
Energy loss | -2820 < p < -2760

Count



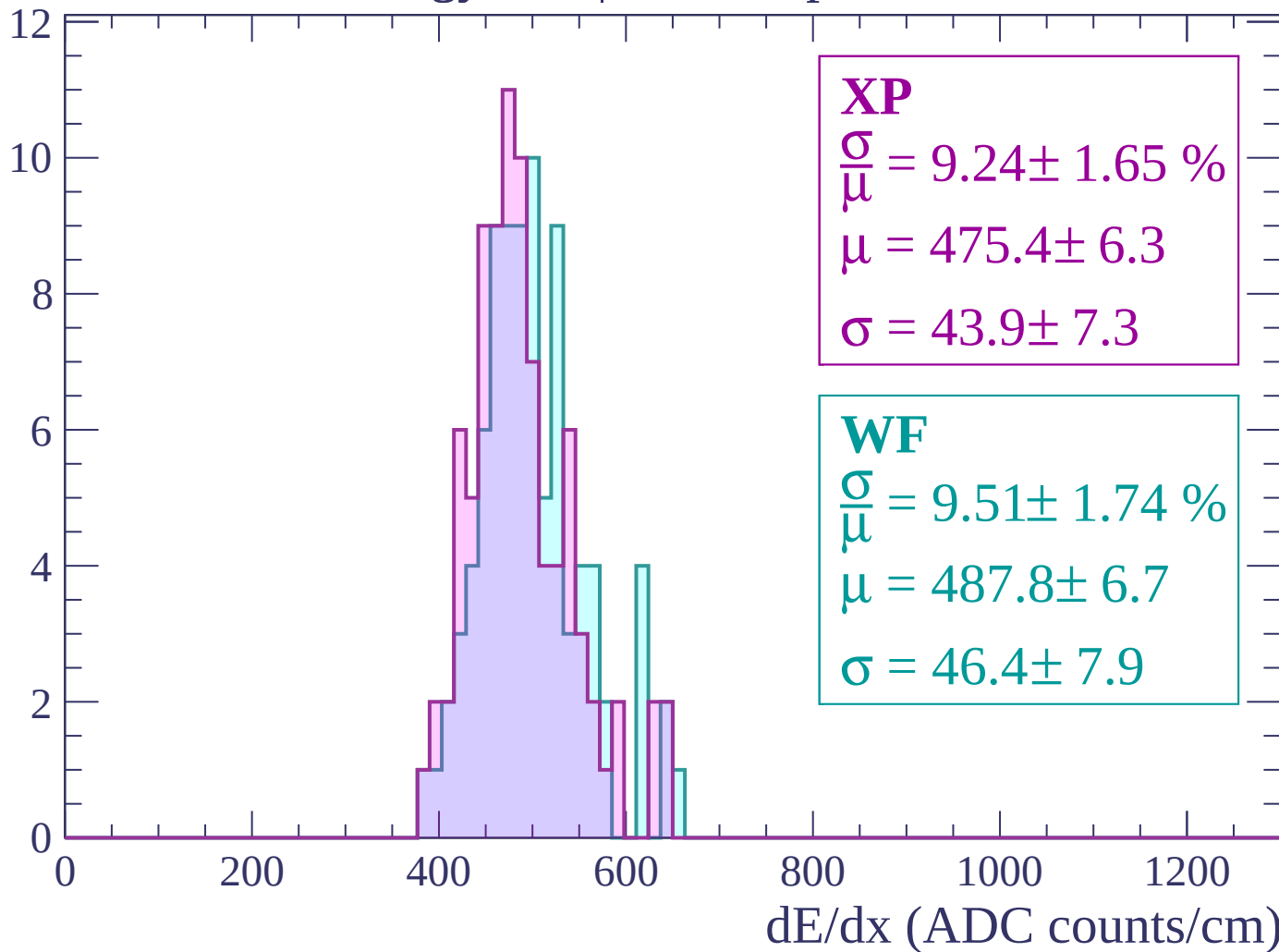
Energy loss | -2760 < p < -2700

Count



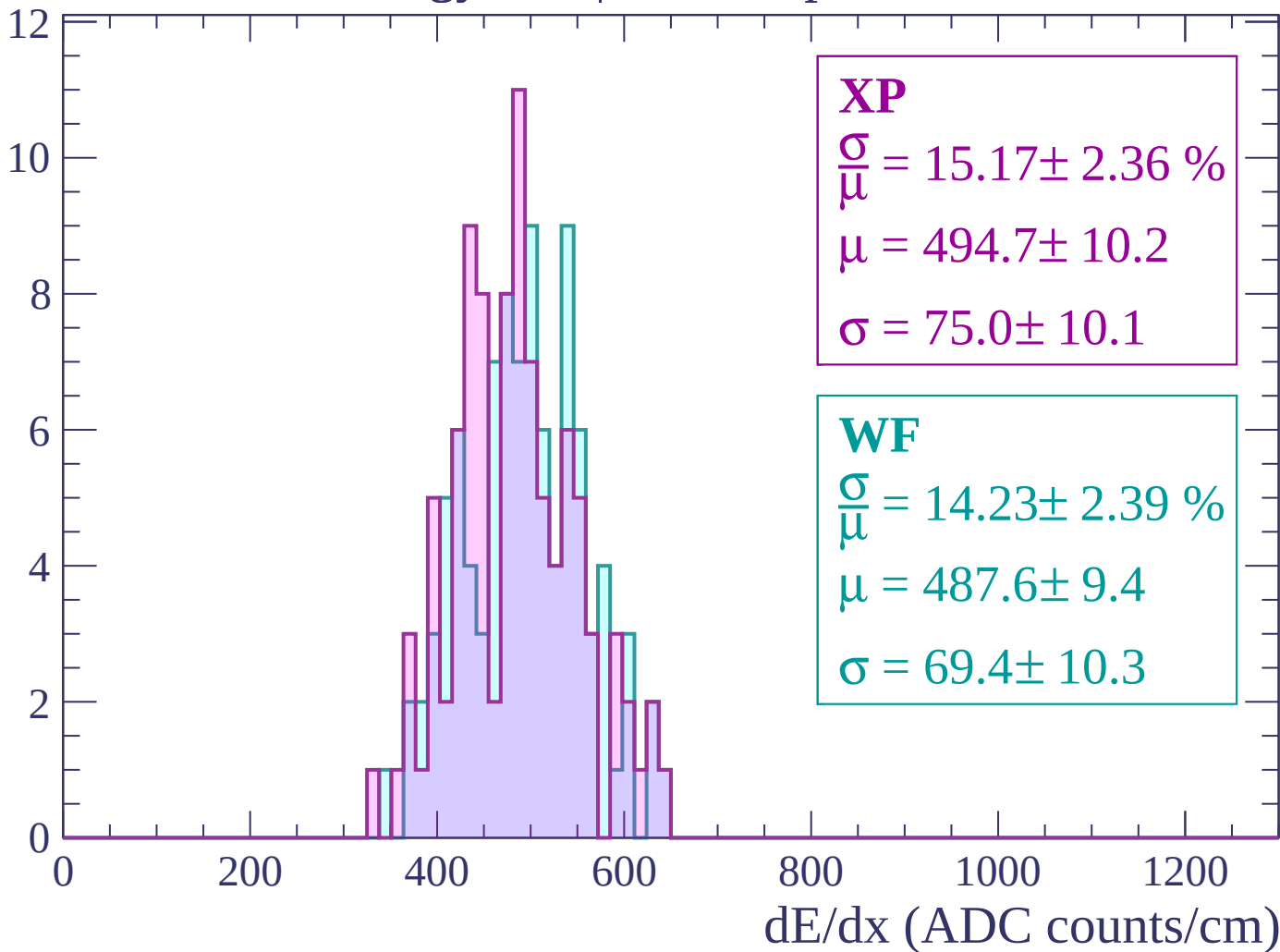
Energy loss | -2700 < p < -2640

Count



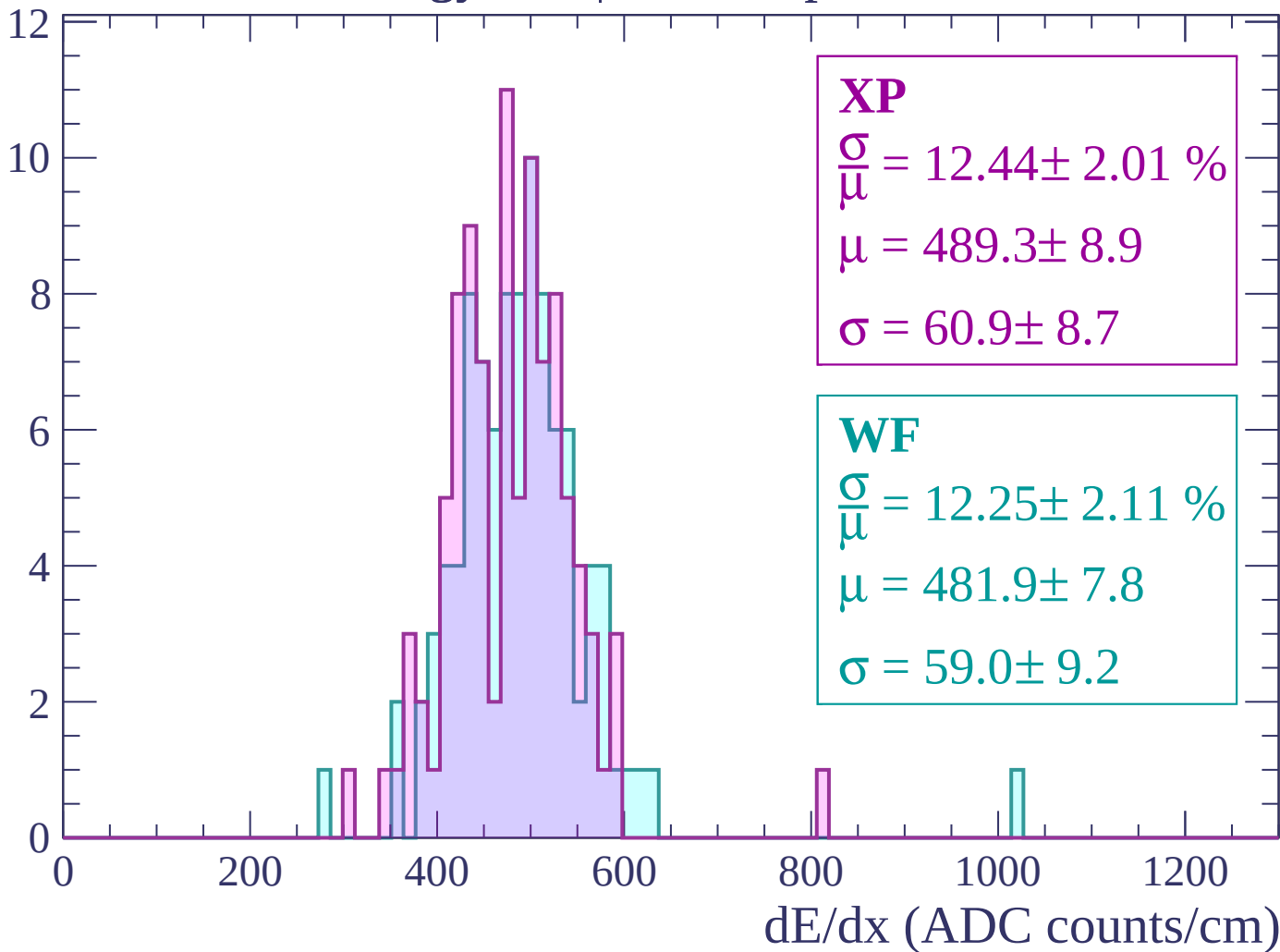
Energy loss | $-2640 < p < -2580$

Count



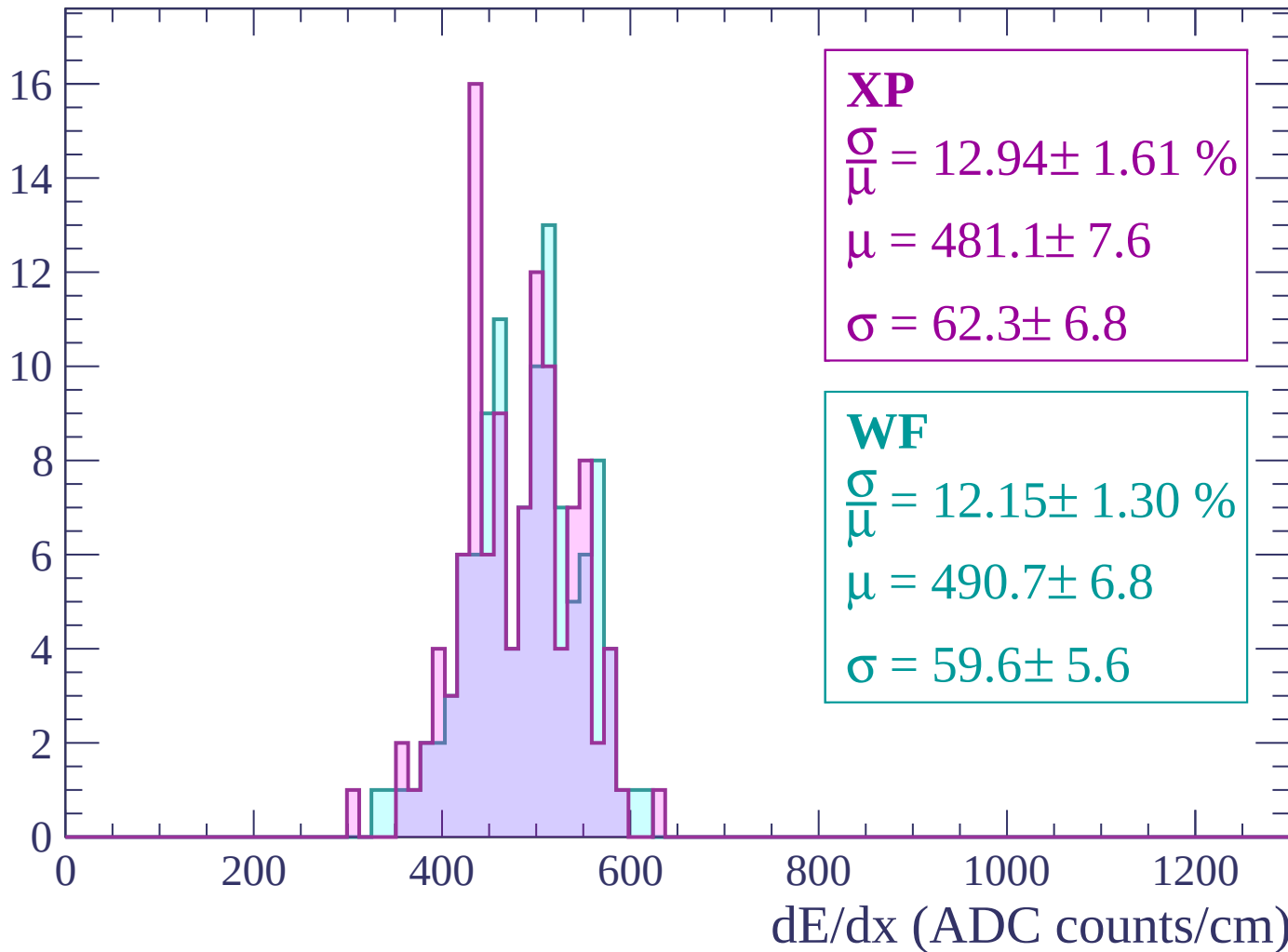
Energy loss | -2580 < p < -2520

Count



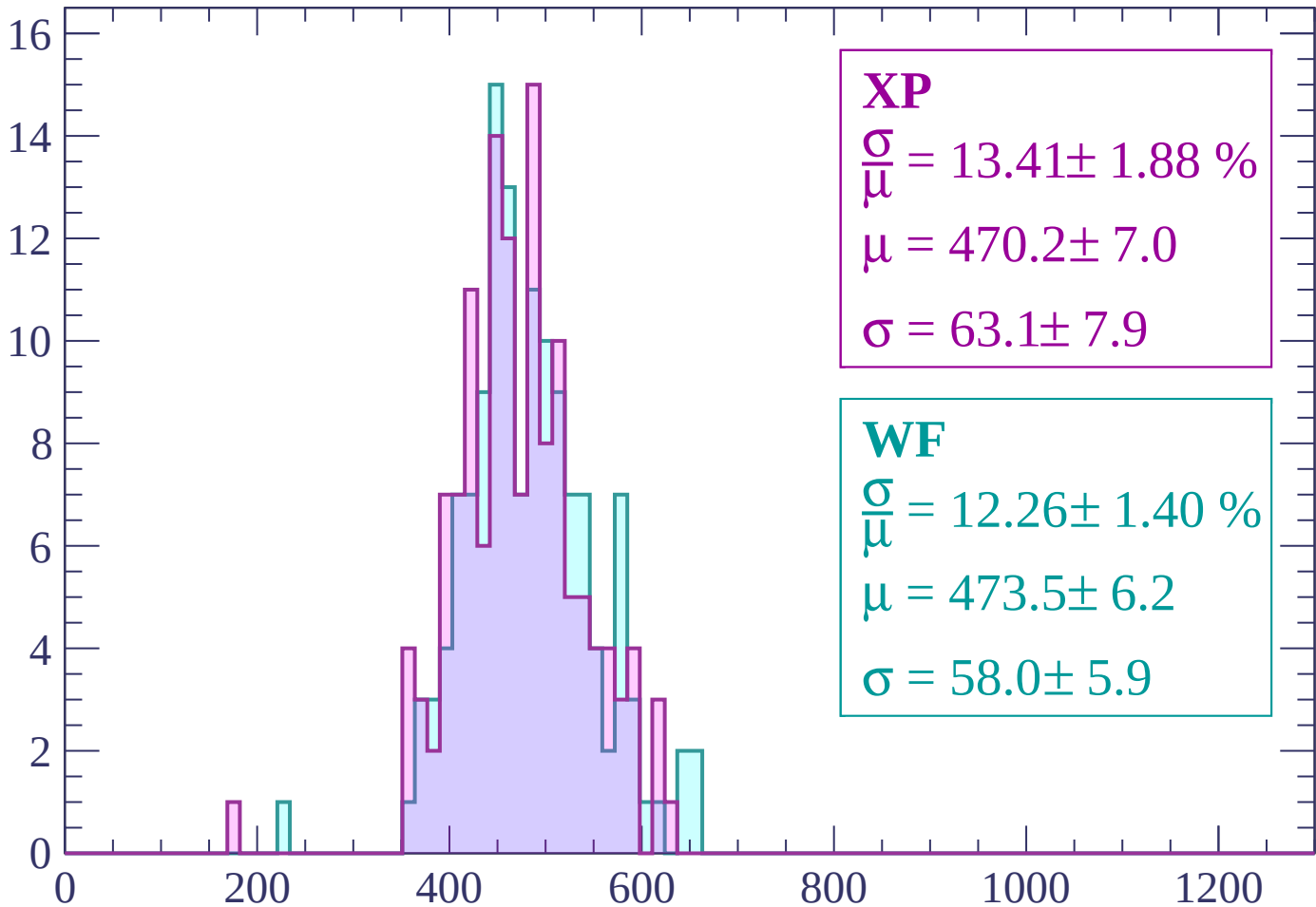
Energy loss | -2520 < p < -2460

Count



Energy loss | -2460 < p < -2400

Count



XP

$$\frac{\sigma}{\mu} = 13.41 \pm 1.88 \%$$

$$\mu = 470.2 \pm 7.0$$

$$\sigma = 63.1 \pm 7.9$$

WF

$$\frac{\sigma}{\mu} = 12.26 \pm 1.40 \%$$

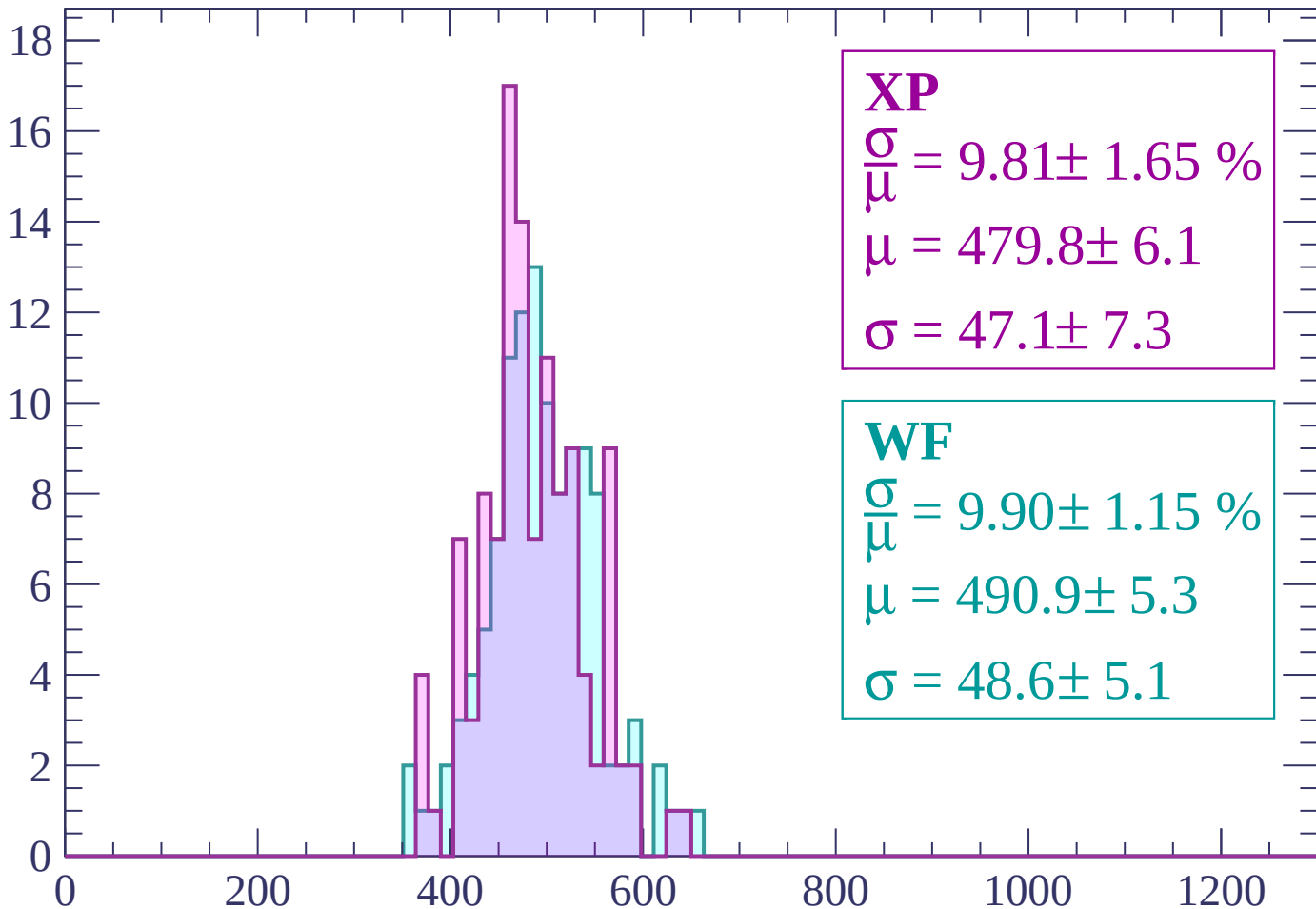
$$\mu = 473.5 \pm 6.2$$

$$\sigma = 58.0 \pm 5.9$$

dE/dx (ADC counts/cm)

Energy loss | -2400 < p < -2340

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.81 \pm 1.65 \%$$

$$\mu = 479.8 \pm 6.1$$

$$\sigma = 47.1 \pm 7.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.90 \pm 1.15 \%$$

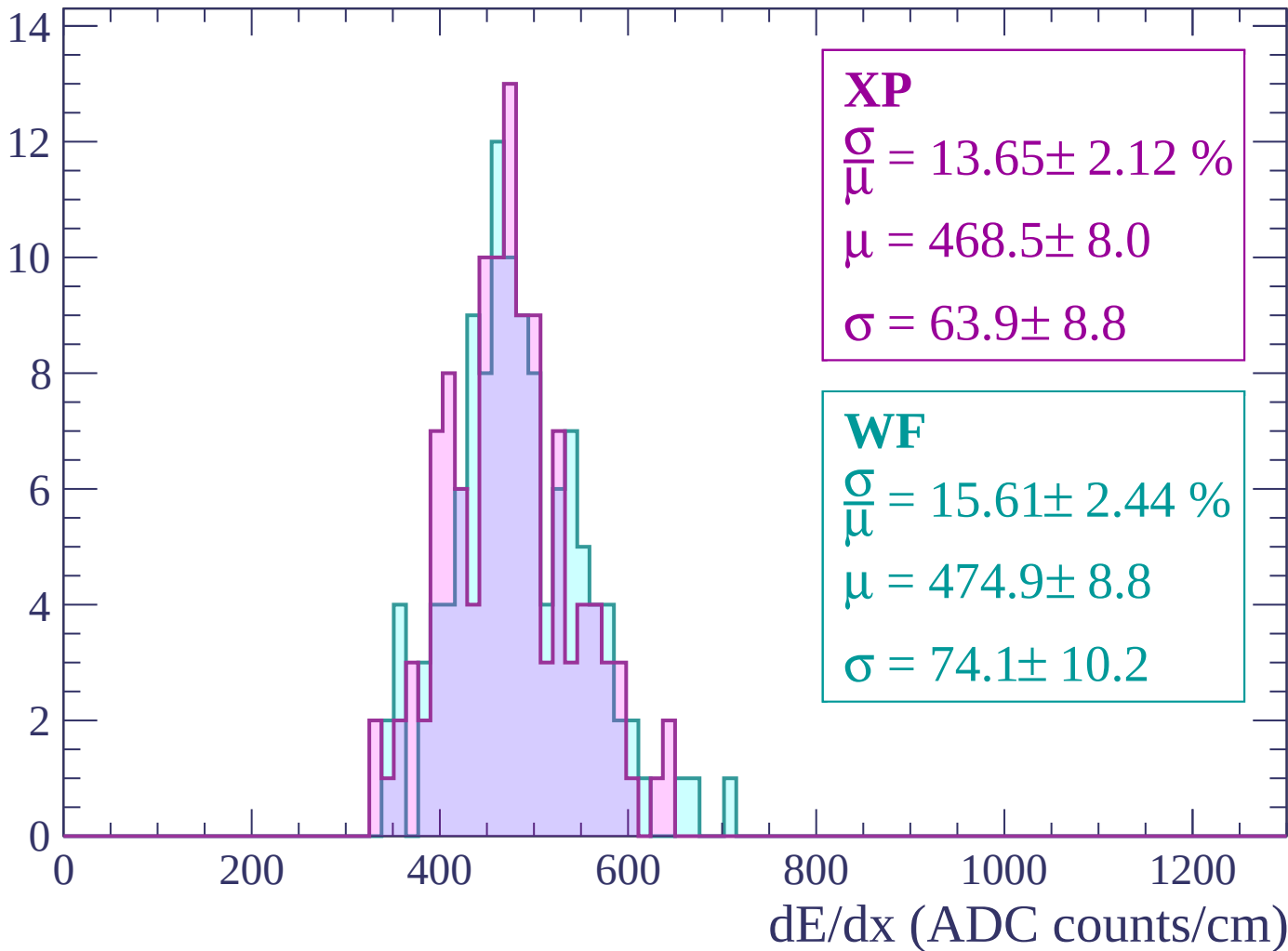
$$\mu = 490.9 \pm 5.3$$

$$\sigma = 48.6 \pm 5.1$$

dE/dx (ADC counts/cm)

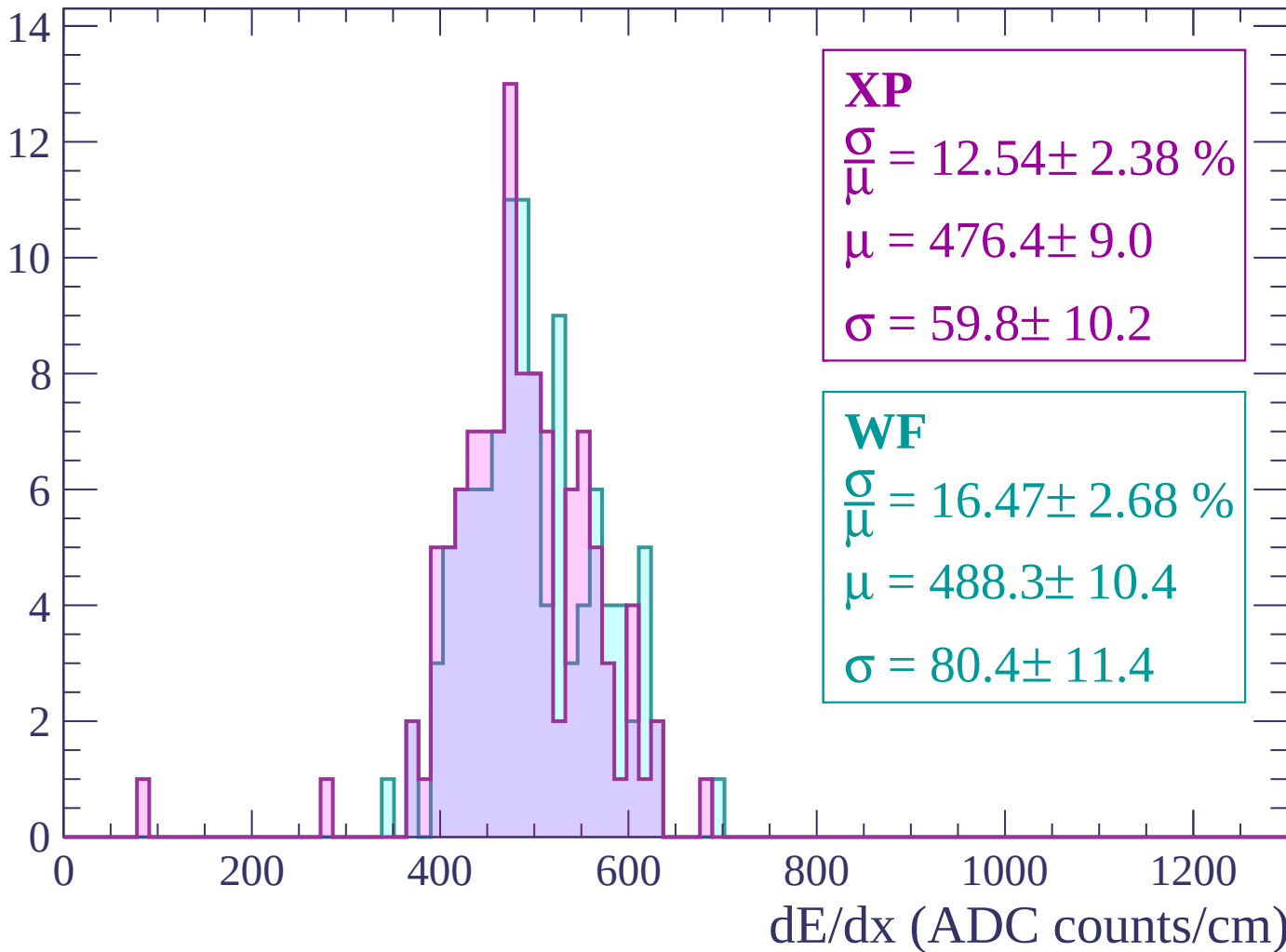
Energy loss | $-2340 < p < -2280$

Count



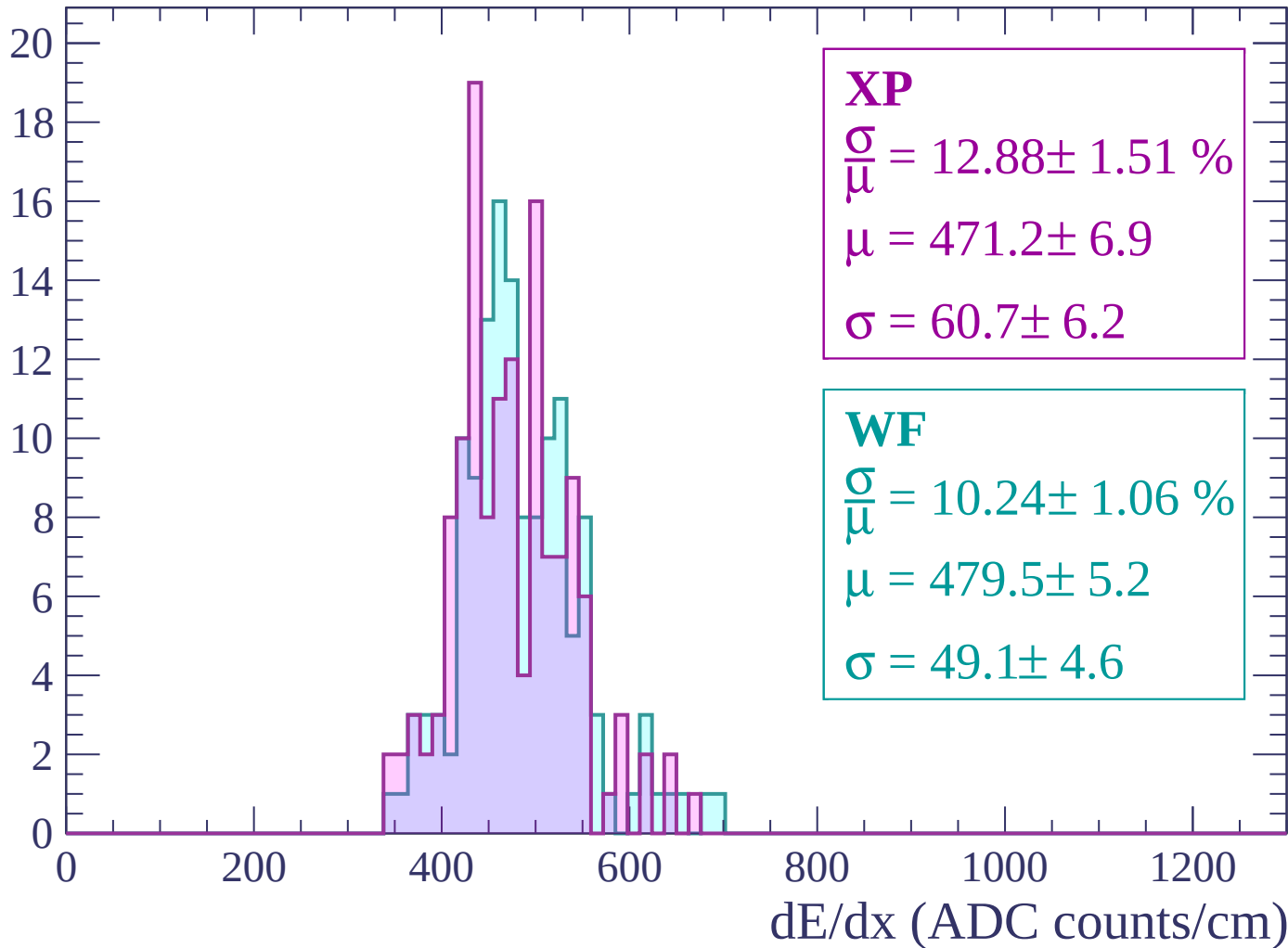
Energy loss | $-2280 < p < -2220$

Count



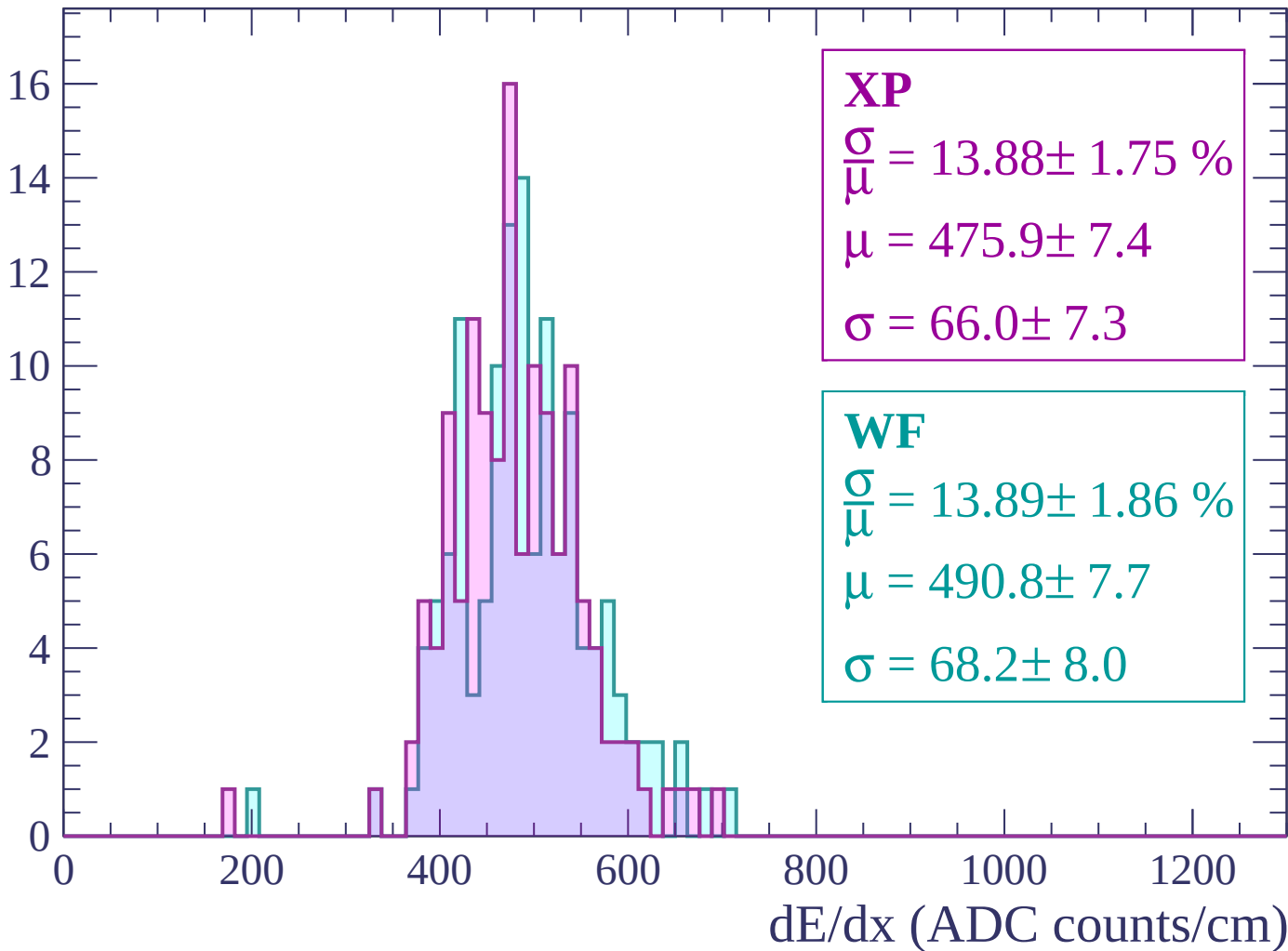
Energy loss | -2220 < p < -2160

Count



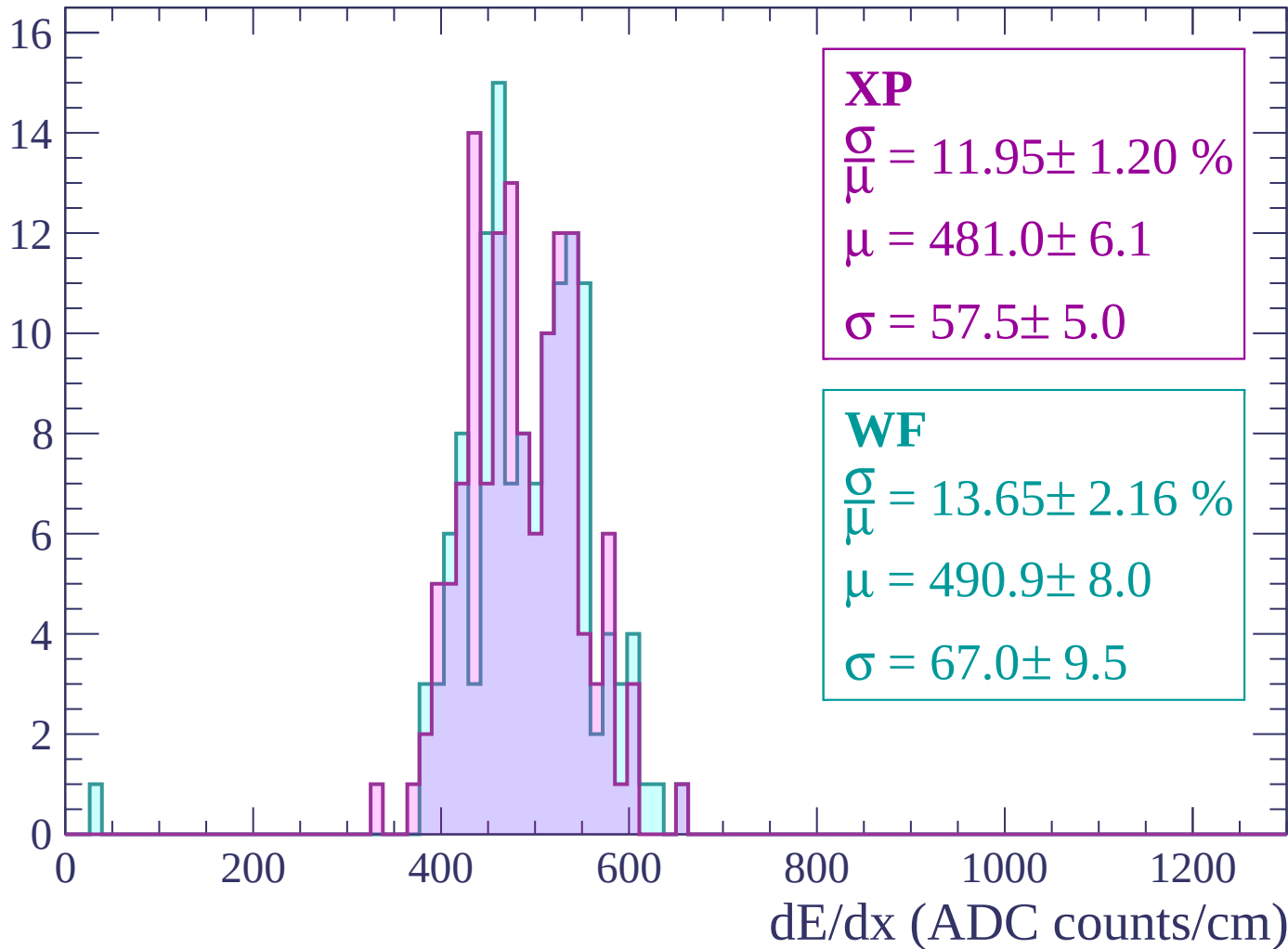
Energy loss | -2160 < p < -2100

Count



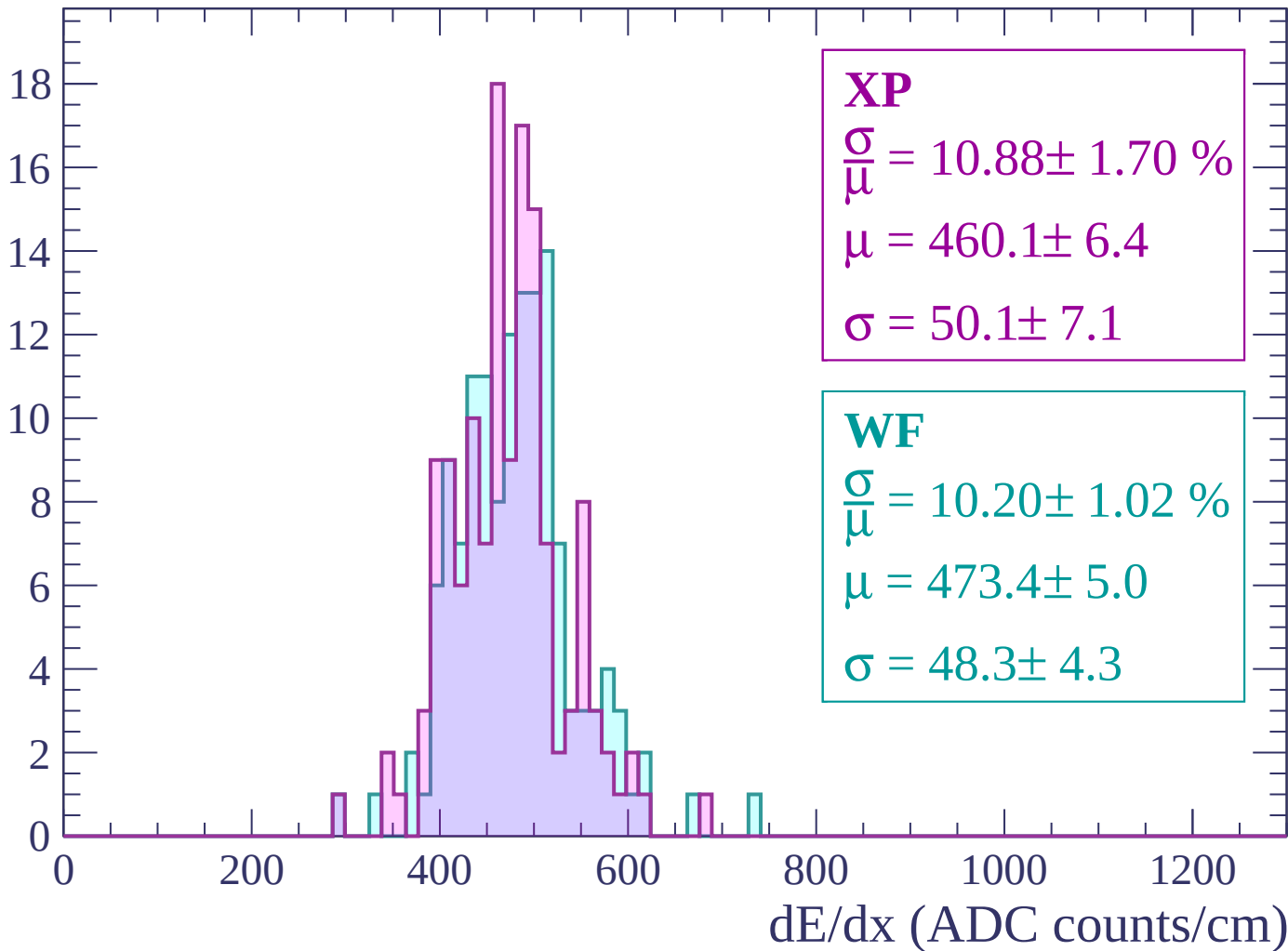
Energy loss | -2100 < p < -2040

Count



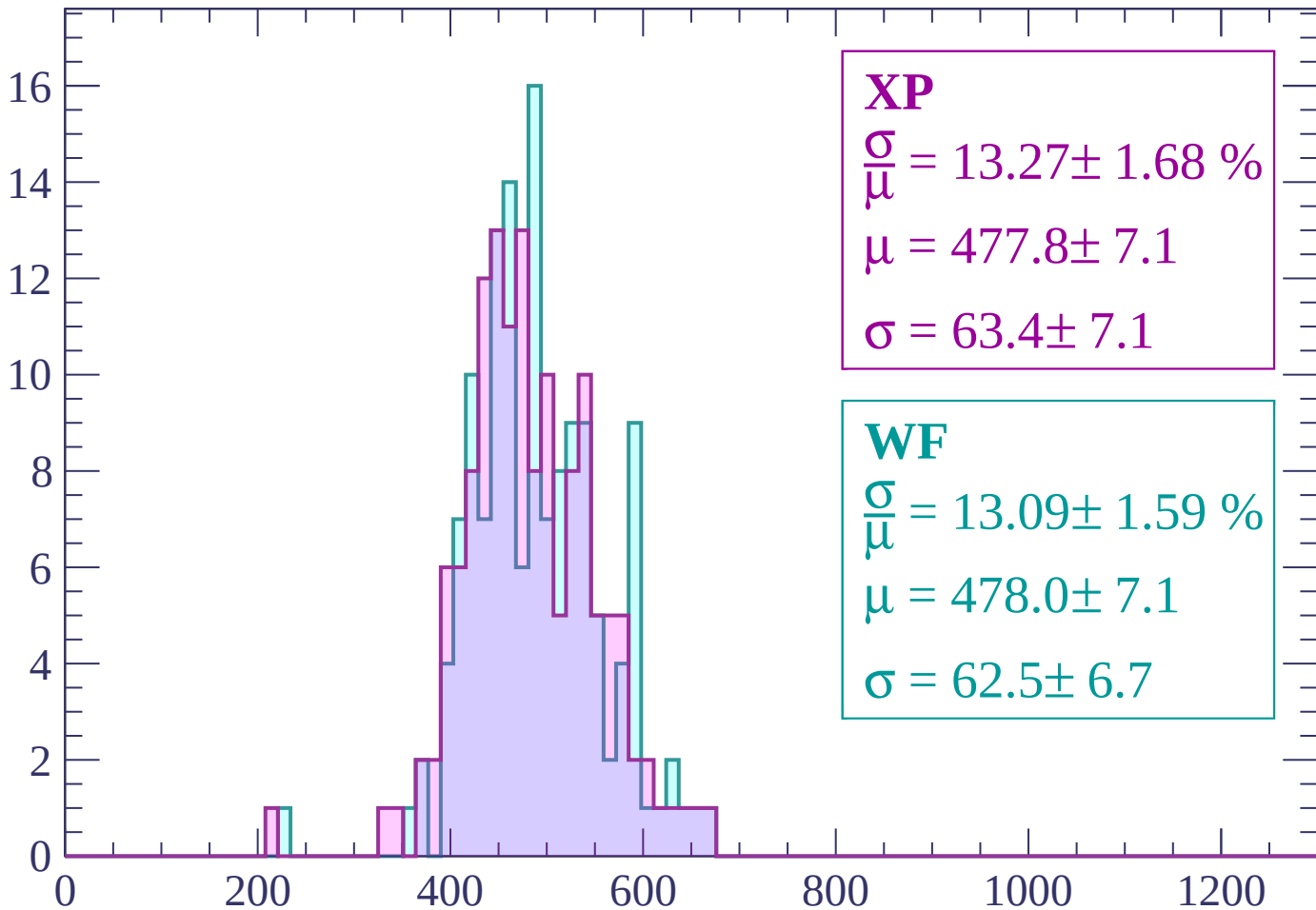
Energy loss | -2040 < p < -1980

Count



Energy loss | -1980 < p < -1920

Count



XP

$$\frac{\sigma}{\mu} = 13.27 \pm 1.68 \%$$

$$\mu = 477.8 \pm 7.1$$

$$\sigma = 63.4 \pm 7.1$$

WF

$$\frac{\sigma}{\mu} = 13.09 \pm 1.59 \%$$

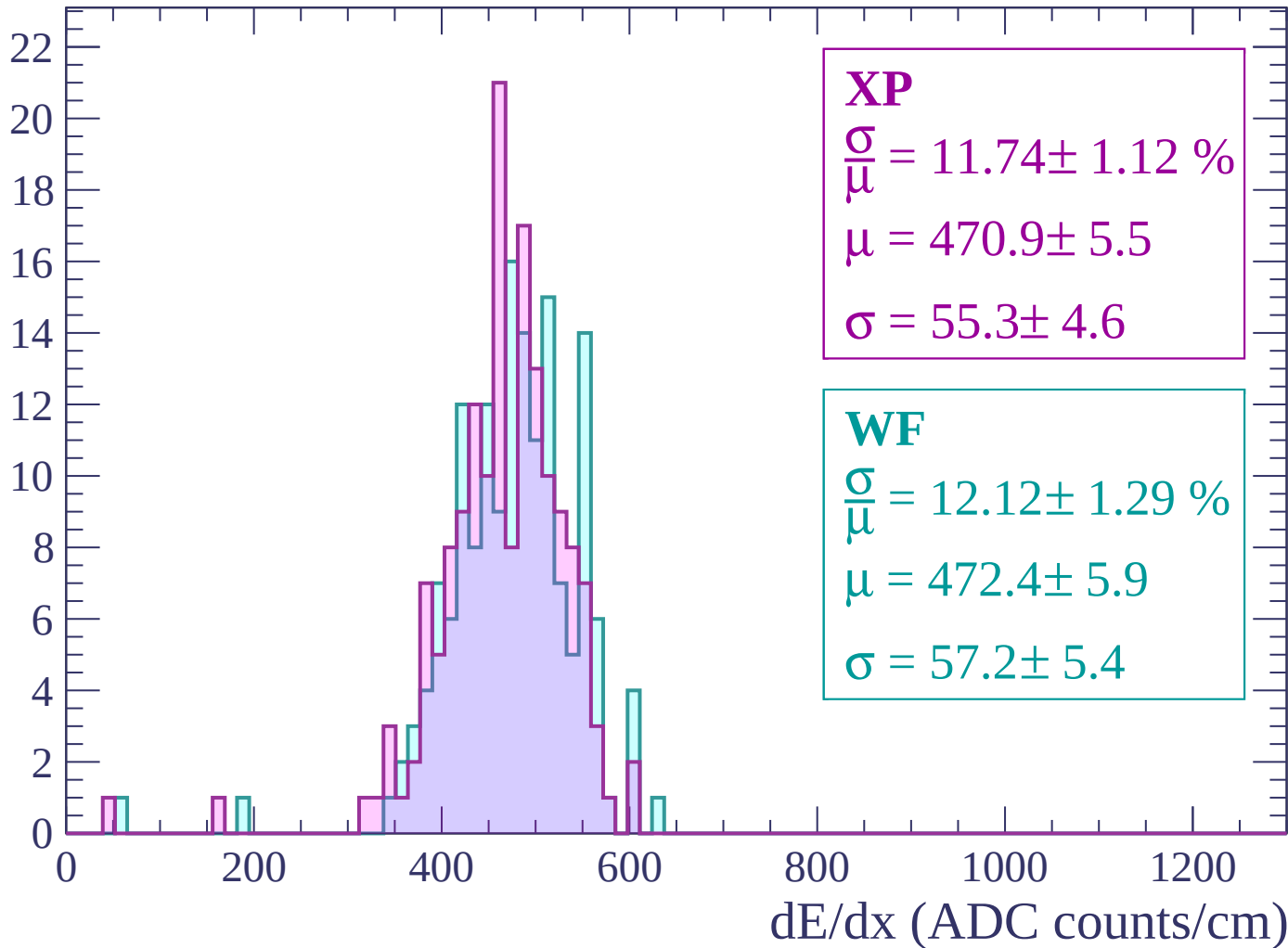
$$\mu = 478.0 \pm 7.1$$

$$\sigma = 62.5 \pm 6.7$$

dE/dx (ADC counts/cm)

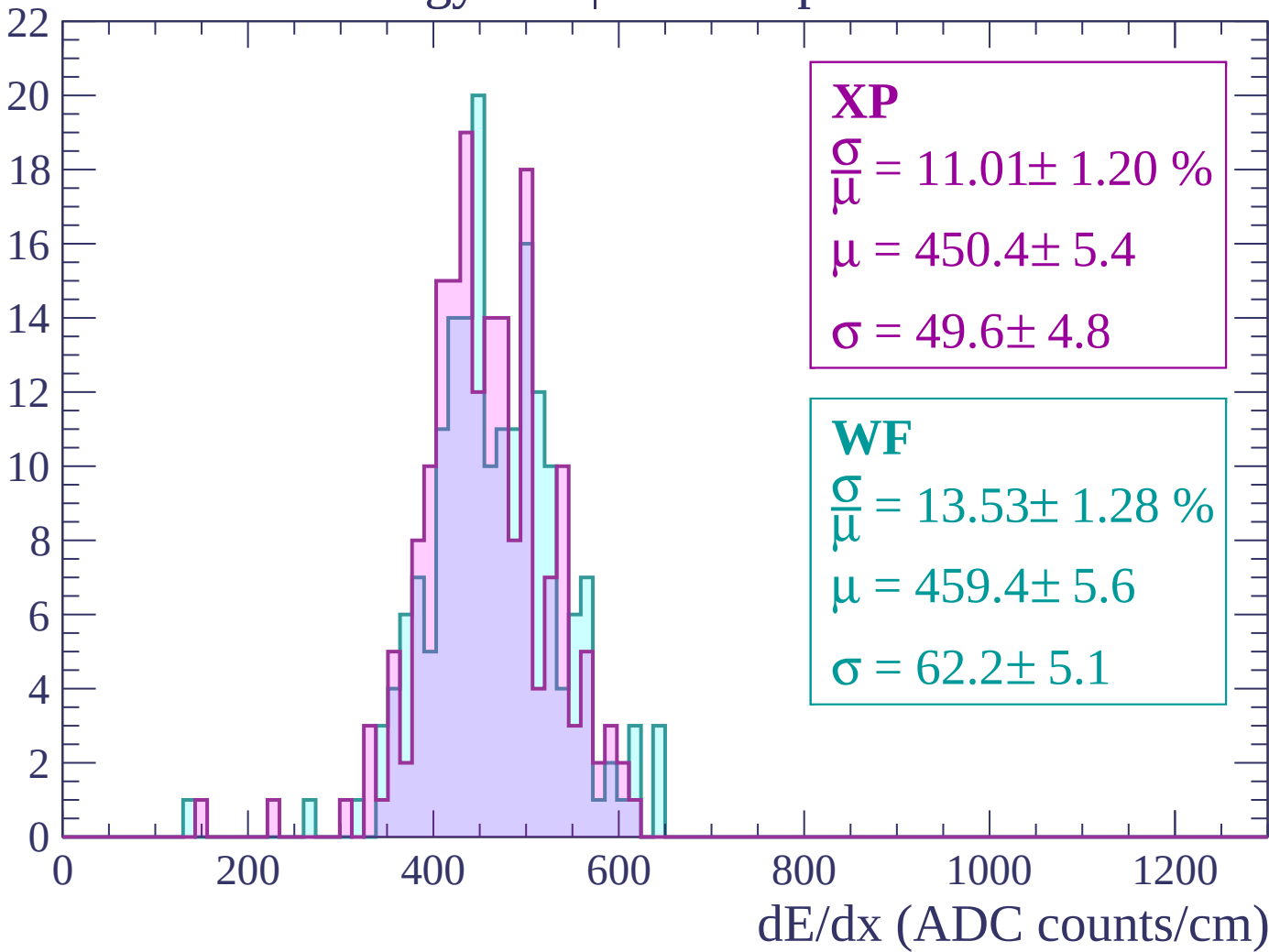
Energy loss | $-1920 < p < -1860$

Count



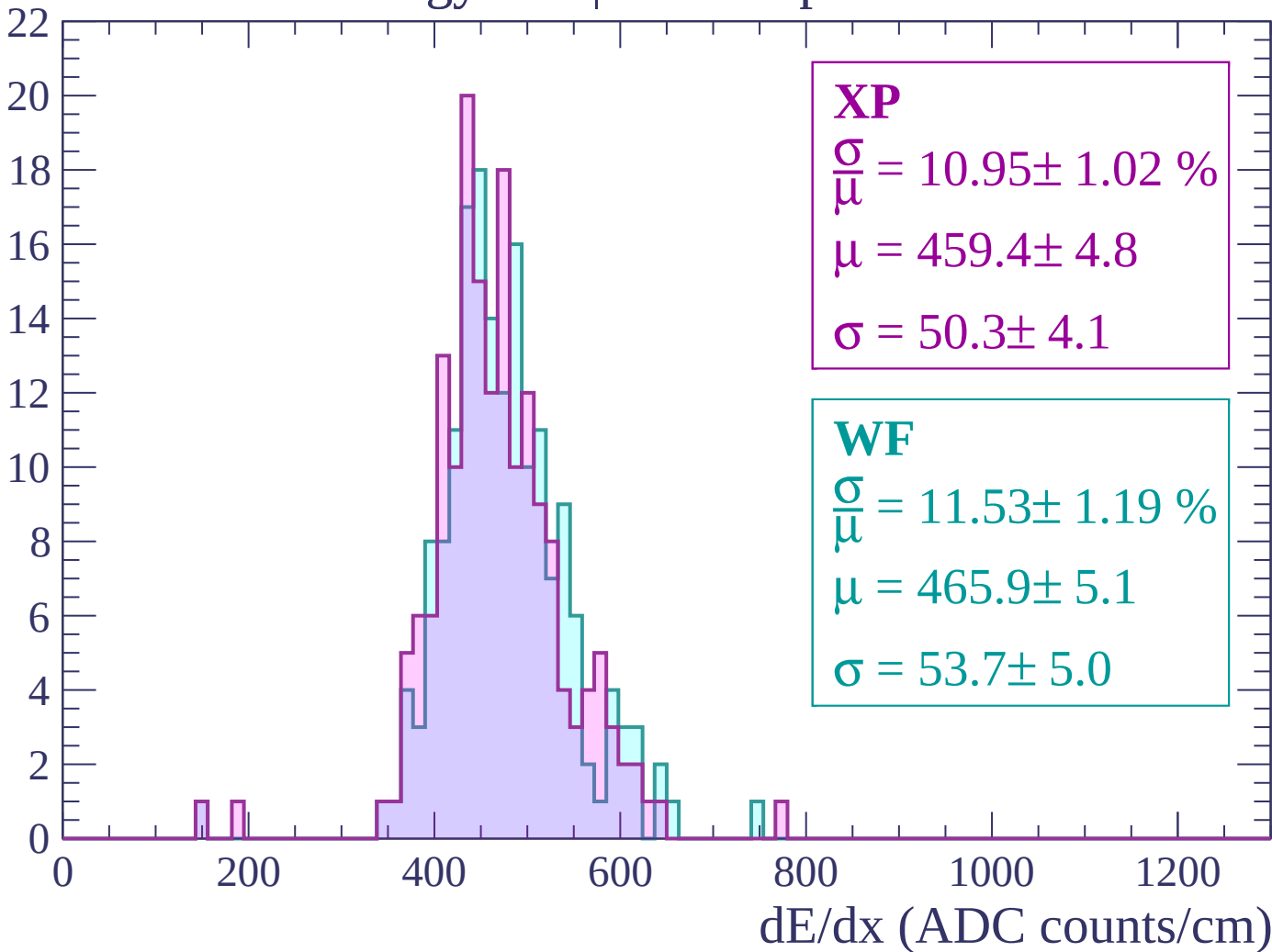
Energy loss | -1860 < p < -1800

Count



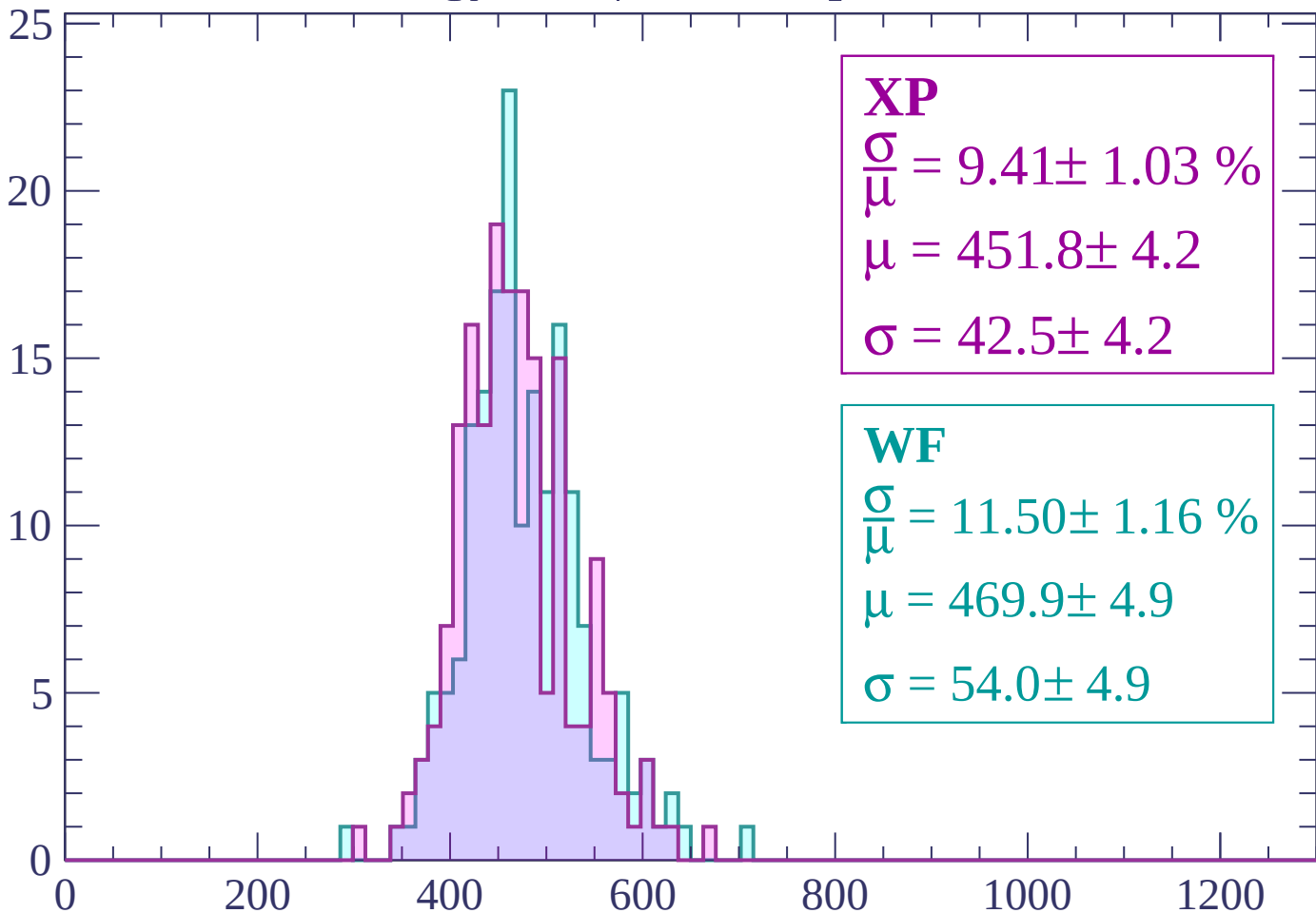
Energy loss | -1800 < p < -1740

Count



Energy loss | $-1740 < p < -1680$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.41 \pm 1.03 \%$$

$$\mu = 451.8 \pm 4.2$$

$$\sigma = 42.5 \pm 4.2$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.50 \pm 1.16 \%$$

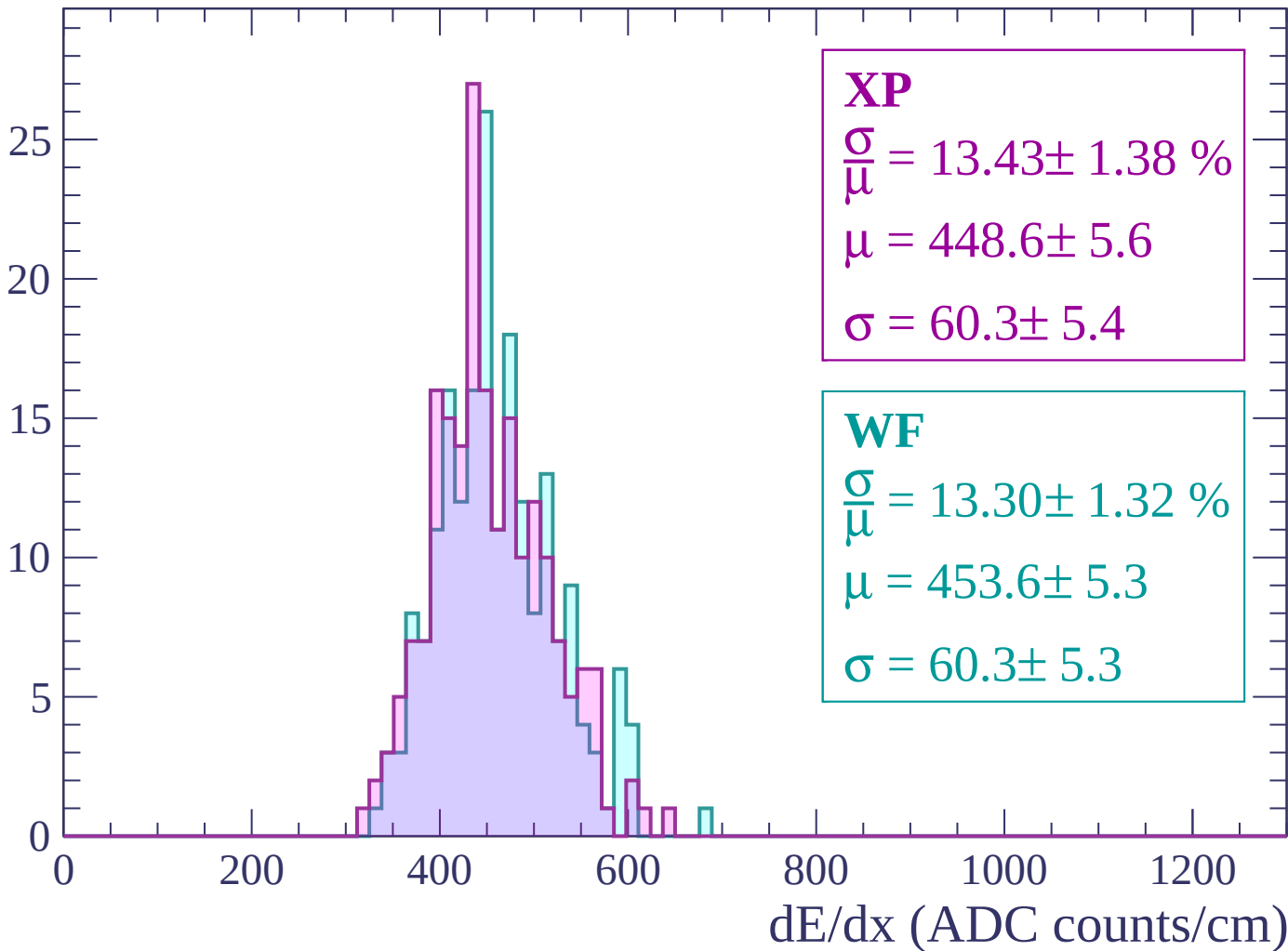
$$\mu = 469.9 \pm 4.9$$

$$\sigma = 54.0 \pm 4.9$$

dE/dx (ADC counts/cm)

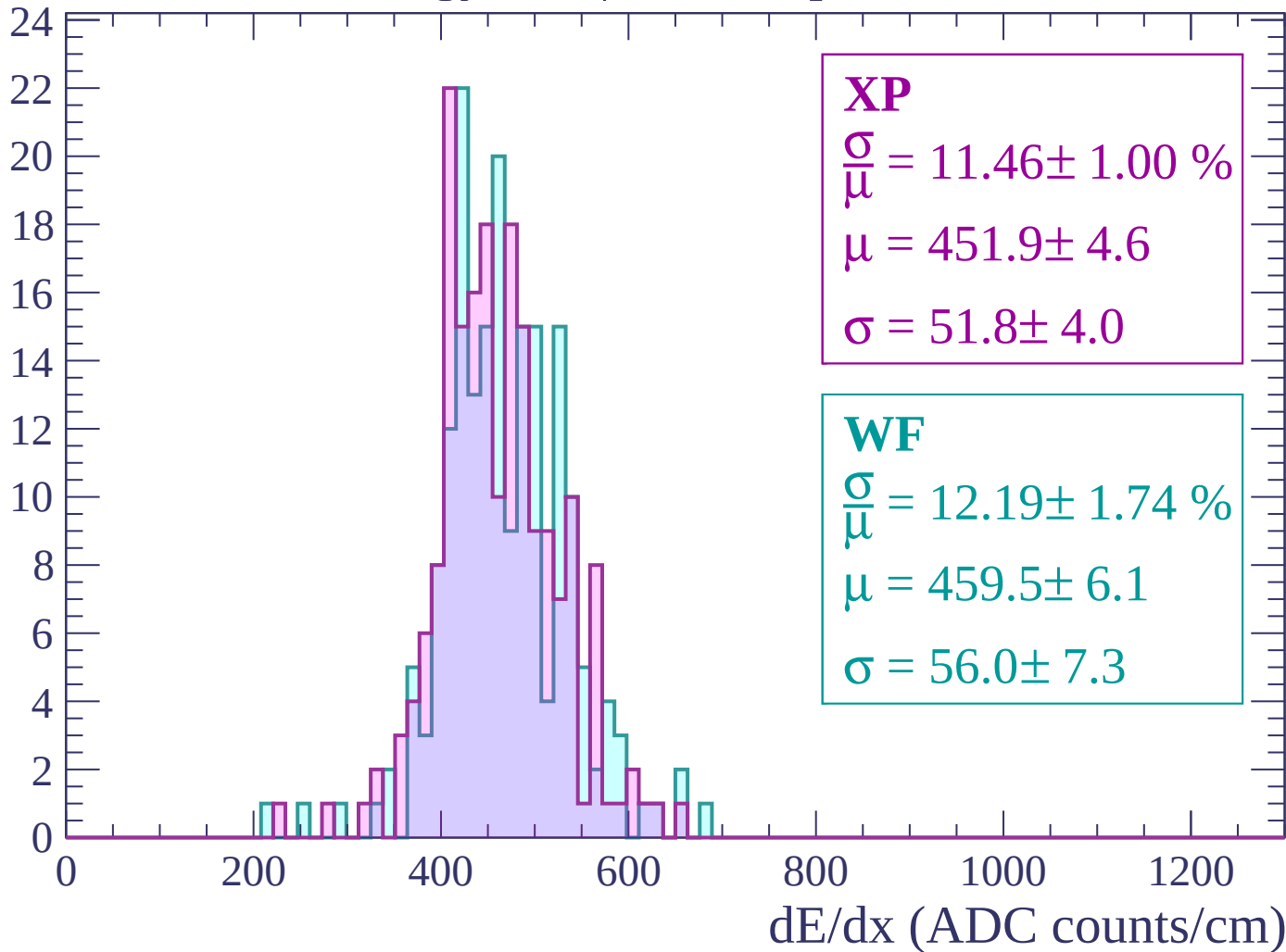
Energy loss | $-1680 < p < -1620$

Count



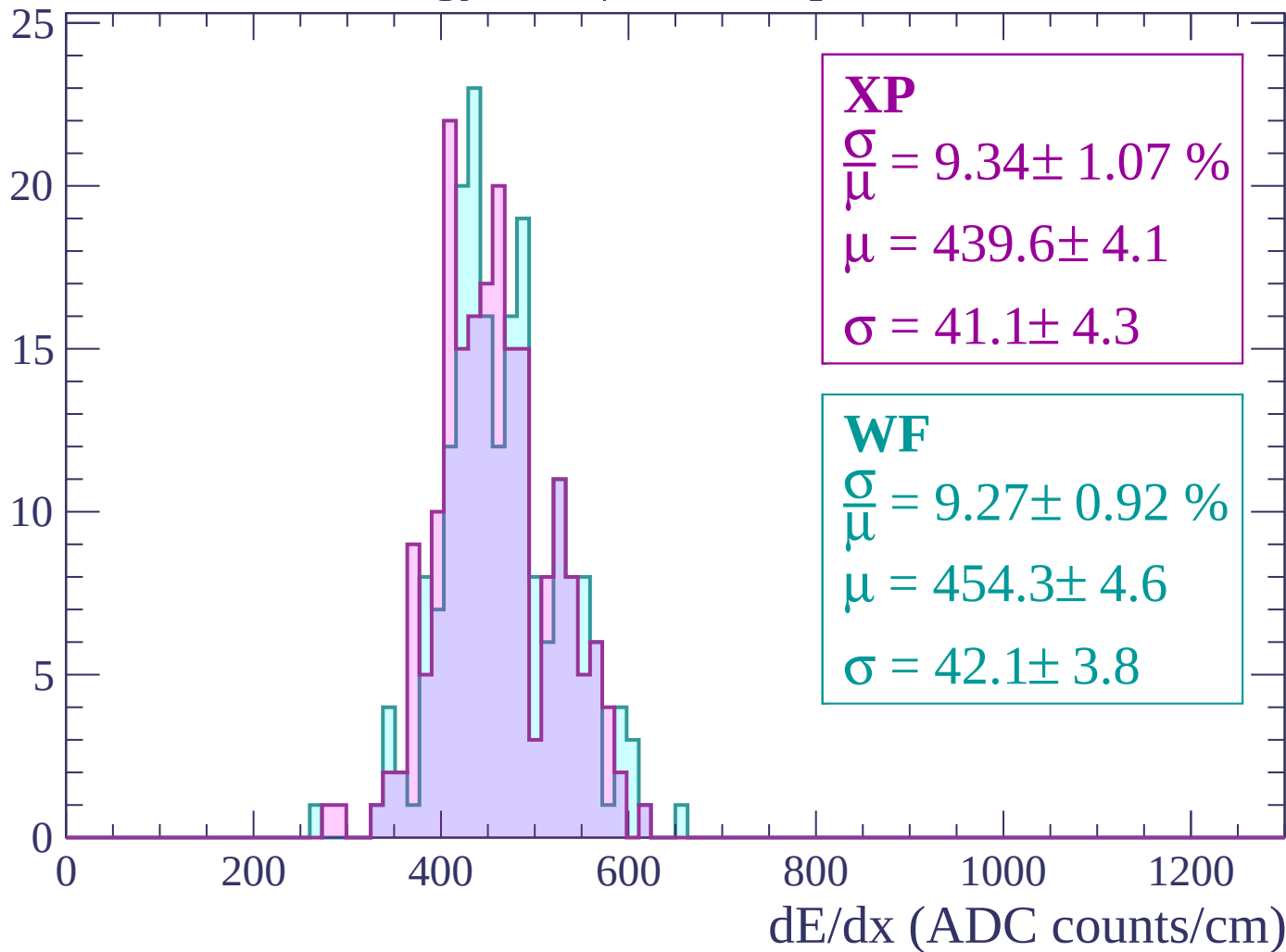
Energy loss | -1620 < p < -1560

Count



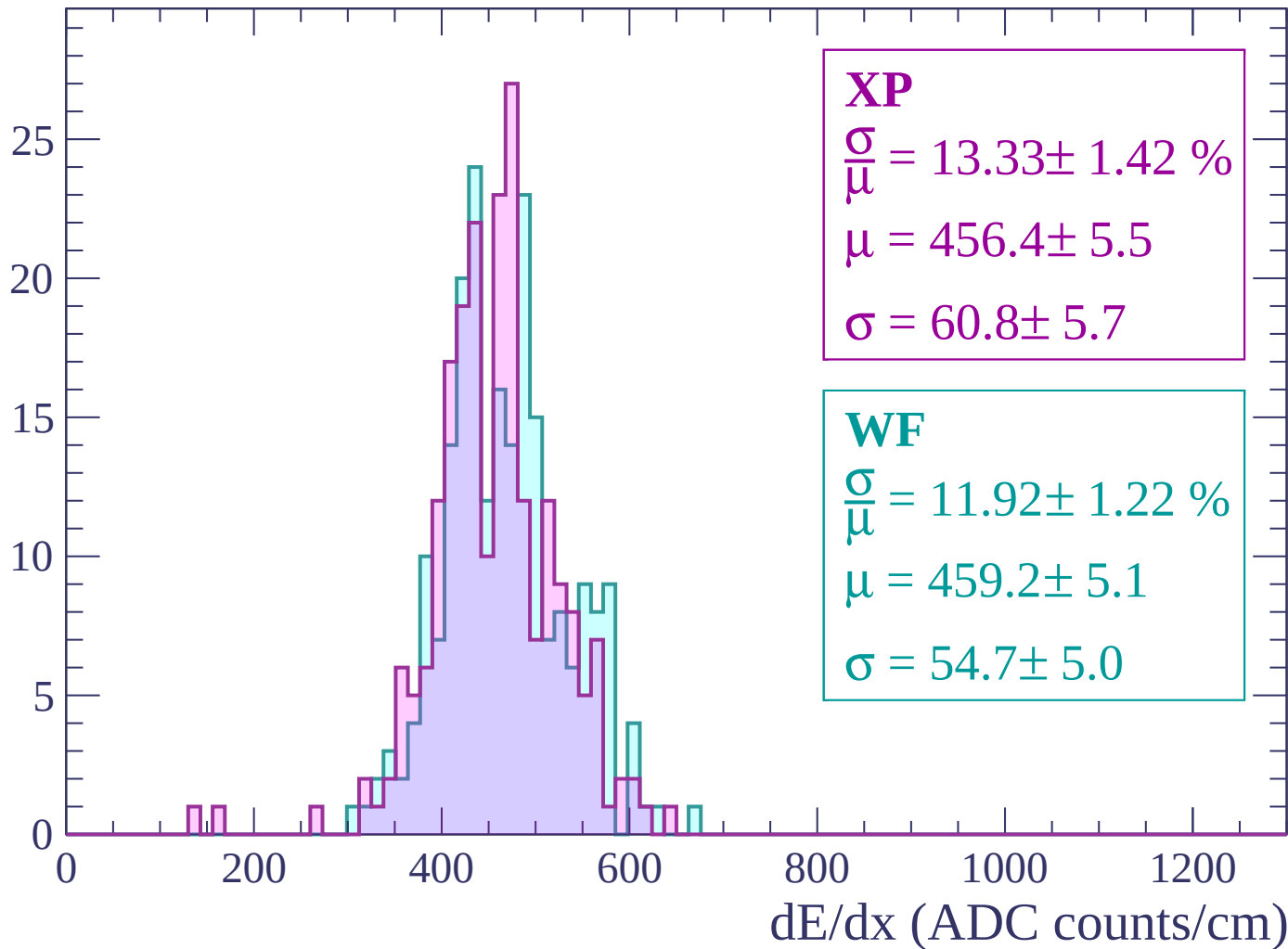
Energy loss | -1560 < p < -1500

Count



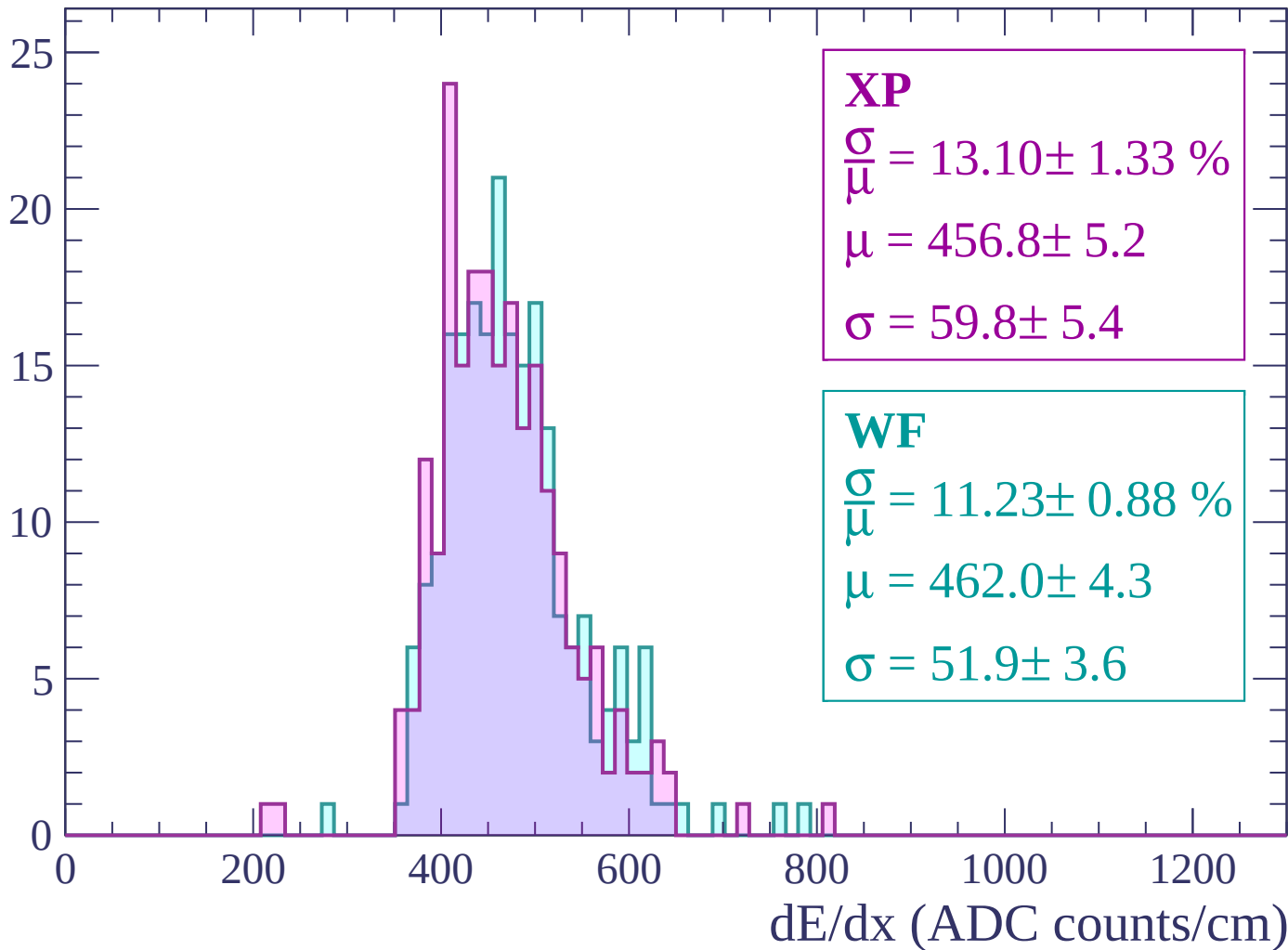
Energy loss | -1500 < p < -1440

Count



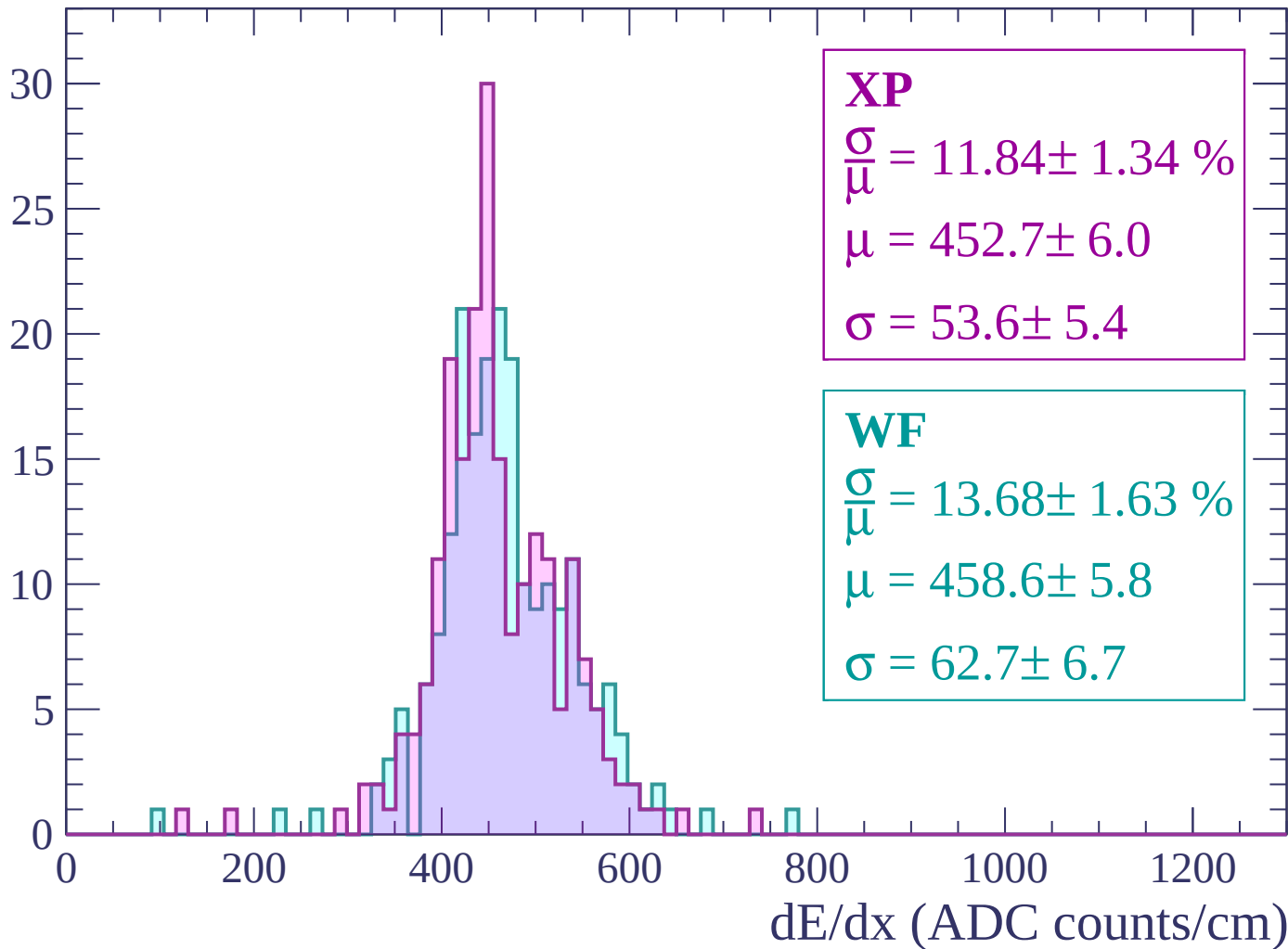
Energy loss | -1440 < p < -1380

Count



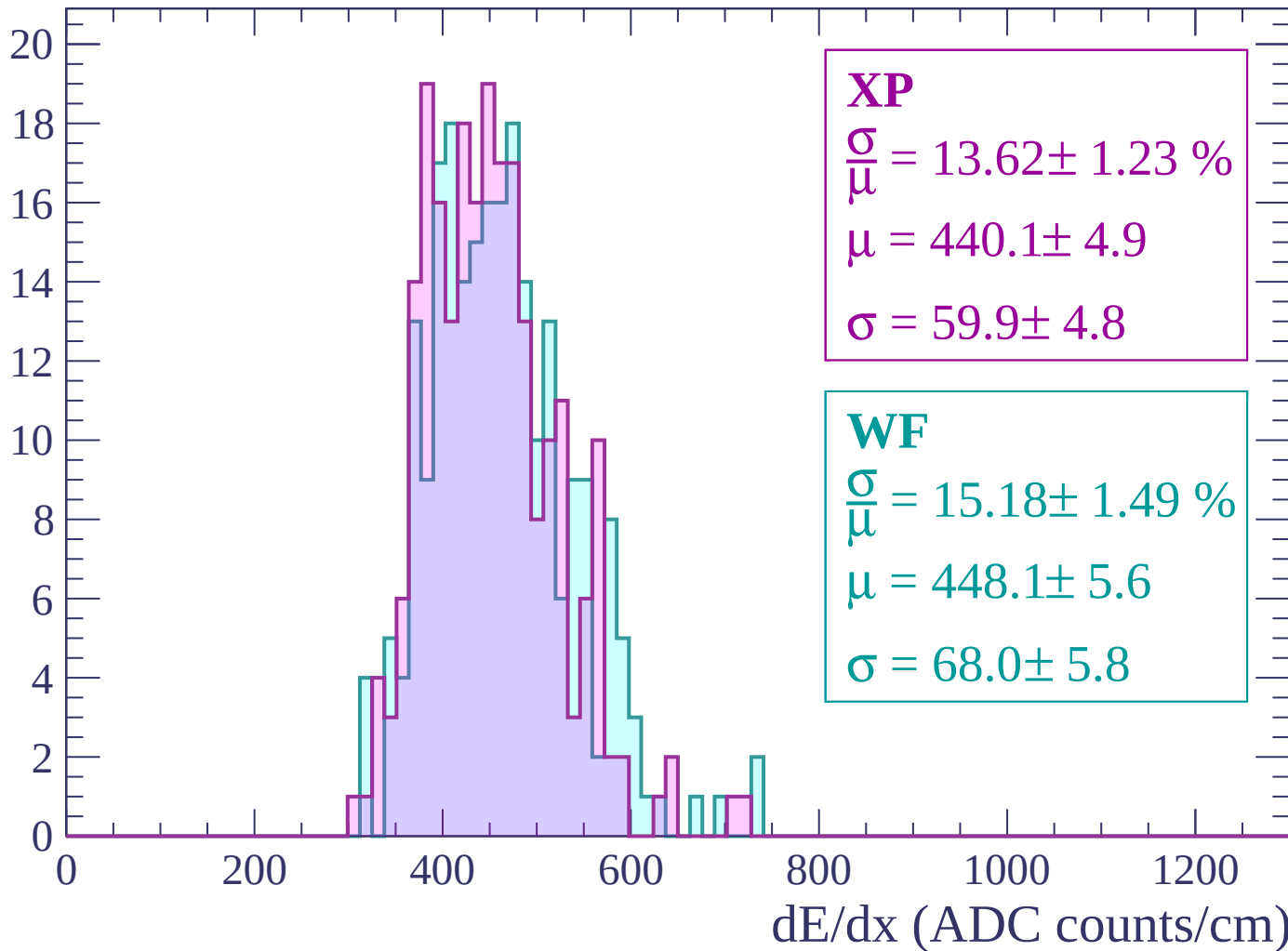
Energy loss | -1380 < p < -1320

Count



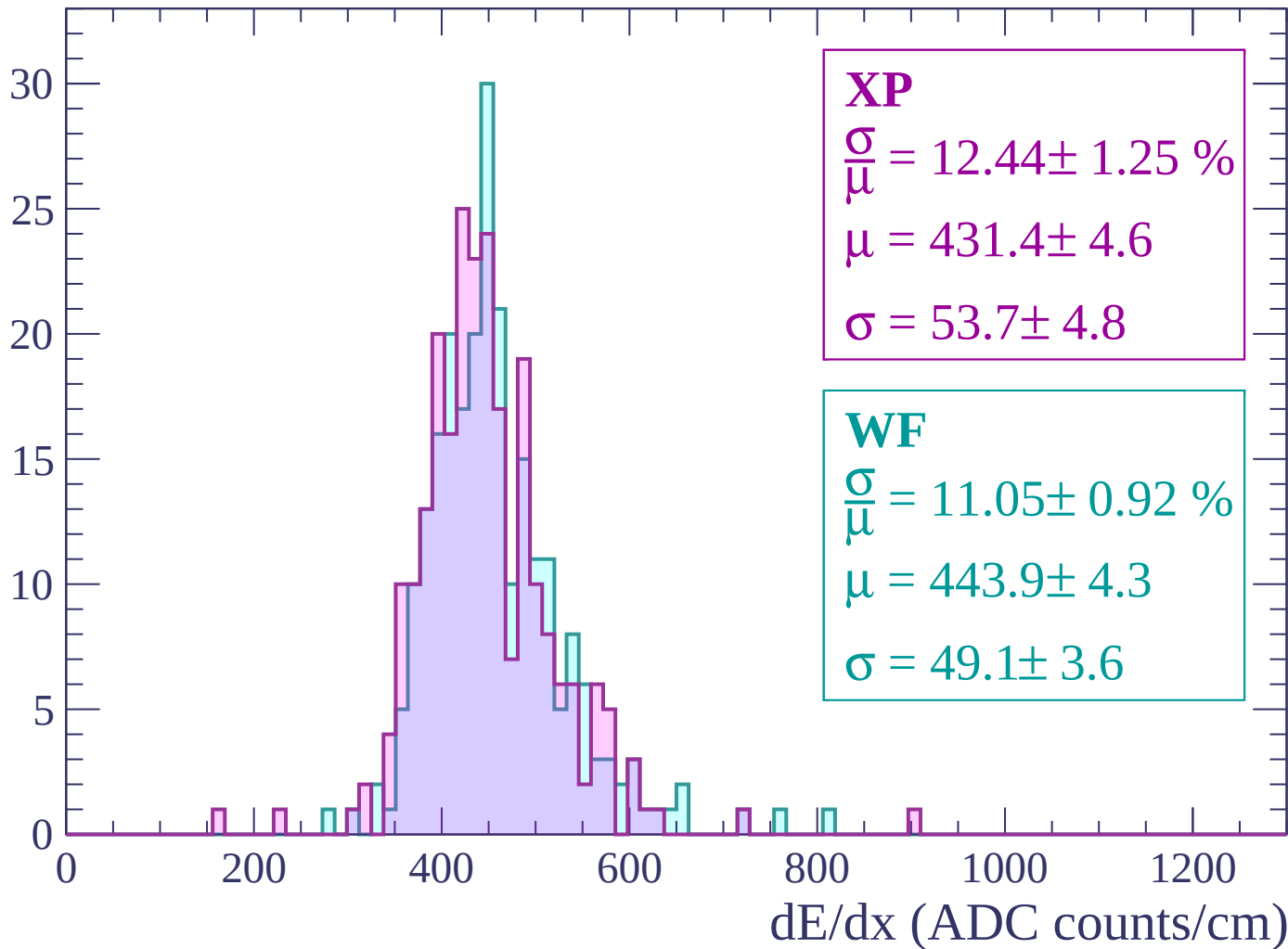
Energy loss | -1320 < p < -1260

Count



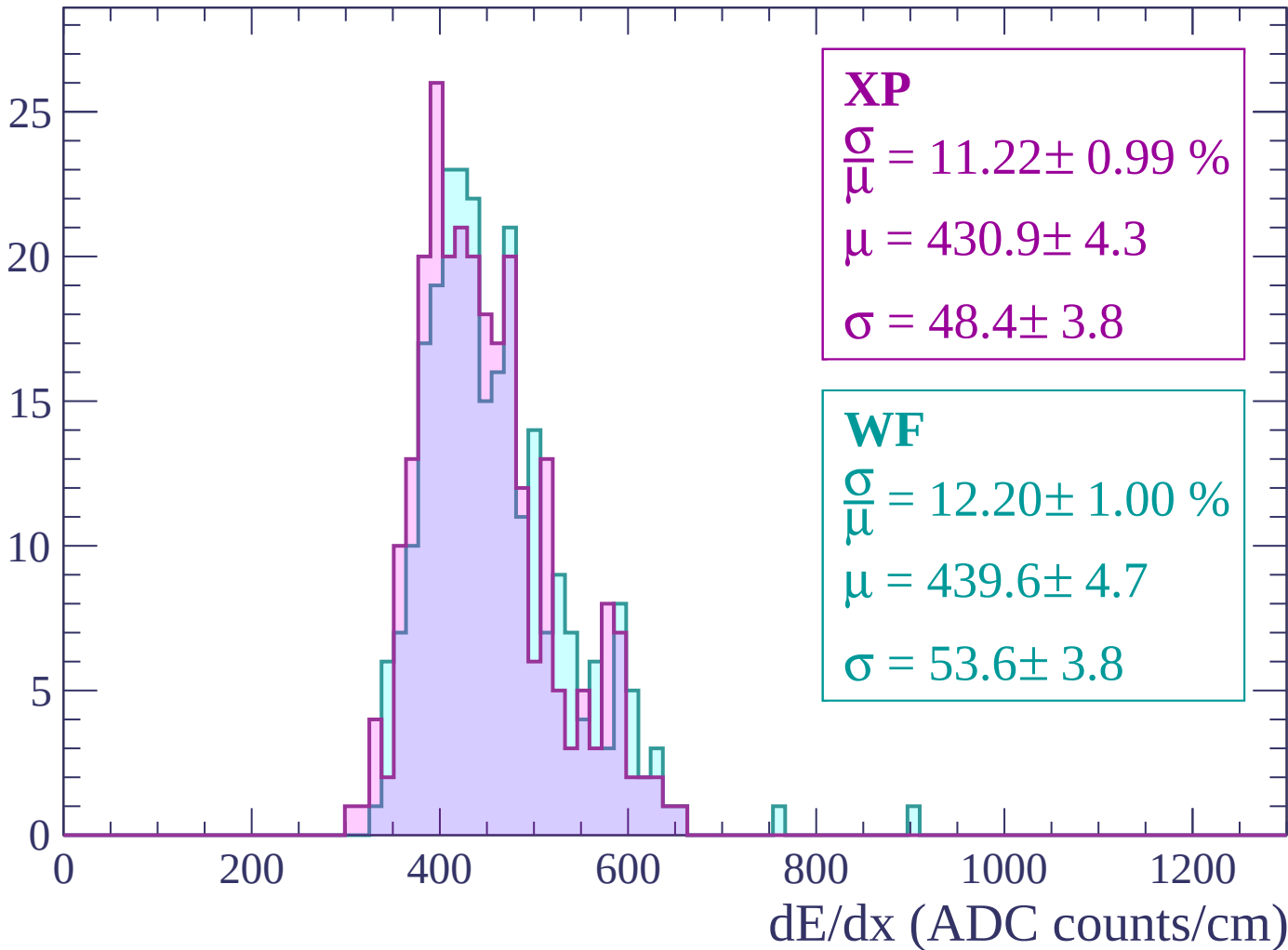
Energy loss | -1260 < p < -1200

Count



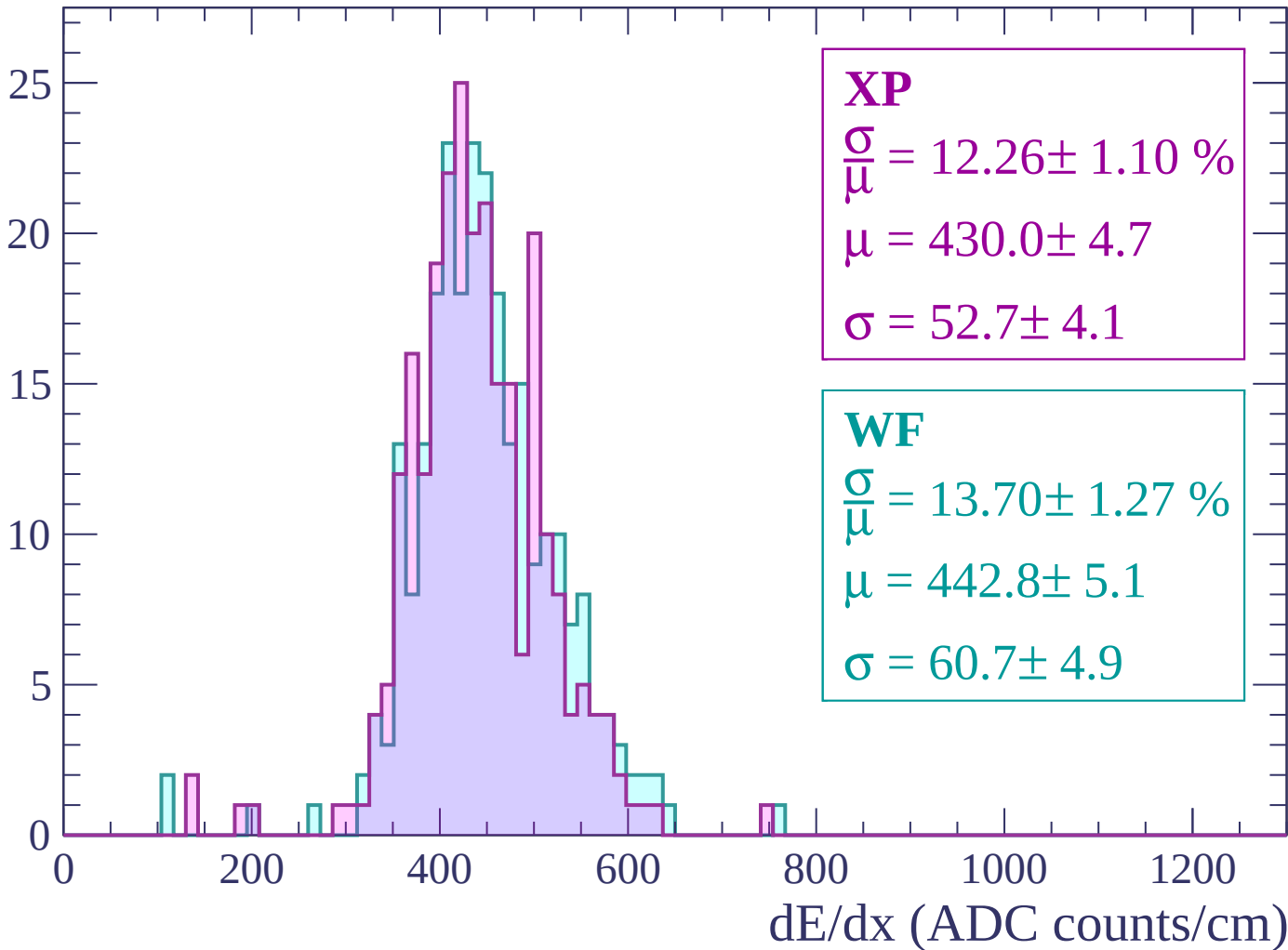
Energy loss | -1200 < p < -1140

Count



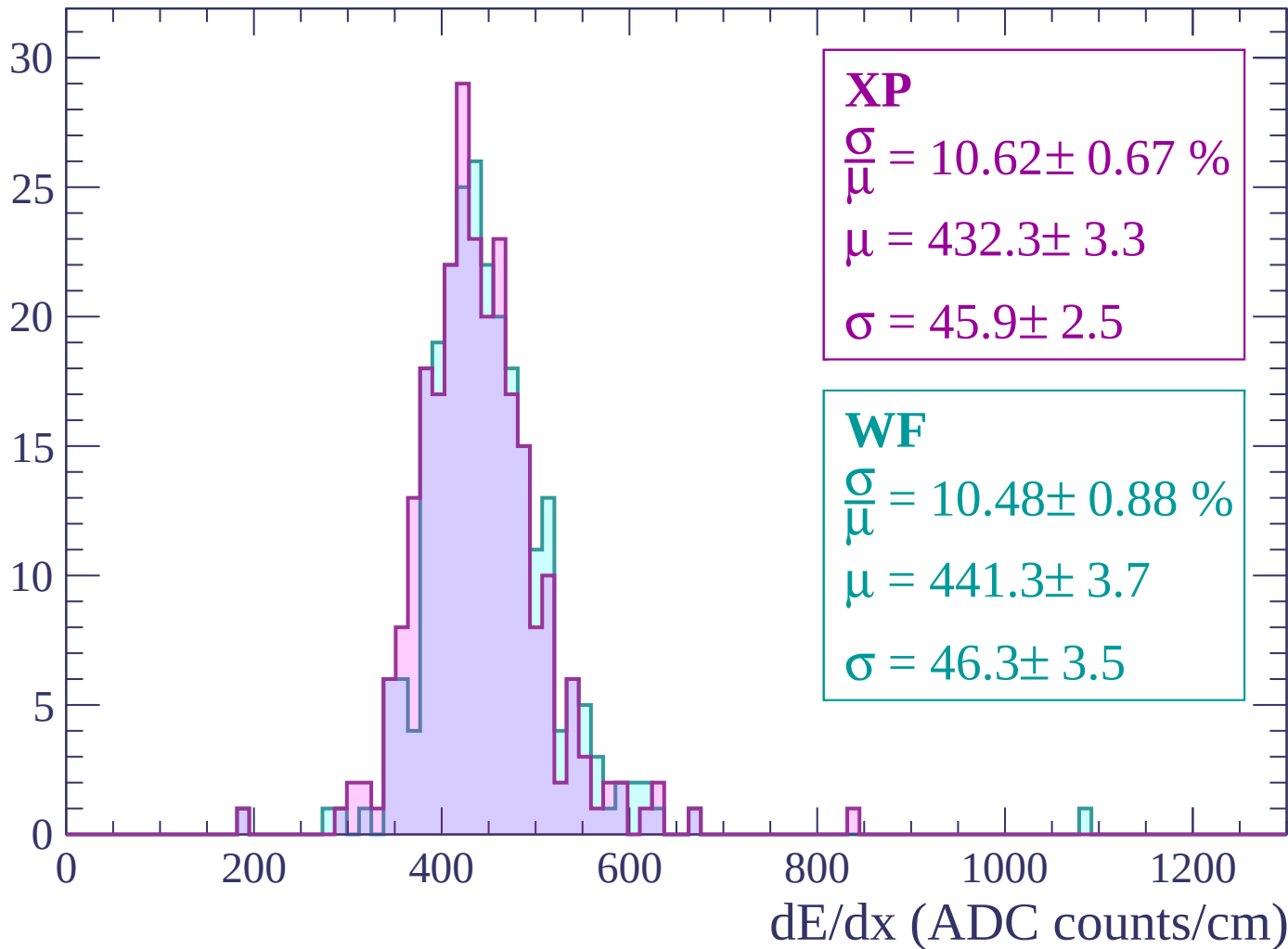
Energy loss | -1140 < p < -1080

Count



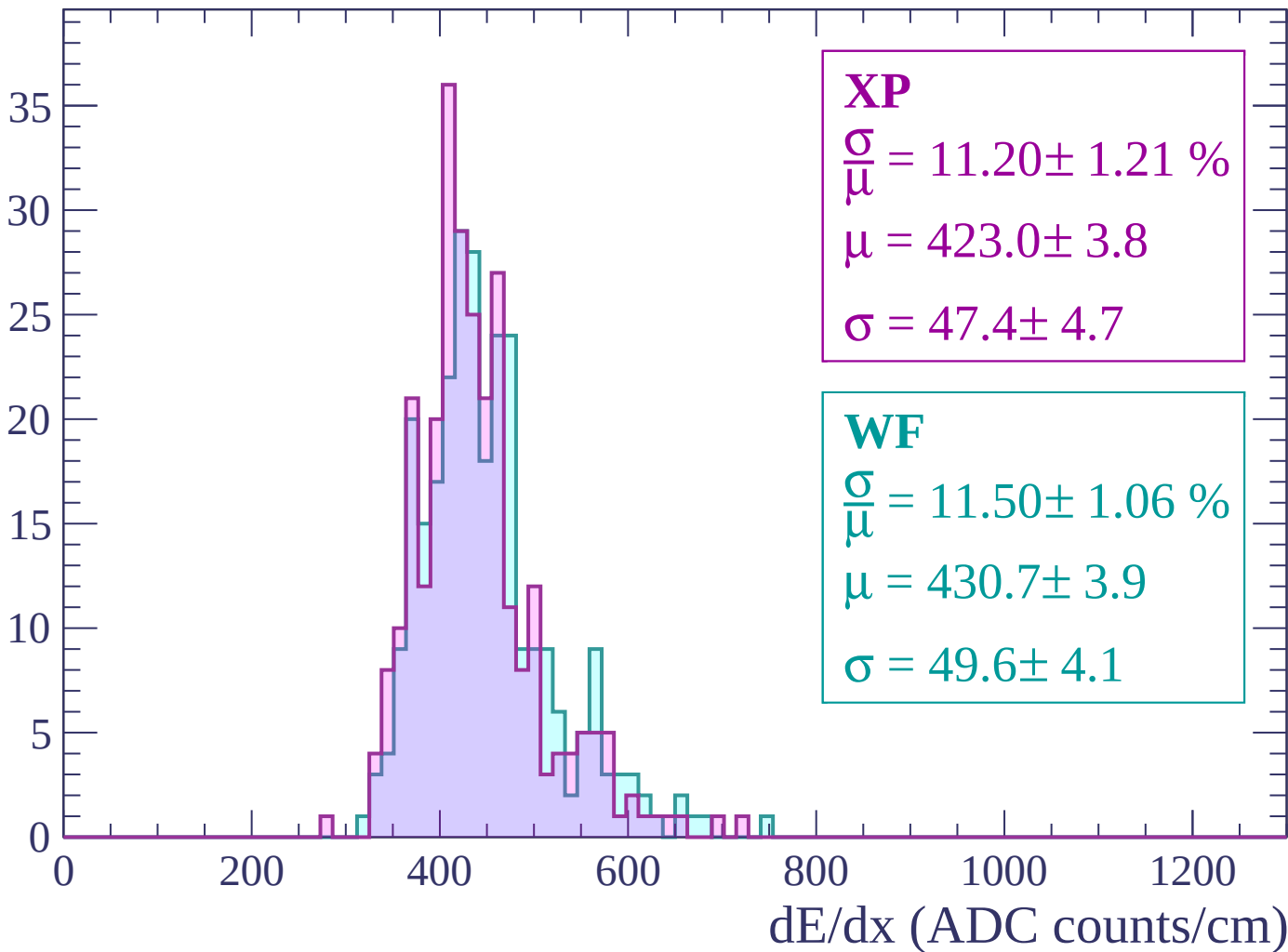
Energy loss | -1080 < p < -1020

Count



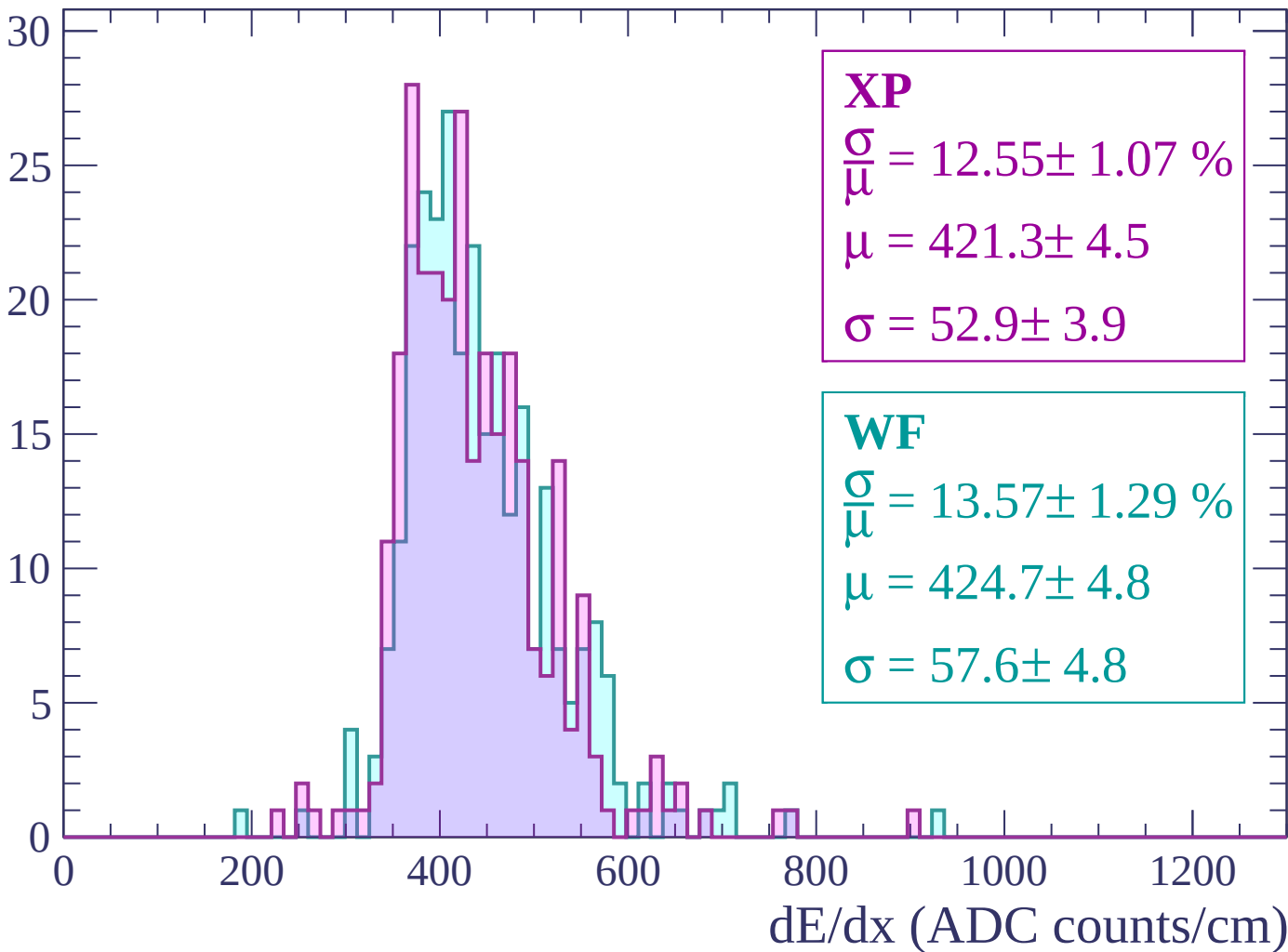
Energy loss | $-1020 < p < -960$

Count



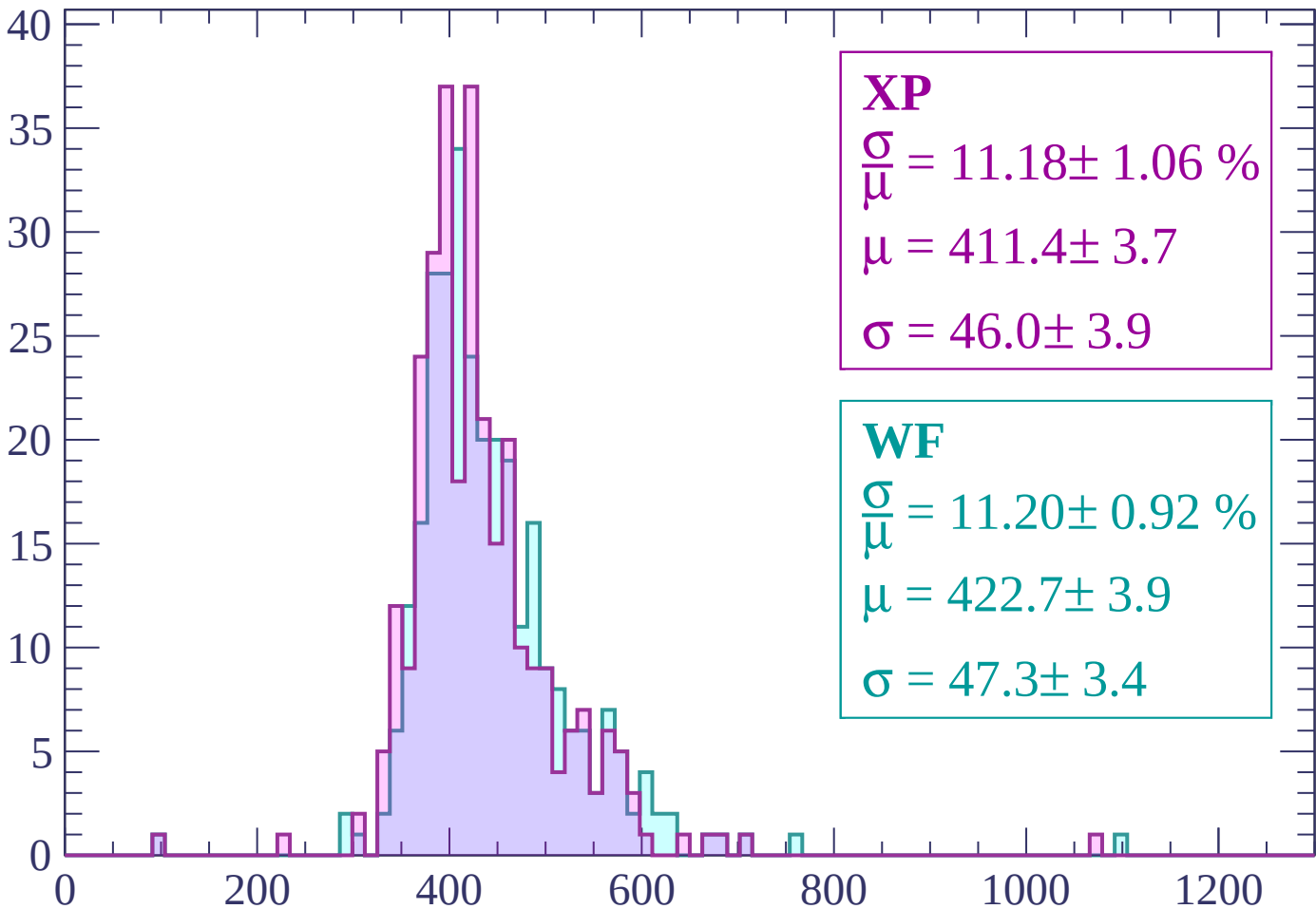
Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 11.18 \pm 1.06 \%$$

$$\mu = 411.4 \pm 3.7$$

$$\sigma = 46.0 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 11.20 \pm 0.92 \%$$

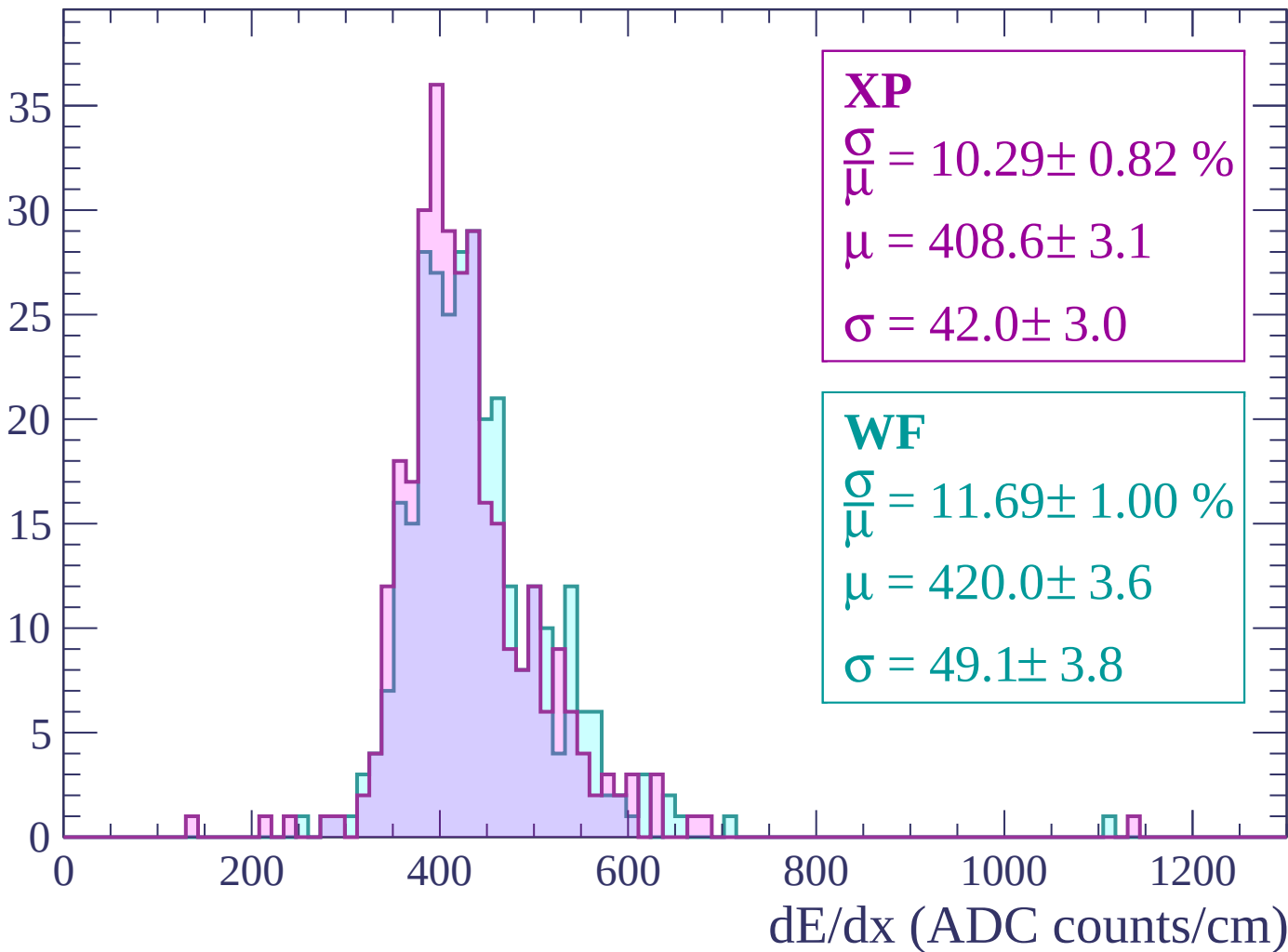
$$\mu = 422.7 \pm 3.9$$

$$\sigma = 47.3 \pm 3.4$$

dE/dx (ADC counts/cm)

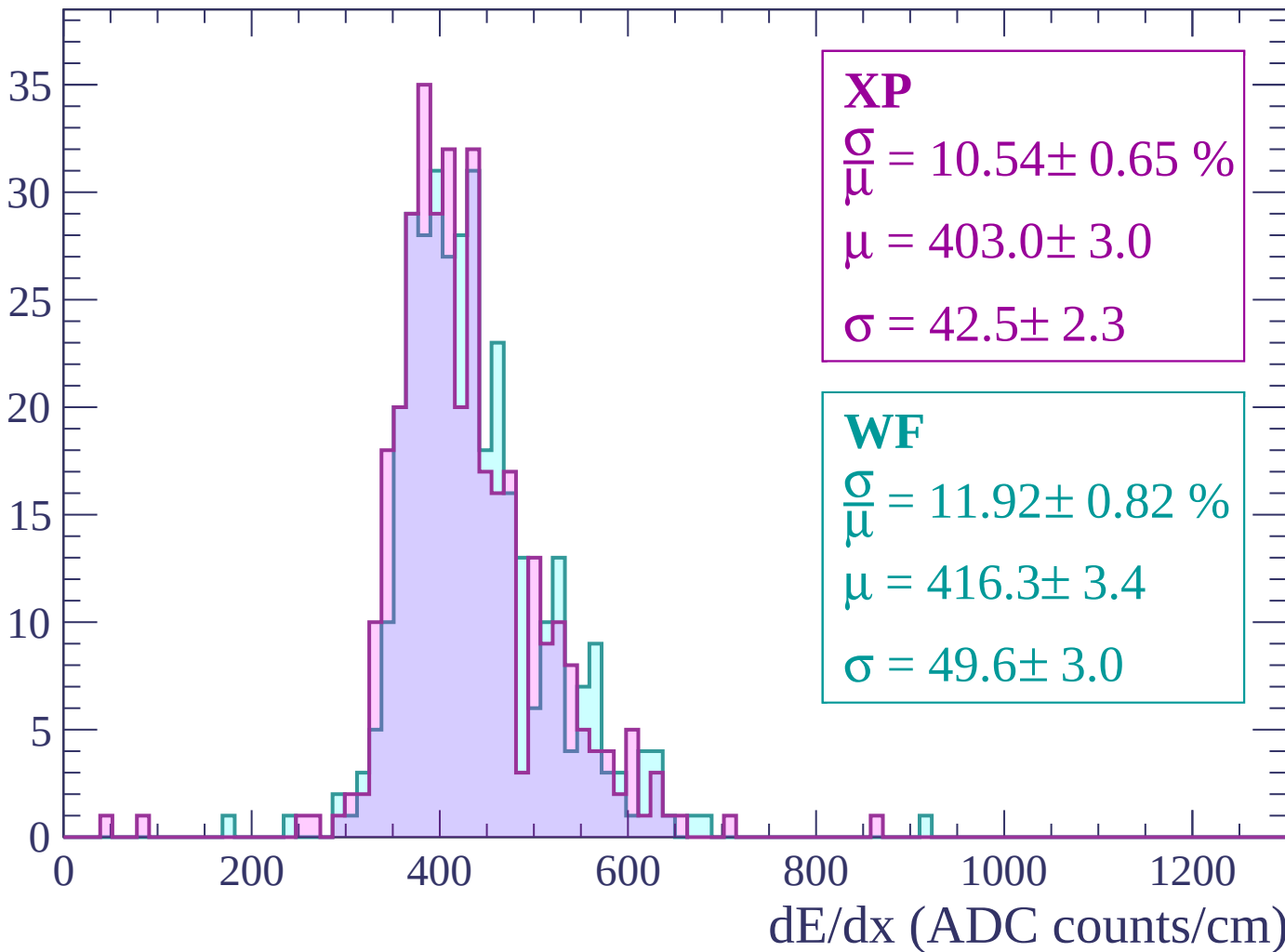
Energy loss | $-840 < p < -780$

Count



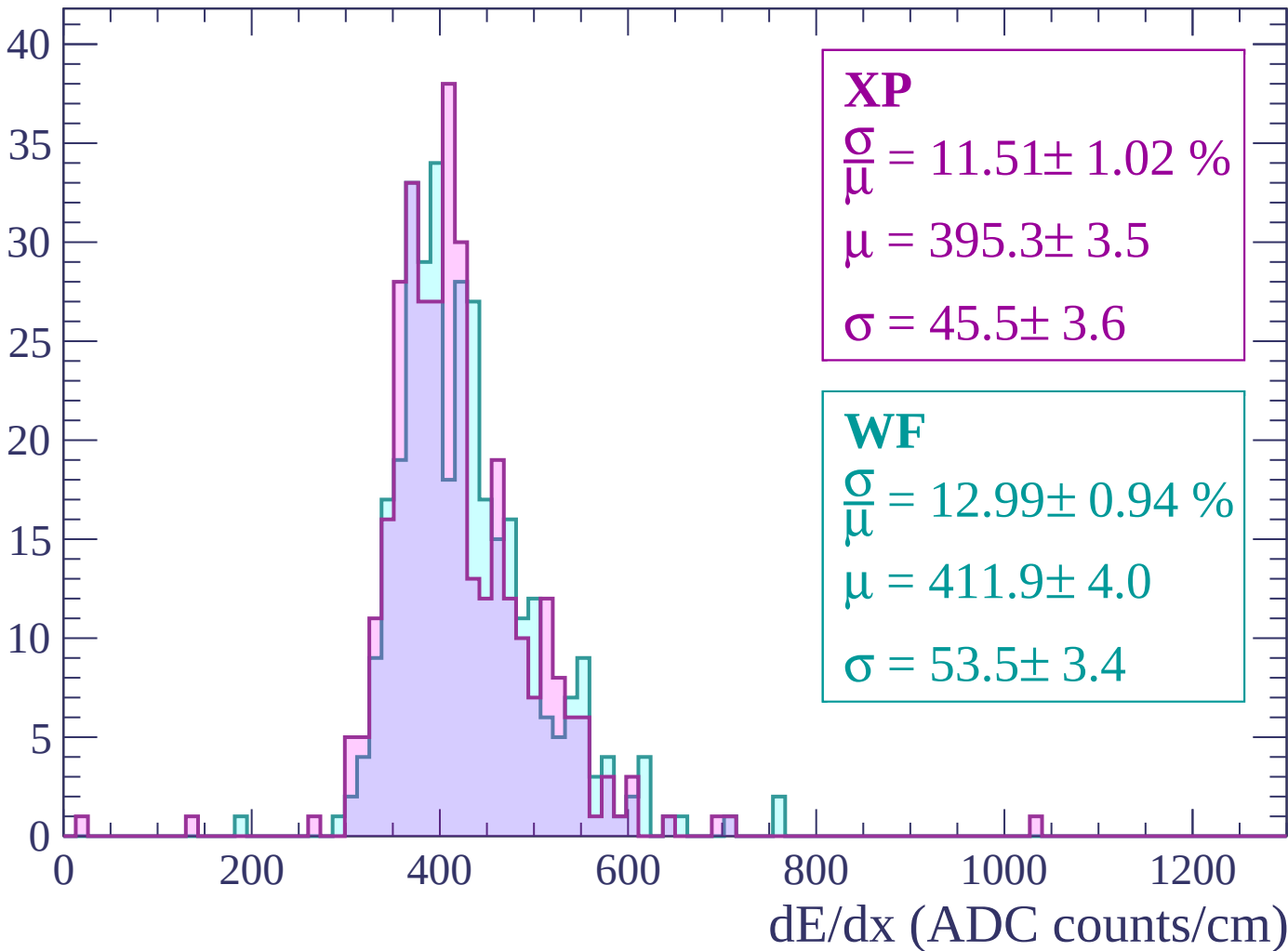
Energy loss | $-780 < p < -720$

Count



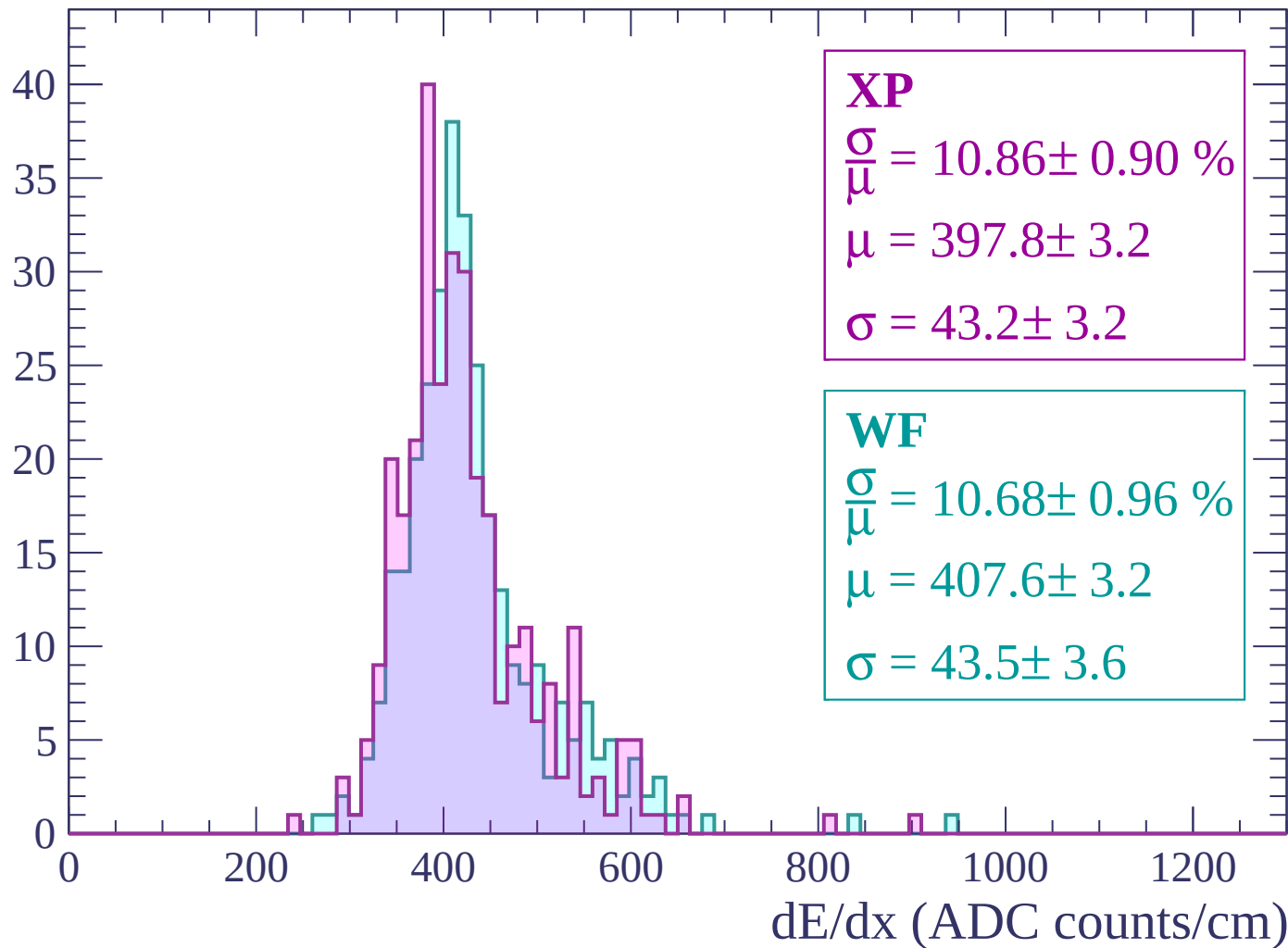
Energy loss | $-720 < p < -660$

Count



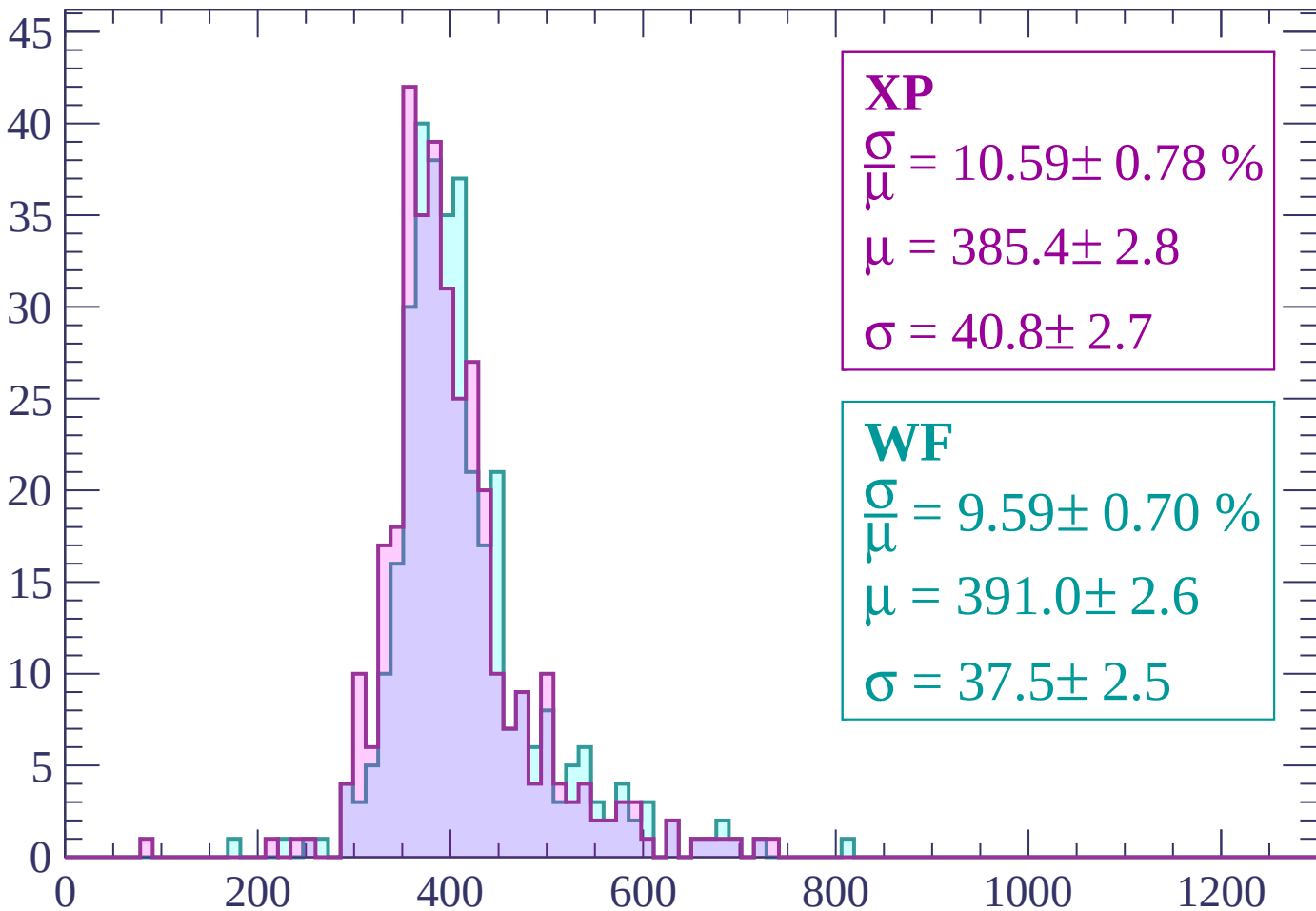
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



XP

$$\frac{\sigma}{\mu} = 10.59 \pm 0.78 \%$$

$$\mu = 385.4 \pm 2.8$$

$$\sigma = 40.8 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.70 \%$$

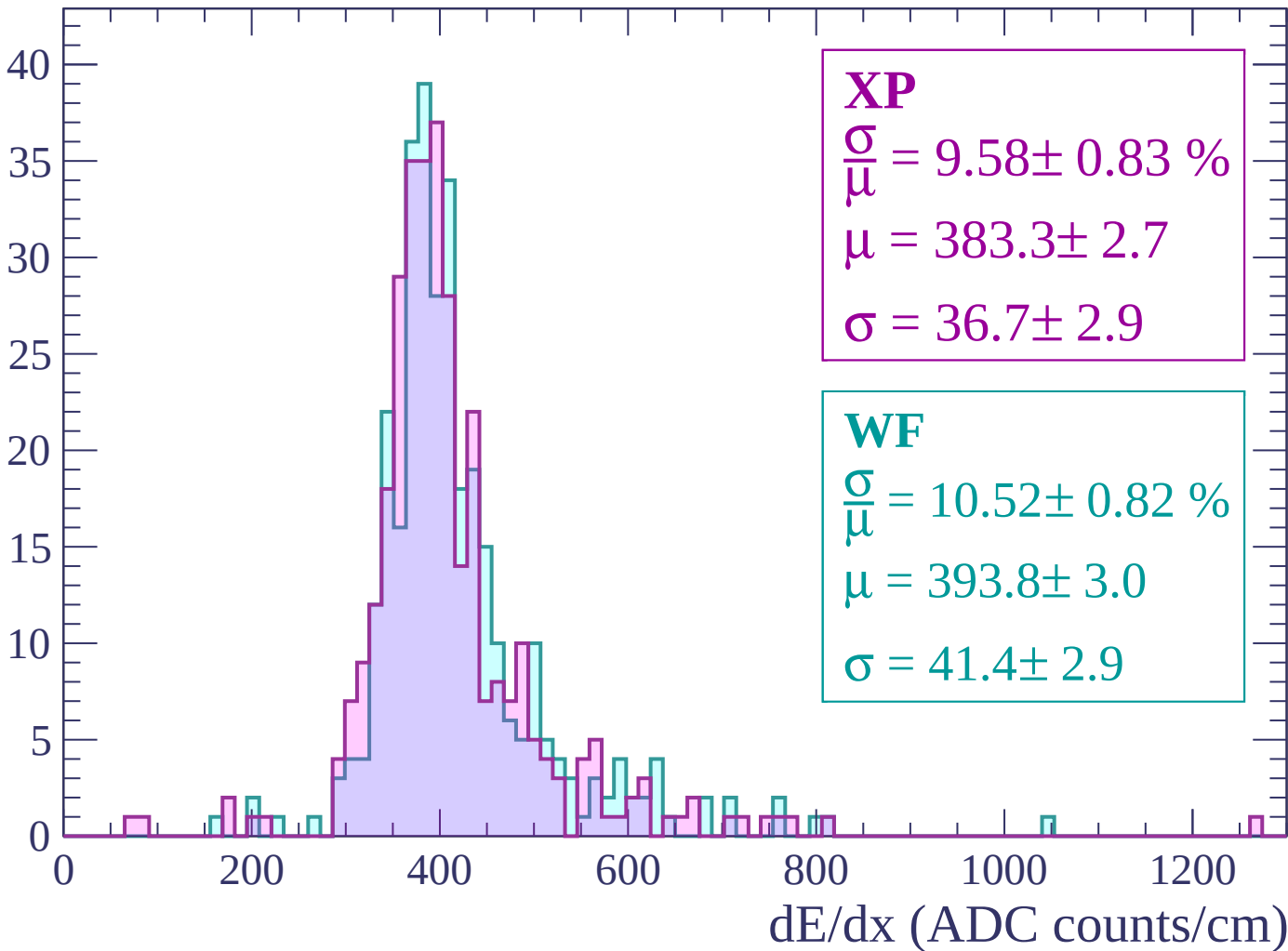
$$\mu = 391.0 \pm 2.6$$

$$\sigma = 37.5 \pm 2.5$$

dE/dx (ADC counts/cm)

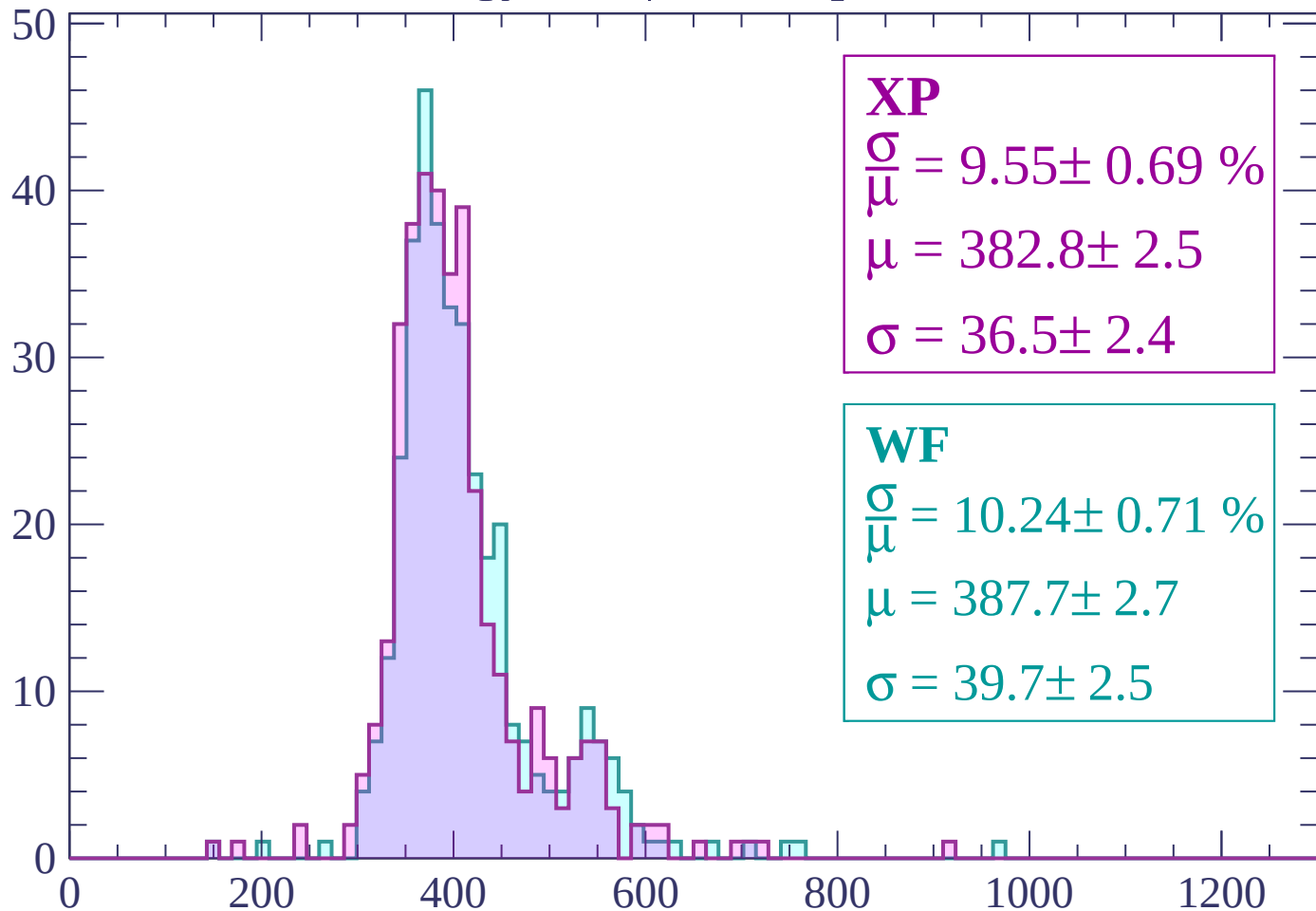
Energy loss | $-540 < p < -480$

Count



Energy loss | $-480 < p < -420$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.69 \%$$

$$\mu = 382.8 \pm 2.5$$

$$\sigma = 36.5 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.71 \%$$

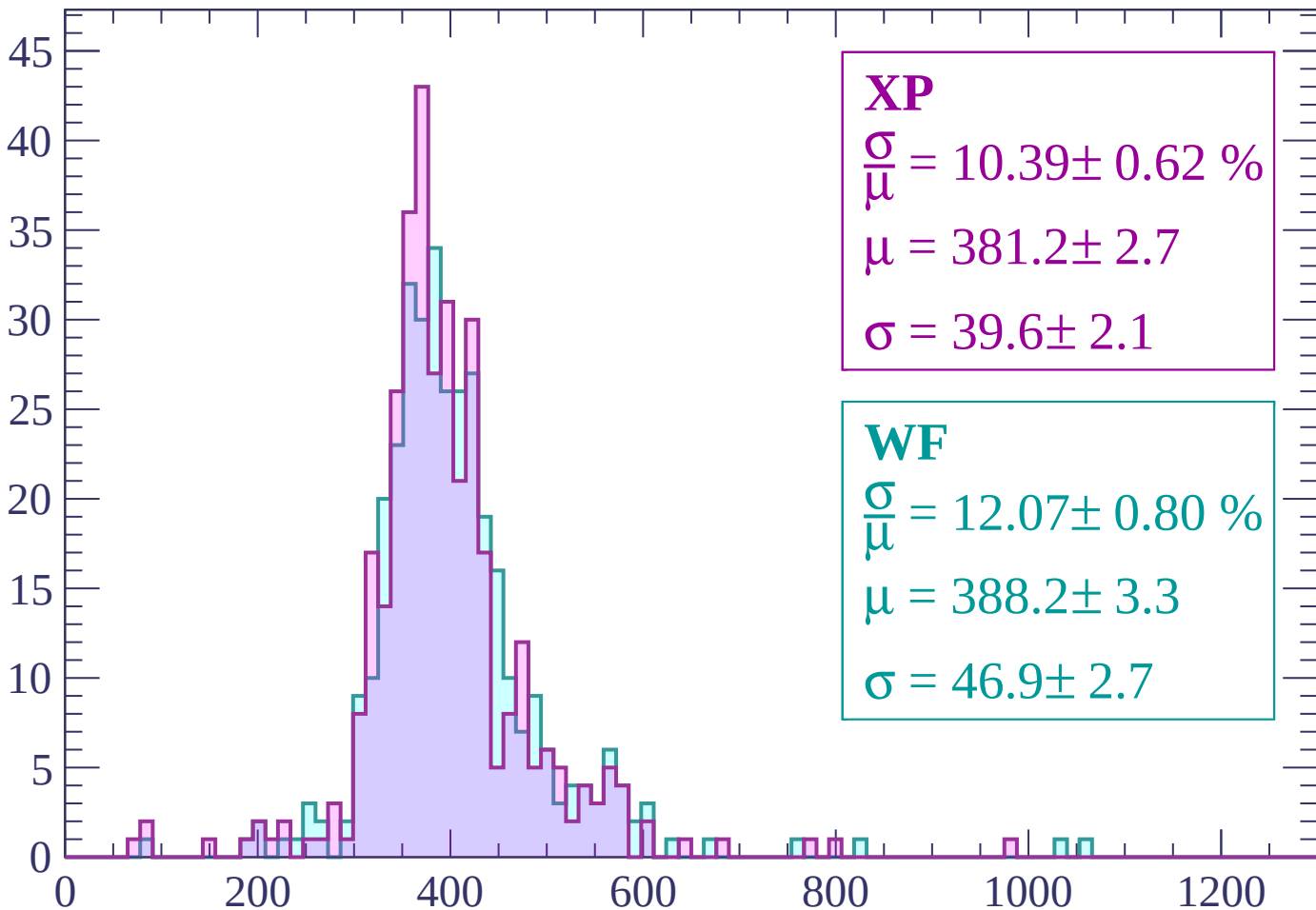
$$\mu = 387.7 \pm 2.7$$

$$\sigma = 39.7 \pm 2.5$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 10.39 \pm 0.62 \%$$

$$\mu = 381.2 \pm 2.7$$

$$\sigma = 39.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 12.07 \pm 0.80 \%$$

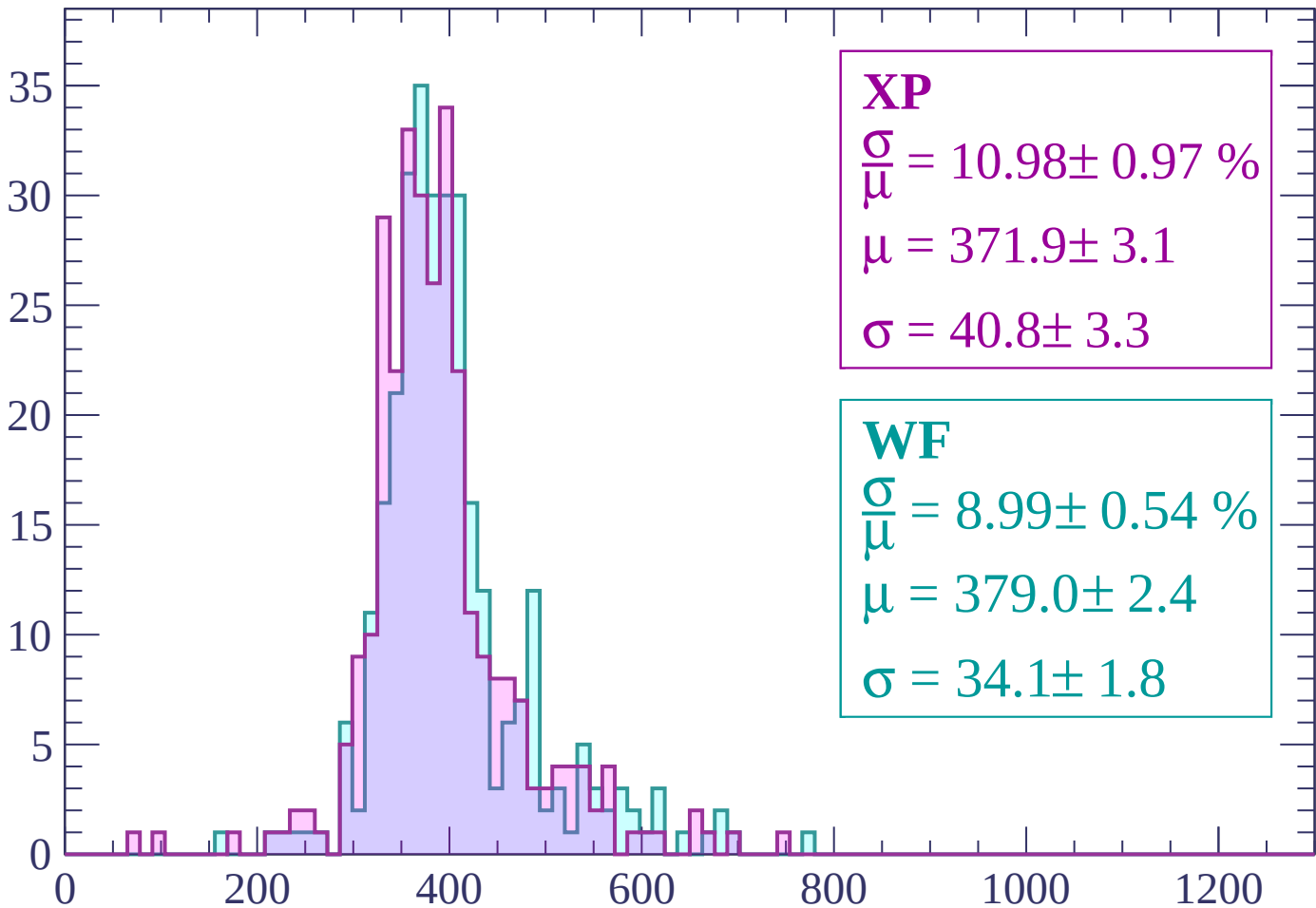
$$\mu = 388.2 \pm 3.3$$

$$\sigma = 46.9 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP

$$\frac{\sigma}{\mu} = 10.98 \pm 0.97 \%$$

$$\mu = 371.9 \pm 3.1$$

$$\sigma = 40.8 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.54 \%$$

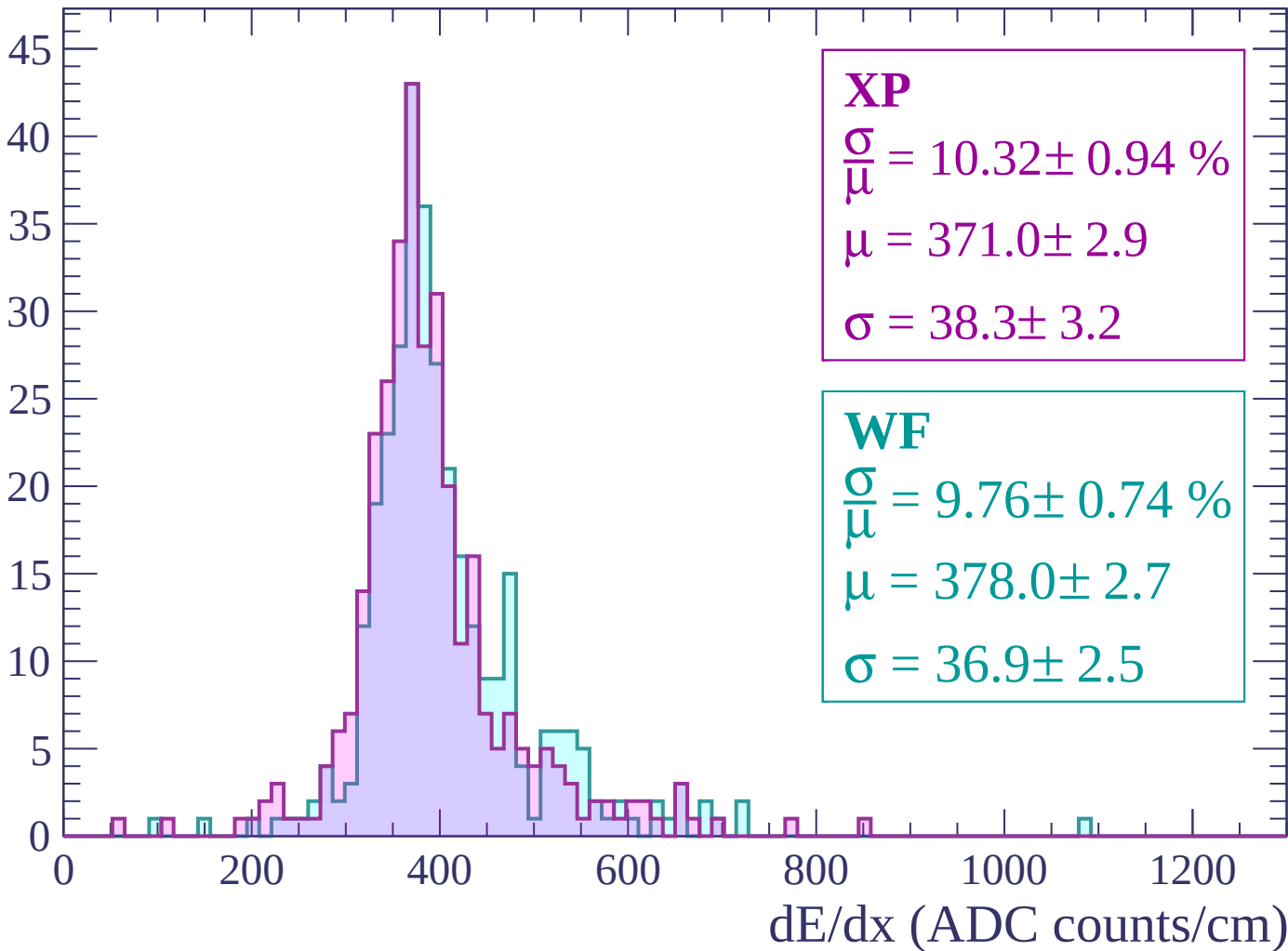
$$\mu = 379.0 \pm 2.4$$

$$\sigma = 34.1 \pm 1.8$$

dE/dx (ADC counts/cm)

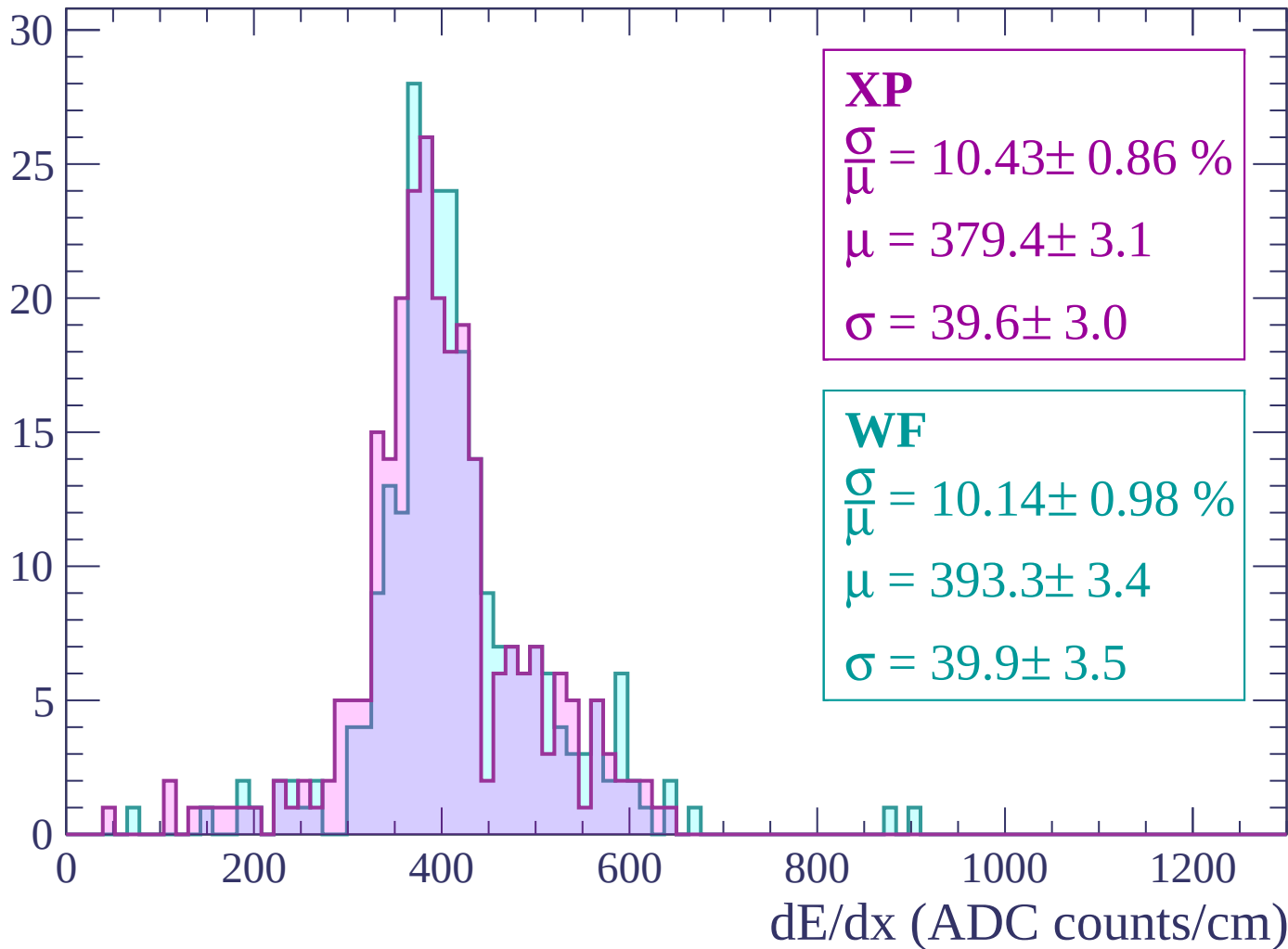
Energy loss | $-300 < p < -240$

Count



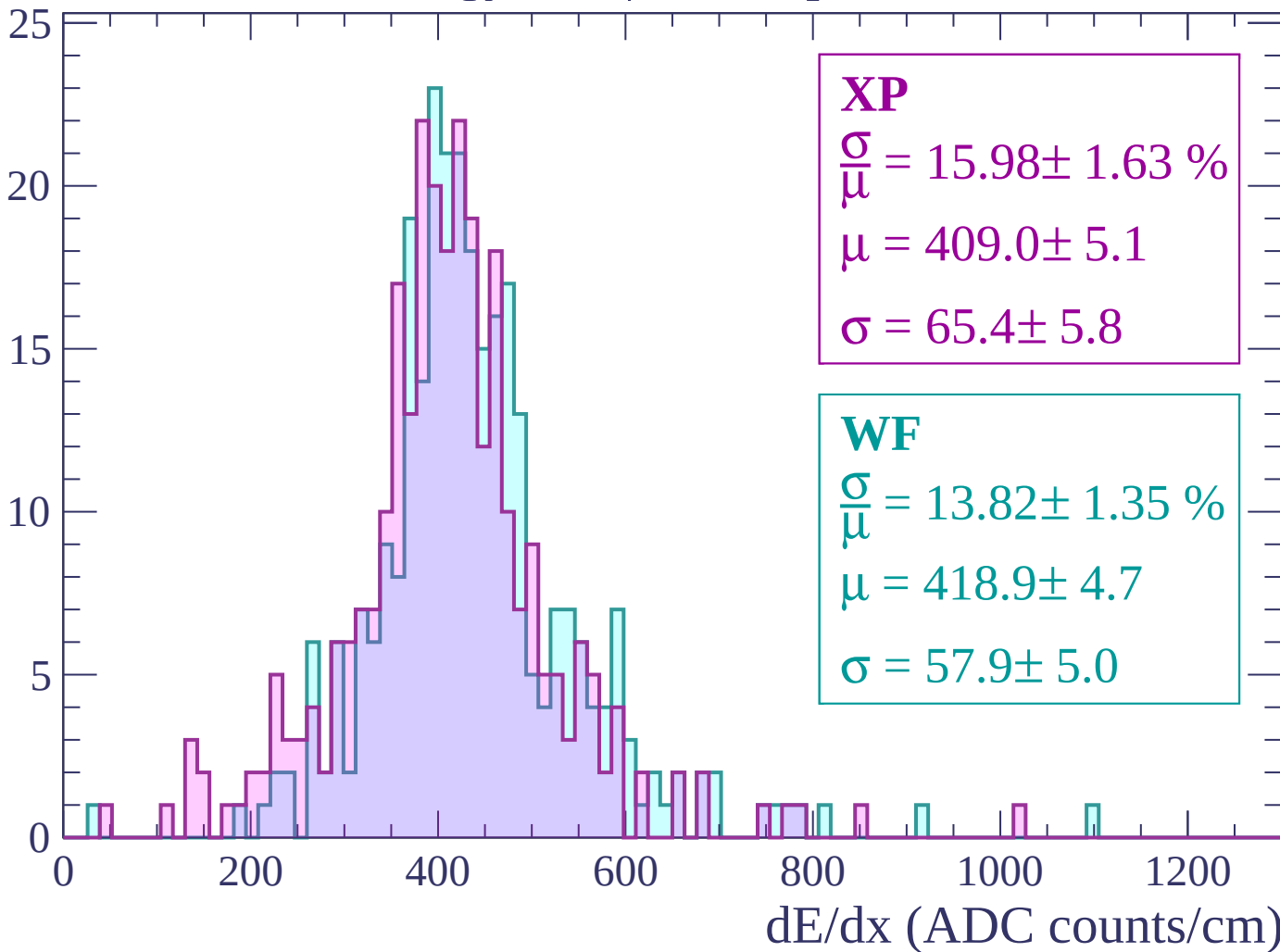
Energy loss | -240 < p < -180

Count



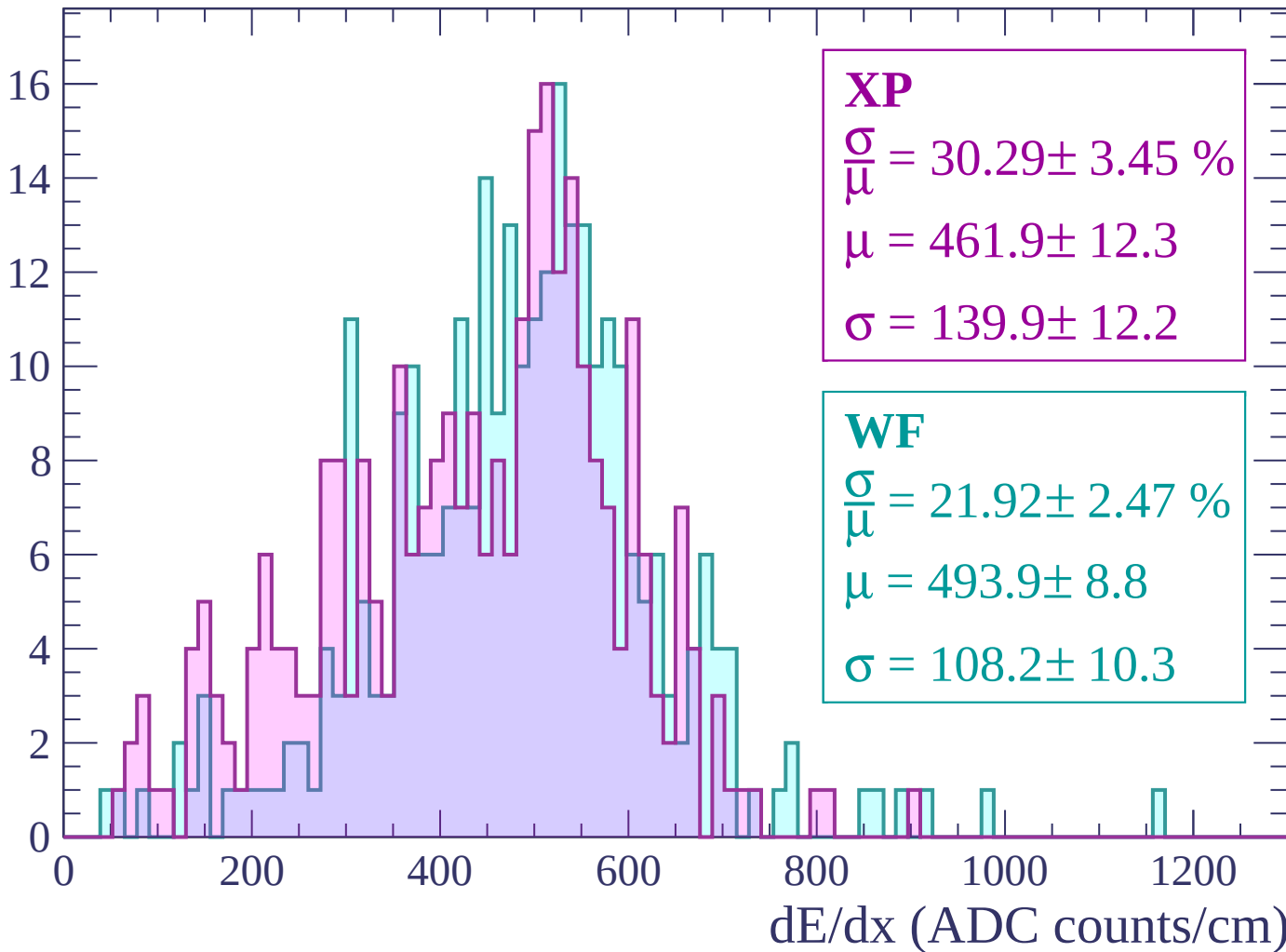
Energy loss | -180 < p < -120

Count



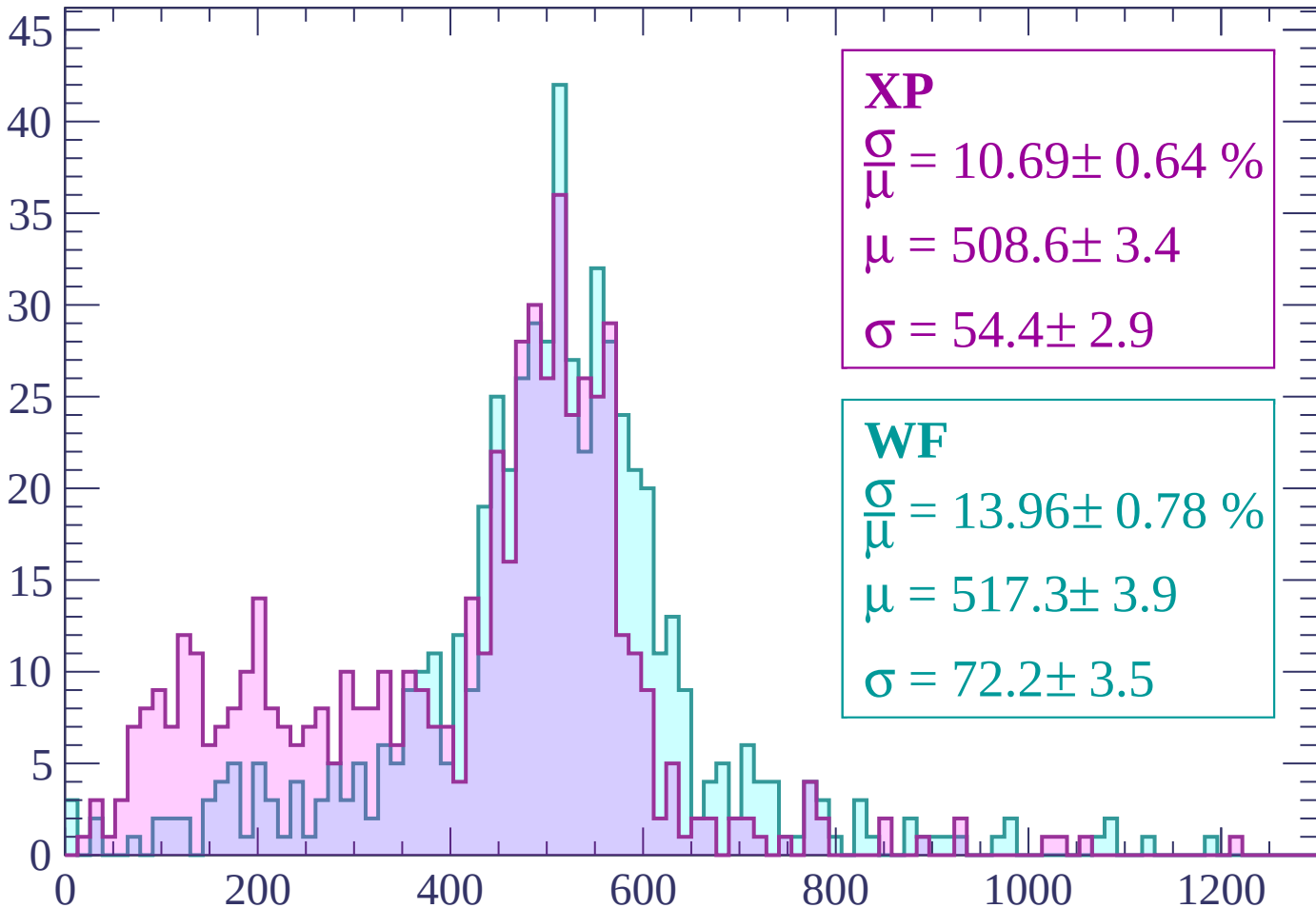
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 10.69 \pm 0.64 \%$$

$$\mu = 508.6 \pm 3.4$$

$$\sigma = 54.4 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 13.96 \pm 0.78 \%$$

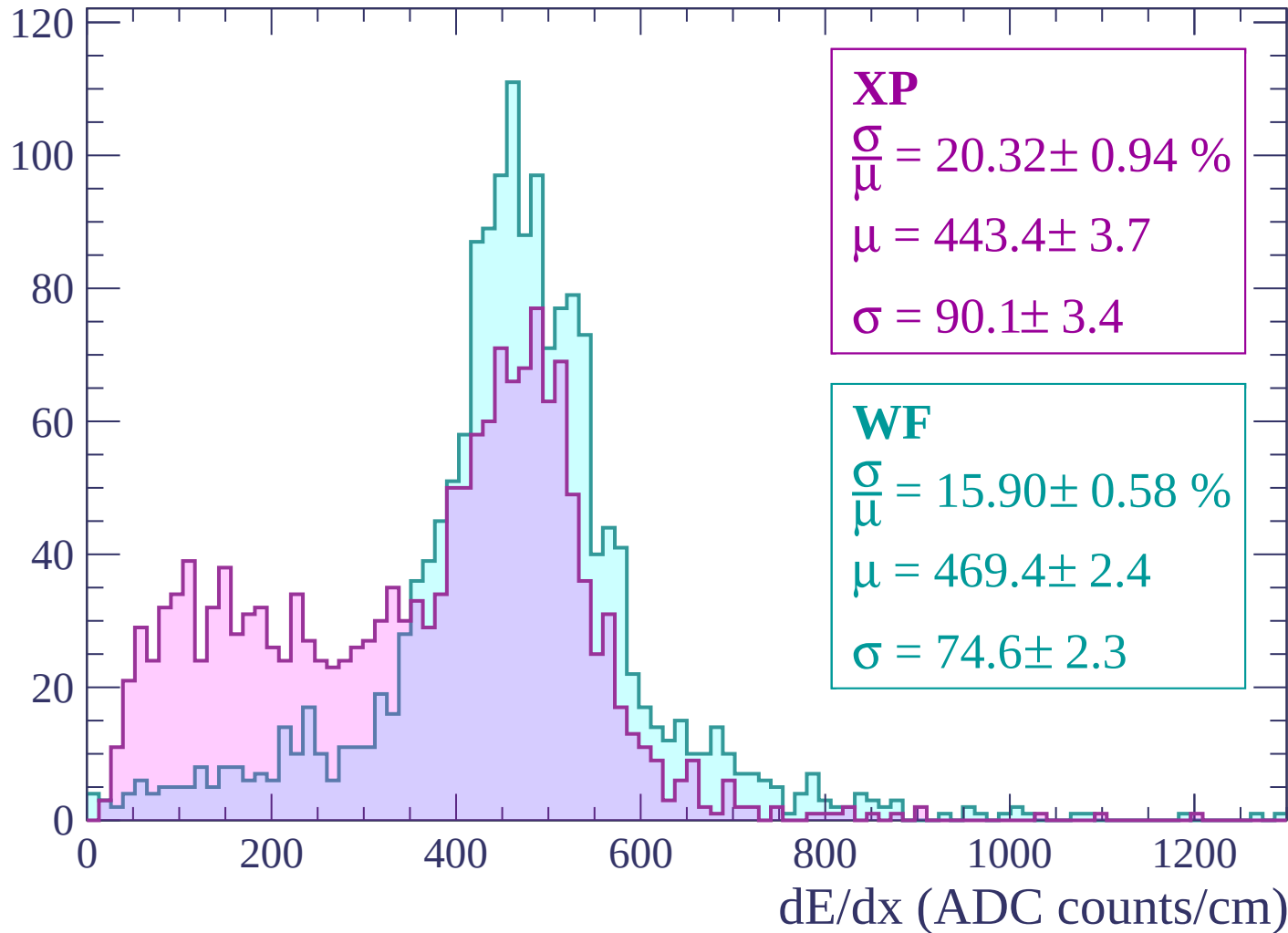
$$\mu = 517.3 \pm 3.9$$

$$\sigma = 72.2 \pm 3.5$$

dE/dx (ADC counts/cm)

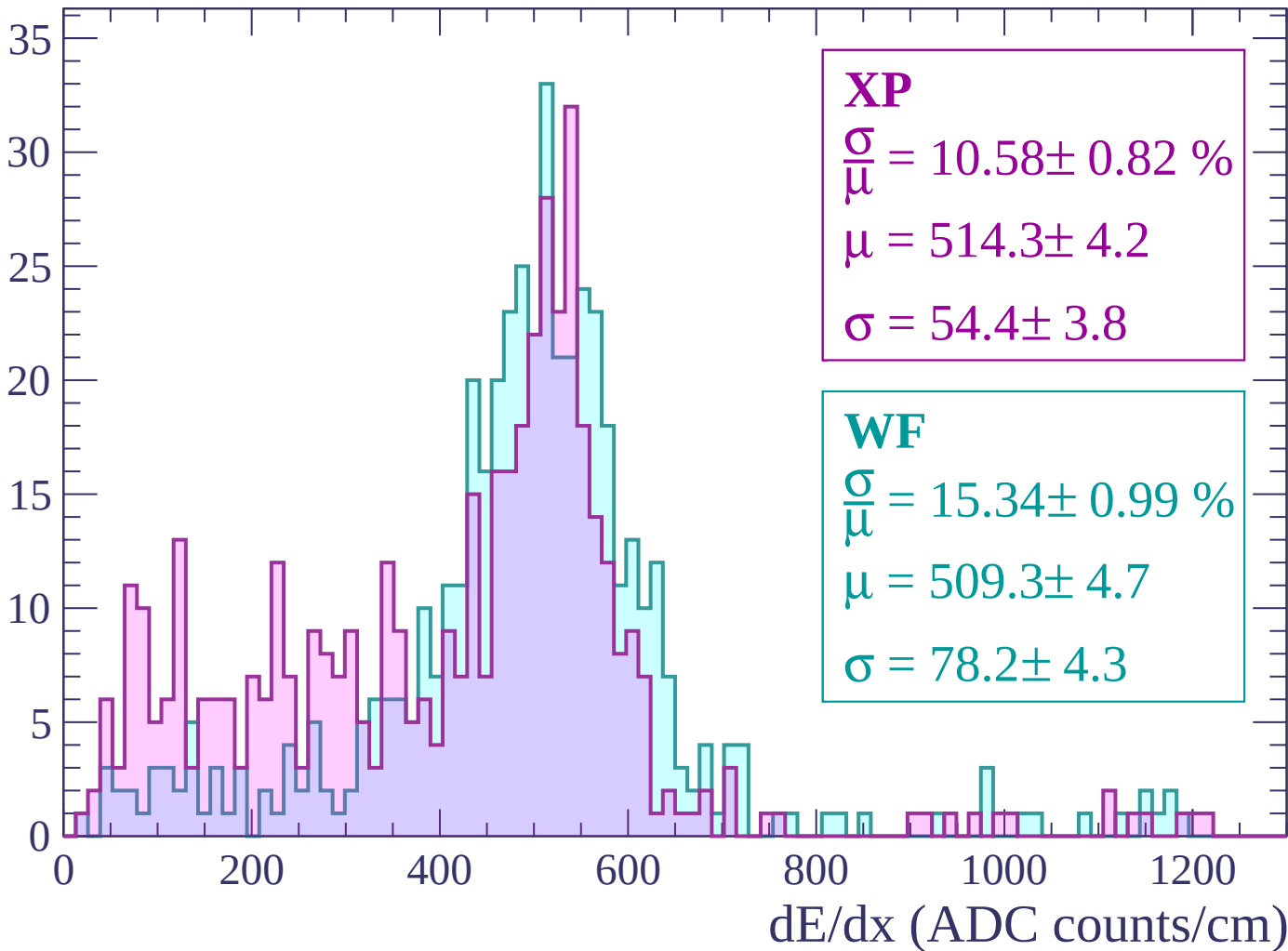
Energy loss | $0 < p < 60$

Count



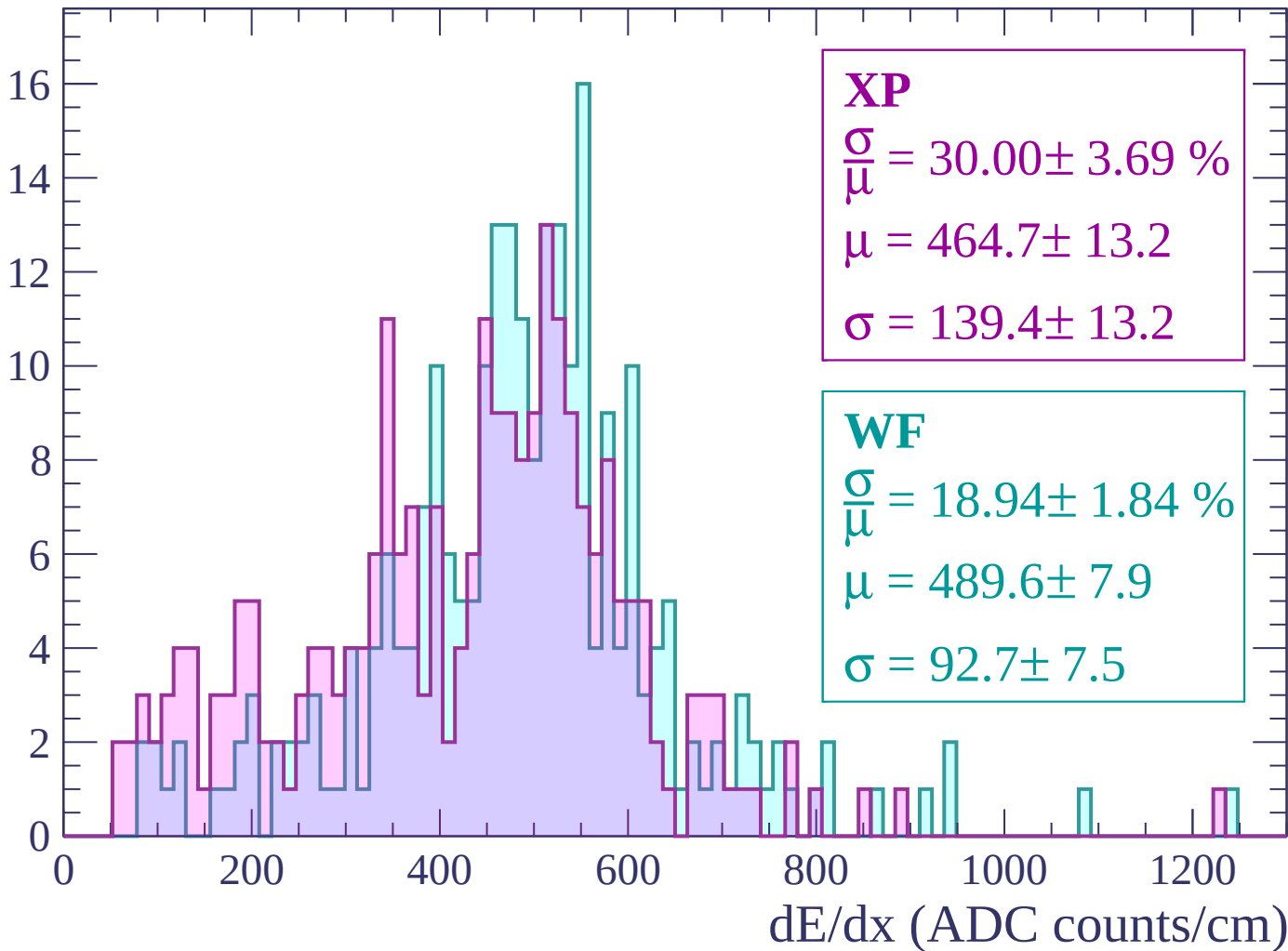
Energy loss | $60 < p < 120$

Count



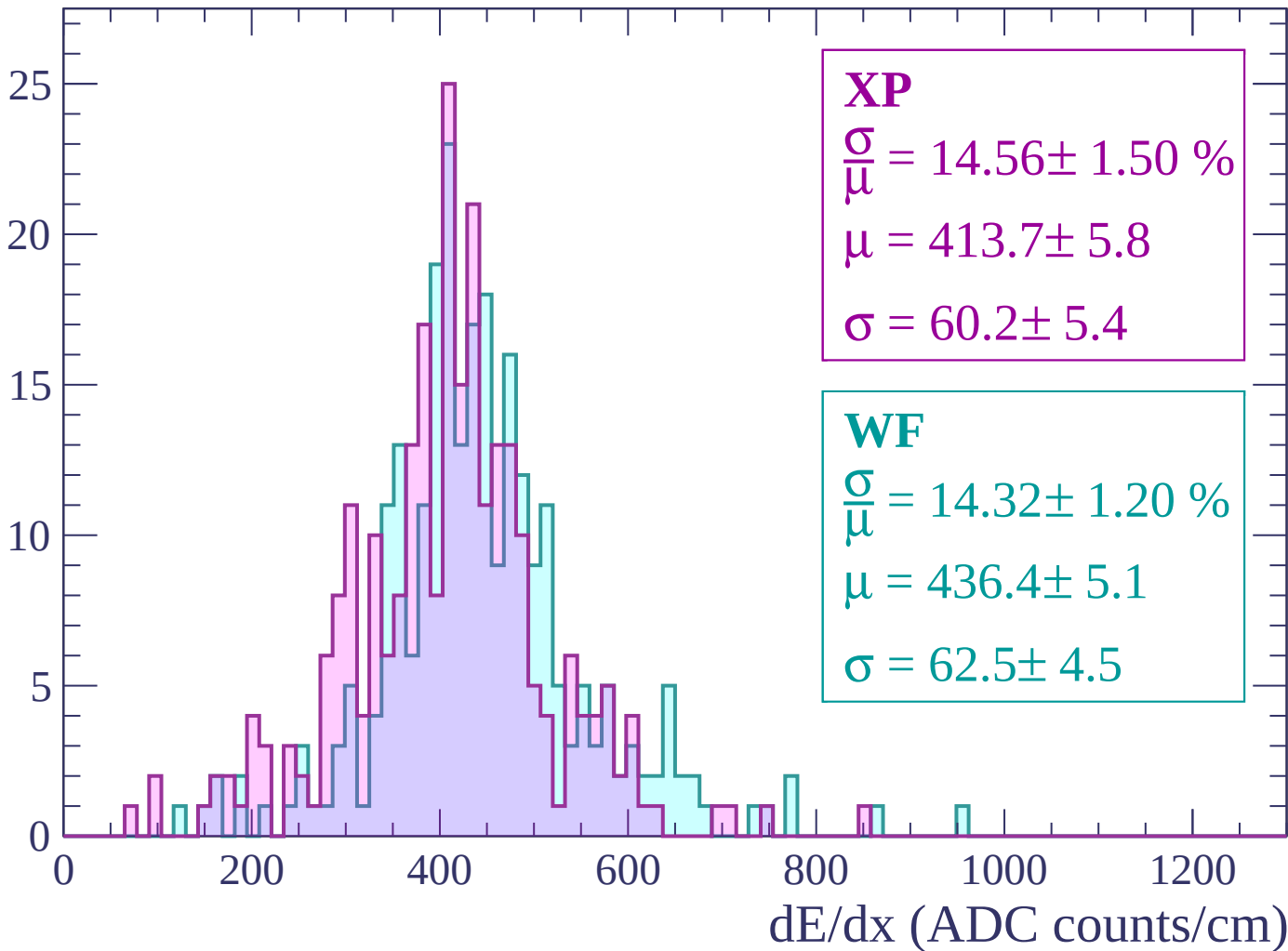
Energy loss | $120 < p < 180$

Count



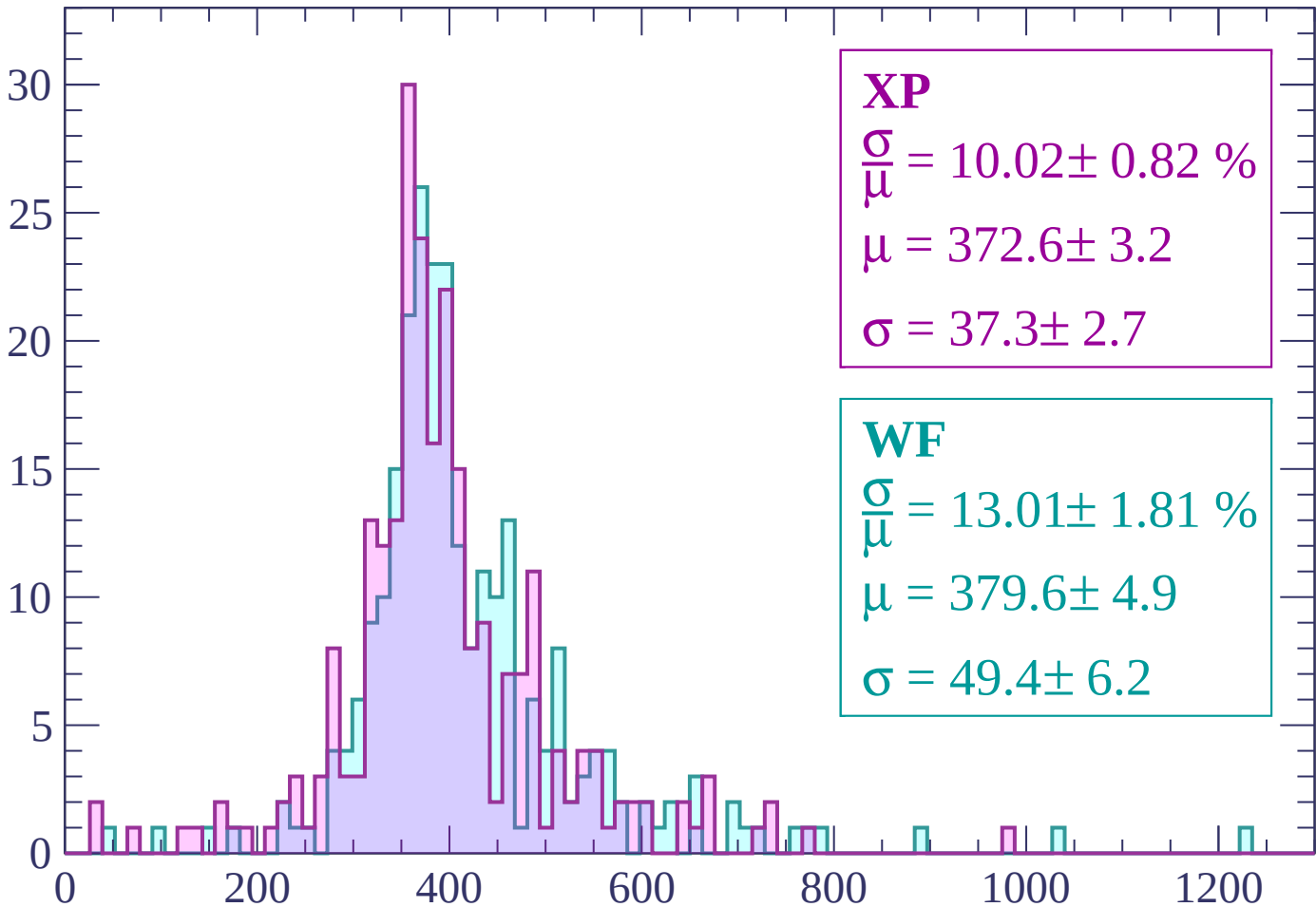
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP

$$\frac{\sigma}{\mu} = 10.02 \pm 0.82 \%$$

$$\mu = 372.6 \pm 3.2$$

$$\sigma = 37.3 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 13.01 \pm 1.81 \%$$

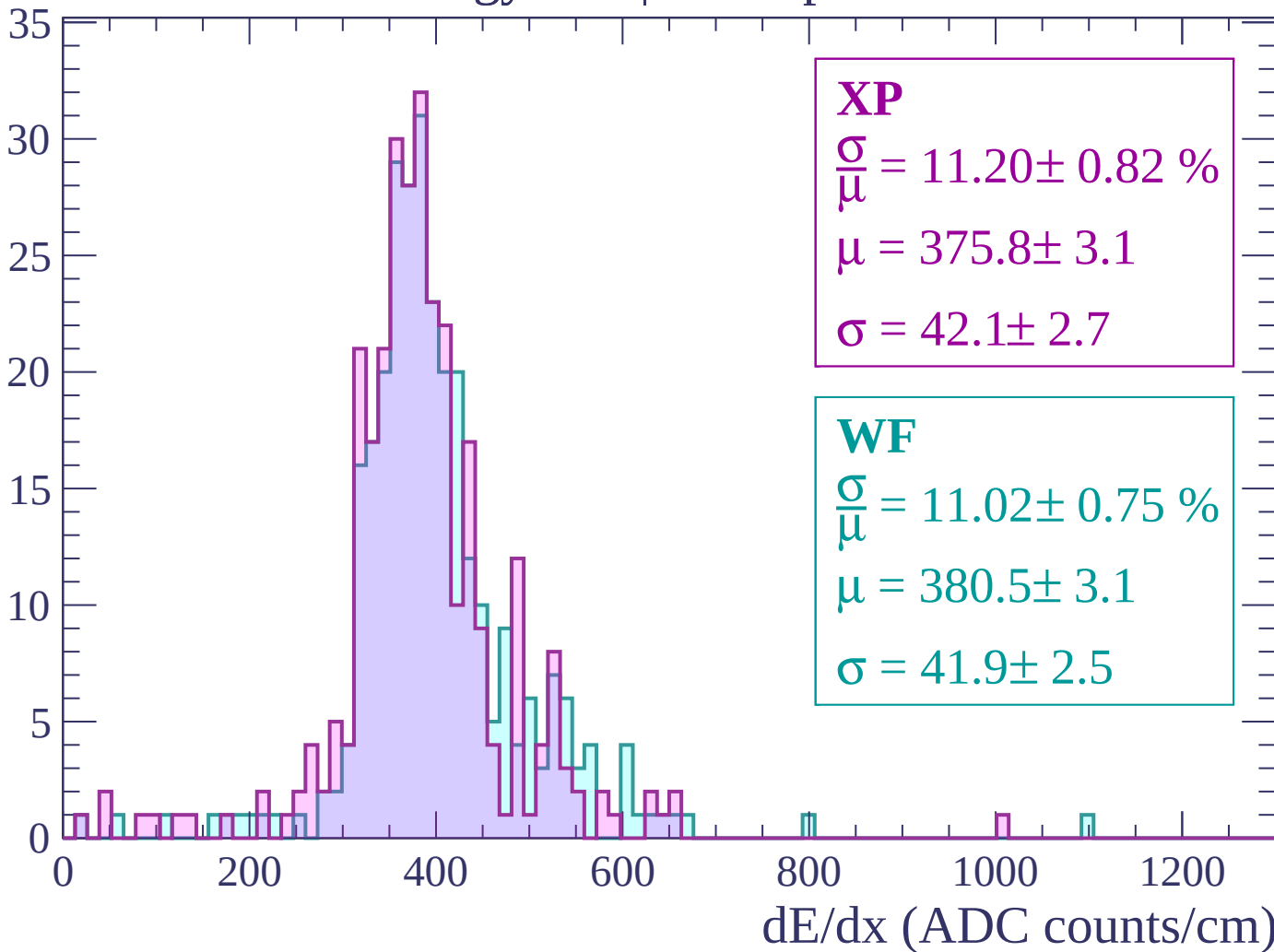
$$\mu = 379.6 \pm 4.9$$

$$\sigma = 49.4 \pm 6.2$$

dE/dx (ADC counts/cm)

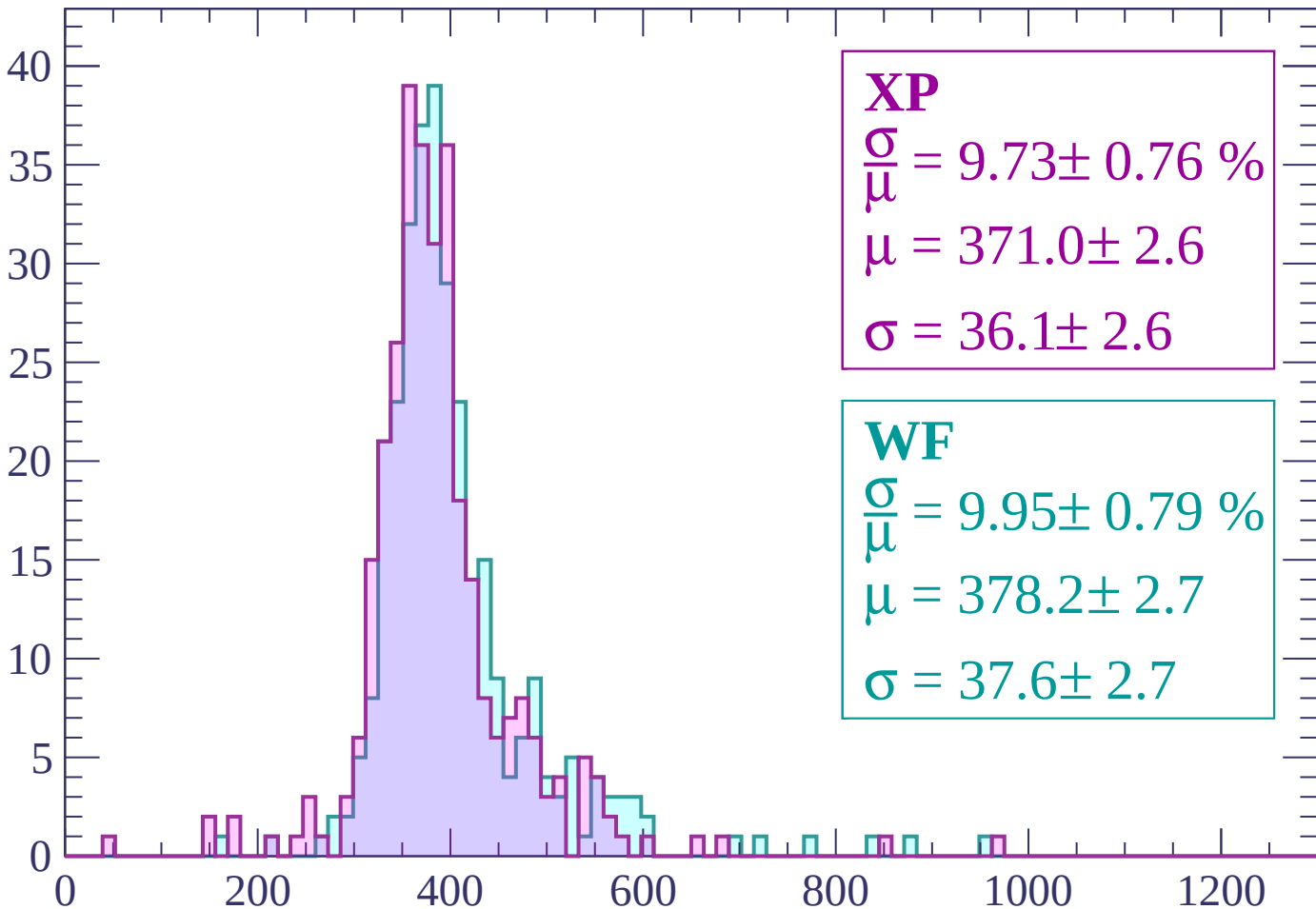
Energy loss | $300 < p < 360$

Count



Energy loss | $360 < p < 420$

Count



XP

$$\frac{\sigma}{\mu} = 9.73 \pm 0.76 \%$$

$$\mu = 371.0 \pm 2.6$$

$$\sigma = 36.1 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 9.95 \pm 0.79 \%$$

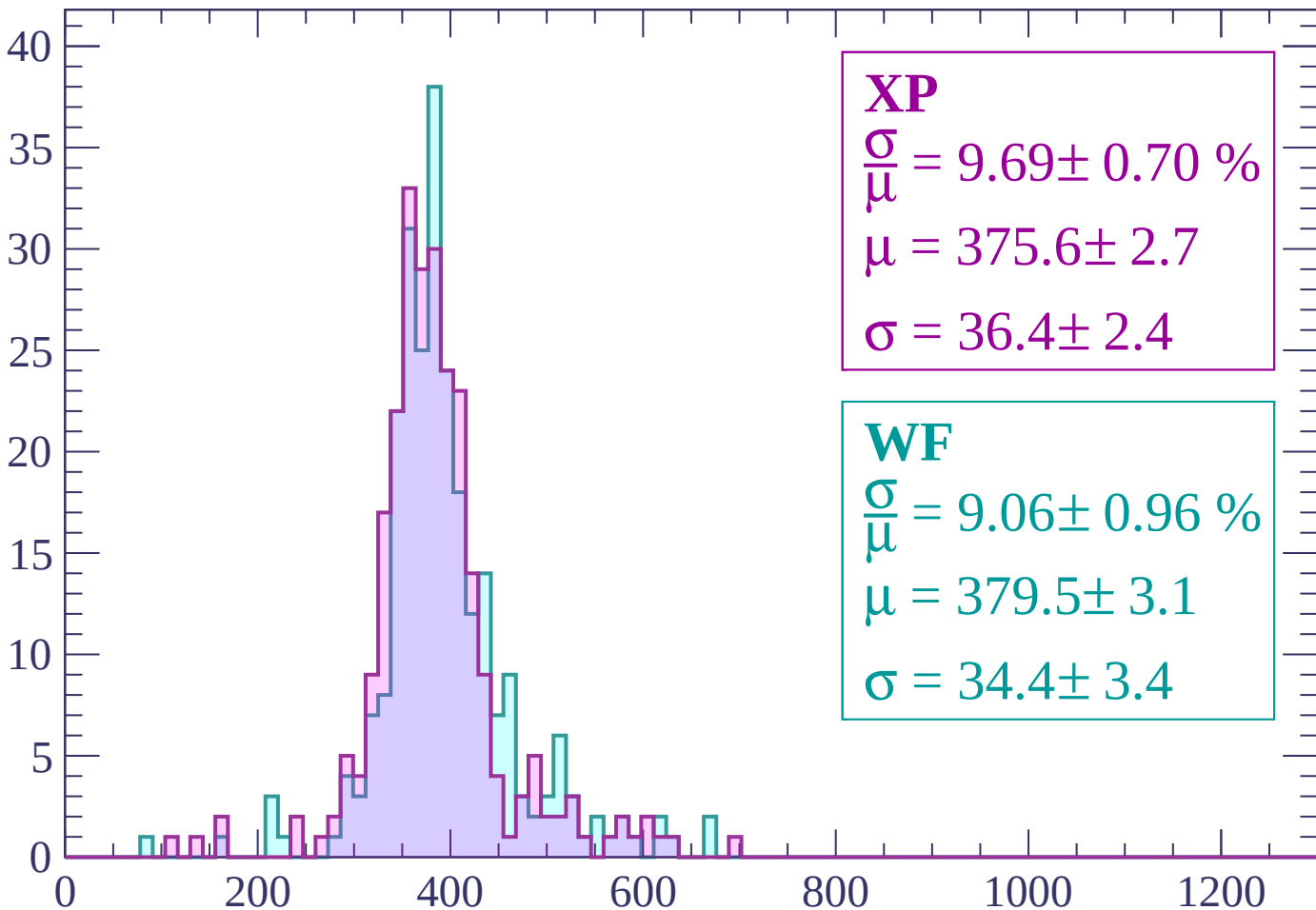
$$\mu = 378.2 \pm 2.7$$

$$\sigma = 37.6 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 9.69 \pm 0.70 \%$$

$$\mu = 375.6 \pm 2.7$$

$$\sigma = 36.4 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.96 \%$$

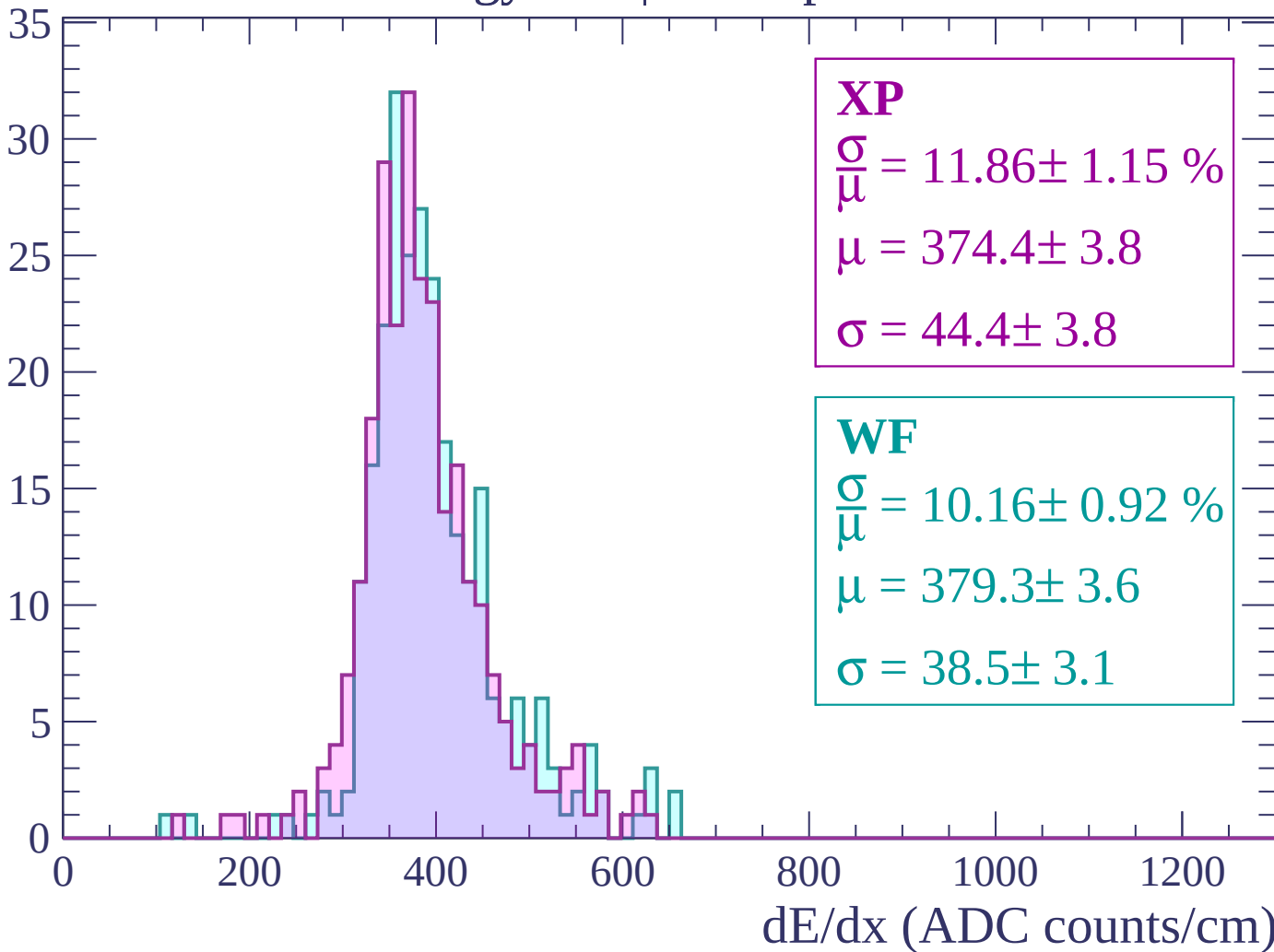
$$\mu = 379.5 \pm 3.1$$

$$\sigma = 34.4 \pm 3.4$$

dE/dx (ADC counts/cm)

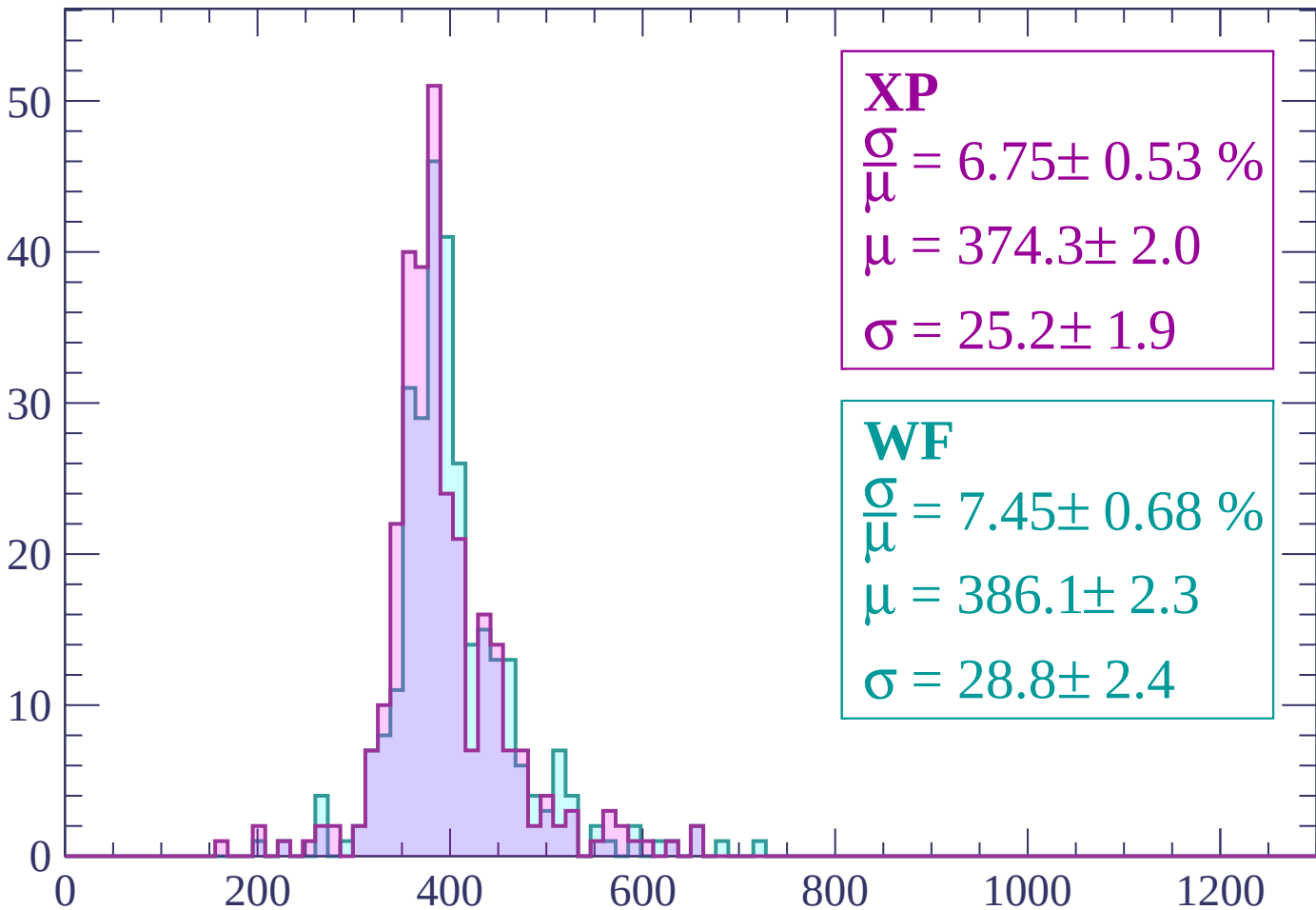
Energy loss | $480 < p < 540$

Count



Energy loss | $540 < p < 600$

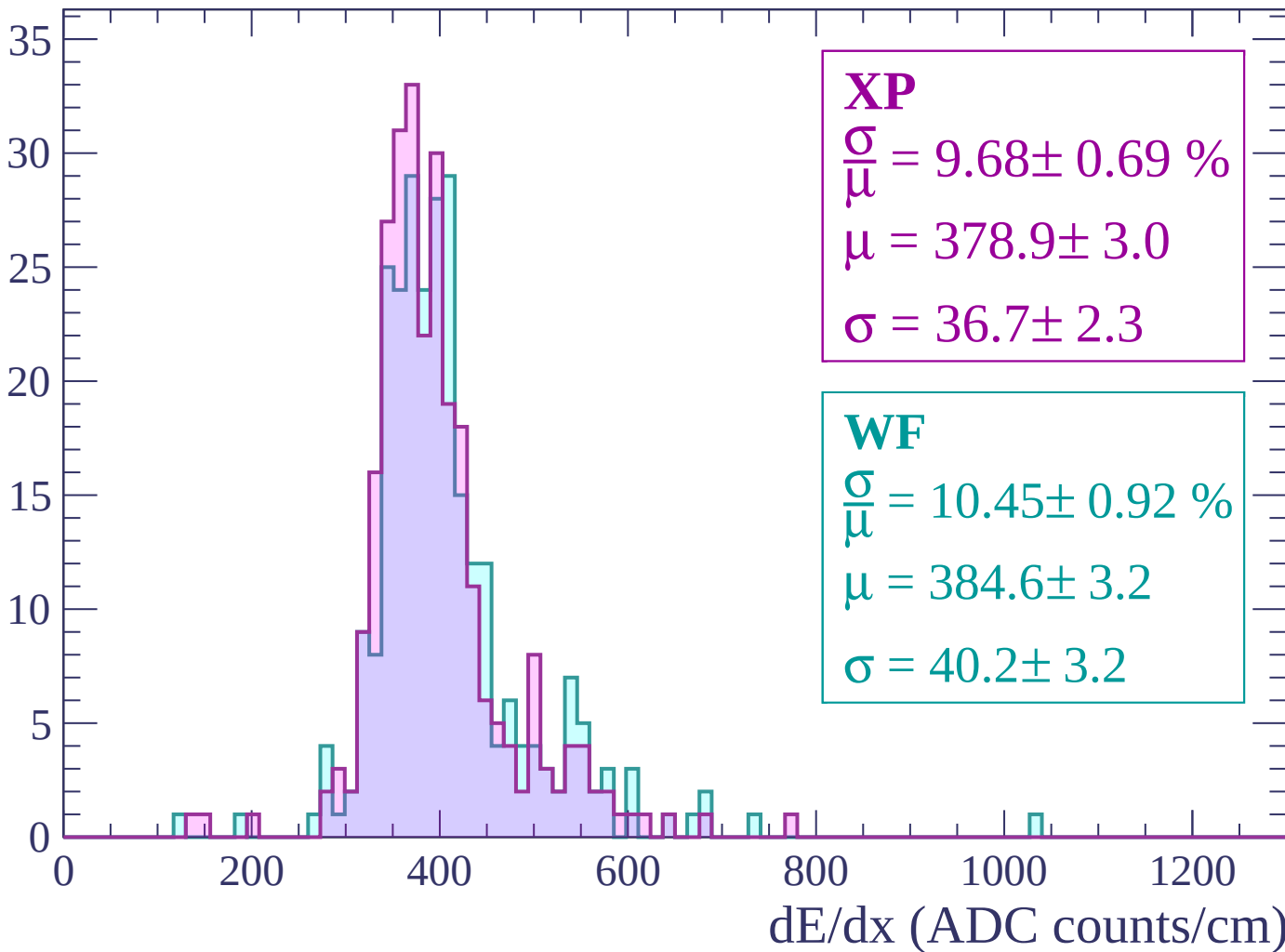
Count



dE/dx (ADC counts/cm)

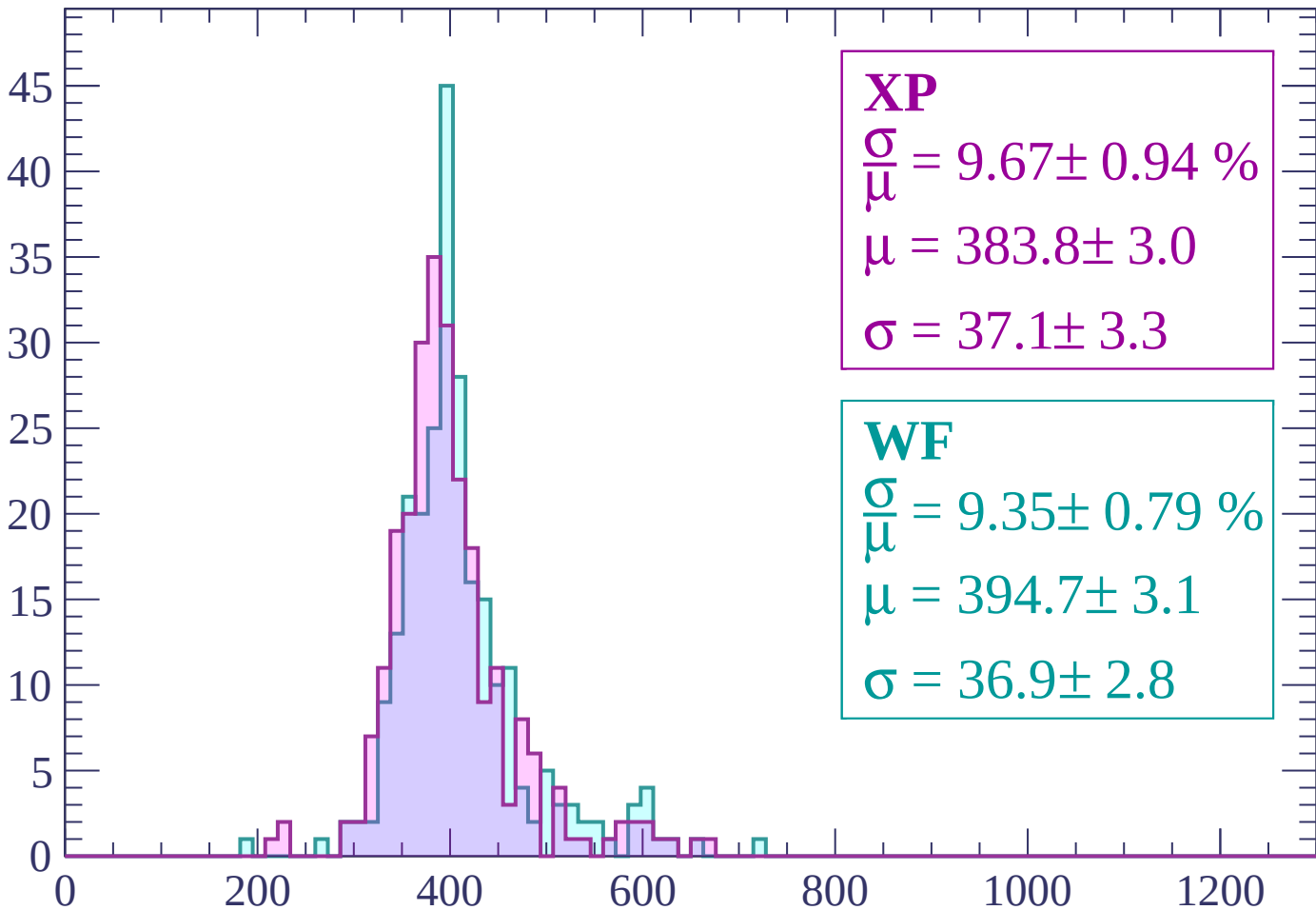
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

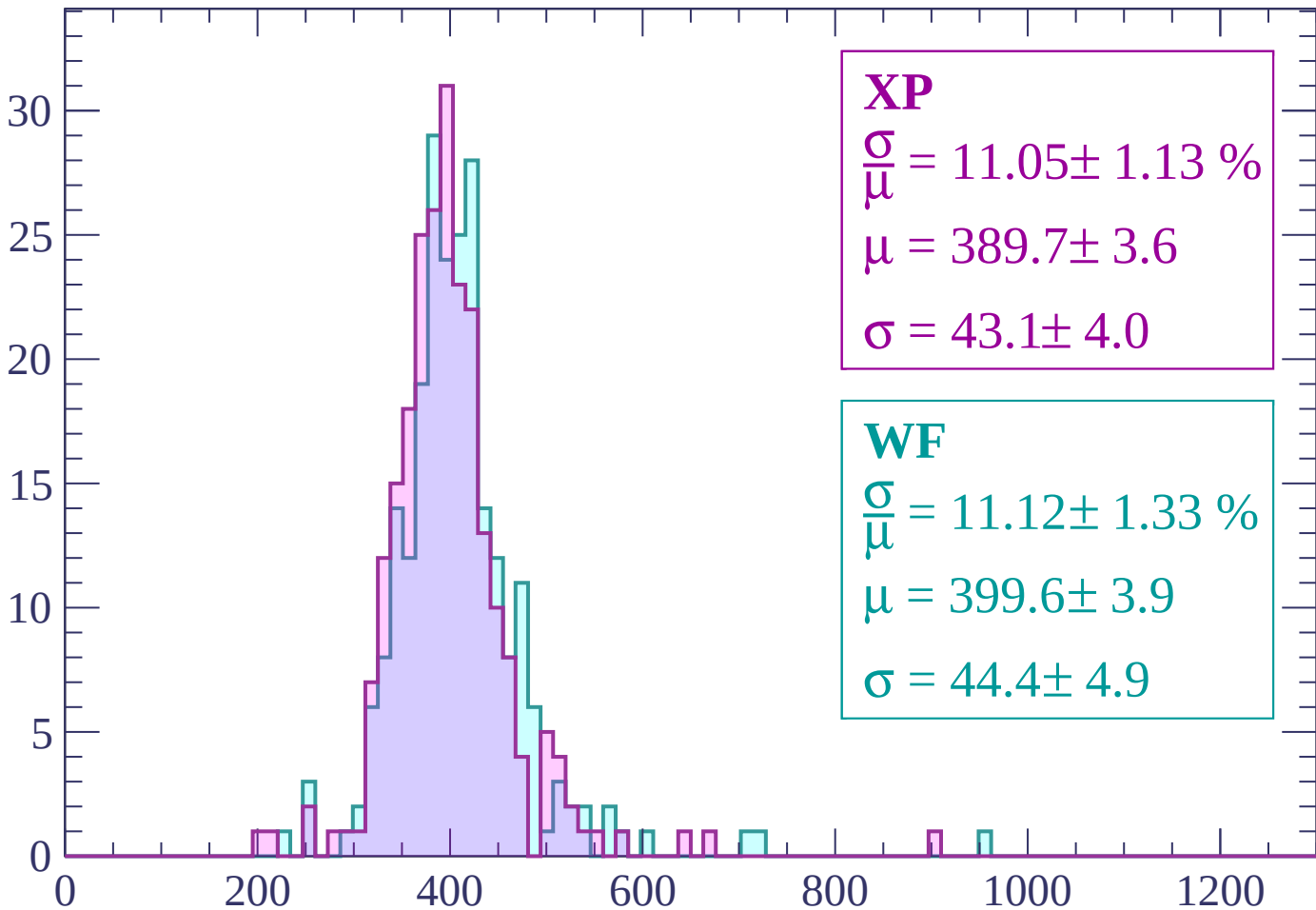
Count



dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP

$$\frac{\sigma}{\mu} = 11.05 \pm 1.13 \%$$

$$\mu = 389.7 \pm 3.6$$

$$\sigma = 43.1 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 11.12 \pm 1.33 \%$$

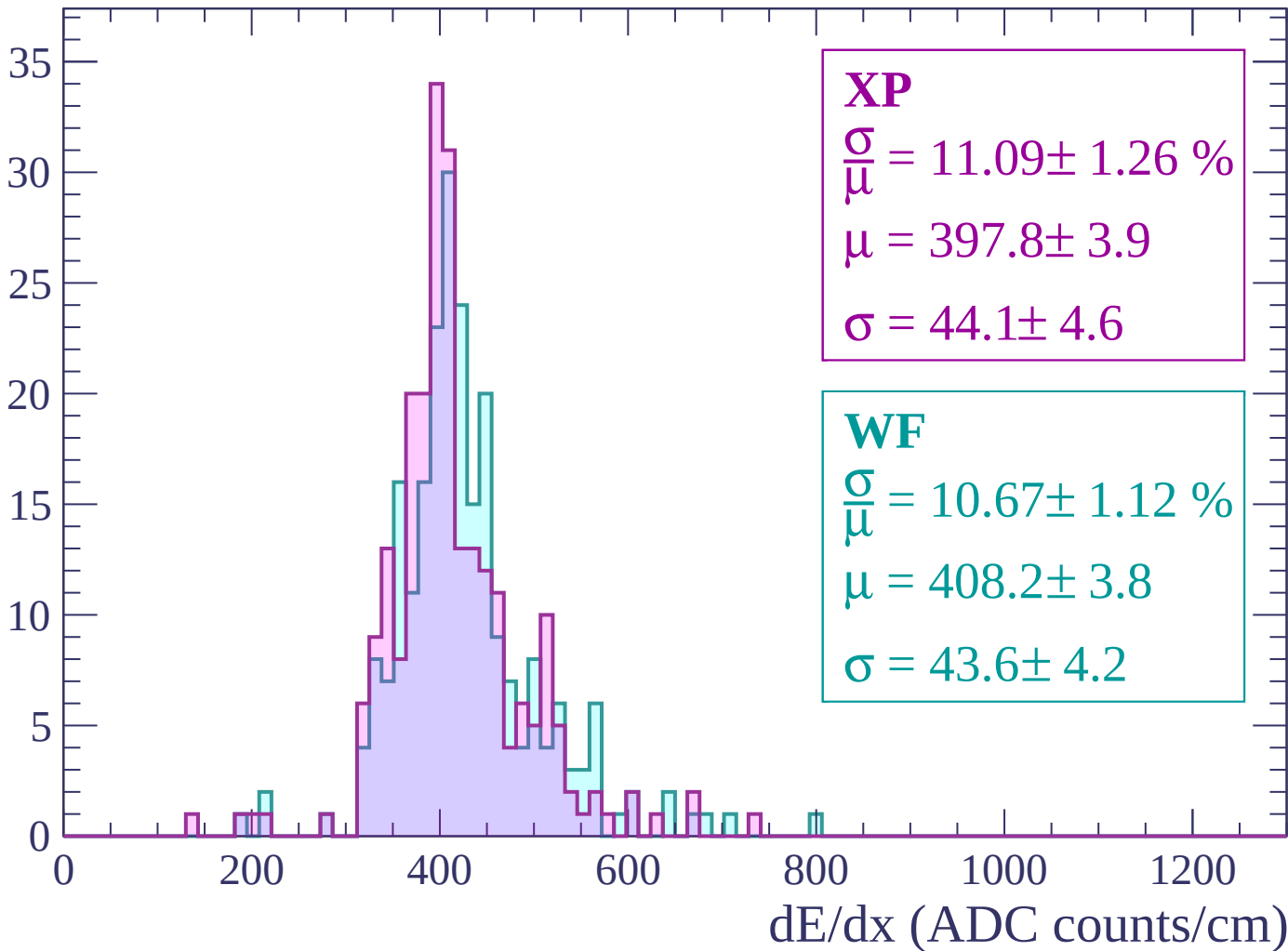
$$\mu = 399.6 \pm 3.9$$

$$\sigma = 44.4 \pm 4.9$$

dE/dx (ADC counts/cm)

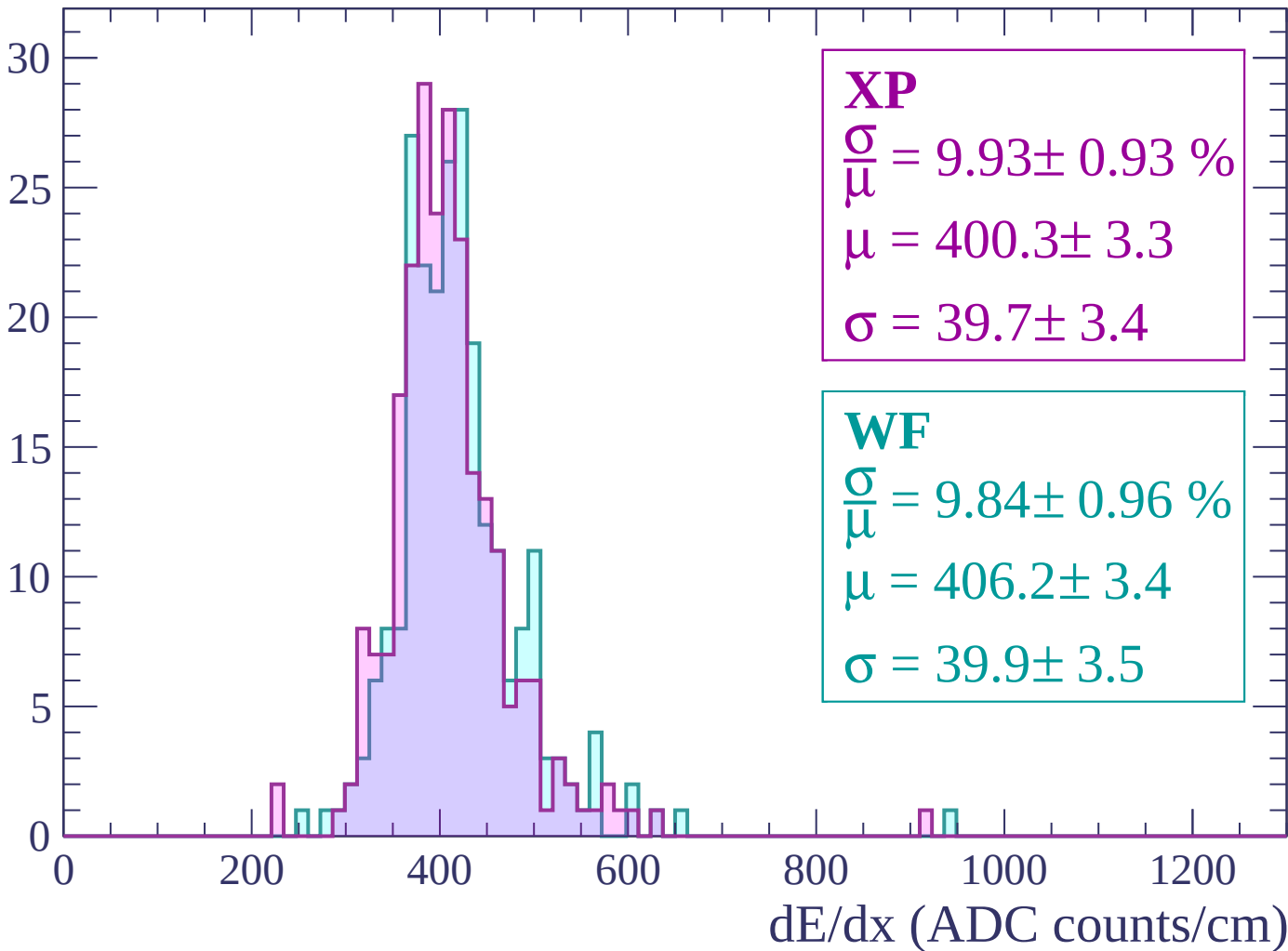
Energy loss | $780 < p < 840$

Count



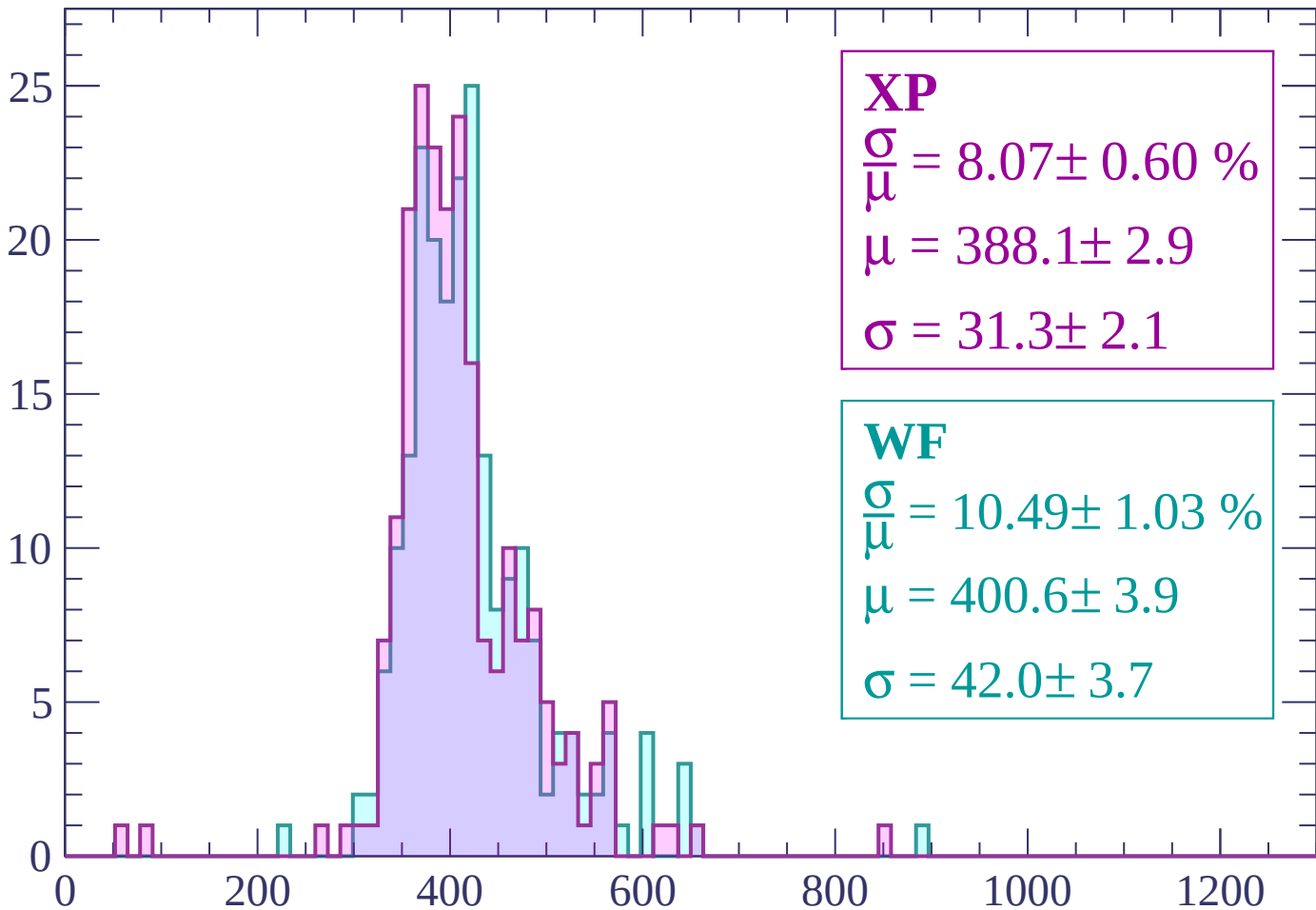
Energy loss | $840 < p < 900$

Count



Energy loss | $900 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.60 \%$$

$$\mu = 388.1 \pm 2.9$$

$$\sigma = 31.3 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.49 \pm 1.03 \%$$

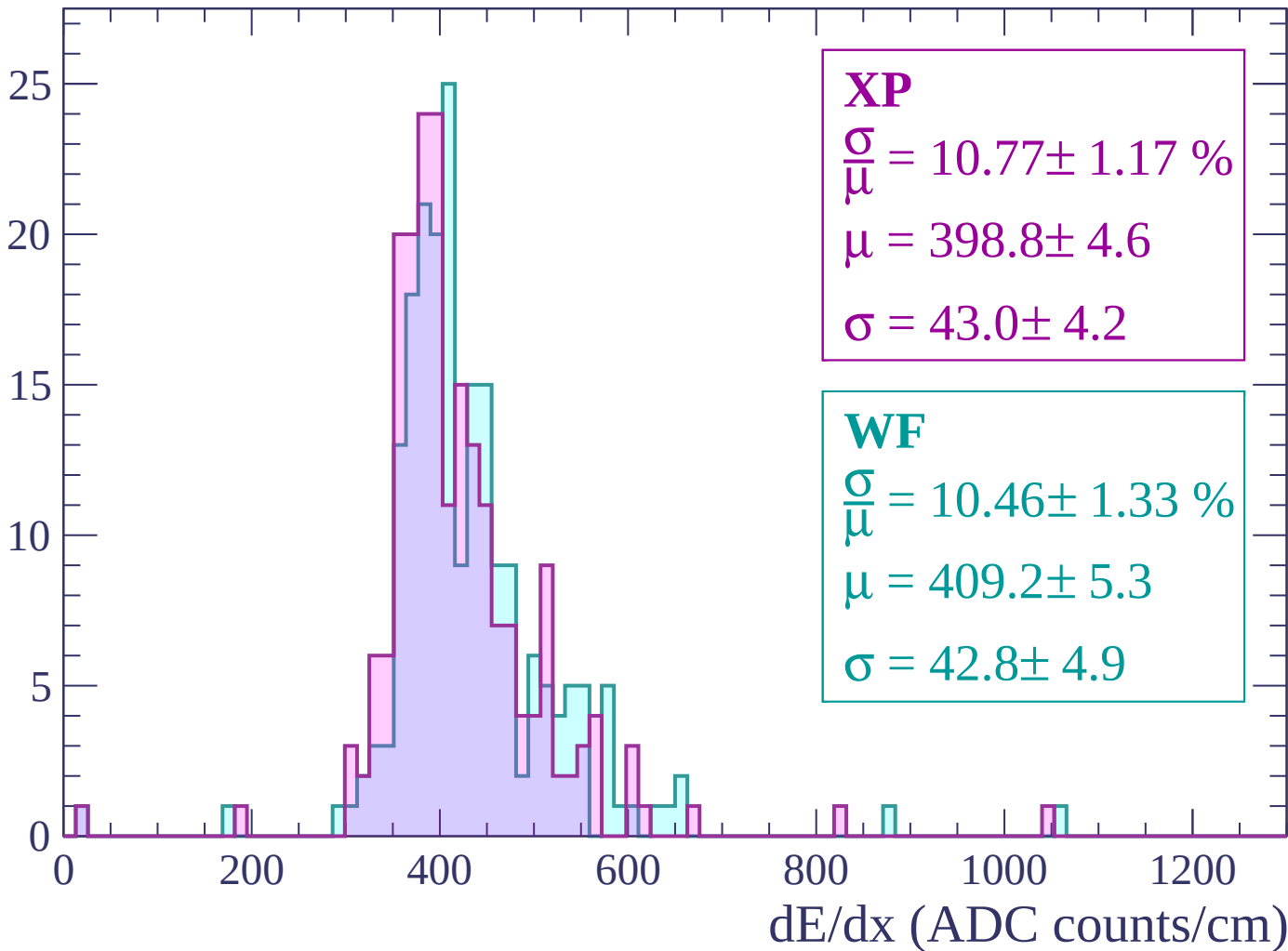
$$\mu = 400.6 \pm 3.9$$

$$\sigma = 42.0 \pm 3.7$$

dE/dx (ADC counts/cm)

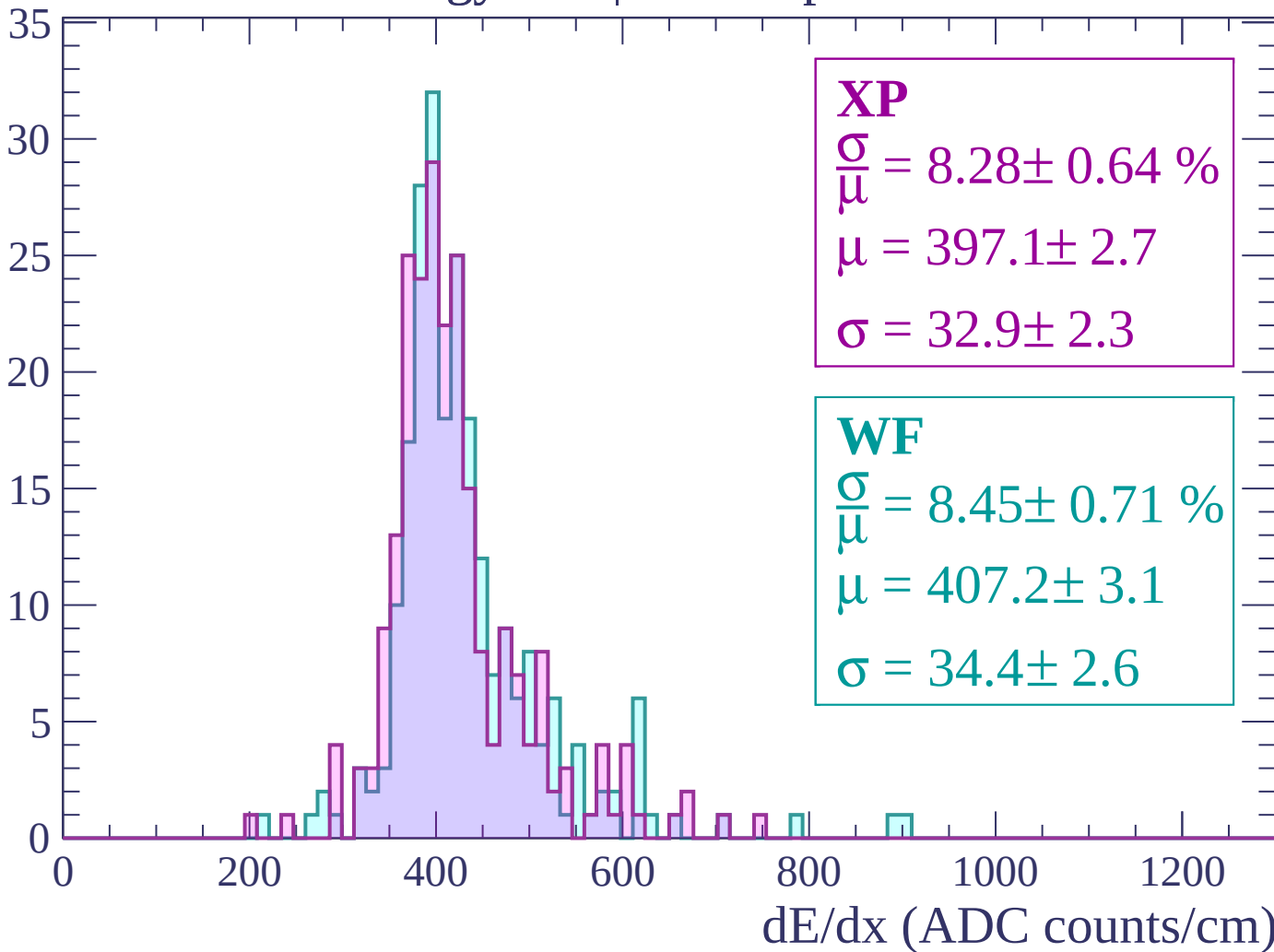
Energy loss | $960 < p < 1020$

Count



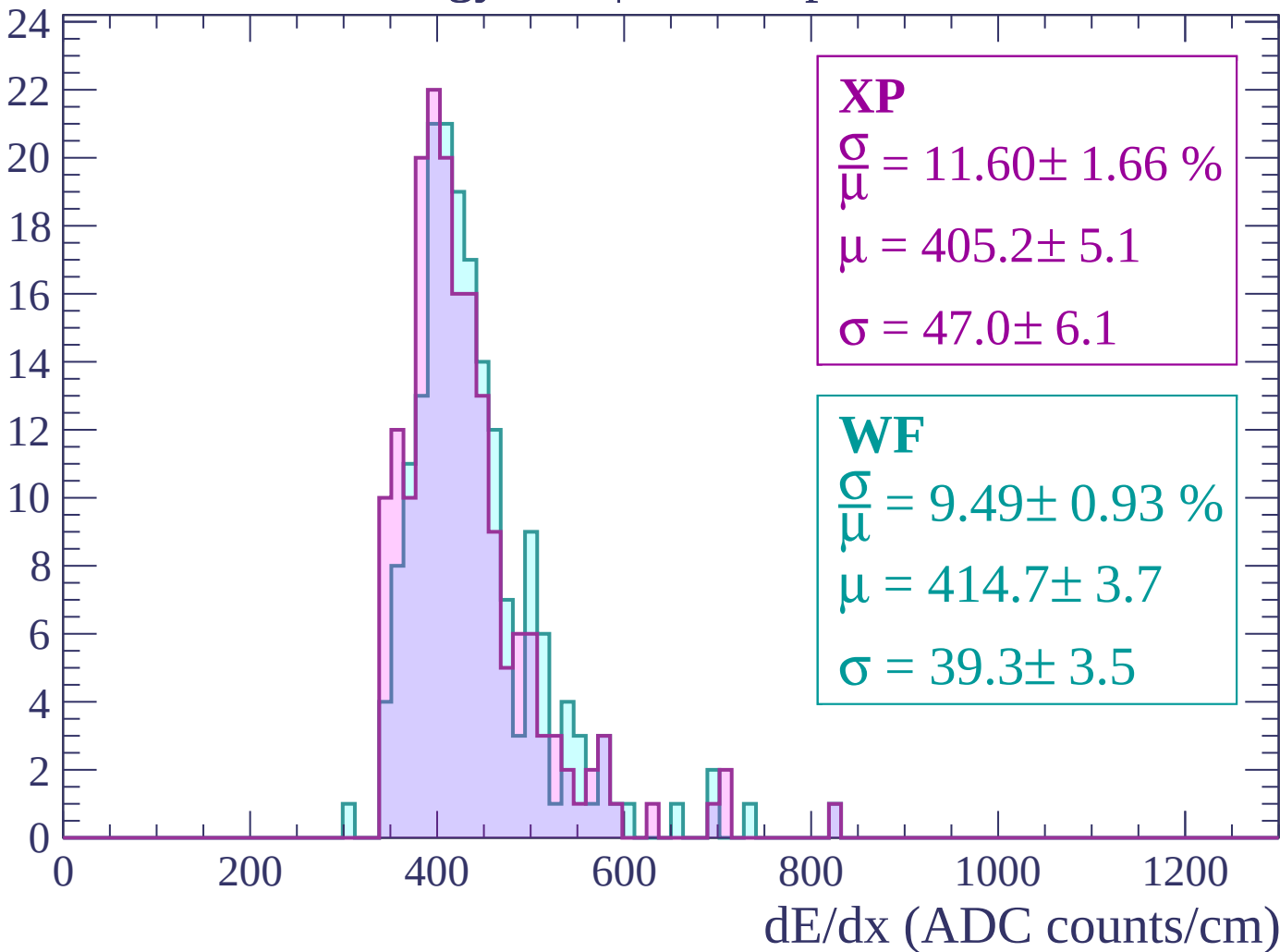
Energy loss | $1020 < p < 1080$

Count



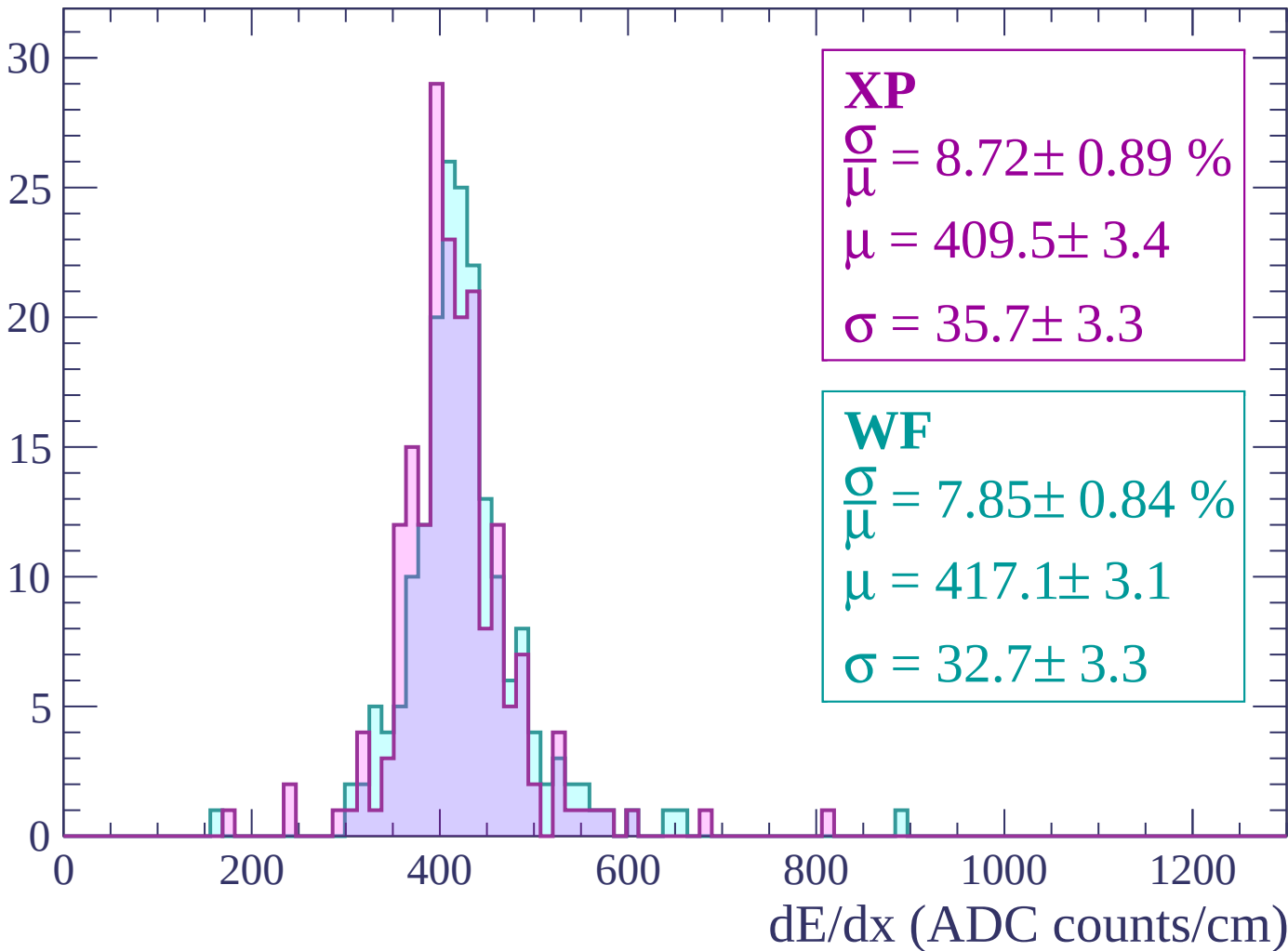
Energy loss | $1080 < p < 1140$

Count



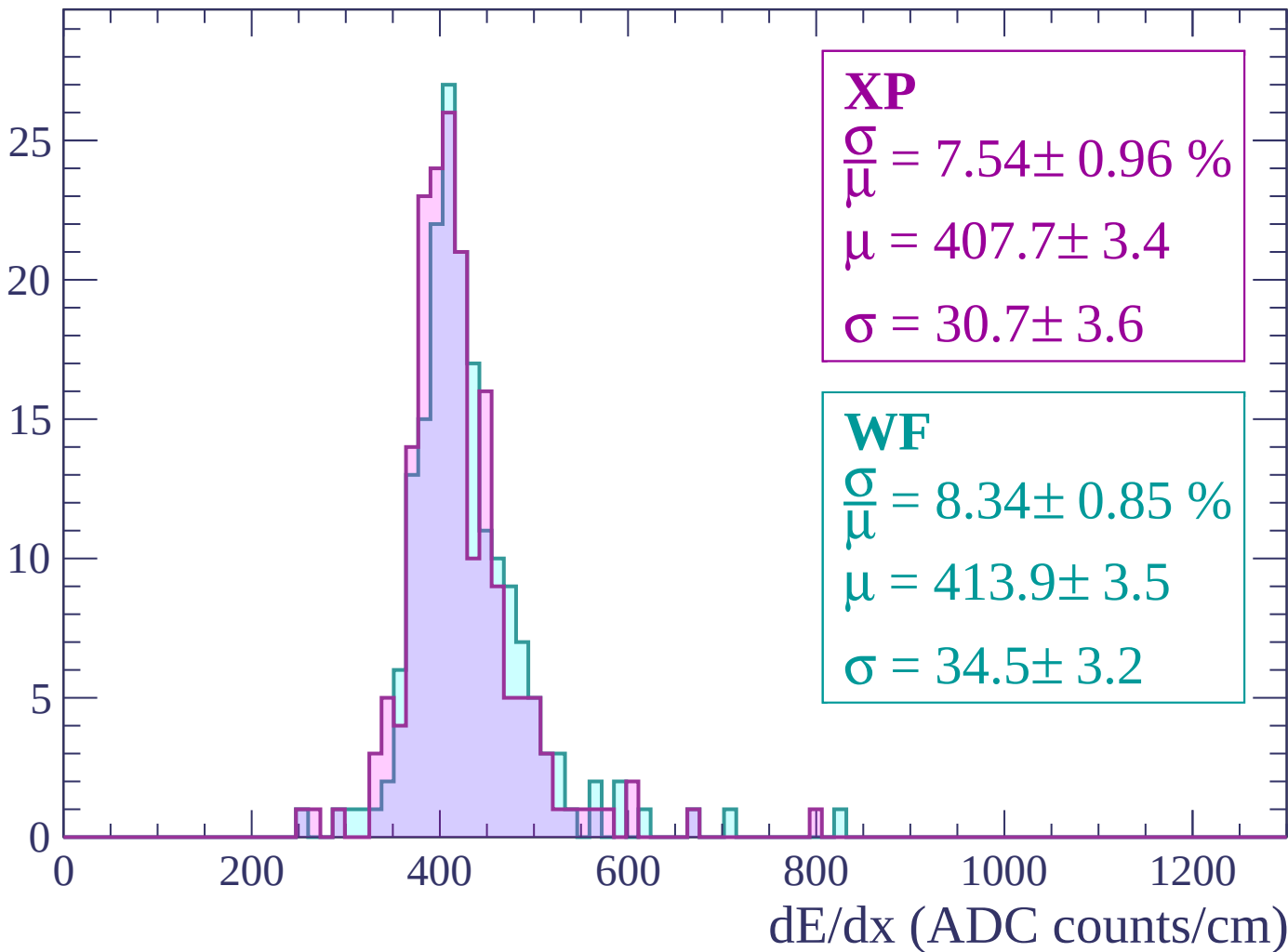
Energy loss | $1140 < p < 1200$

Count



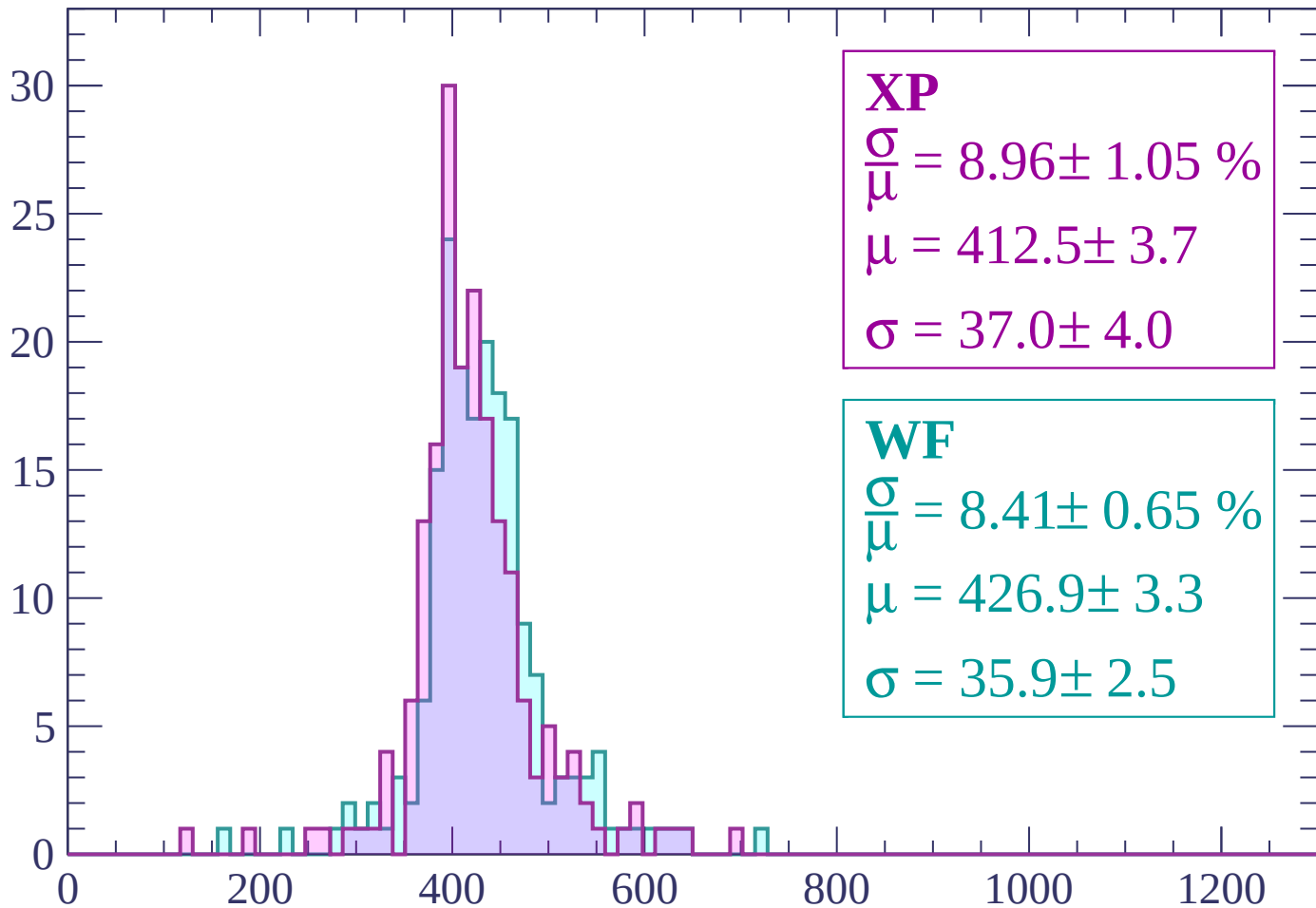
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 8.96 \pm 1.05 \%$$

$$\mu = 412.5 \pm 3.7$$

$$\sigma = 37.0 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 8.41 \pm 0.65 \%$$

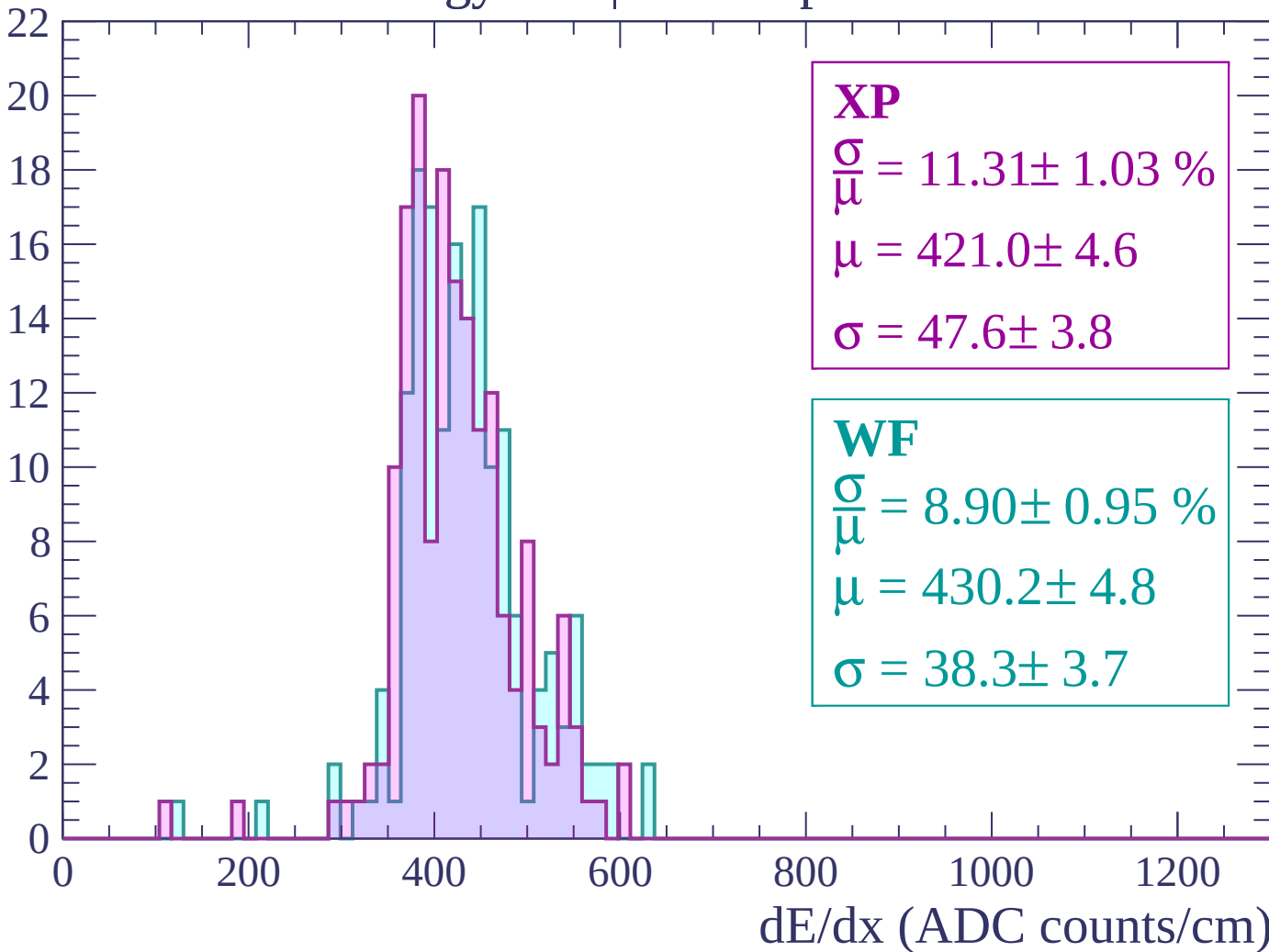
$$\mu = 426.9 \pm 3.3$$

$$\sigma = 35.9 \pm 2.5$$

dE/dx (ADC counts/cm)

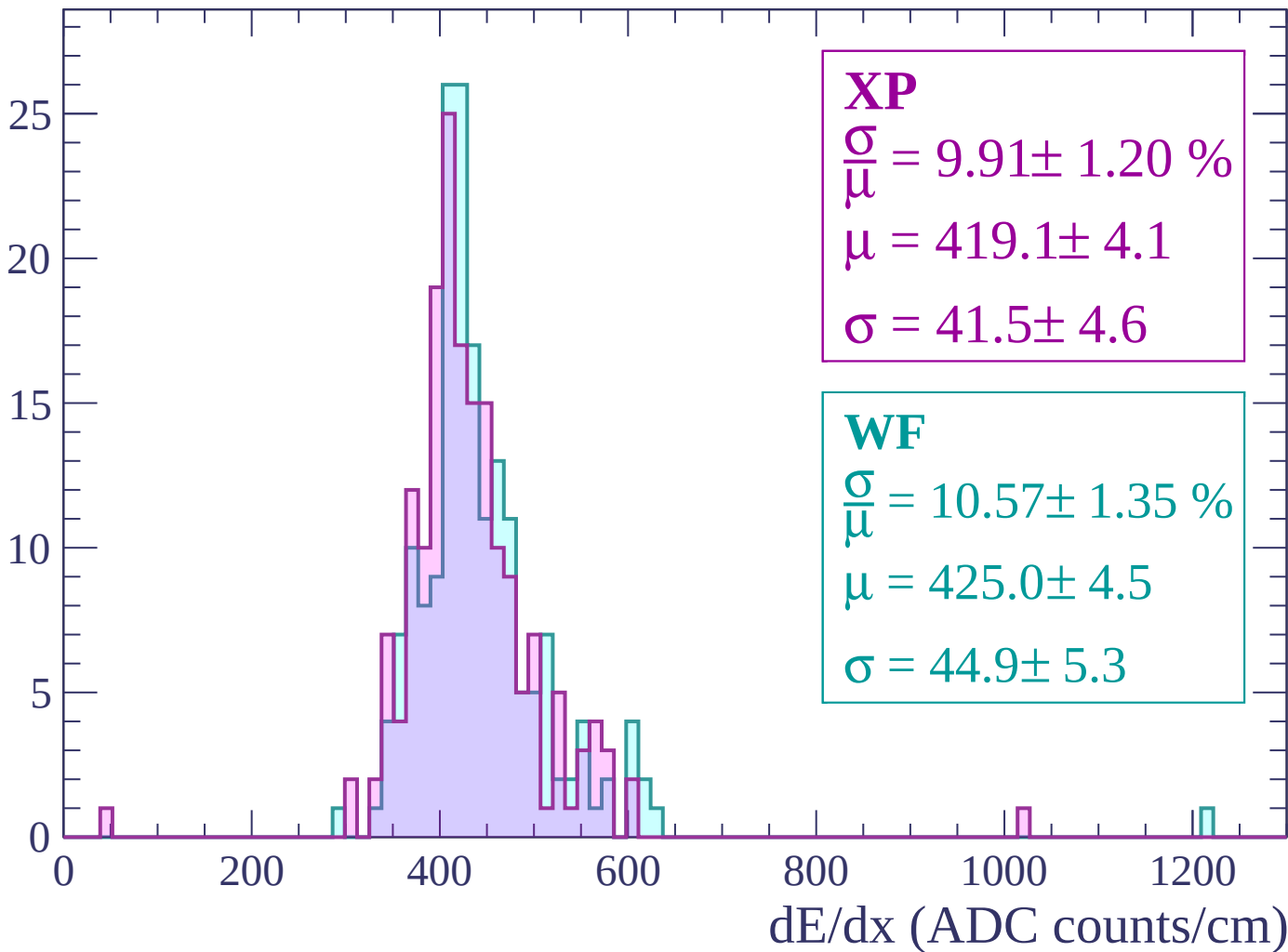
Energy loss | $1320 < p < 1380$

Count



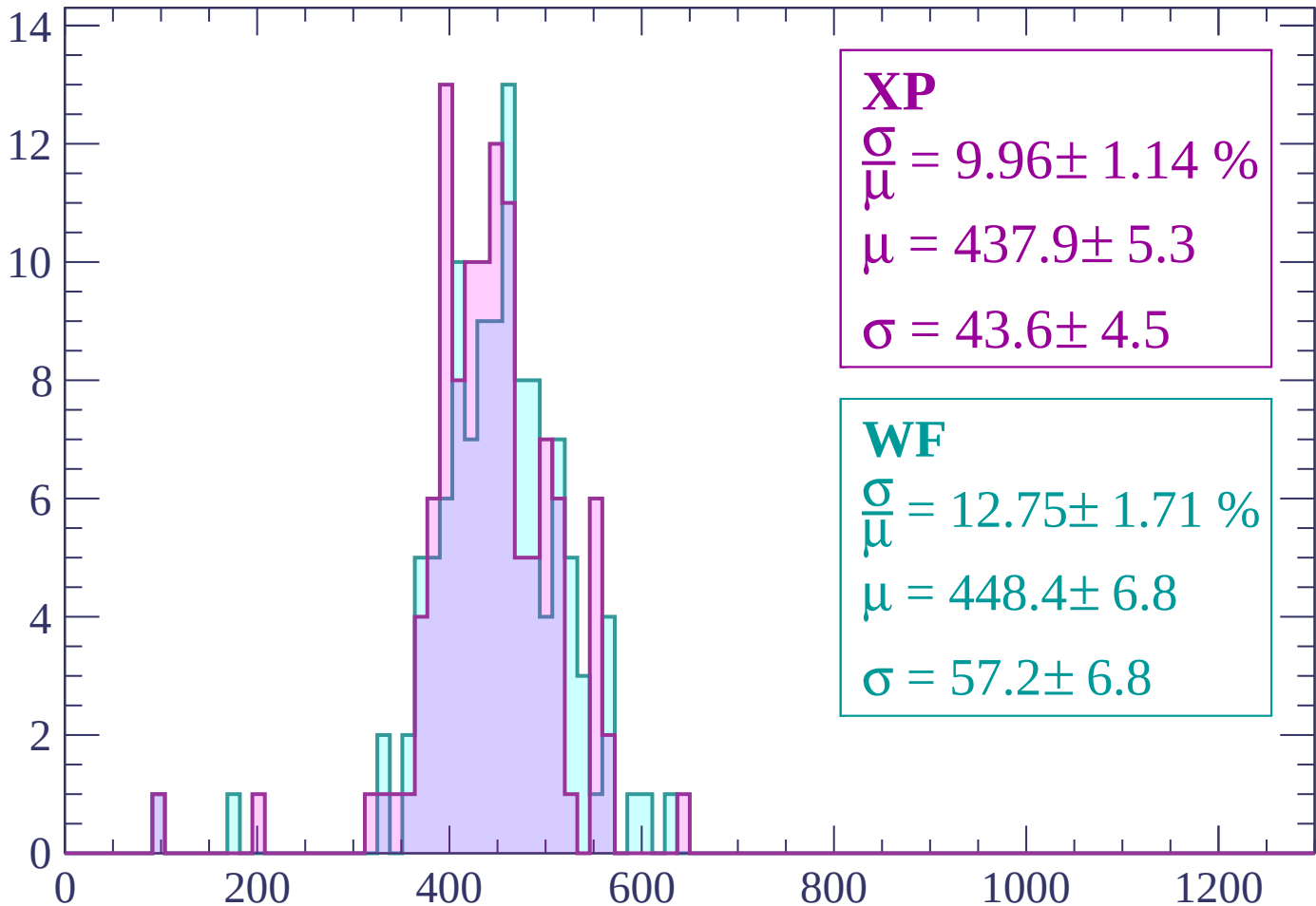
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.96 \pm 1.14 \%$$

$$\mu = 437.9 \pm 5.3$$

$$\sigma = 43.6 \pm 4.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 12.75 \pm 1.71 \%$$

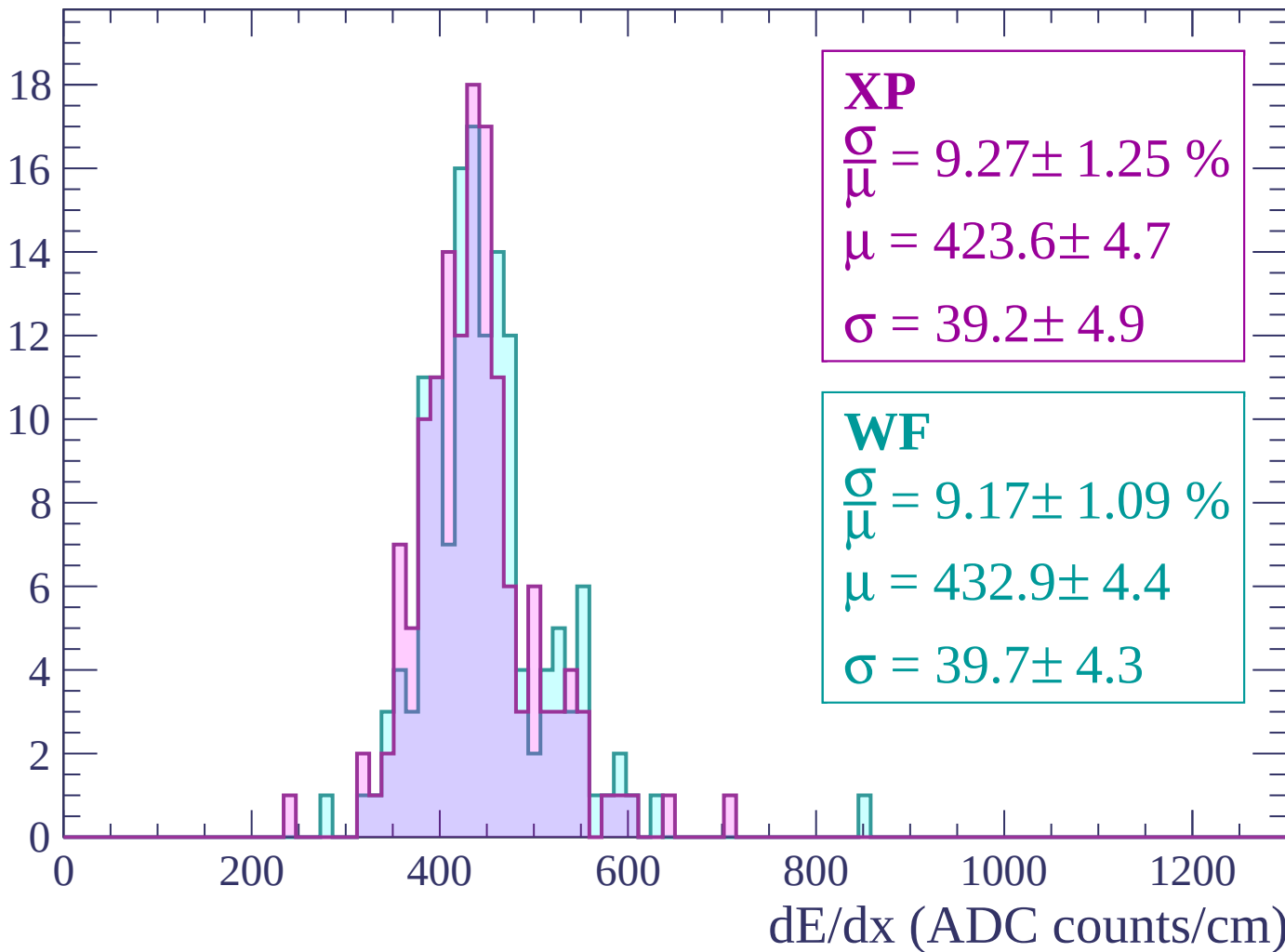
$$\mu = 448.4 \pm 6.8$$

$$\sigma = 57.2 \pm 6.8$$

dE/dx (ADC counts/cm)

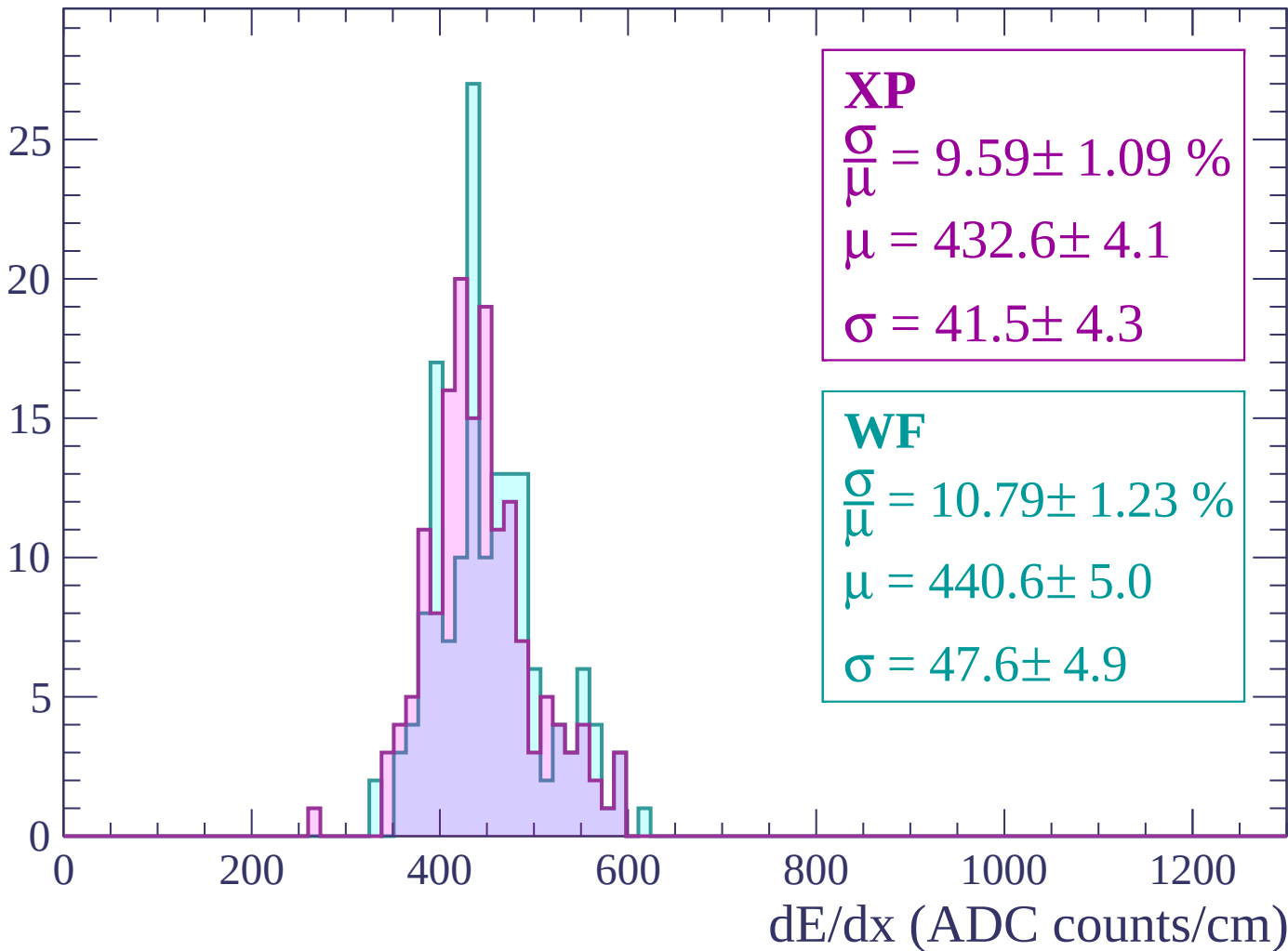
Energy loss | $1500 < p < 1560$

Count



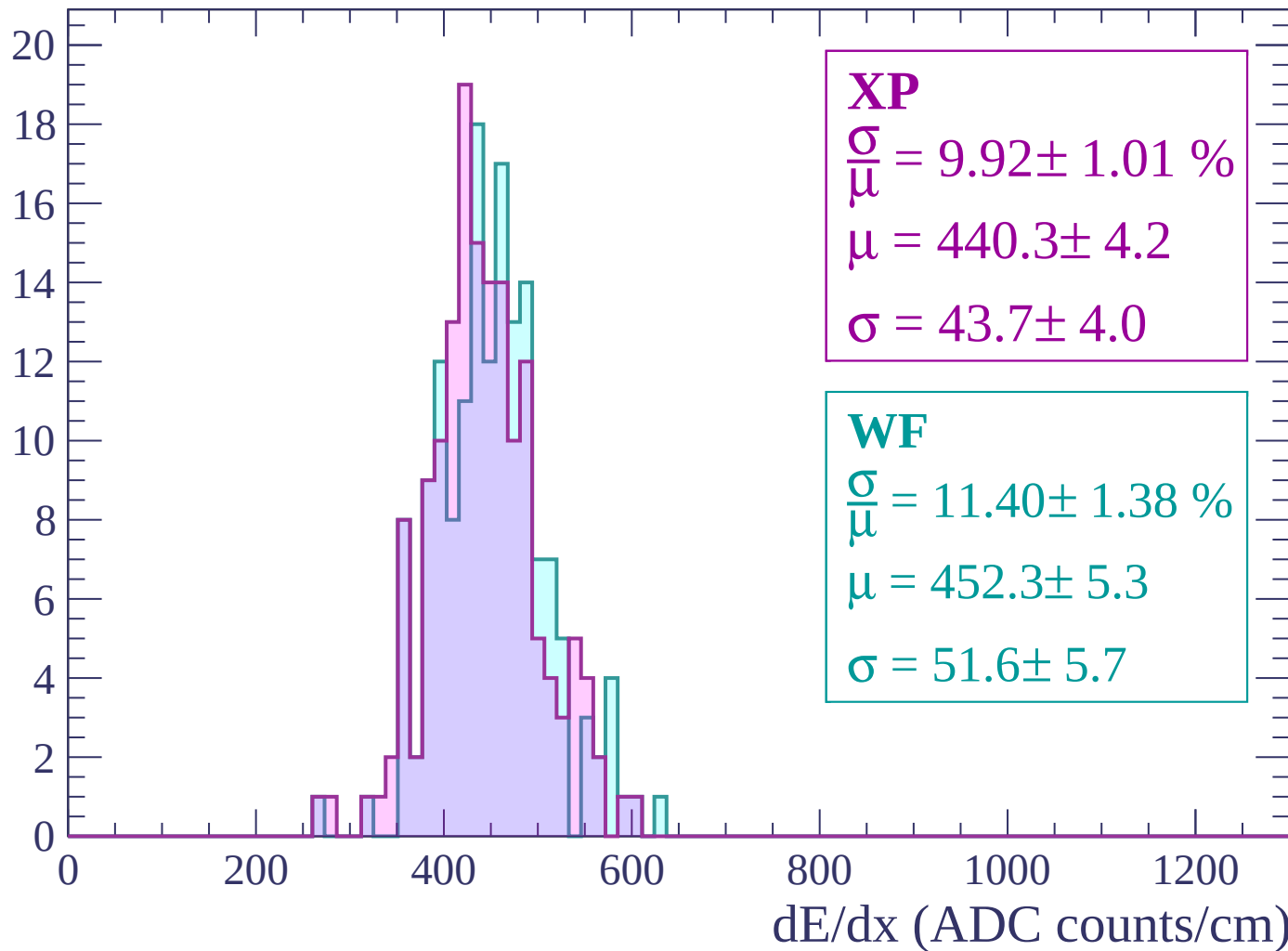
Energy loss | $1560 < p < 1620$

Count



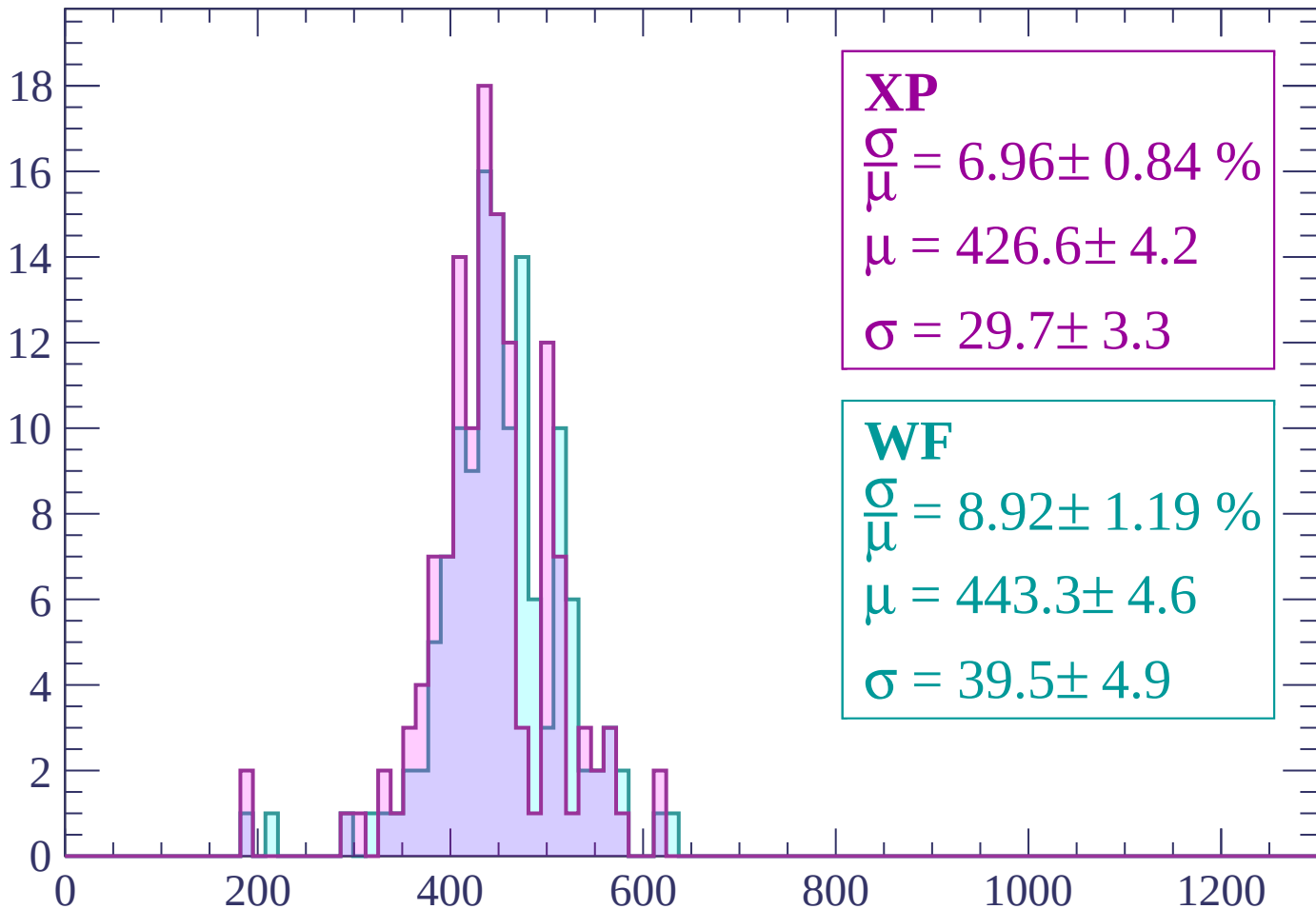
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP

$$\frac{\sigma}{\mu} = 6.96 \pm 0.84 \%$$

$$\mu = 426.6 \pm 4.2$$

$$\sigma = 29.7 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 8.92 \pm 1.19 \%$$

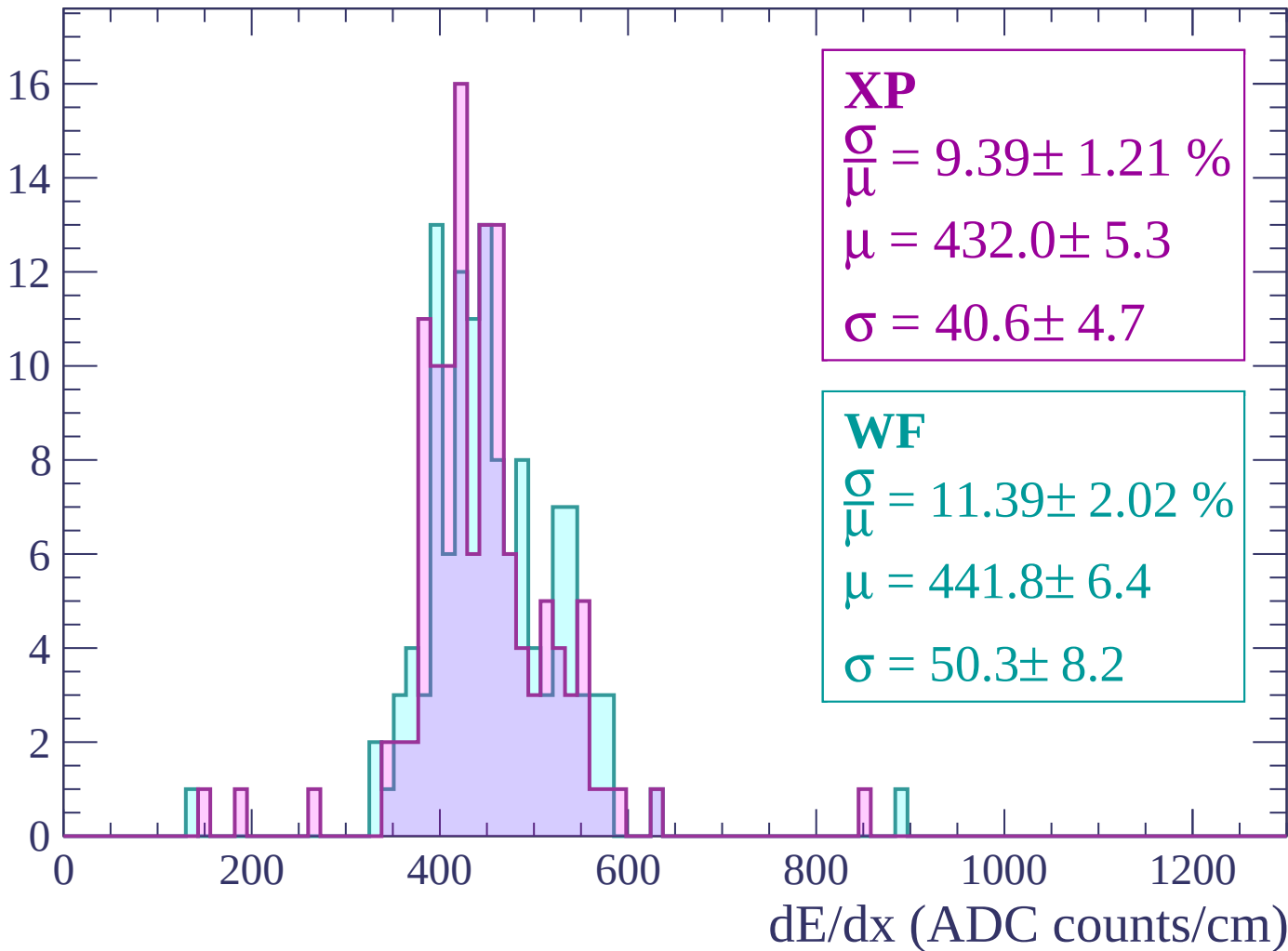
$$\mu = 443.3 \pm 4.6$$

$$\sigma = 39.5 \pm 4.9$$

dE/dx (ADC counts/cm)

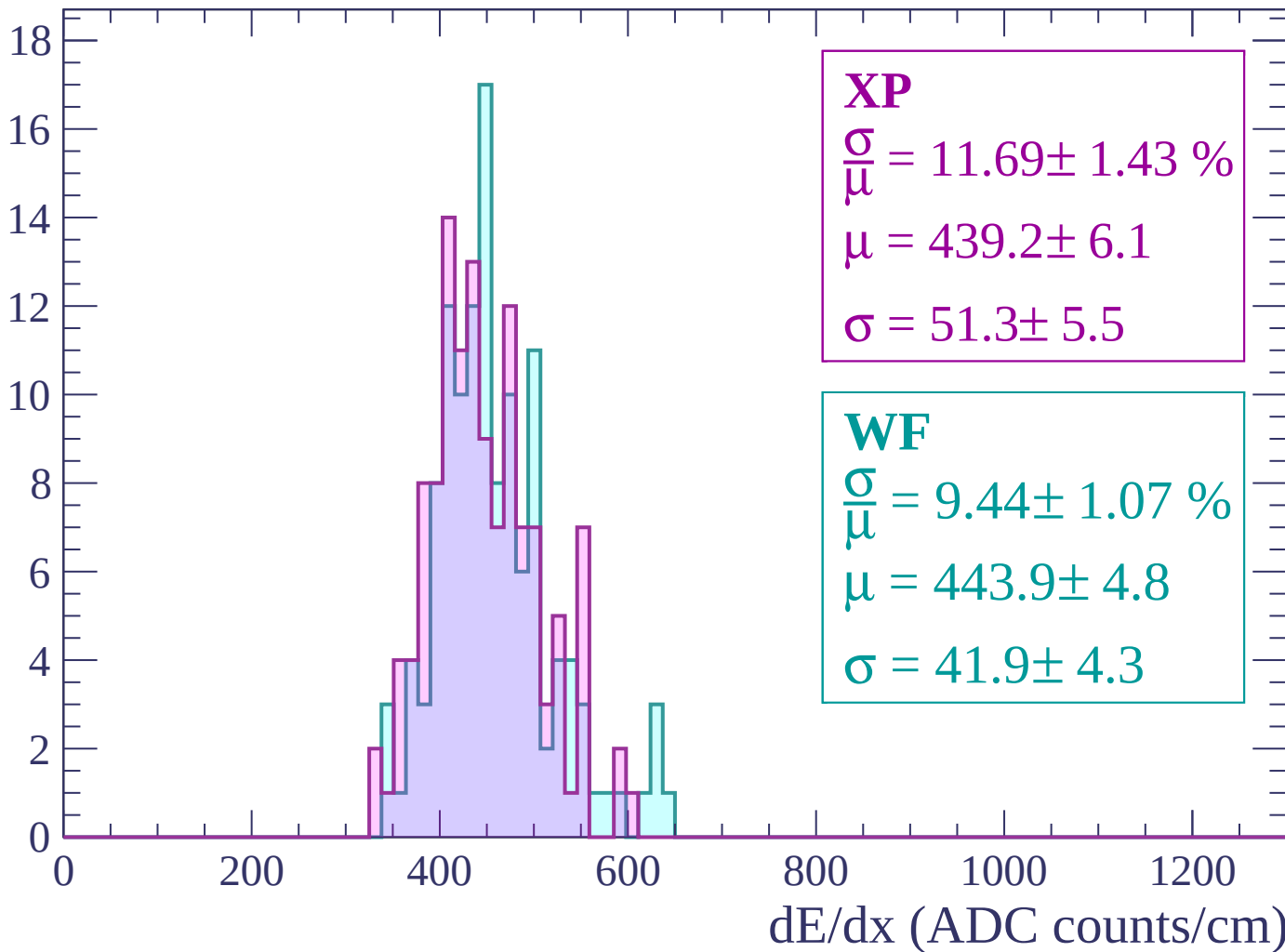
Energy loss | $1740 < p < 1800$

Count



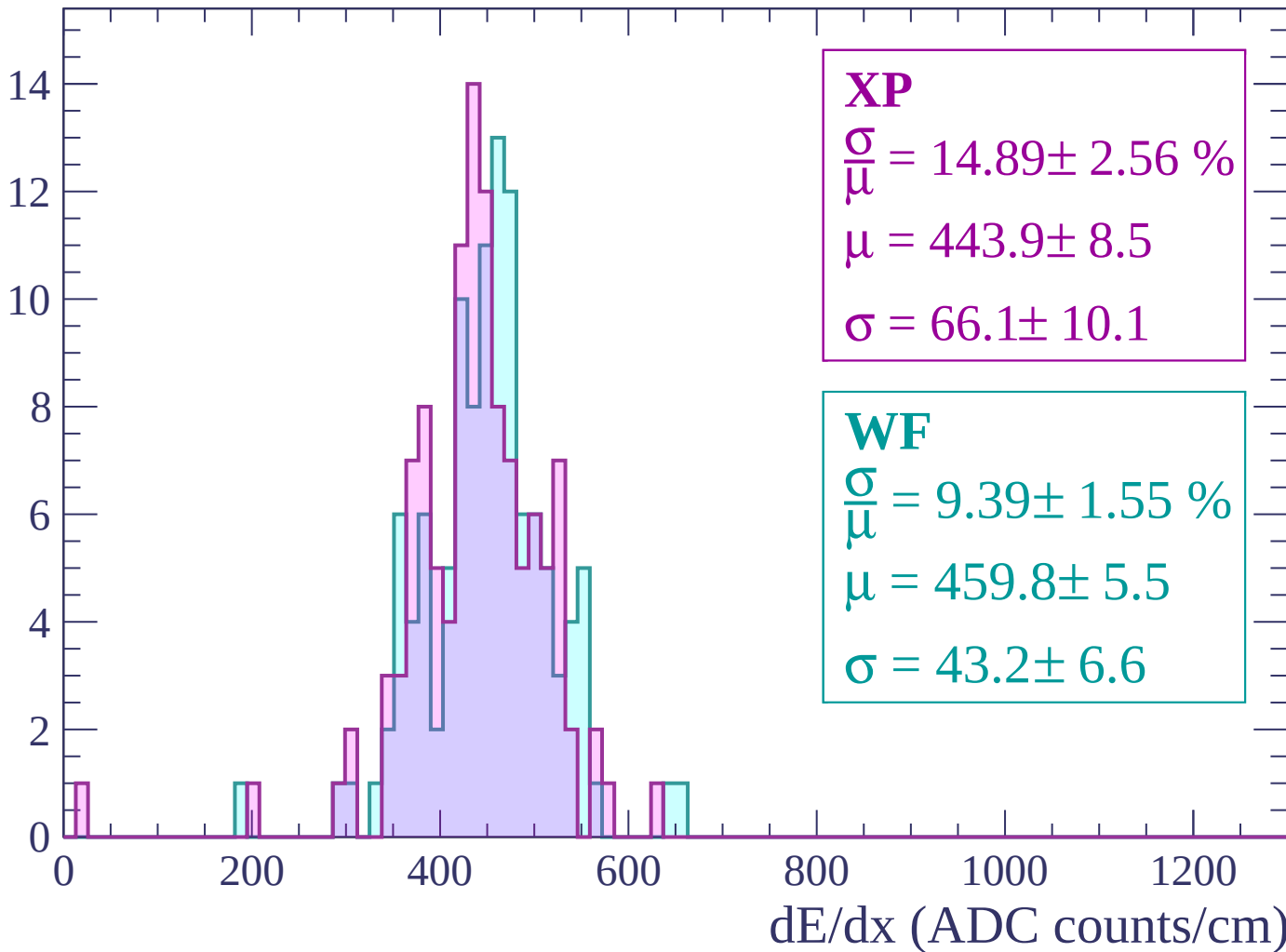
Energy loss | $1800 < p < 1860$

Count



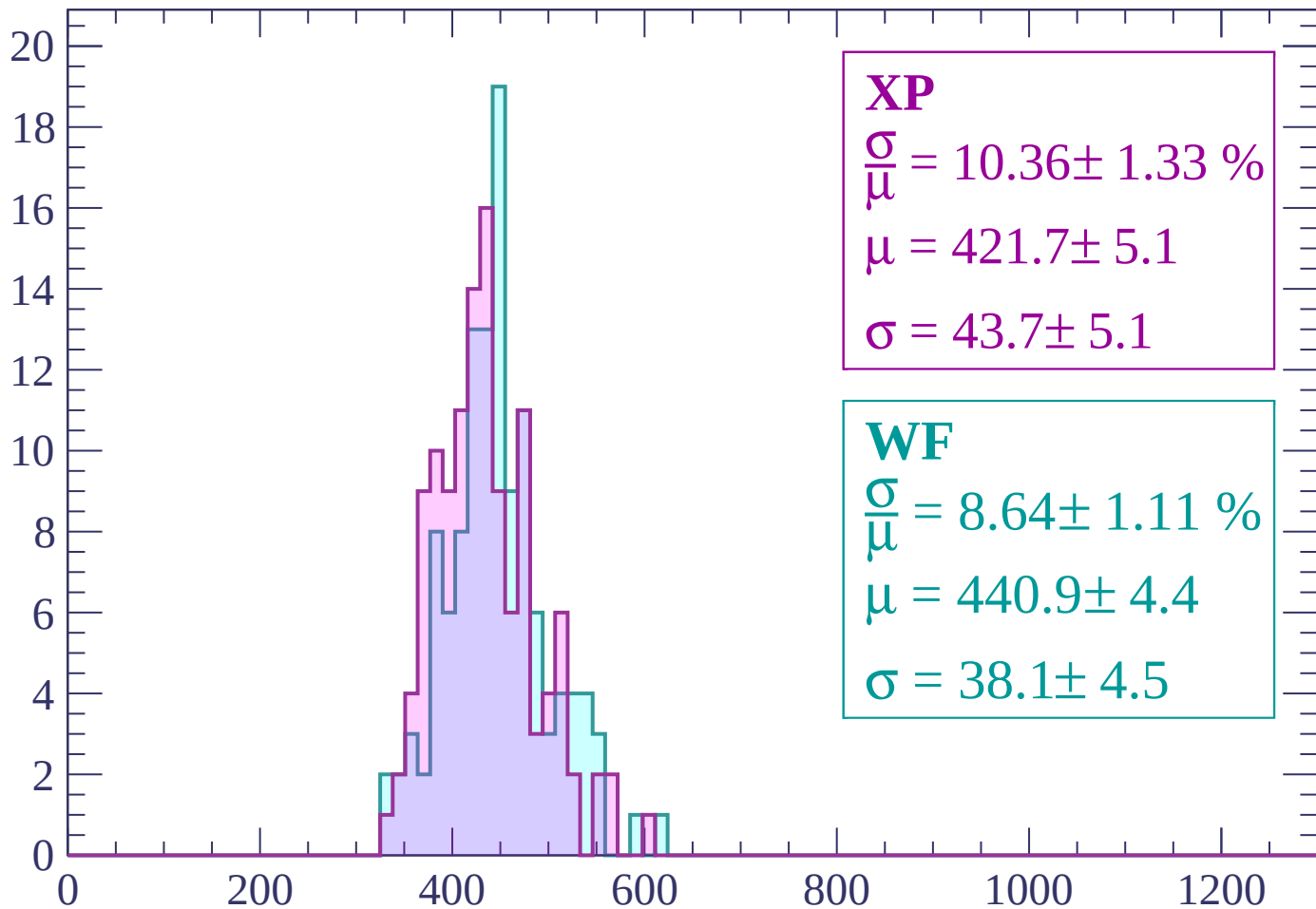
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP

$$\frac{\sigma}{\mu} = 10.36 \pm 1.33 \%$$

$$\mu = 421.7 \pm 5.1$$

$$\sigma = 43.7 \pm 5.1$$

WF

$$\frac{\sigma}{\mu} = 8.64 \pm 1.11 \%$$

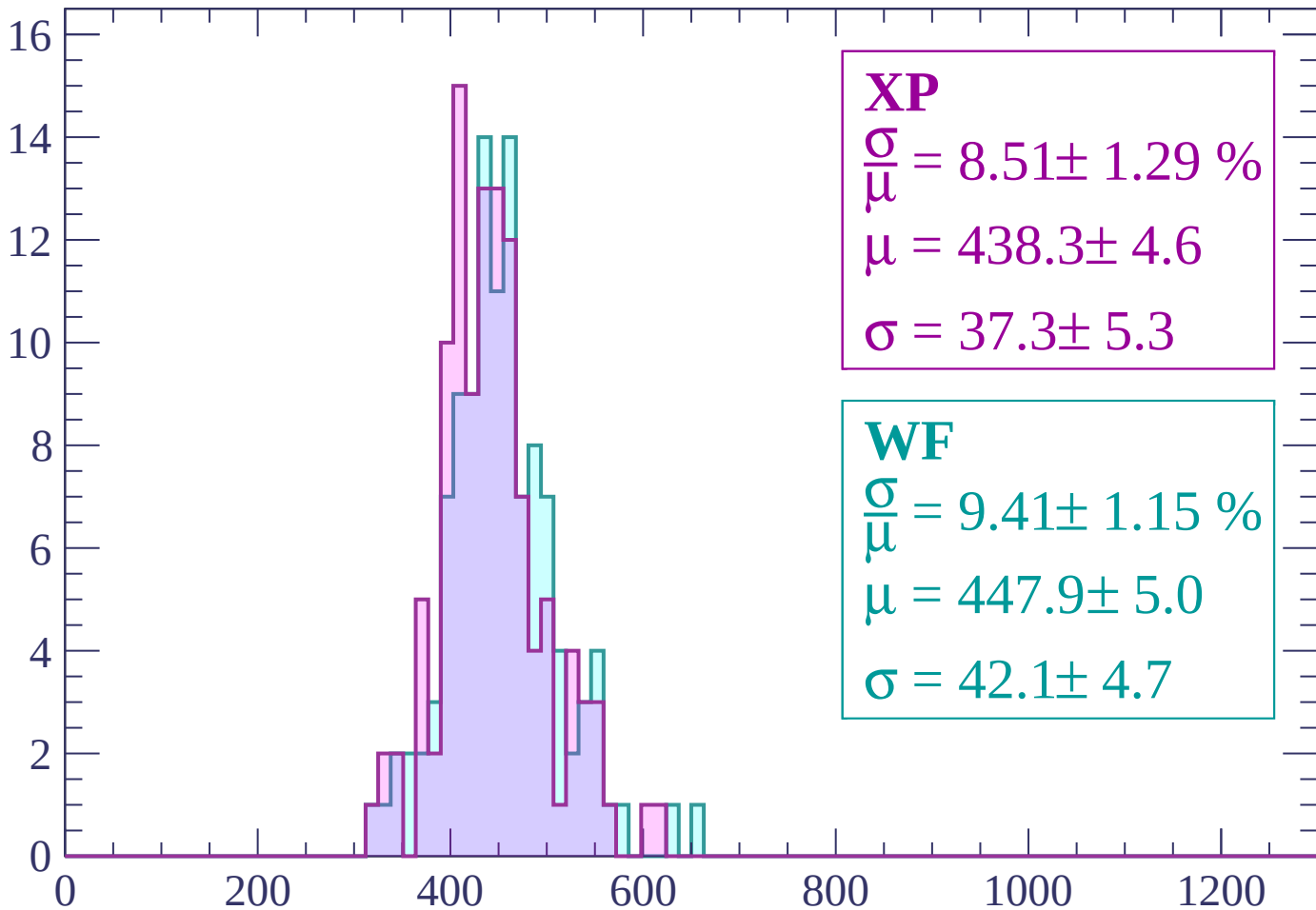
$$\mu = 440.9 \pm 4.4$$

$$\sigma = 38.1 \pm 4.5$$

dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count



XP

$$\frac{\sigma}{\mu} = 8.51 \pm 1.29 \%$$

$$\mu = 438.3 \pm 4.6$$

$$\sigma = 37.3 \pm 5.3$$

WF

$$\frac{\sigma}{\mu} = 9.41 \pm 1.15 \%$$

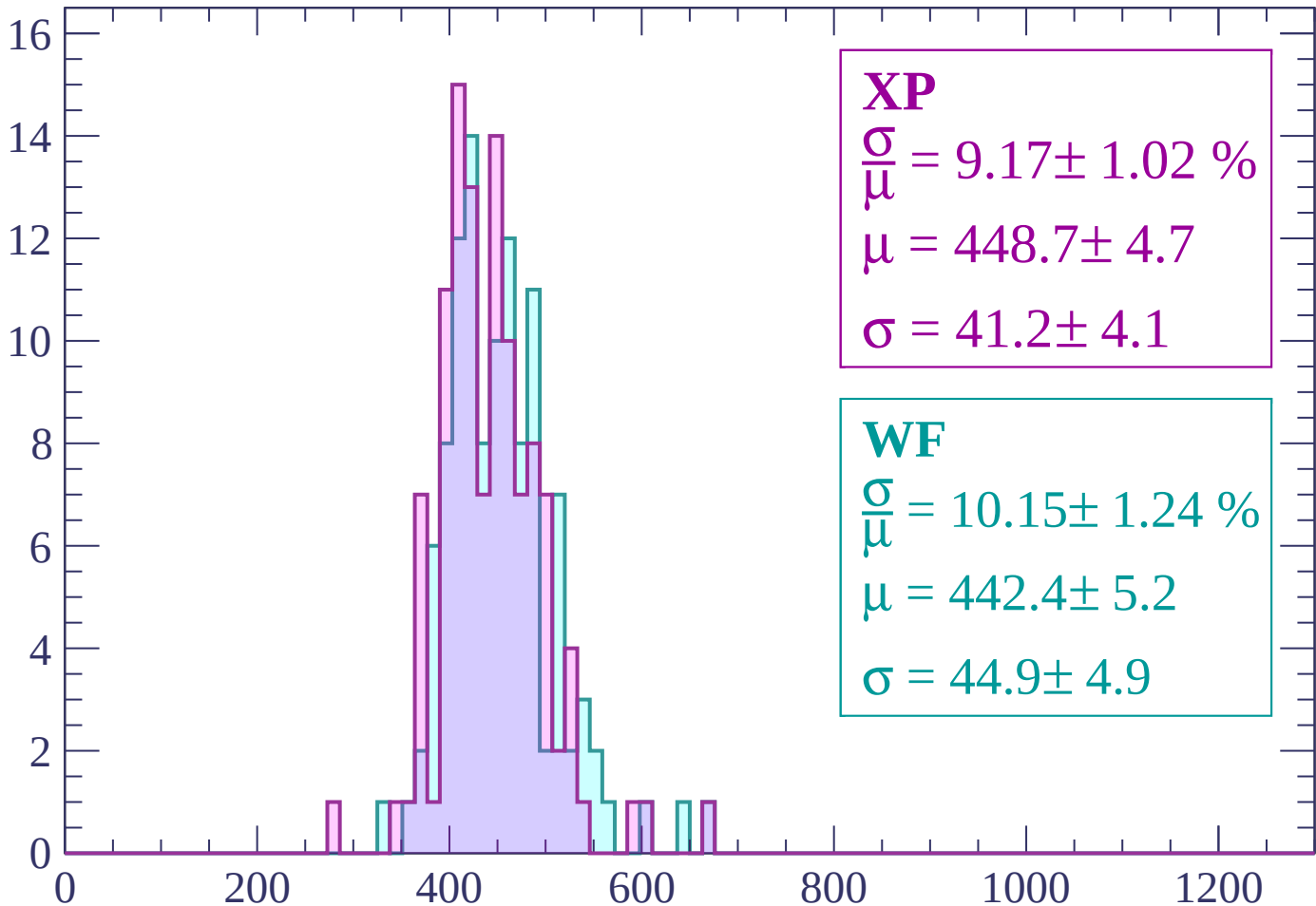
$$\mu = 447.9 \pm 5.0$$

$$\sigma = 42.1 \pm 4.7$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP

$$\frac{\sigma}{\mu} = 9.17 \pm 1.02 \%$$

$$\mu = 448.7 \pm 4.7$$

$$\sigma = 41.2 \pm 4.1$$

WF

$$\frac{\sigma}{\mu} = 10.15 \pm 1.24 \%$$

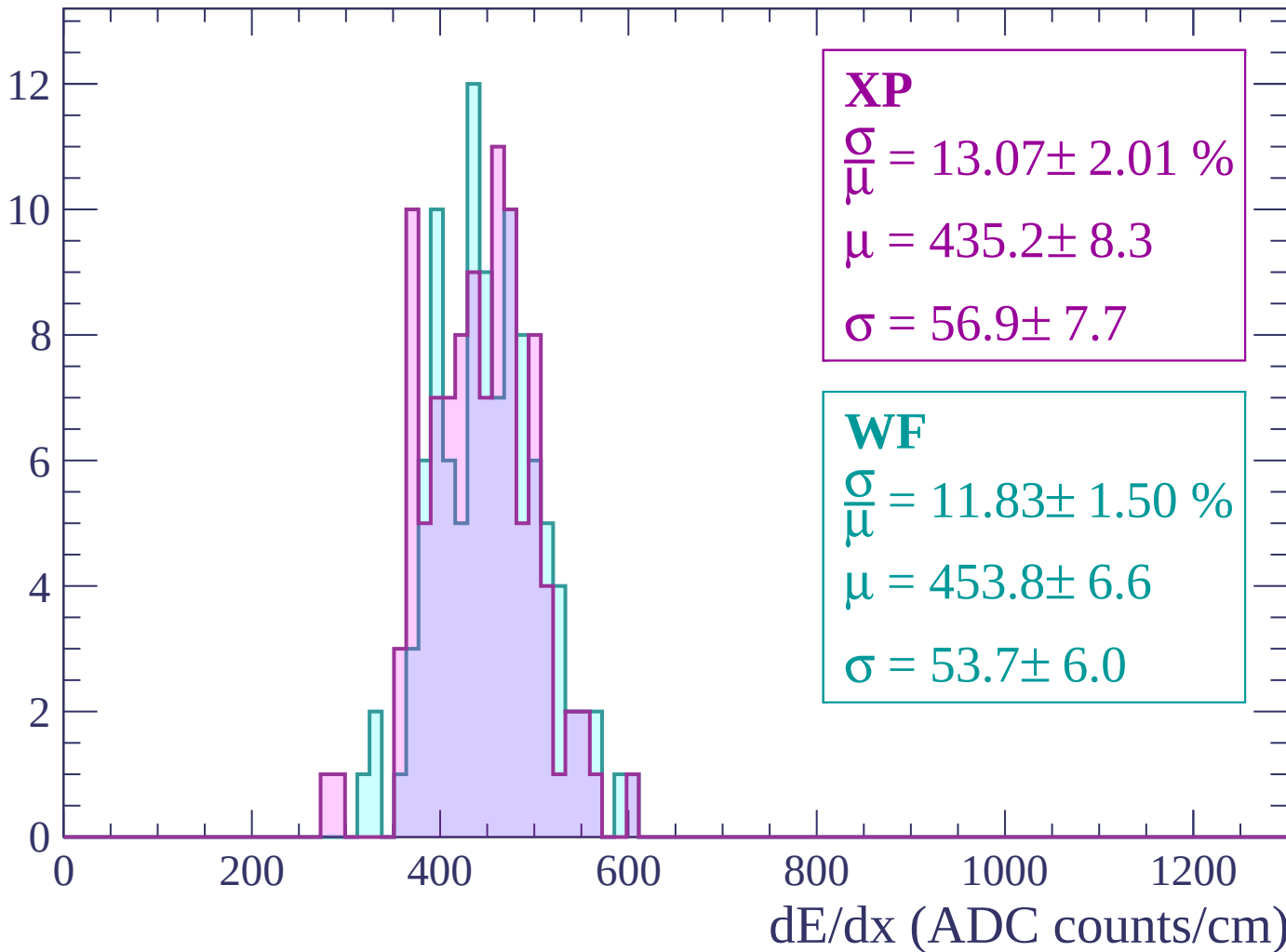
$$\mu = 442.4 \pm 5.2$$

$$\sigma = 44.9 \pm 4.9$$

dE/dx (ADC counts/cm)

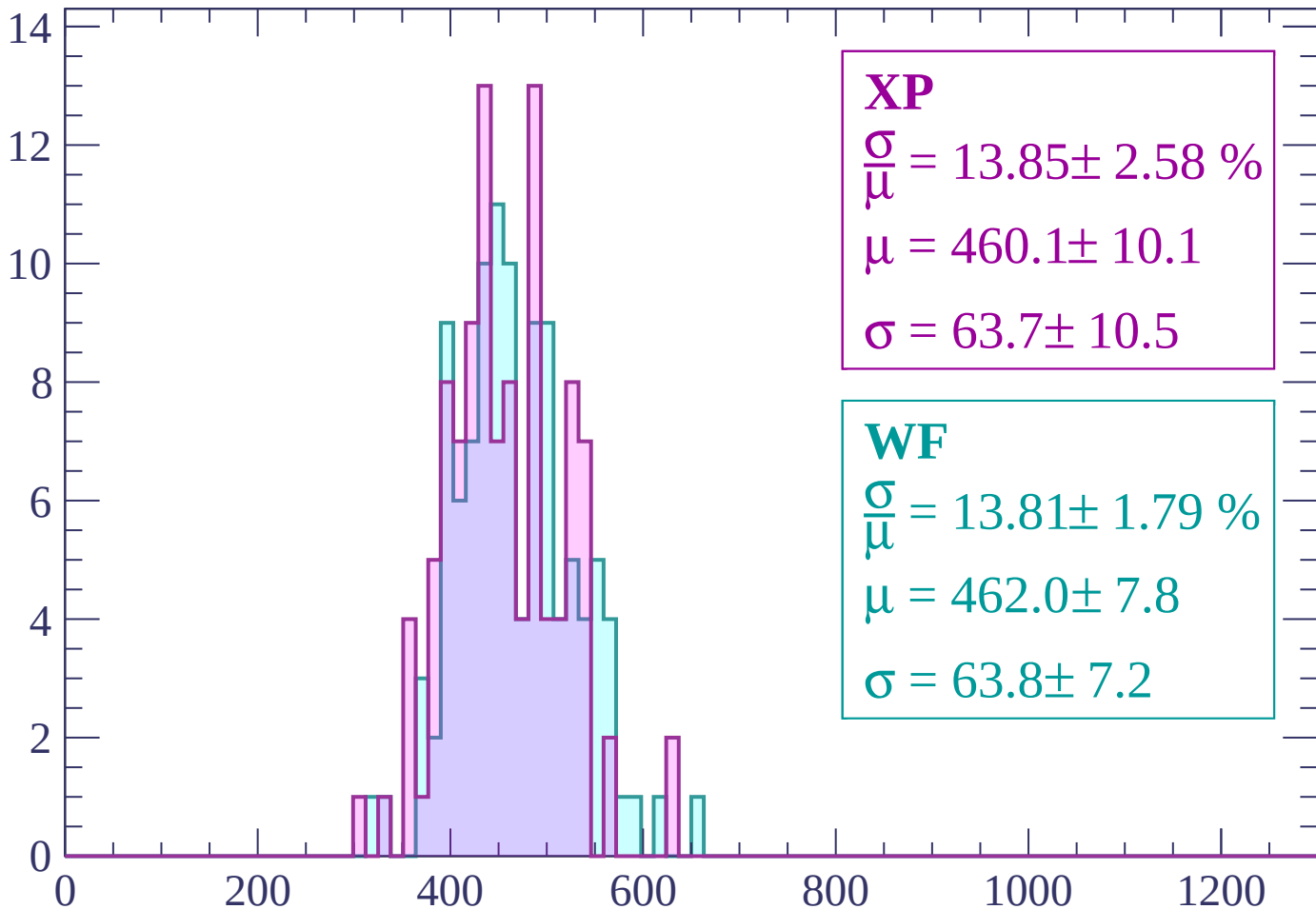
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP

$$\frac{\sigma}{\mu} = 13.85 \pm 2.58 \%$$

$$\mu = 460.1 \pm 10.1$$

$$\sigma = 63.7 \pm 10.5$$

WF

$$\frac{\sigma}{\mu} = 13.81 \pm 1.79 \%$$

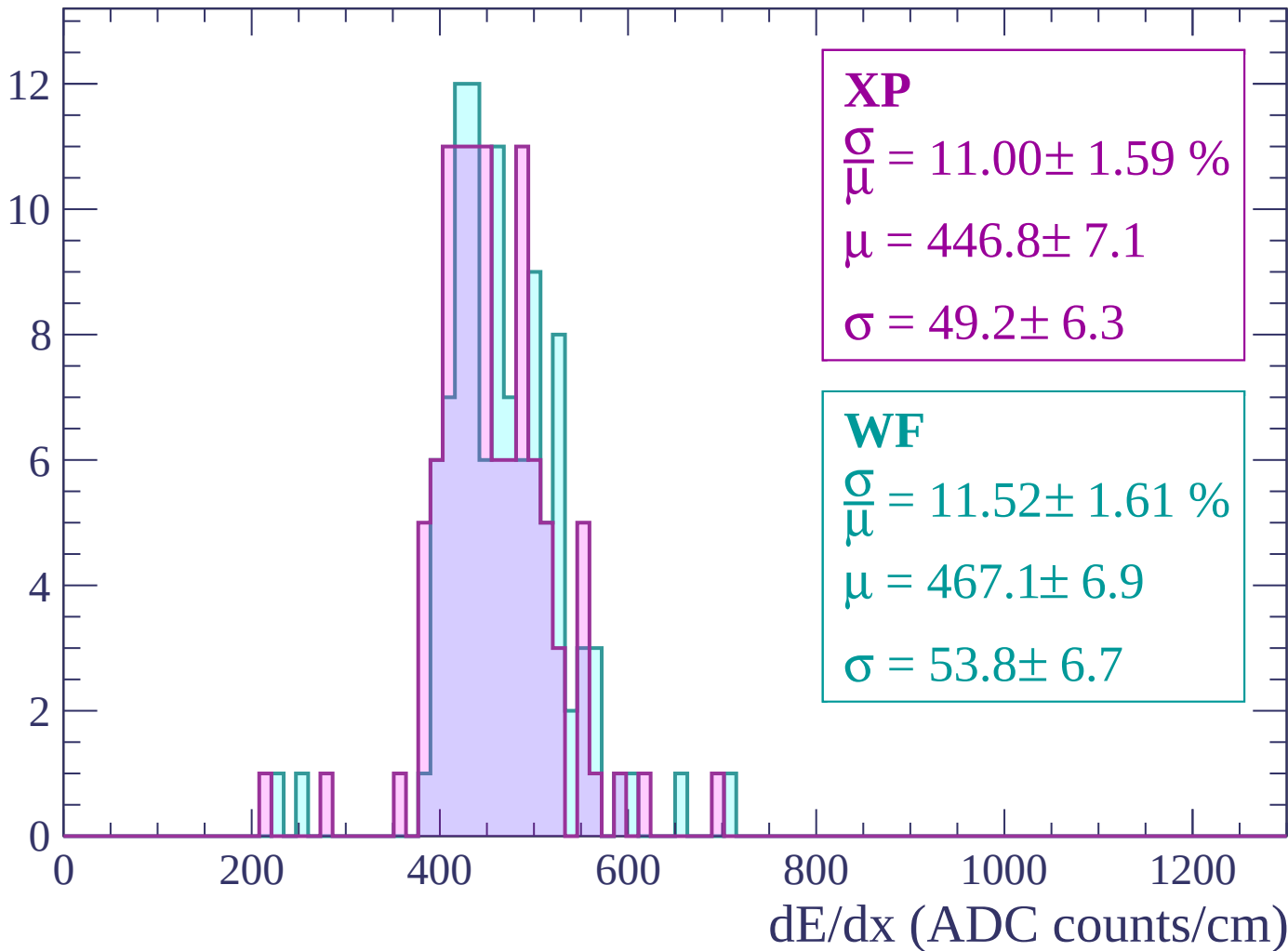
$$\mu = 462.0 \pm 7.8$$

$$\sigma = 63.8 \pm 7.2$$

dE/dx (ADC counts/cm)

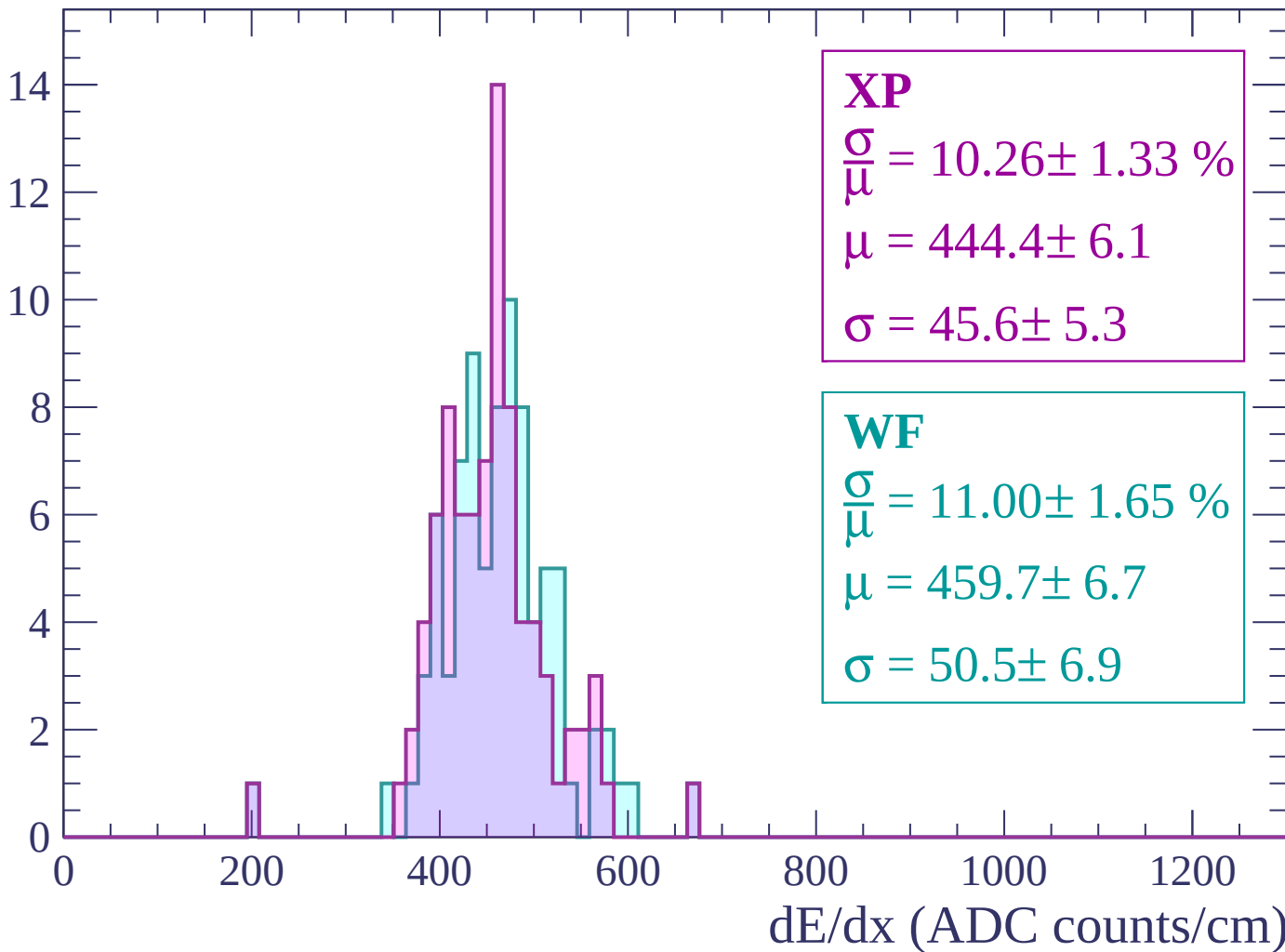
Energy loss | $2220 < p < 2280$

Count



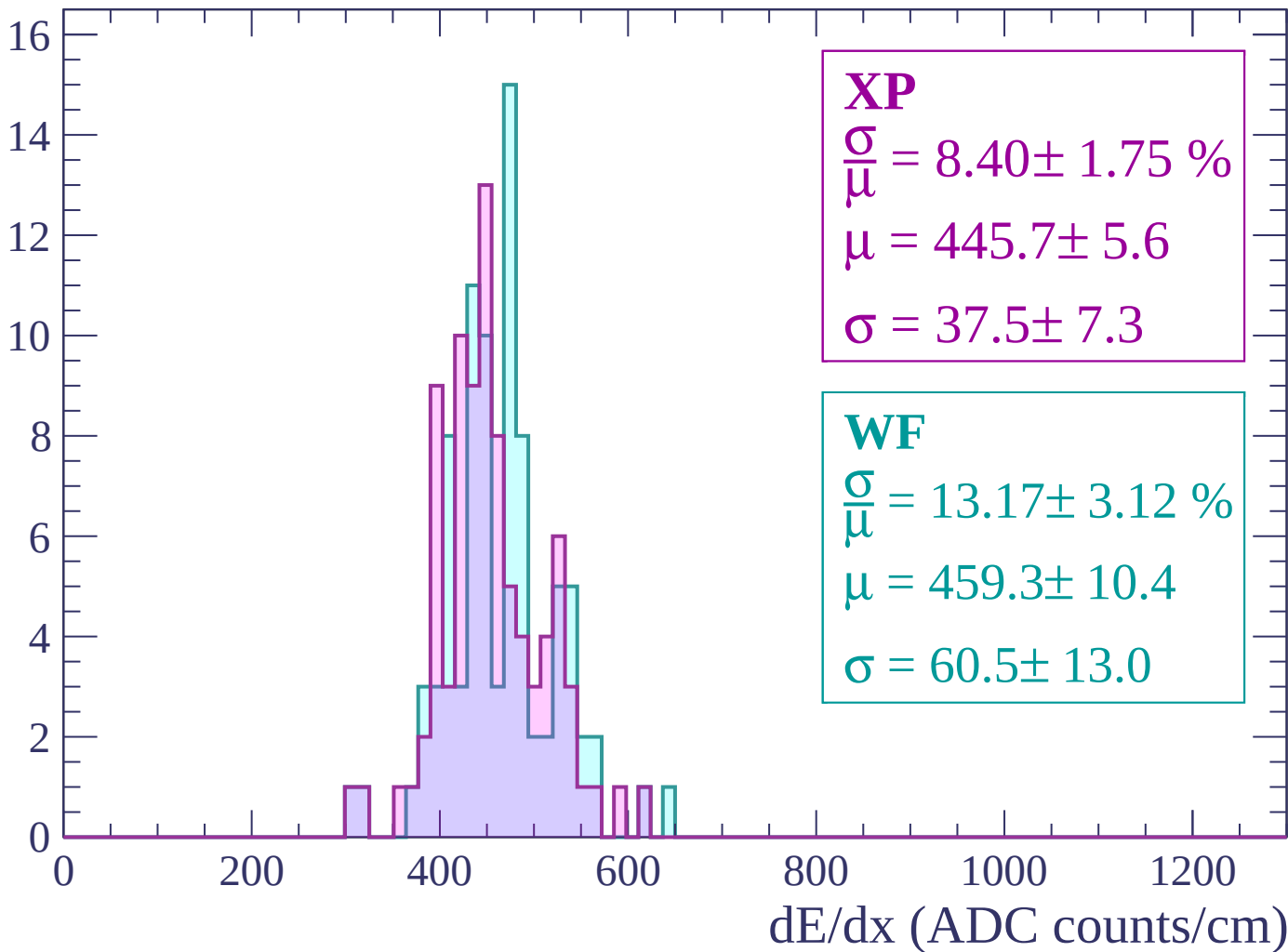
Energy loss | $2280 < p < 2340$

Count



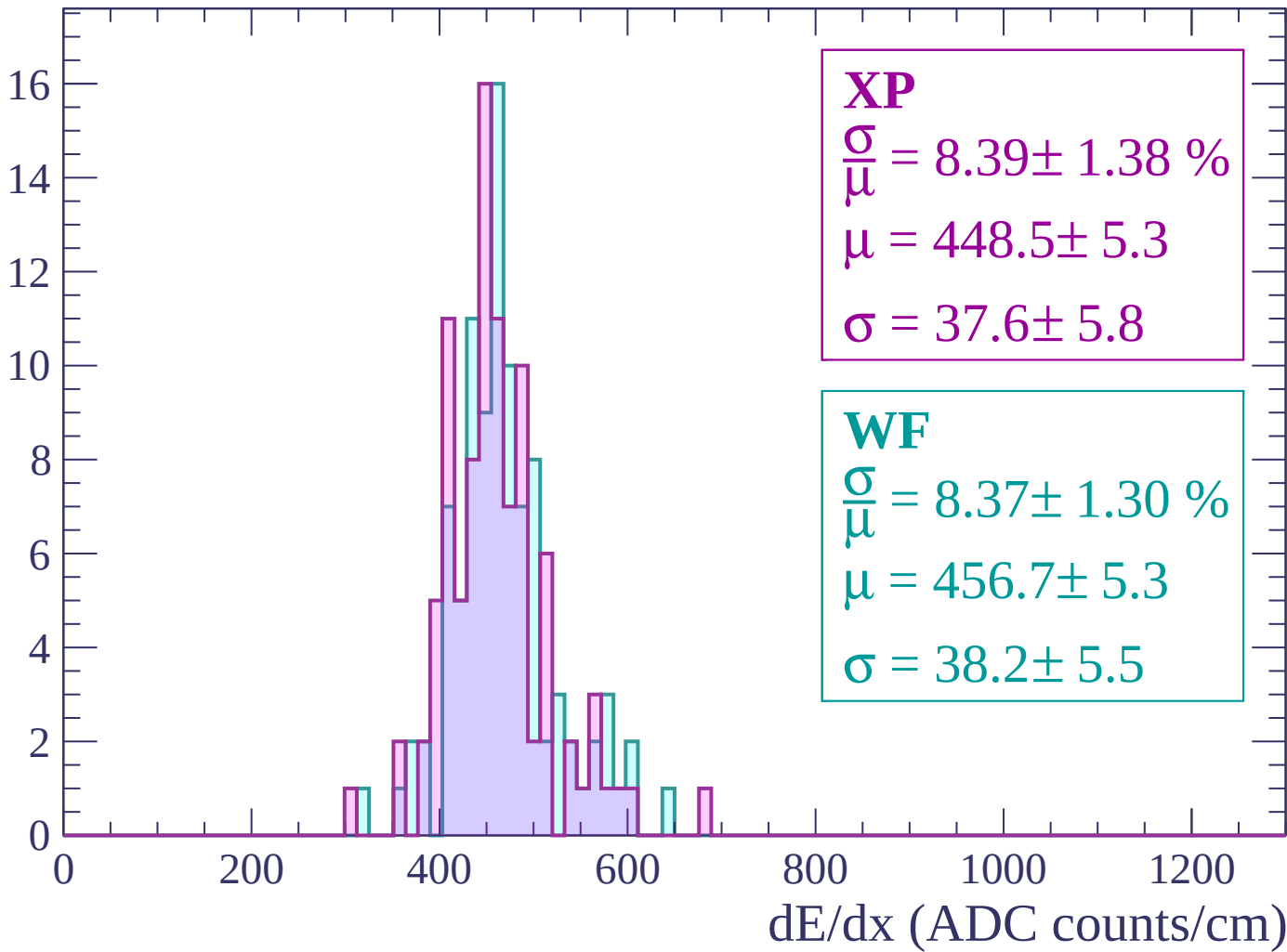
Energy loss | $2340 < p < 2400$

Count



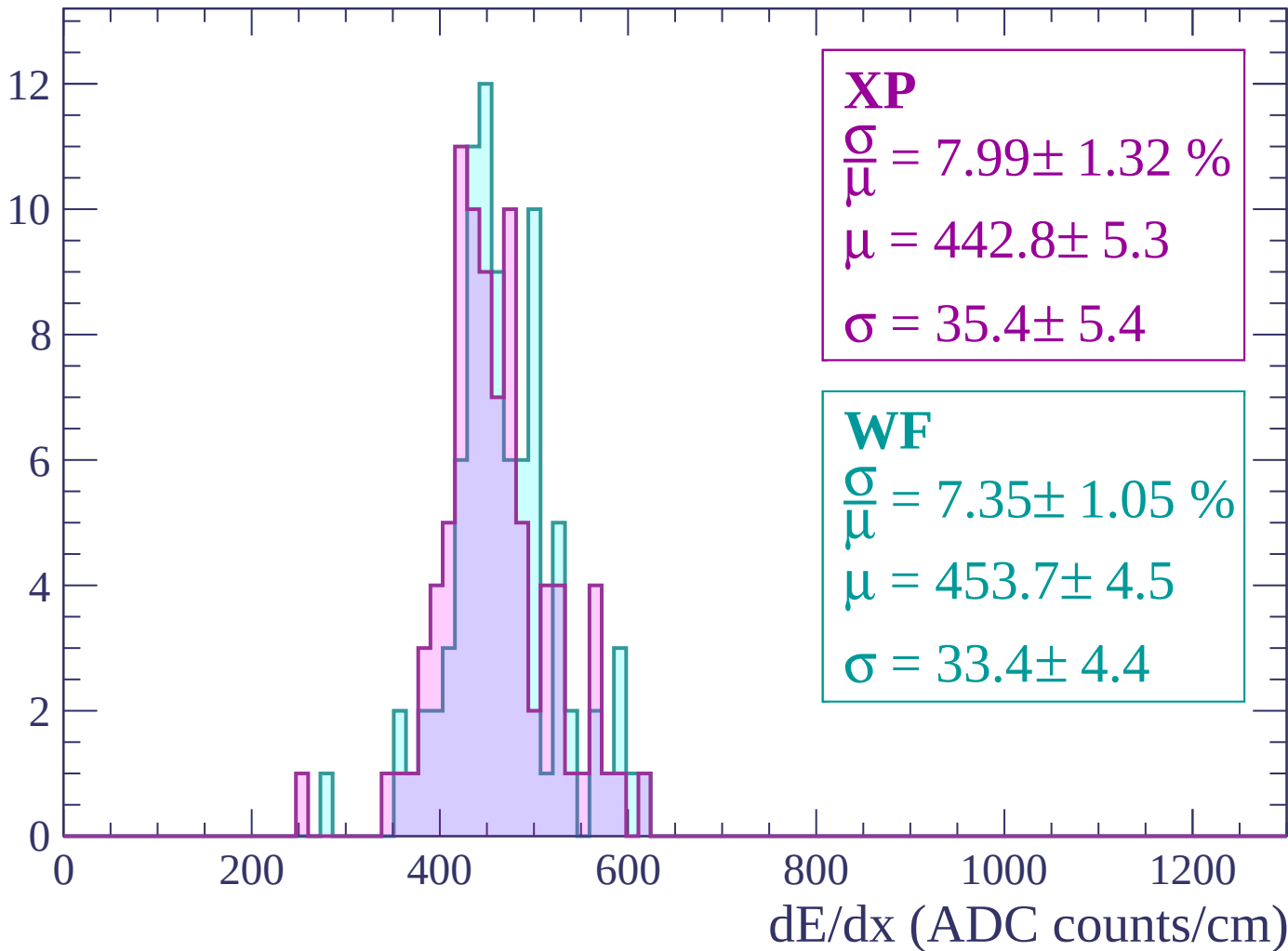
Energy loss | $2400 < p < 2460$

Count



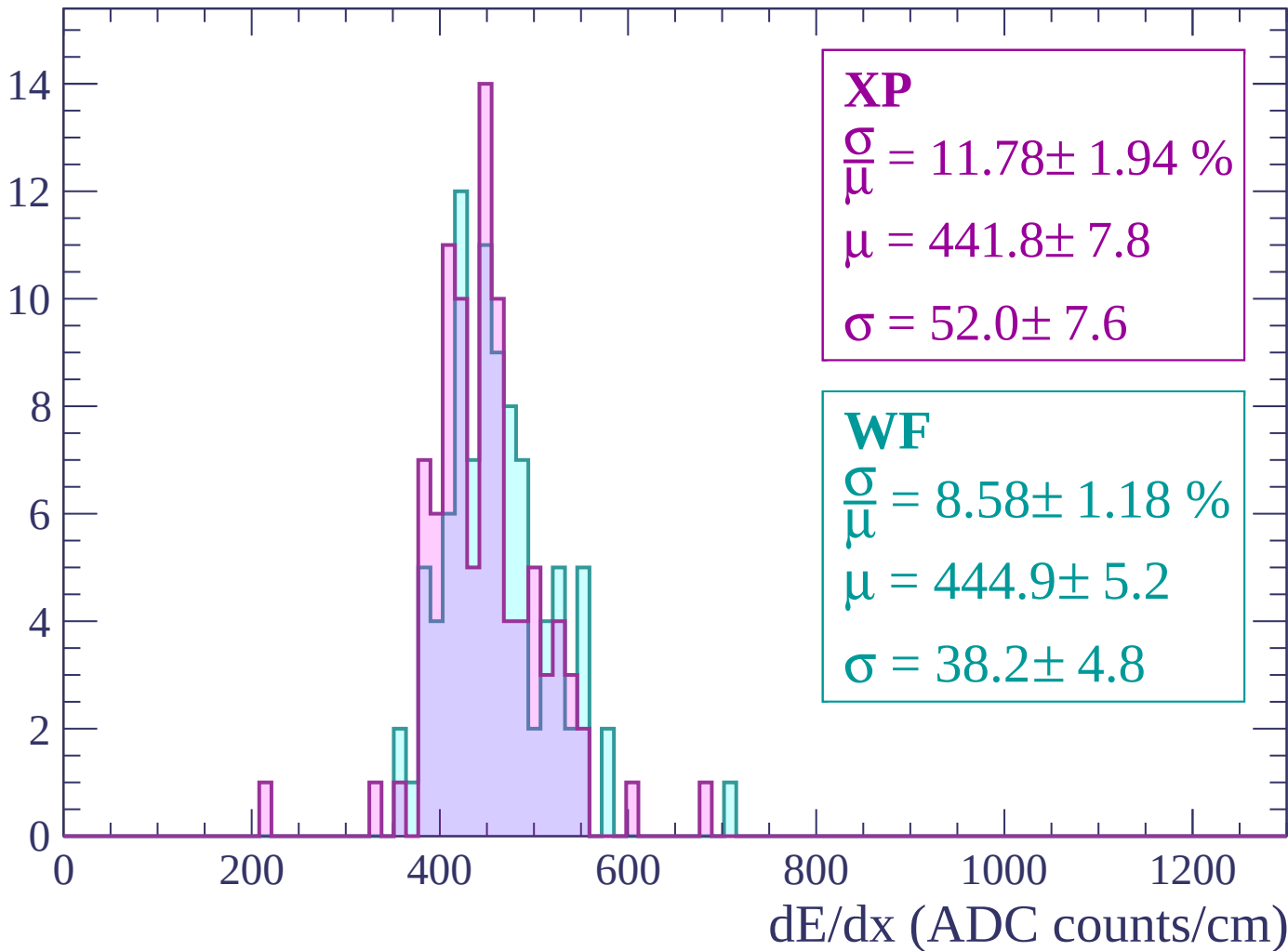
Energy loss | $2460 < p < 2520$

Count



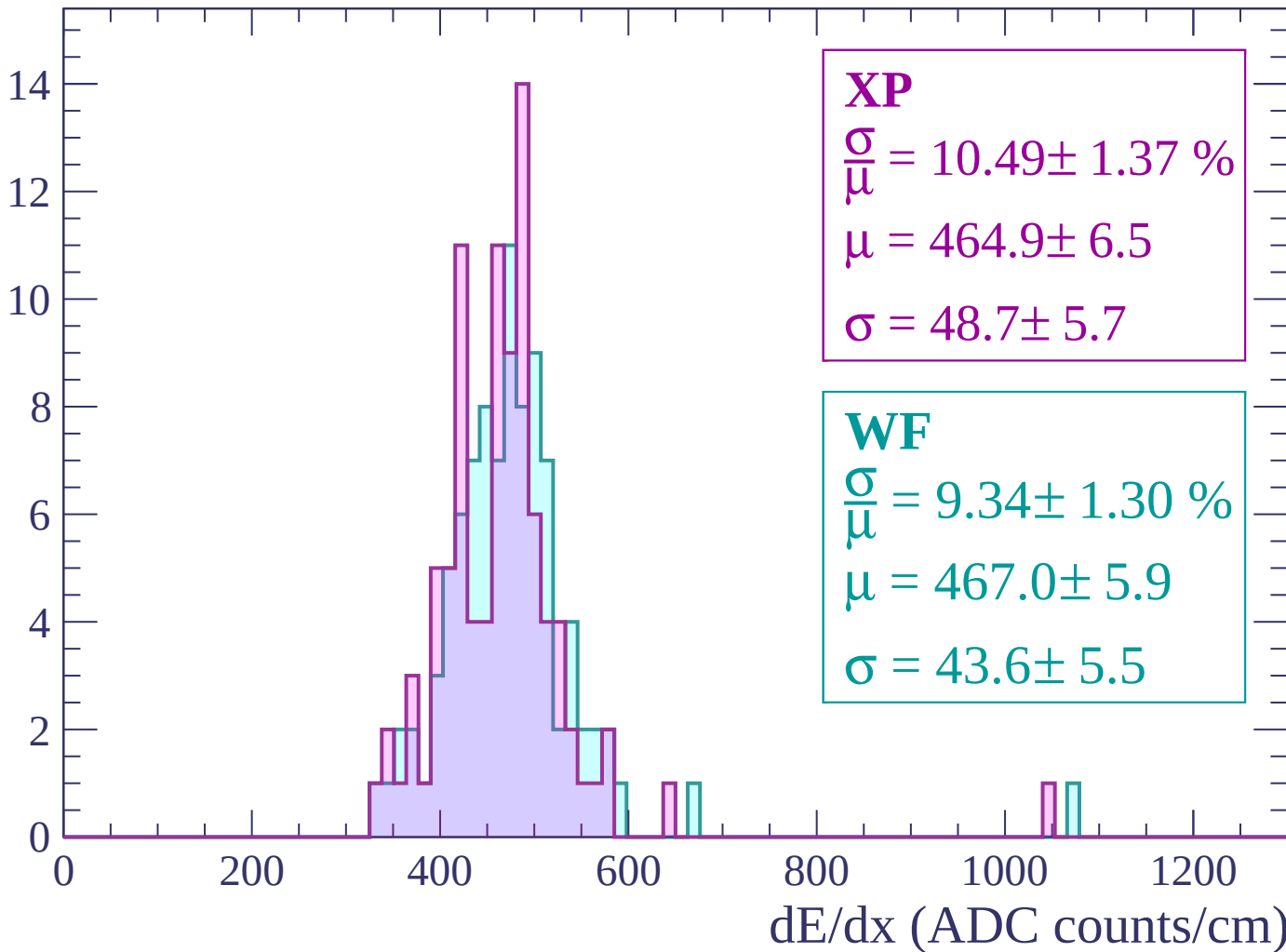
Energy loss | $2520 < p < 2580$

Count



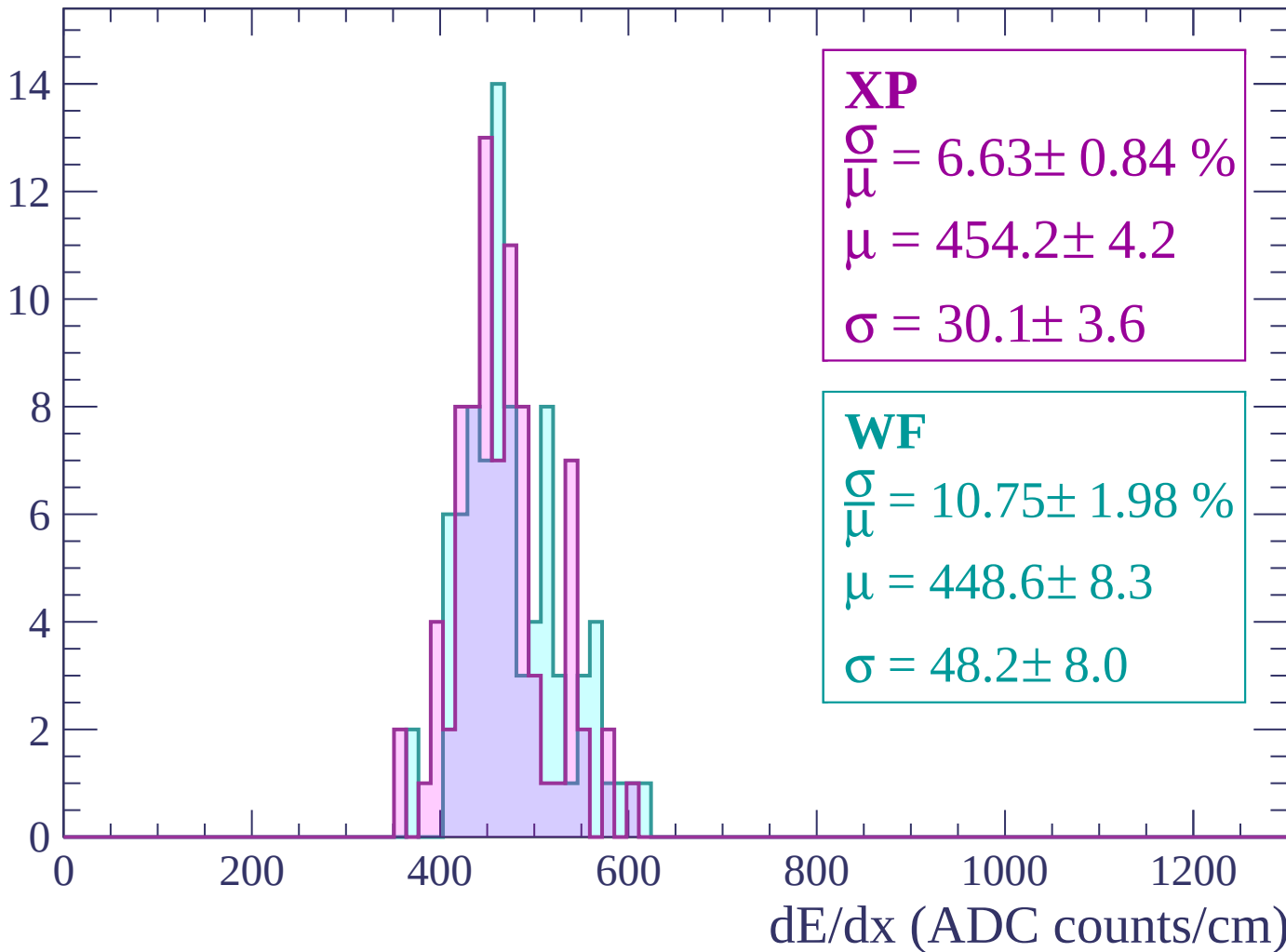
Energy loss | $2580 < p < 2640$

Count



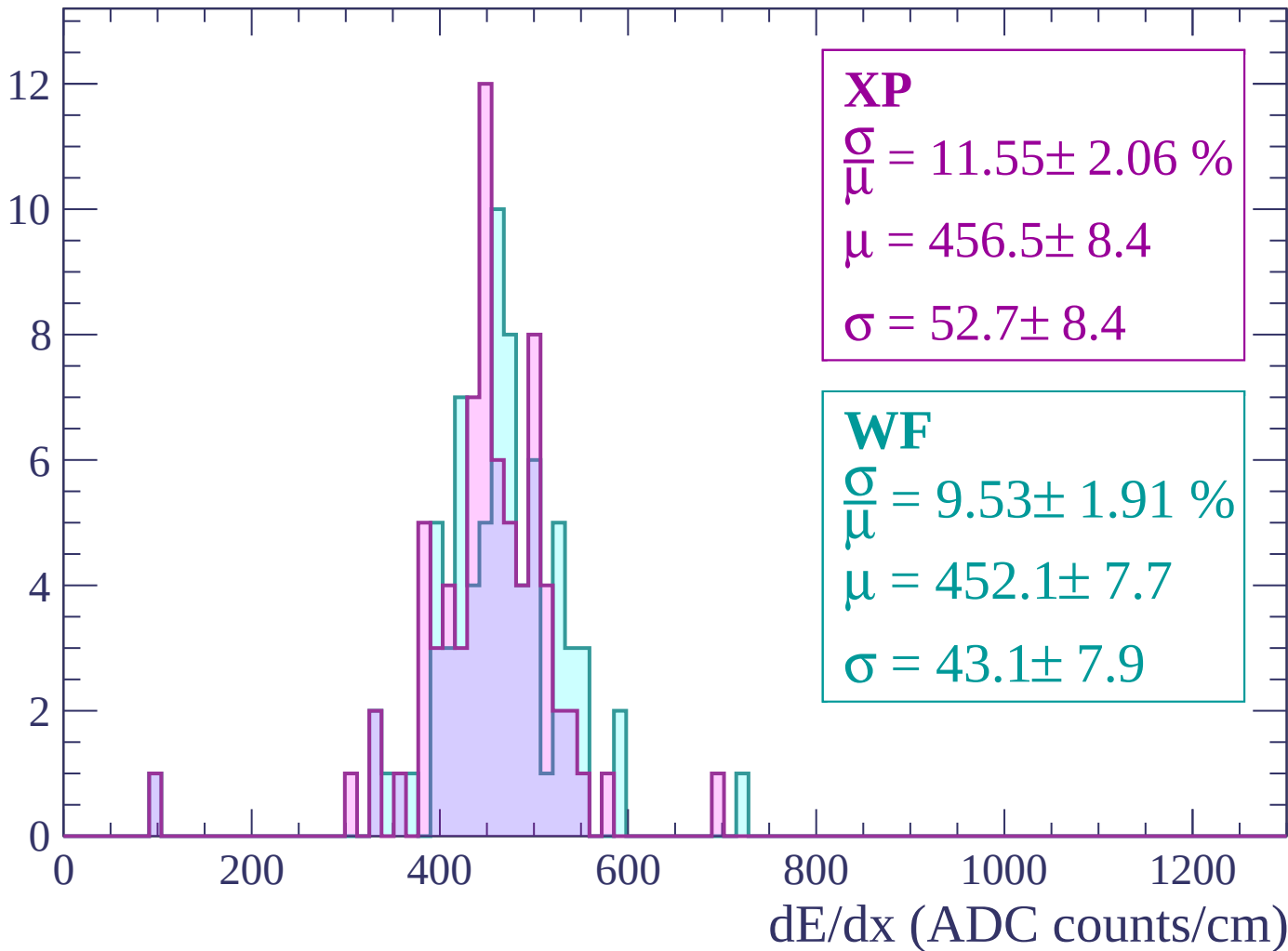
Energy loss | $2640 < p < 2700$

Count



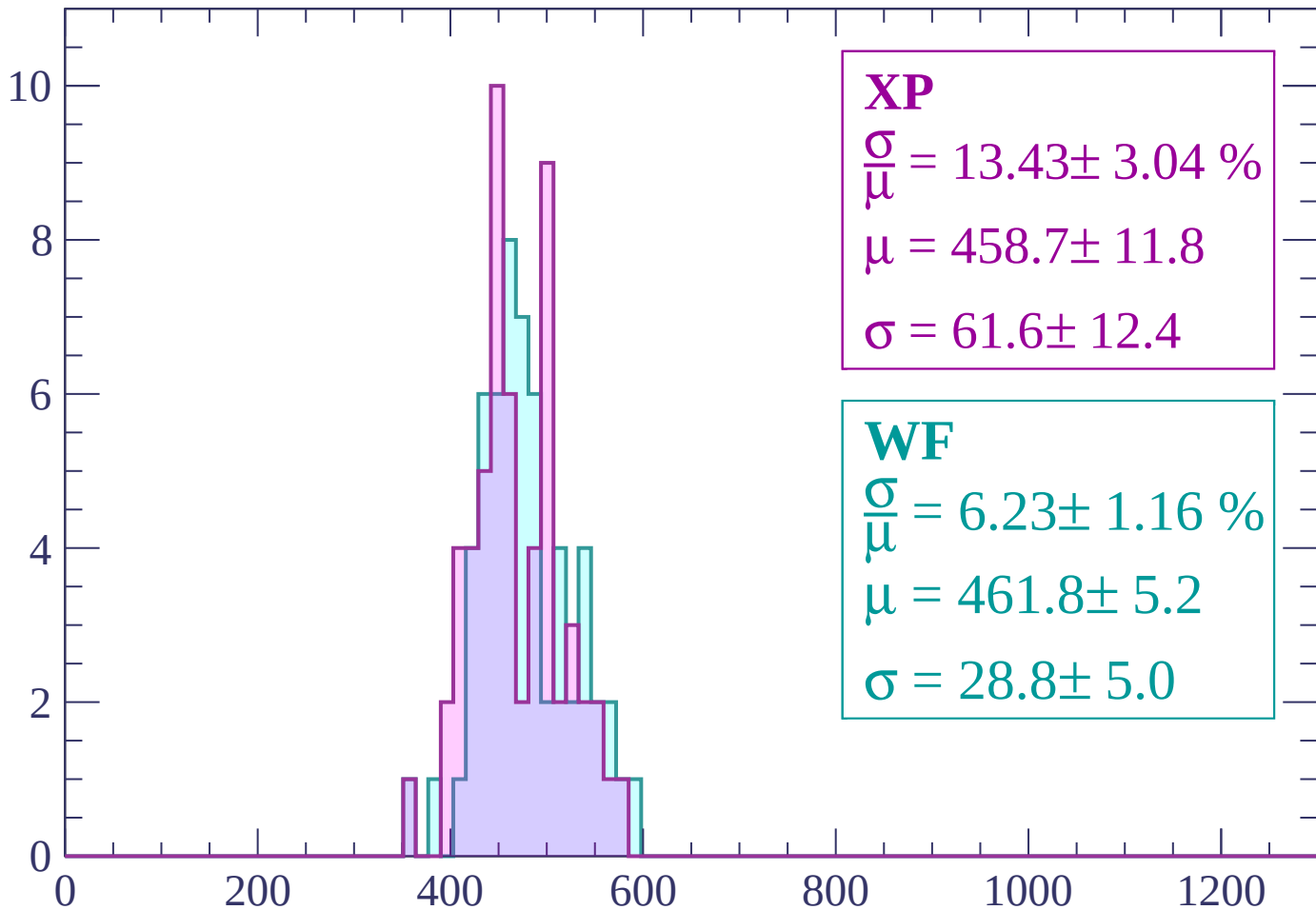
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP

$$\frac{\sigma}{\mu} = 13.43 \pm 3.04 \%$$

$$\mu = 458.7 \pm 11.8$$

$$\sigma = 61.6 \pm 12.4$$

WF

$$\frac{\sigma}{\mu} = 6.23 \pm 1.16 \%$$

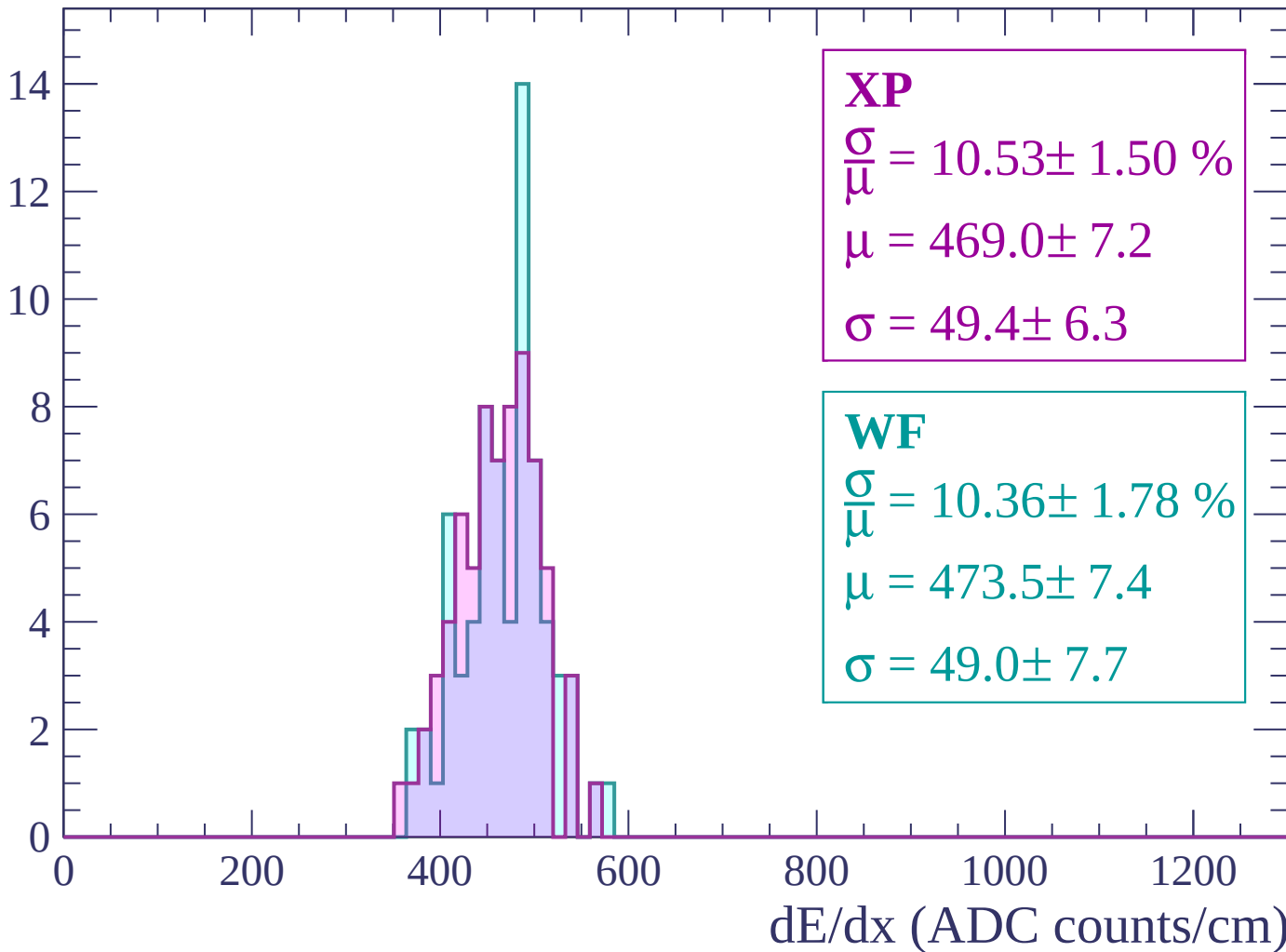
$$\mu = 461.8 \pm 5.2$$

$$\sigma = 28.8 \pm 5.0$$

dE/dx (ADC counts/cm)

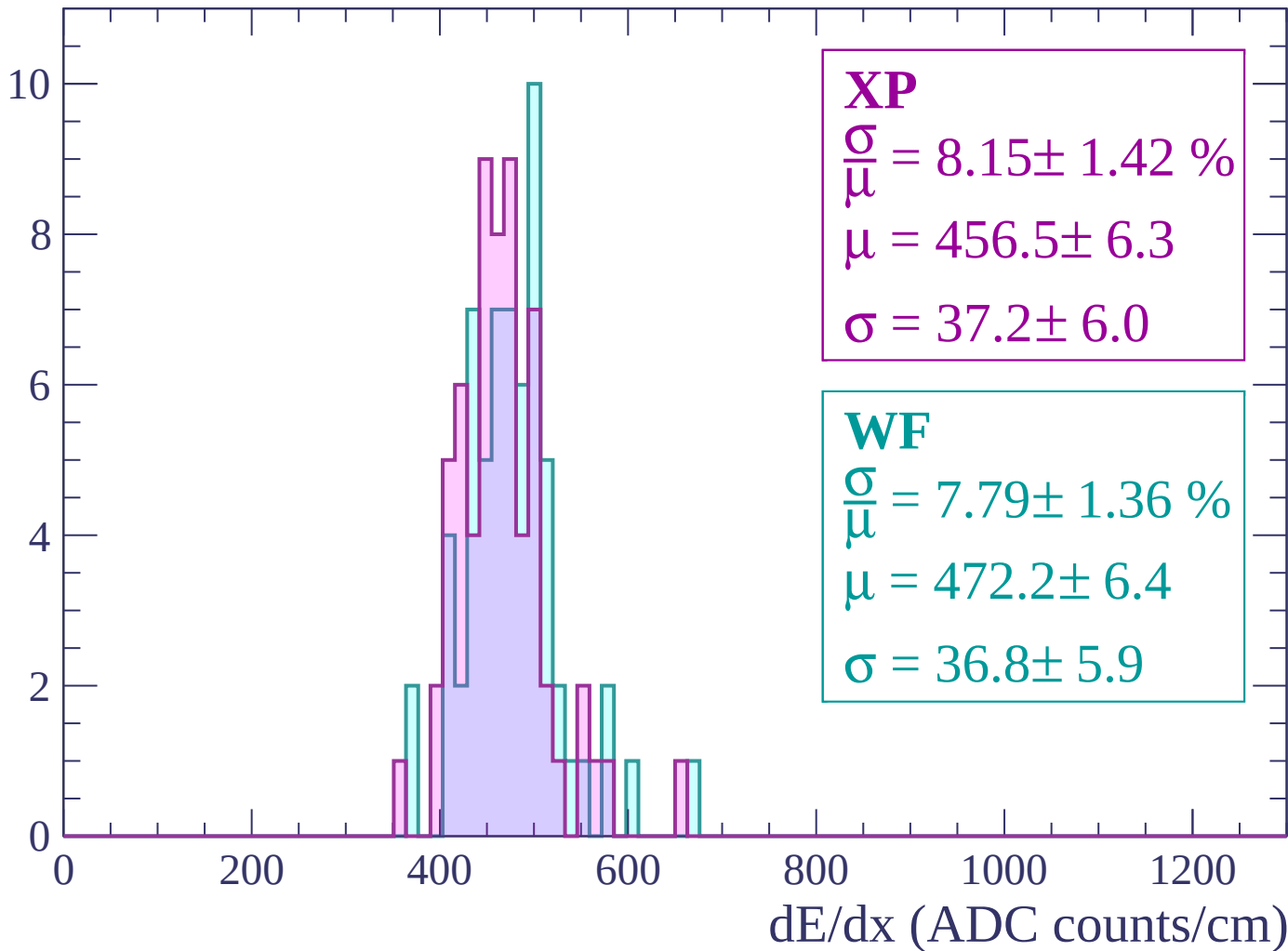
Energy loss | $2820 < p < 2880$

Count



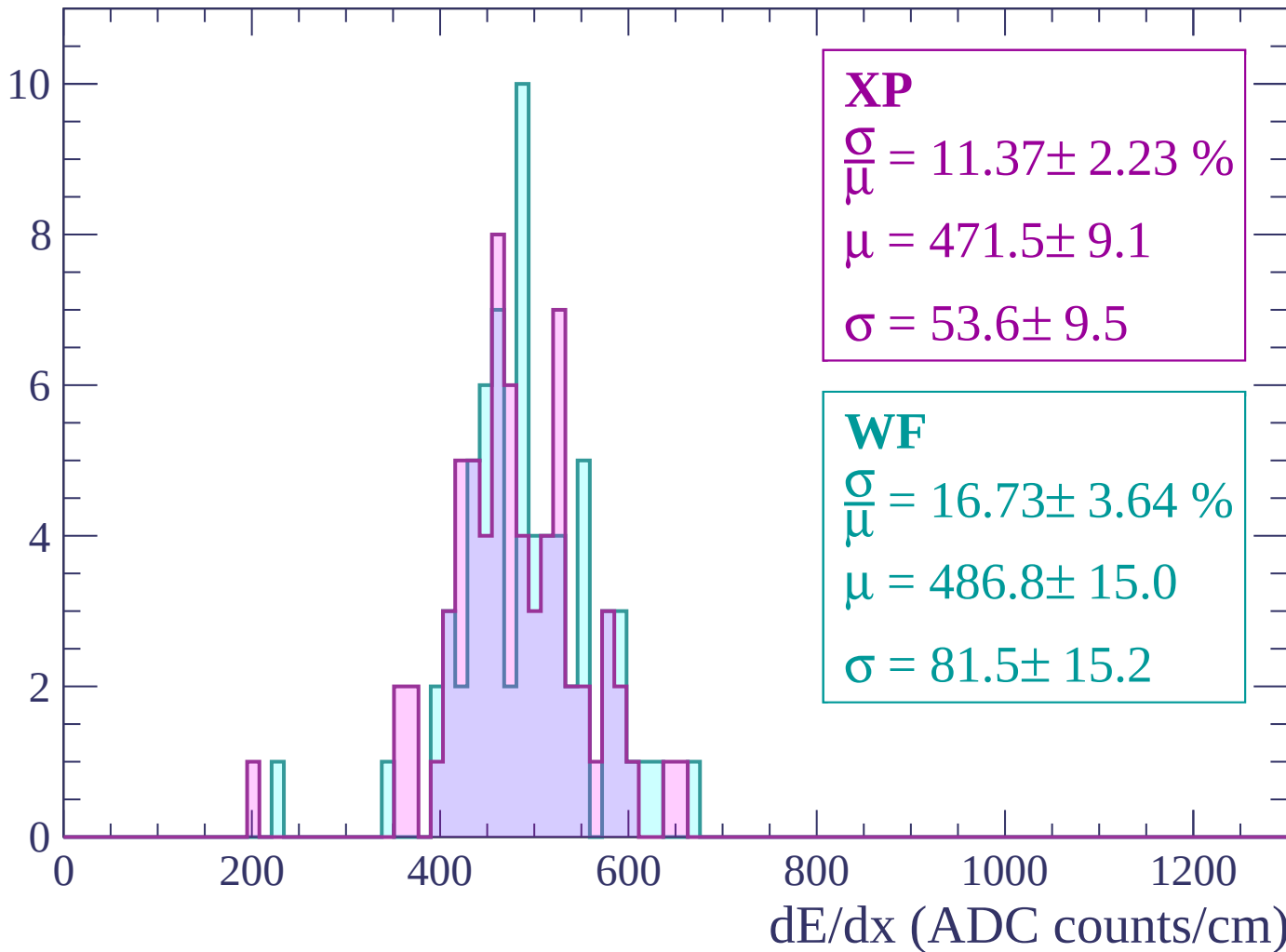
Energy loss | $2880 < p < 2940$

Count



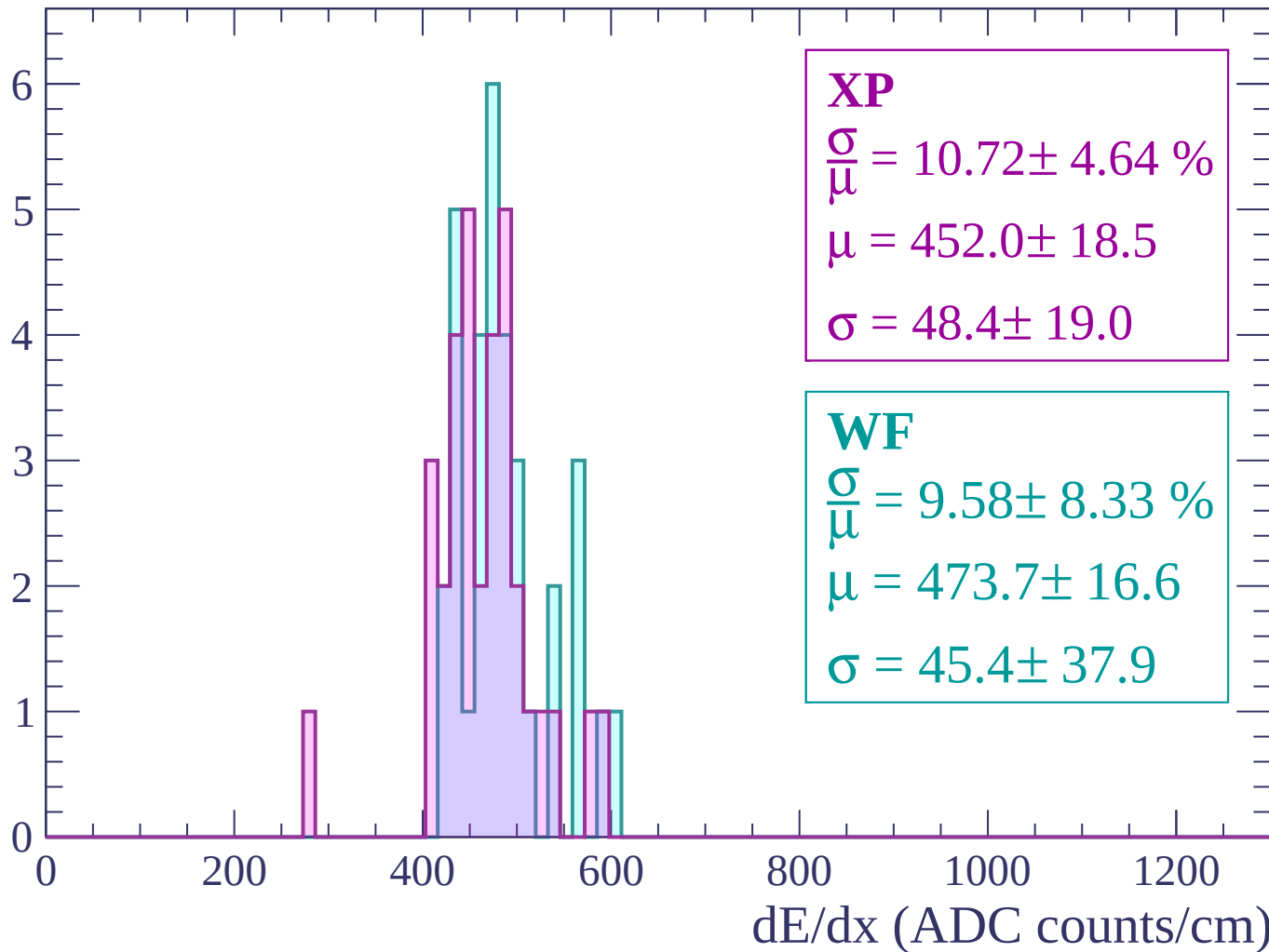
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

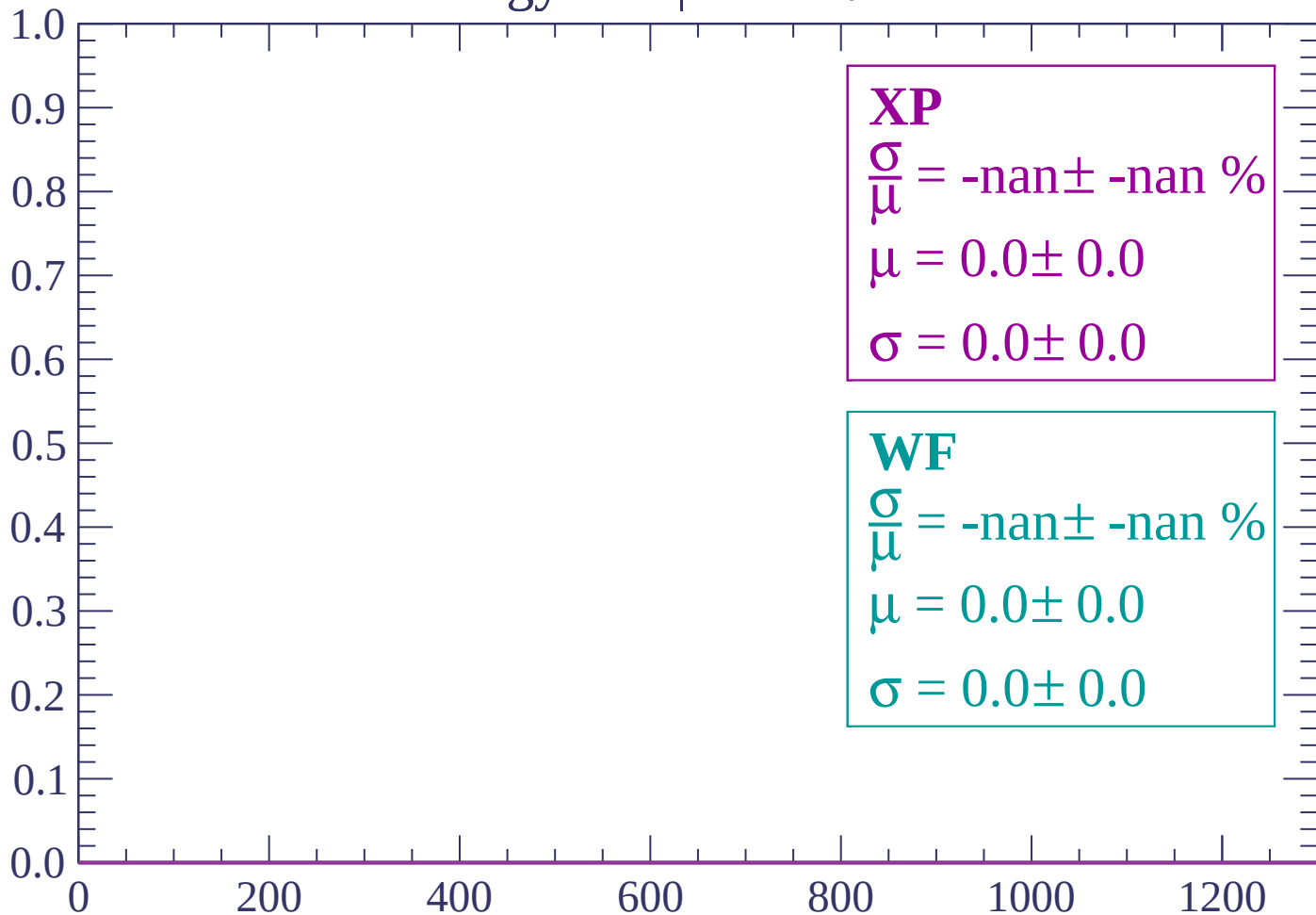
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

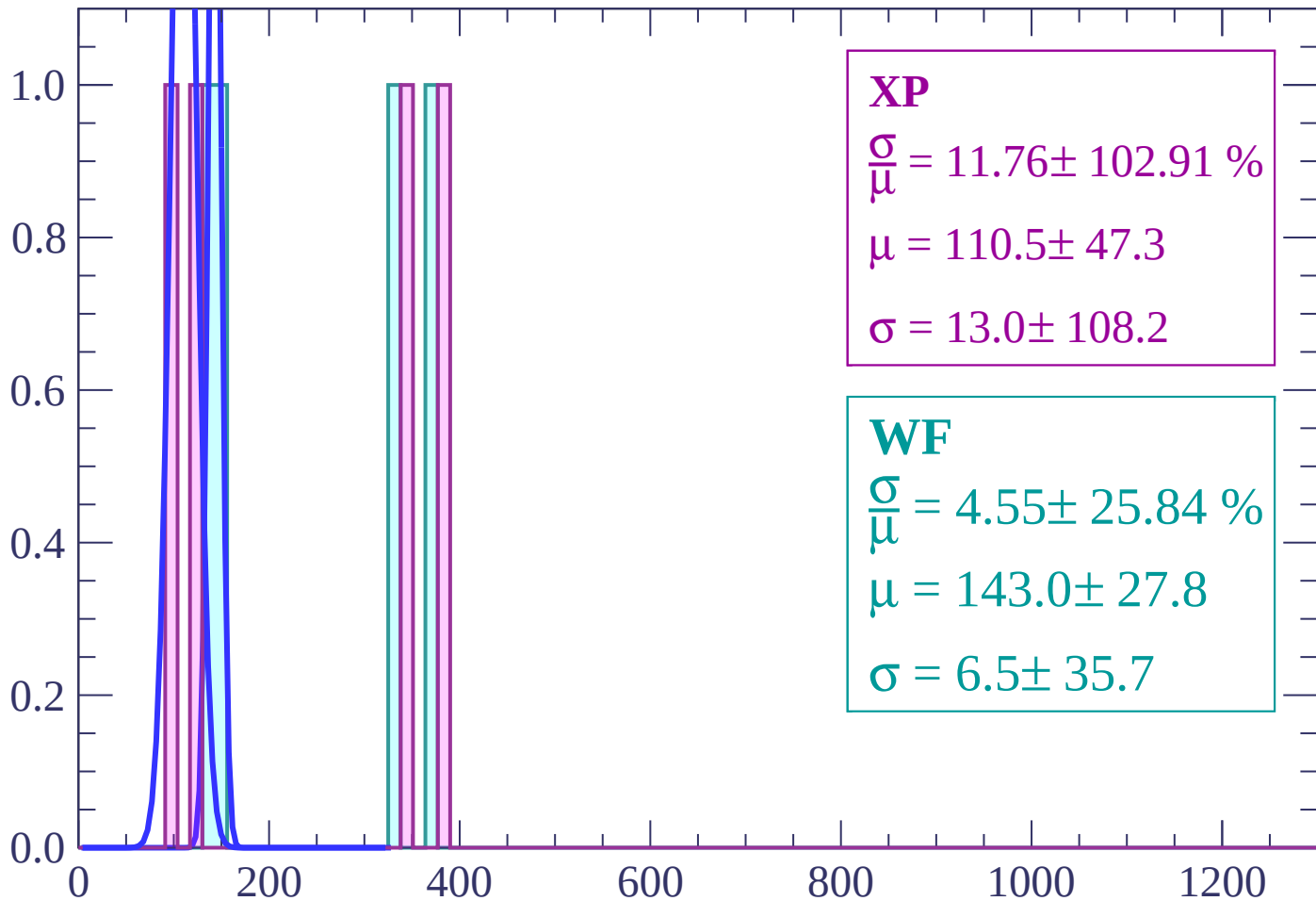
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-76 < \theta < -74$

Count



XP

$$\frac{\sigma}{\mu} = 11.76 \pm 102.91 \%$$

$$\mu = 110.5 \pm 47.3$$

$$\sigma = 13.0 \pm 108.2$$

WF

$$\frac{\sigma}{\mu} = 4.55 \pm 25.84 \%$$

$$\mu = 143.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

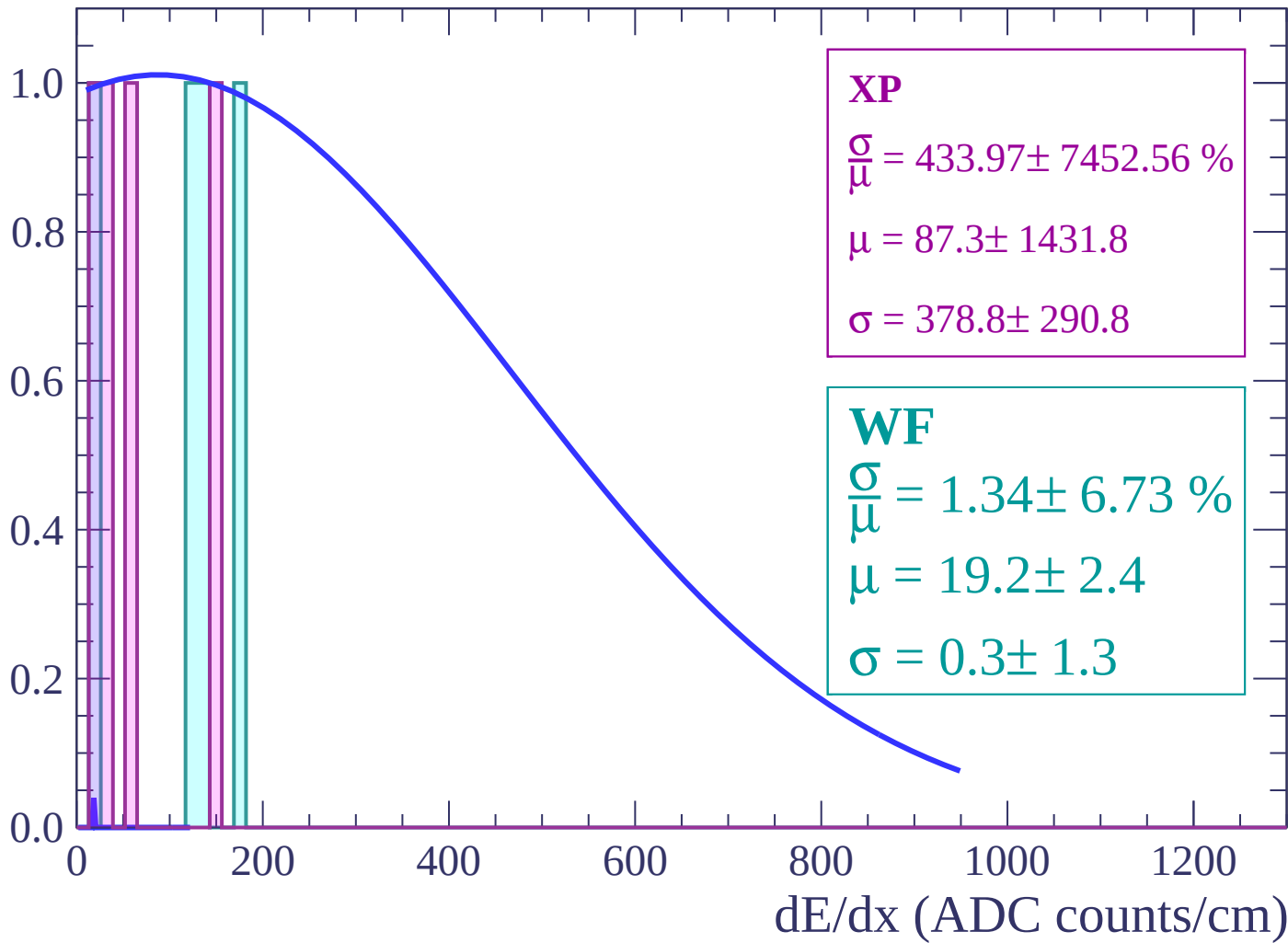
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

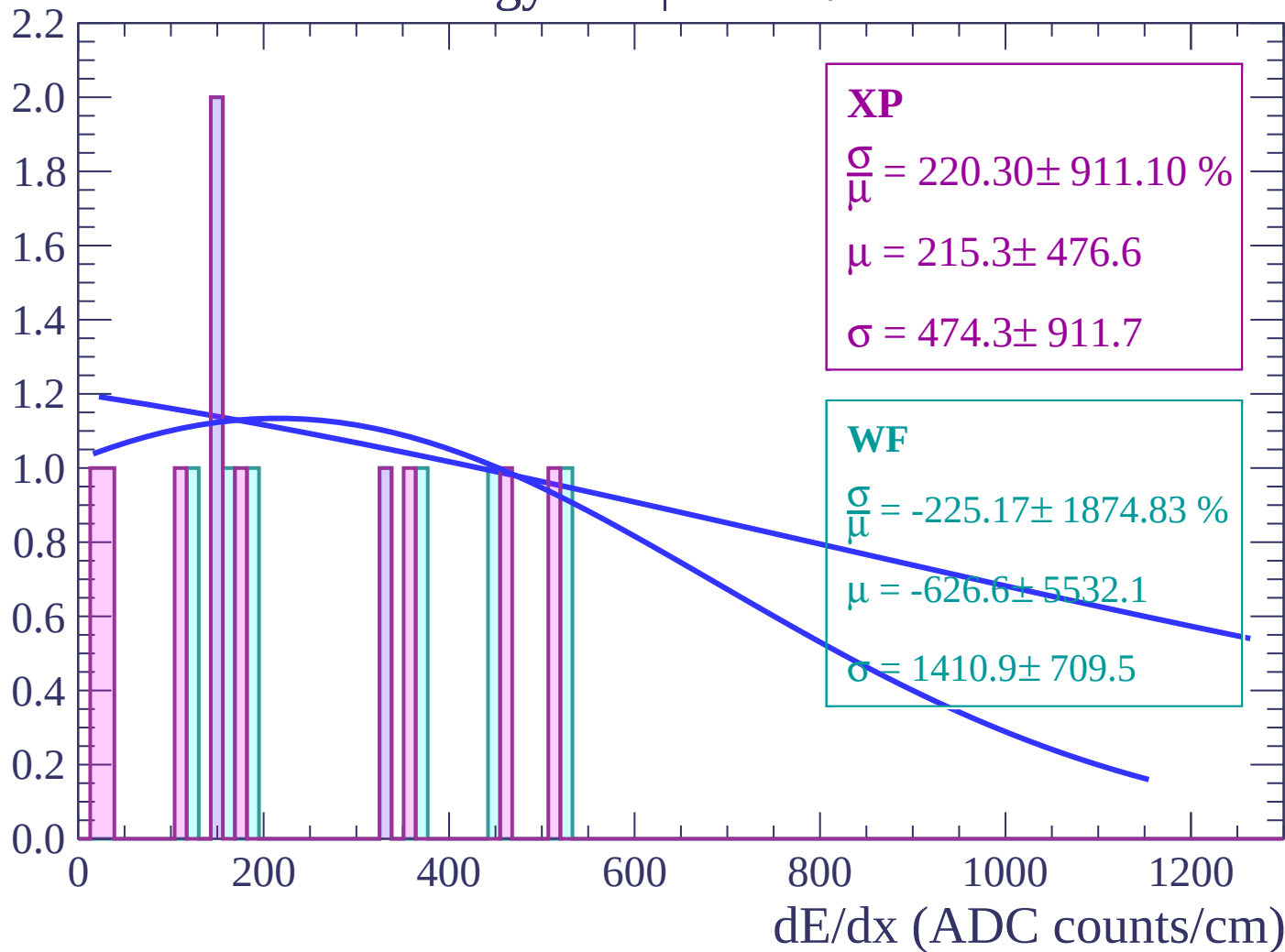
Energy loss | $-72 < \theta < -70$

Count



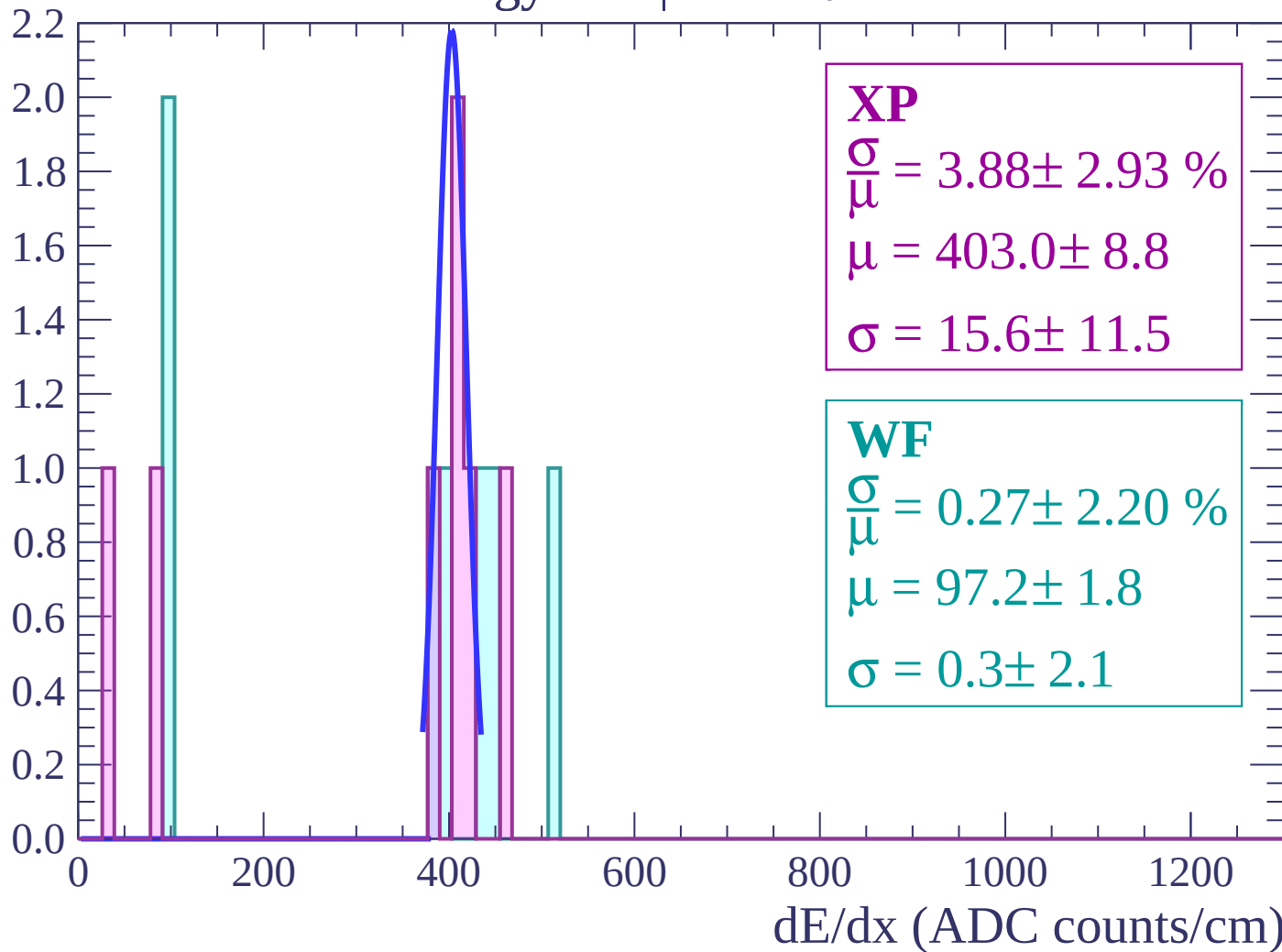
Energy loss | $-70 < \theta < -68$

Count



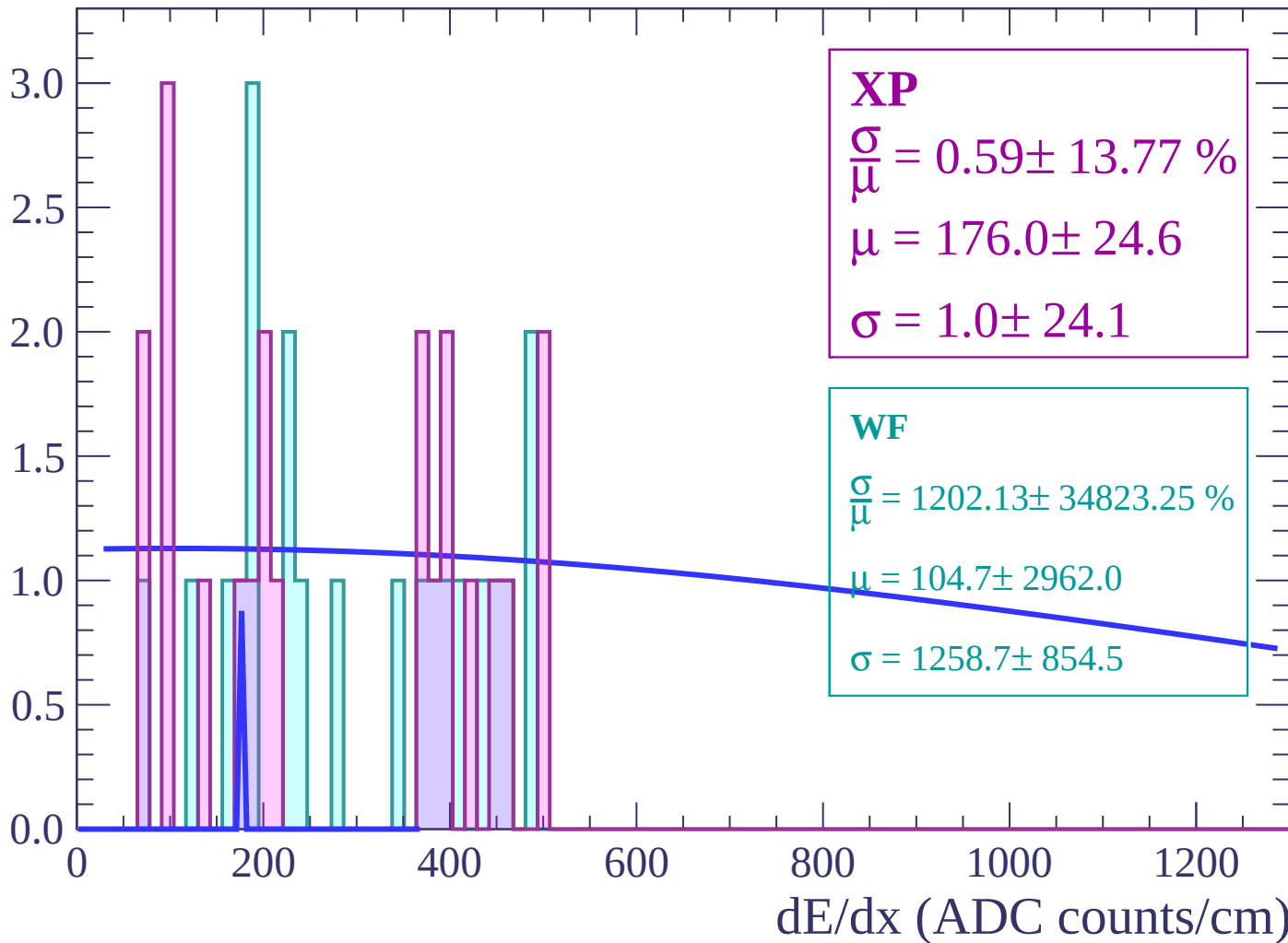
Energy loss | $-68 < \theta < -66$

Count



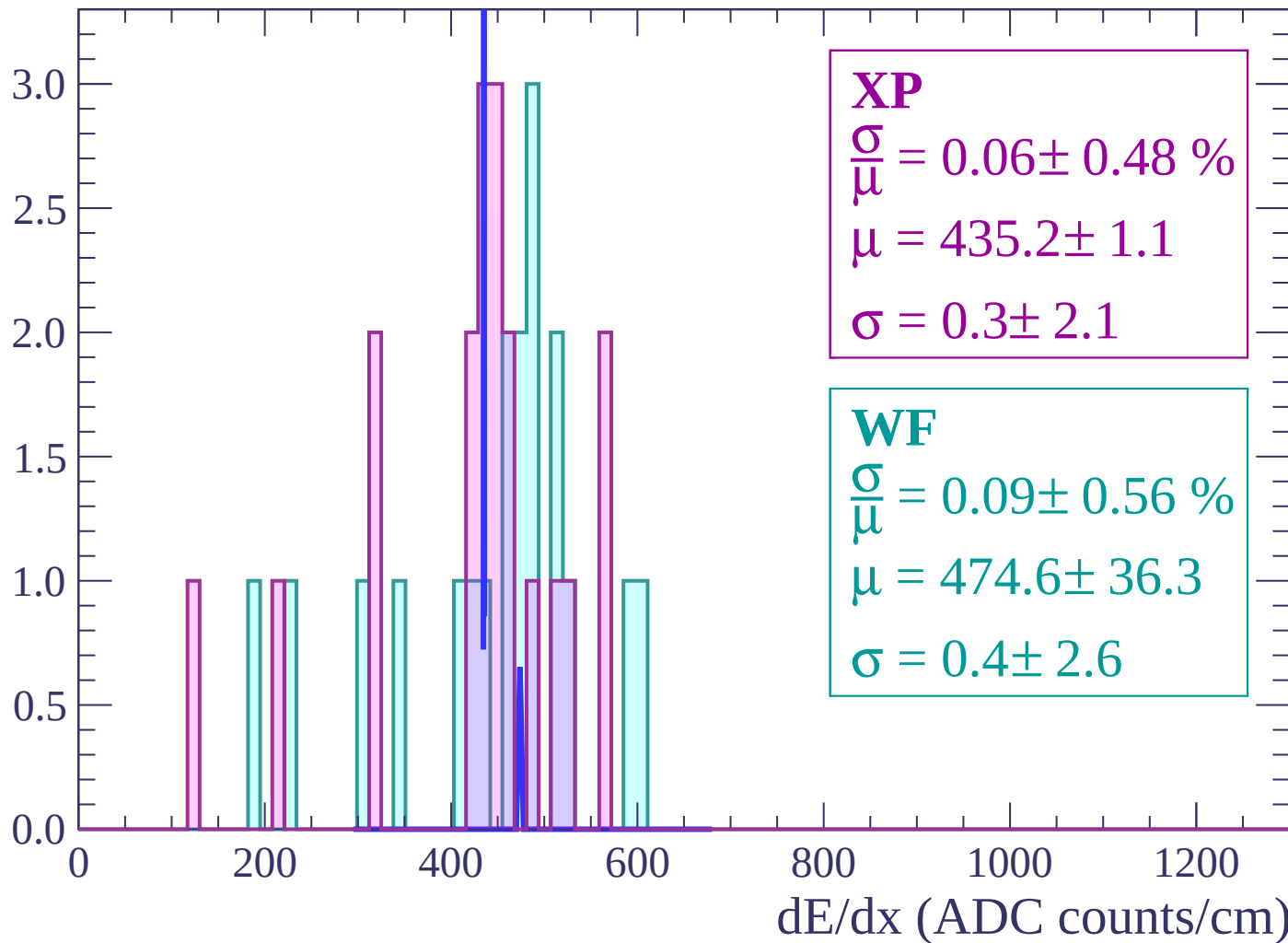
Energy loss | $-66 < \theta < -64$

Count



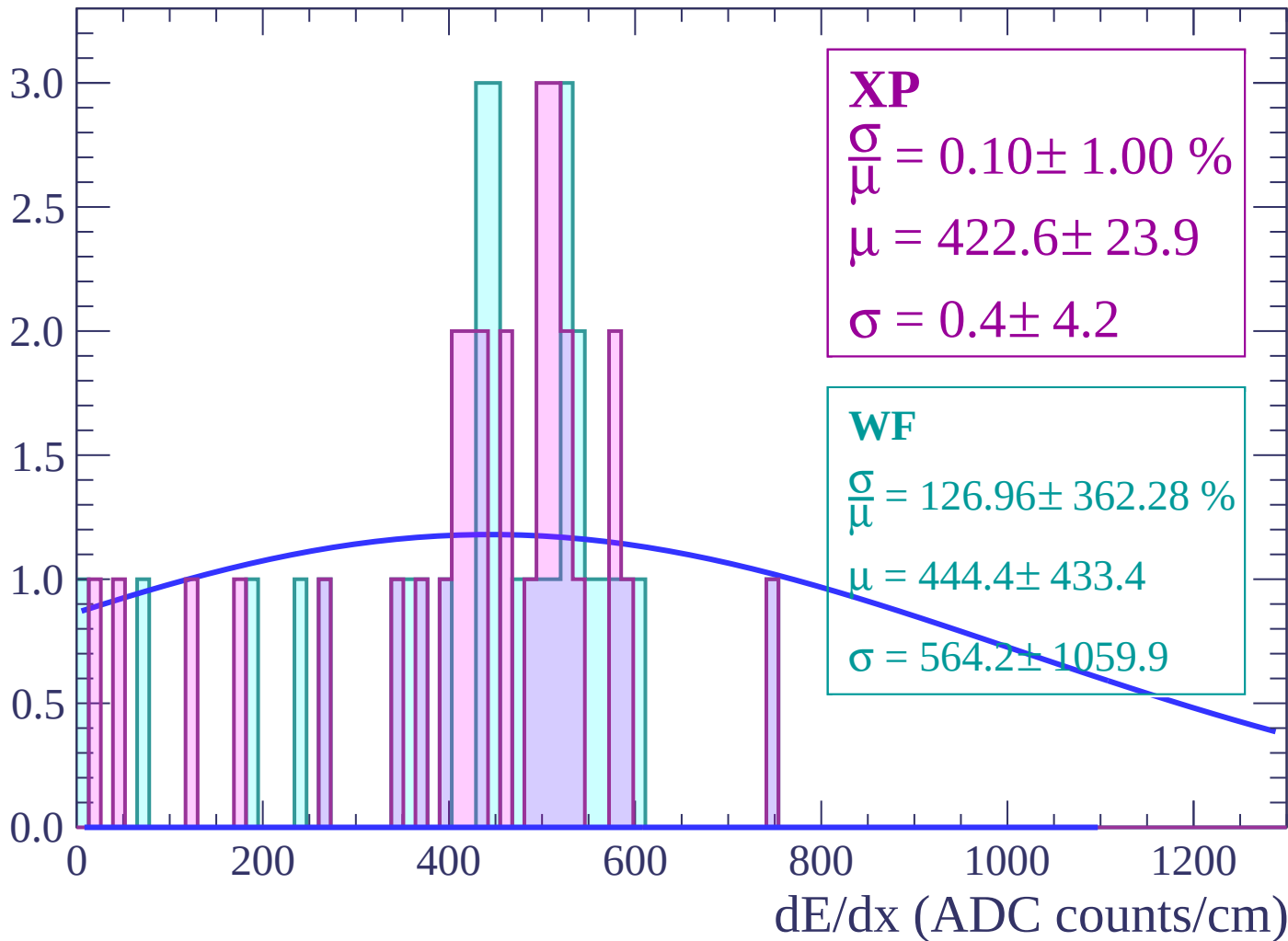
Energy loss | $-64 < \theta < -62$

Count



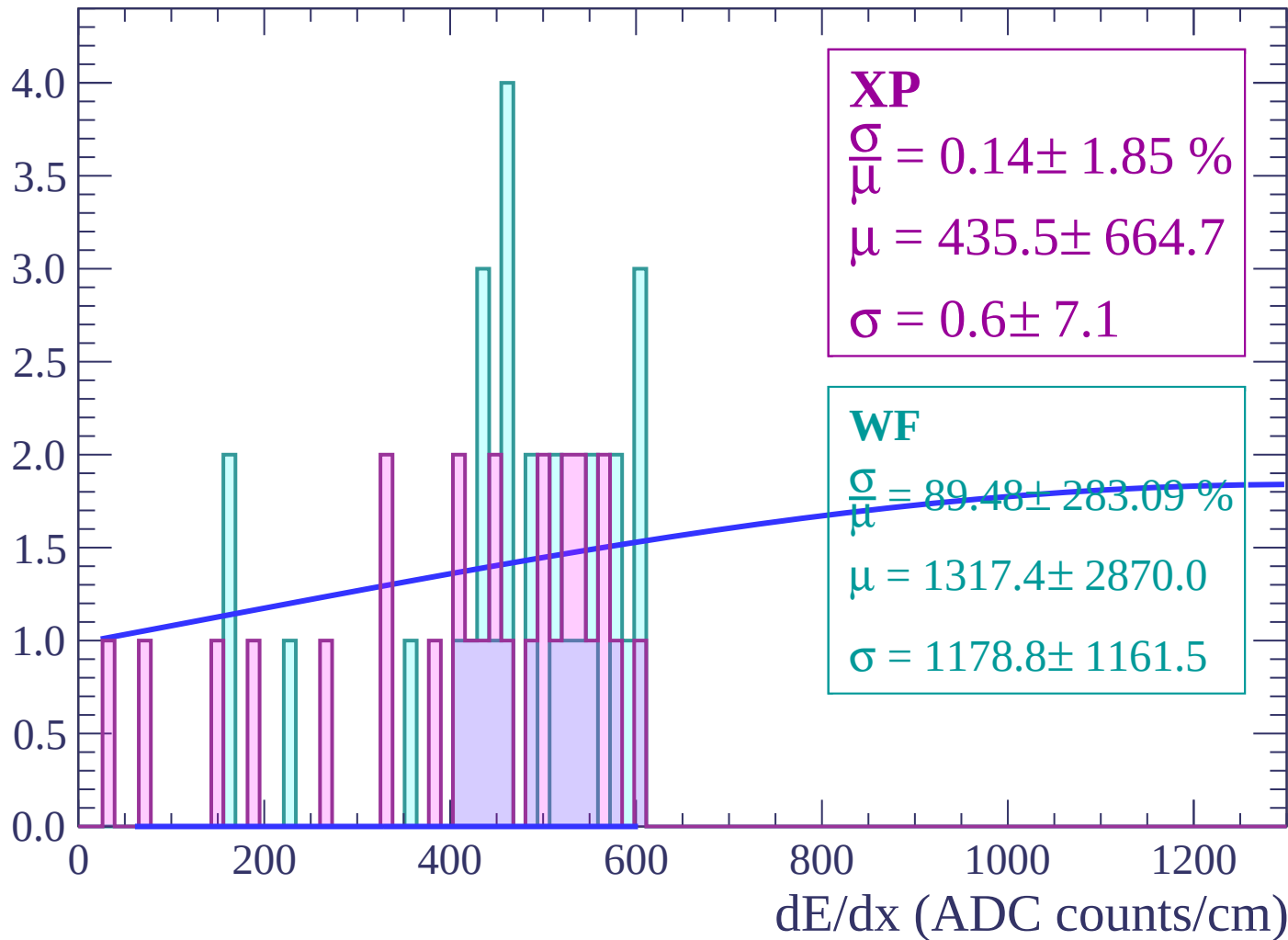
Energy loss | $-62 < \theta < -60$

Count



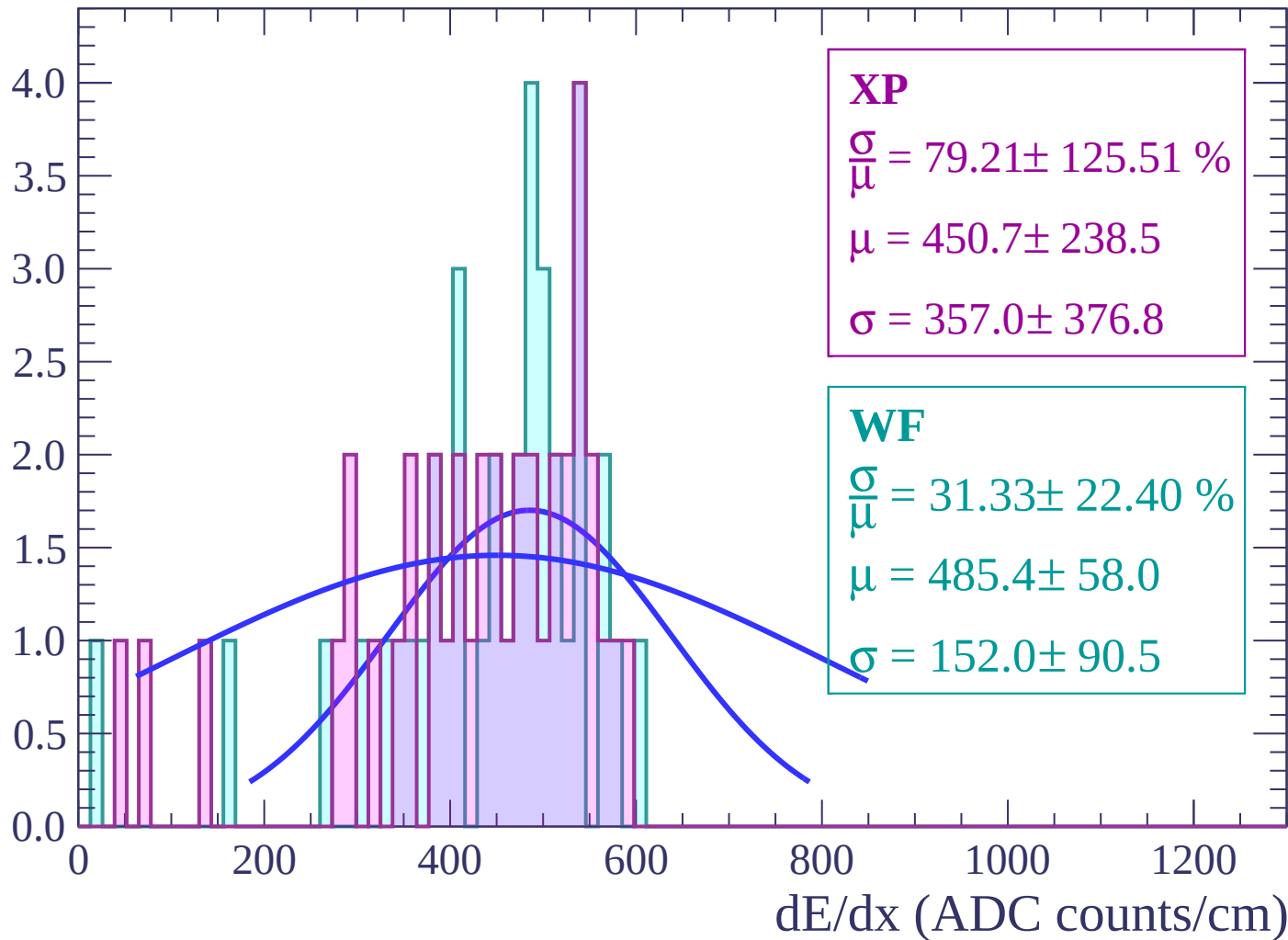
Energy loss | $-60 < \theta < -58$

Count



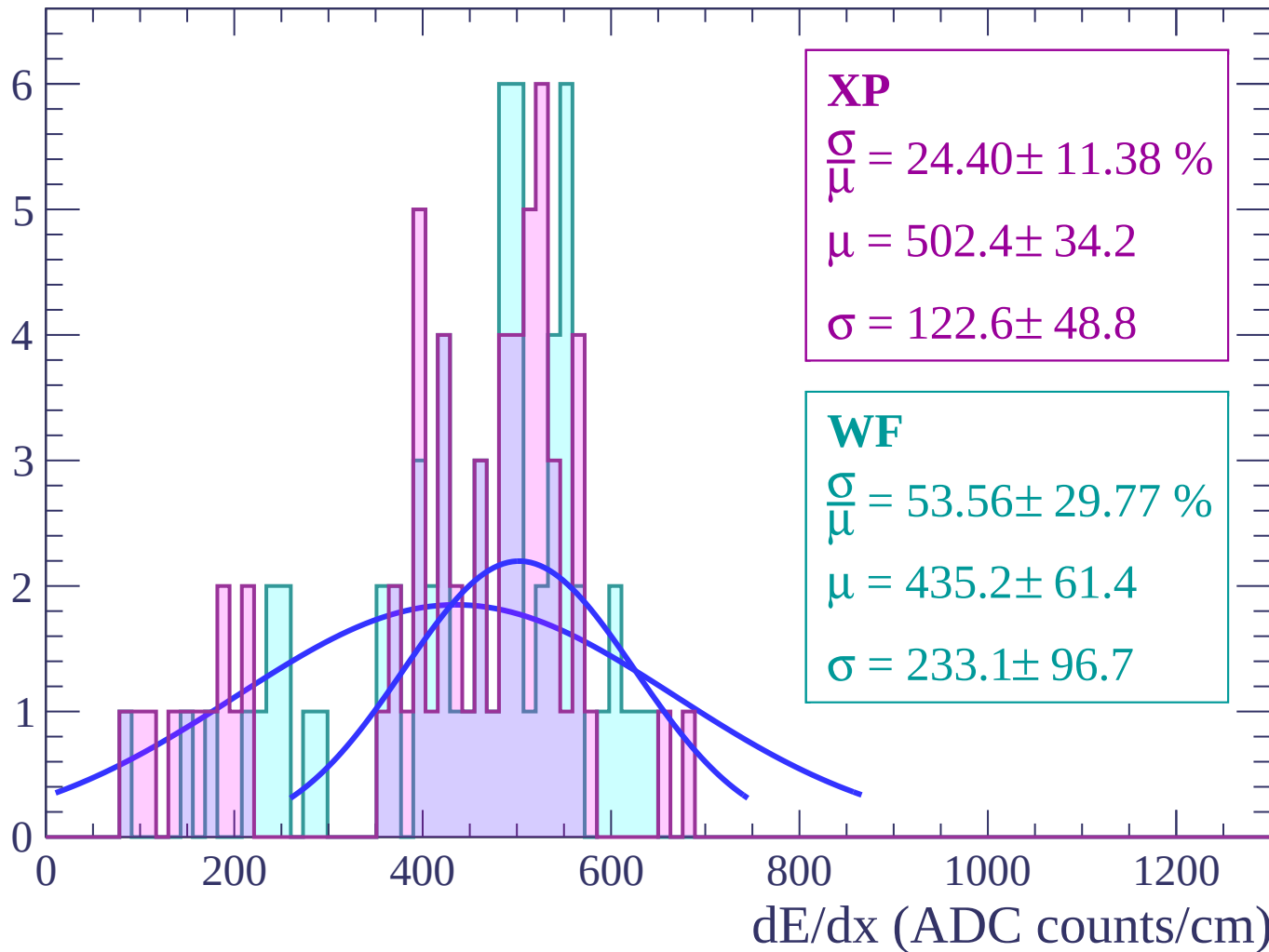
Energy loss | $-58 < \theta < -56$

Count



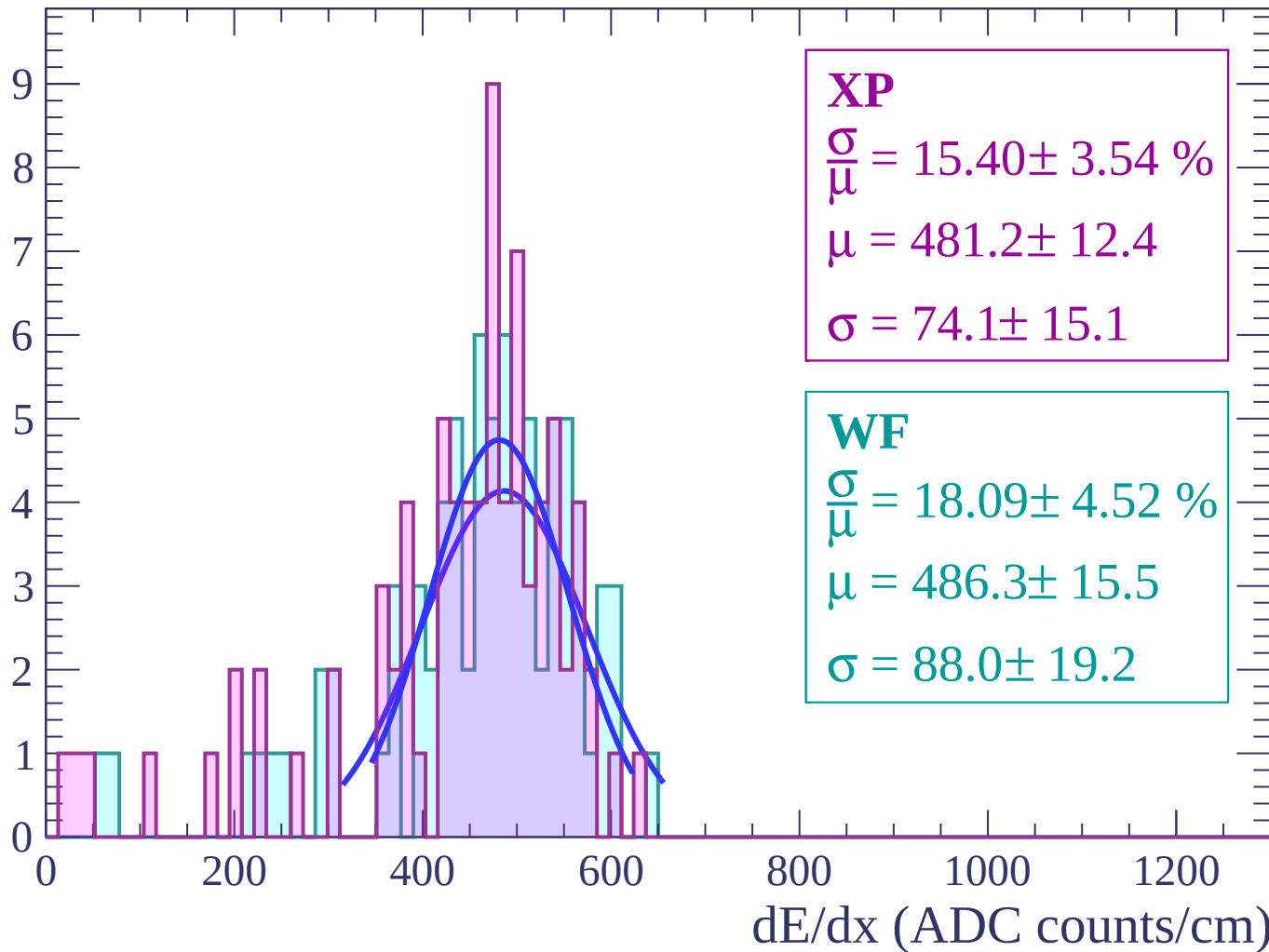
Energy loss | $-56 < \theta < -54$

Count



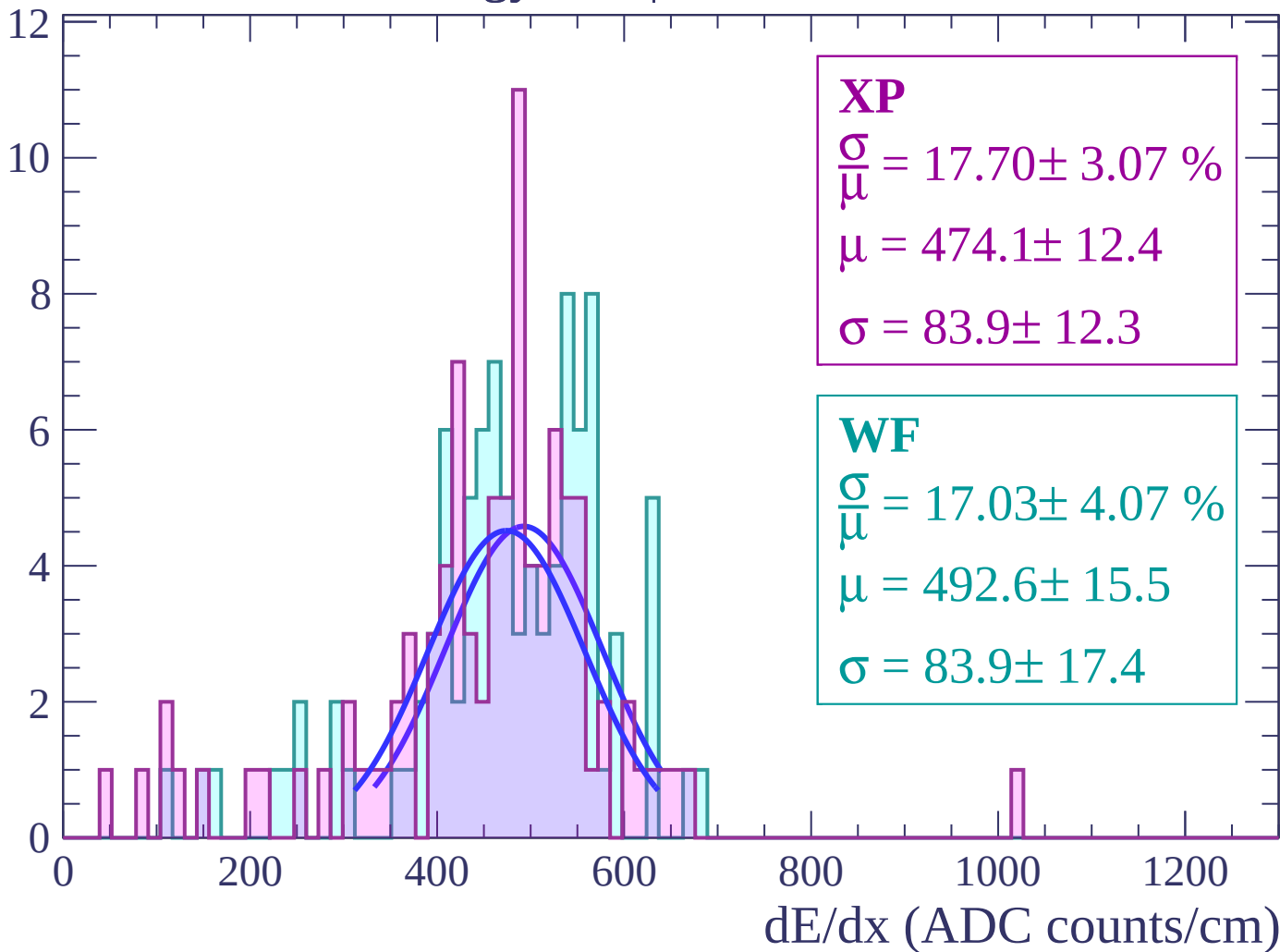
Energy loss | $-54 < \theta < -52$

Count



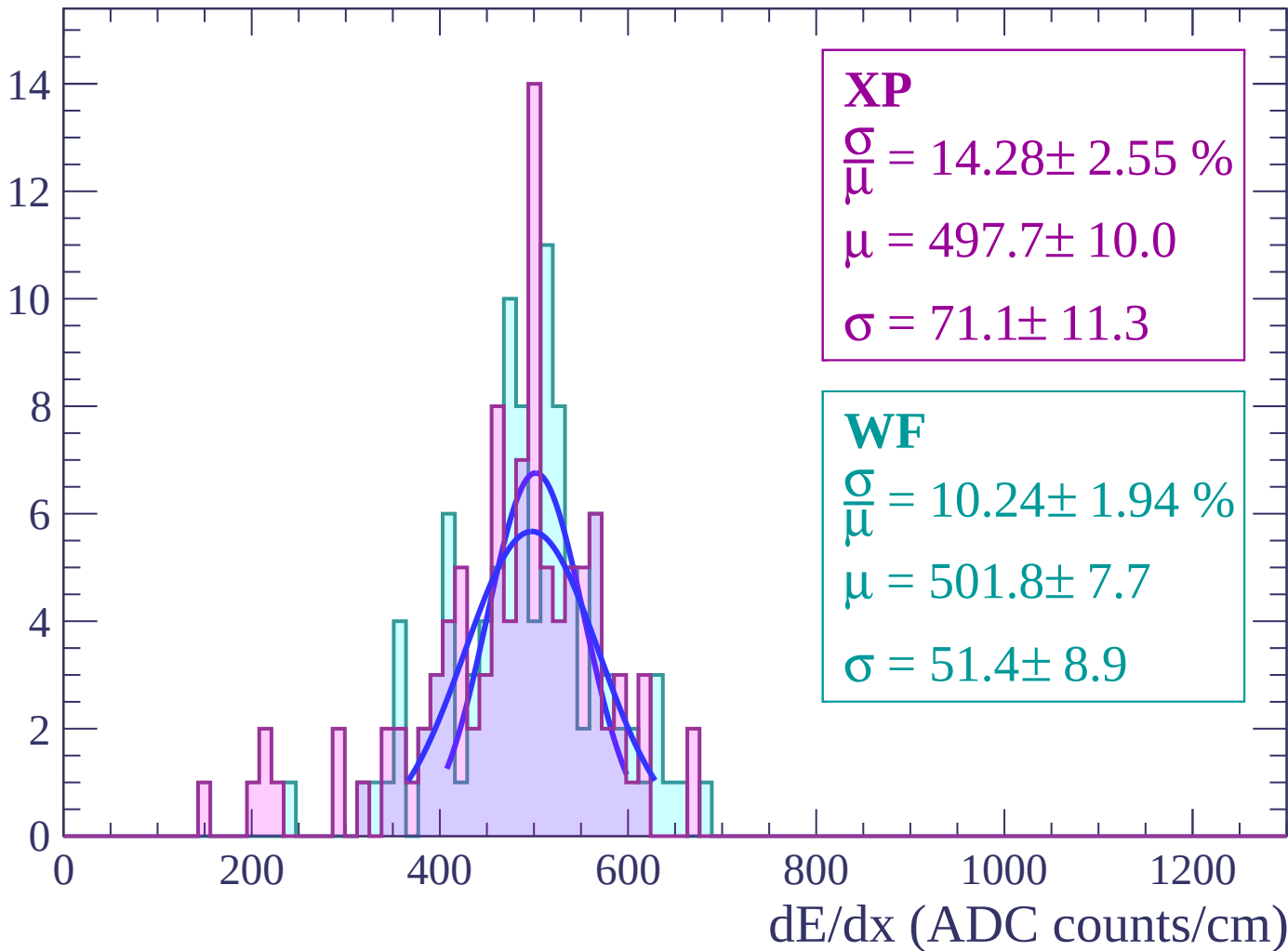
Energy loss | $-52 < \theta < -50$

Count



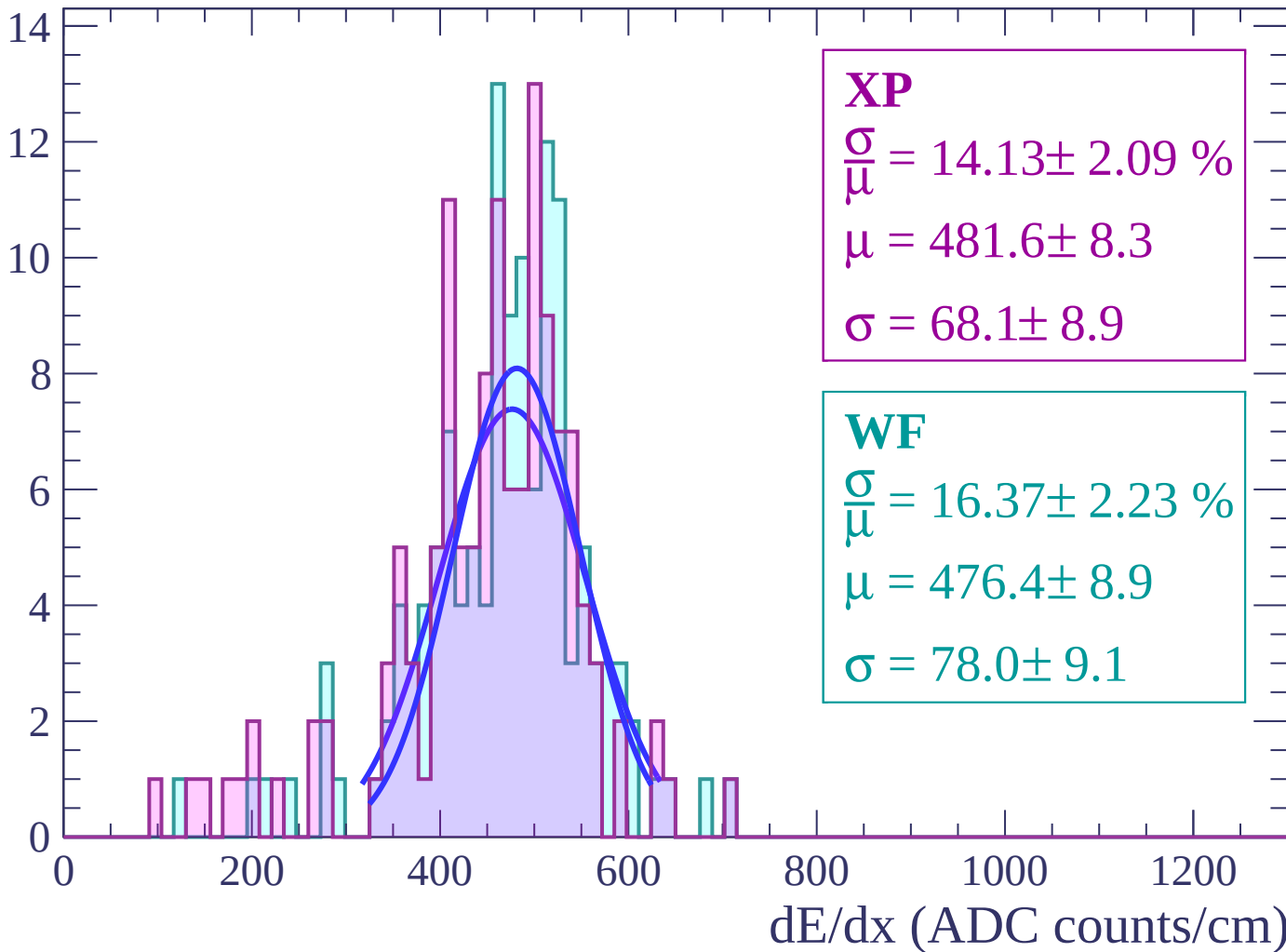
Energy loss | $-50 < \theta < -48$

Count



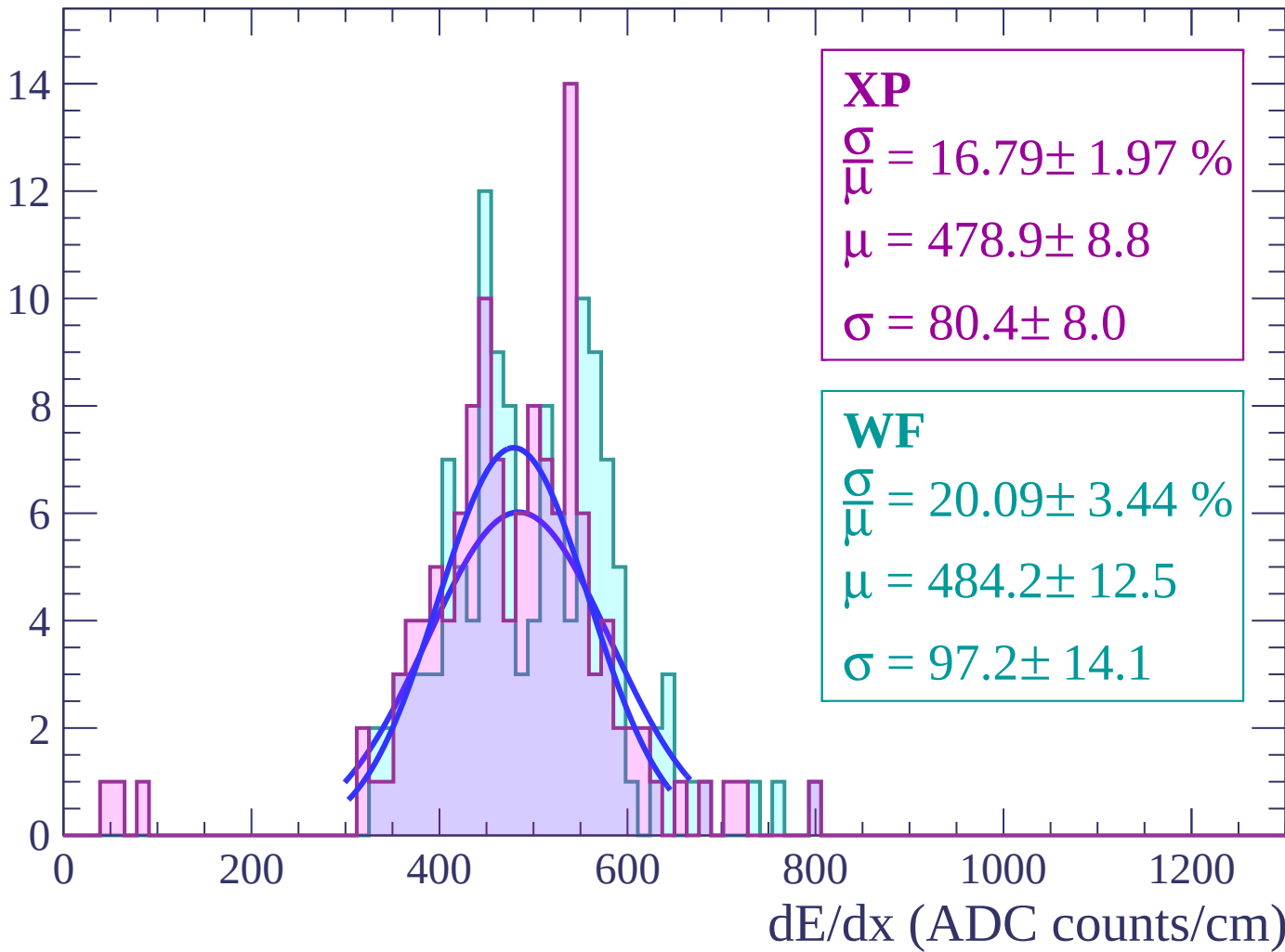
Energy loss | $-48 < \theta < -46$

Count



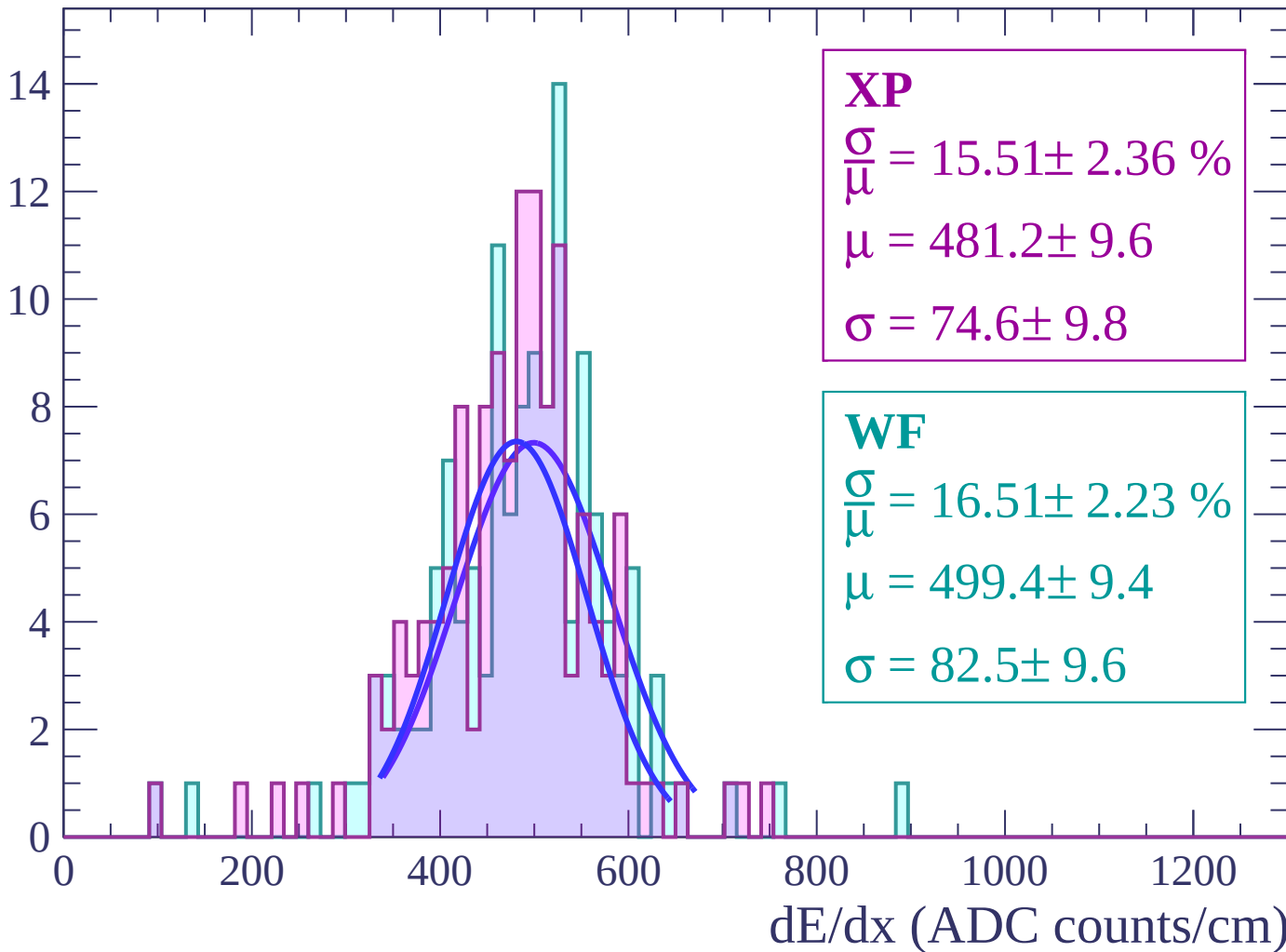
Energy loss | $-46 < \theta < -44$

Count



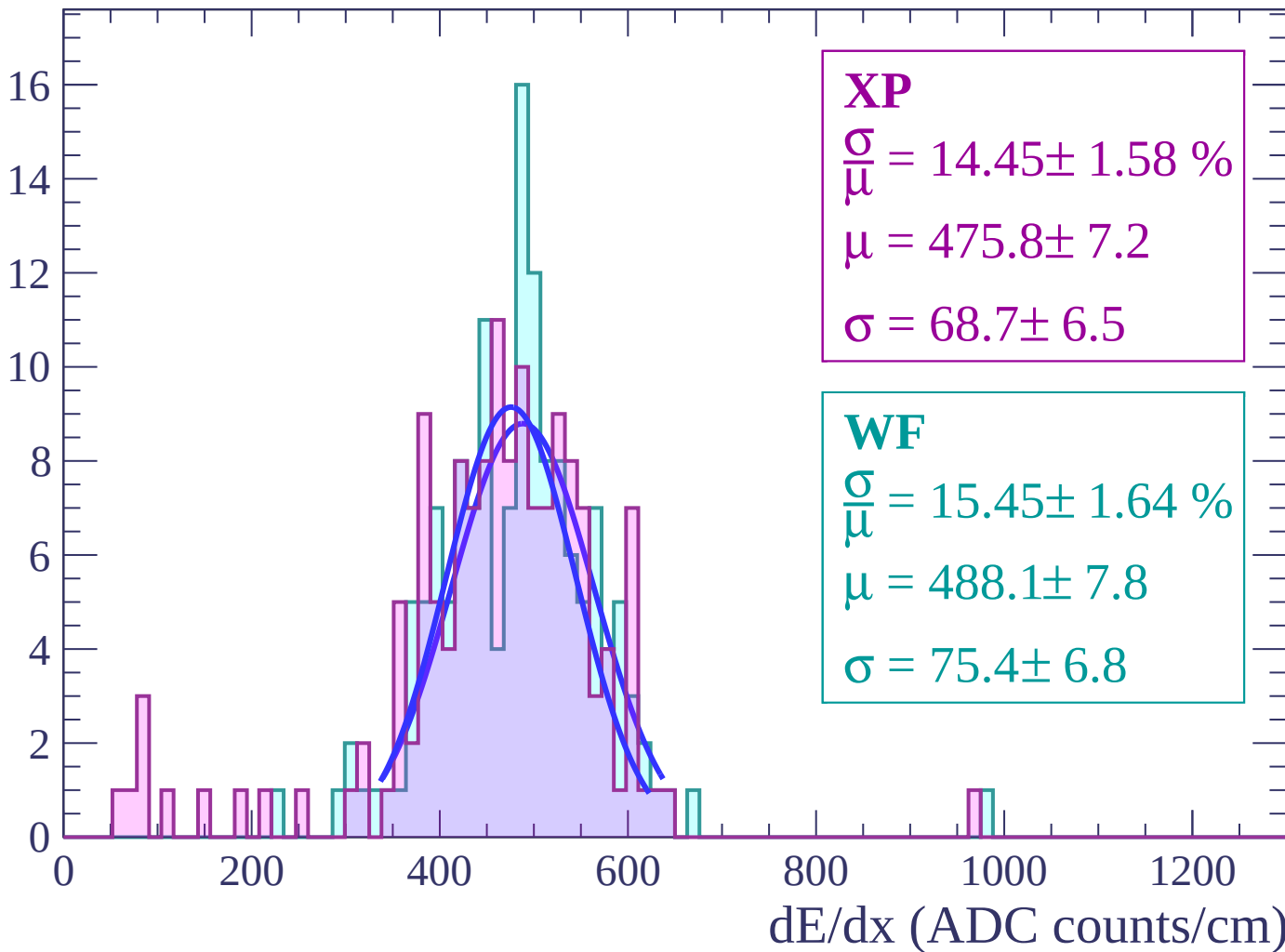
Energy loss | $-44 < \theta < -42$

Count



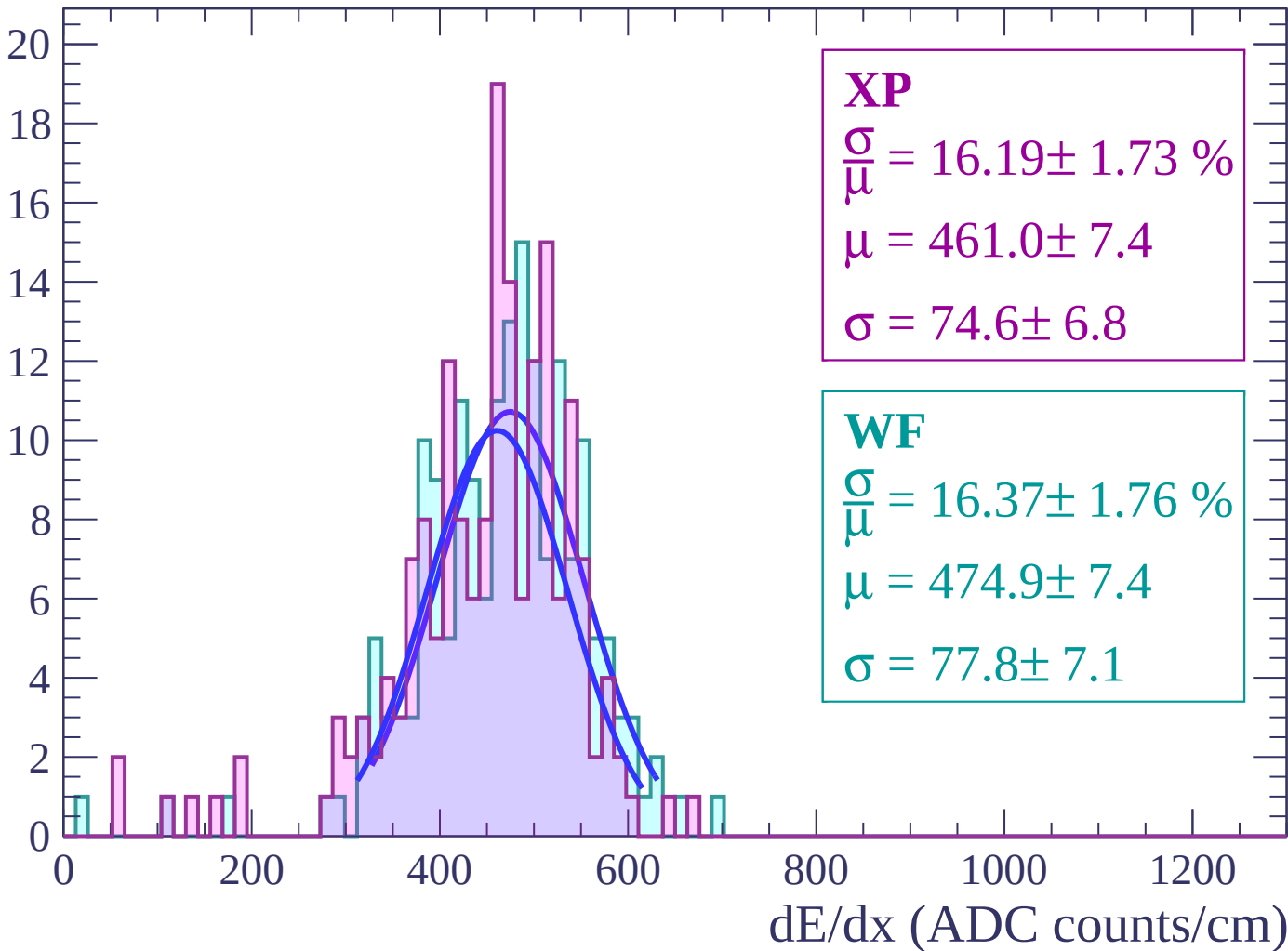
Energy loss | $-42 < \theta < -40$

Count



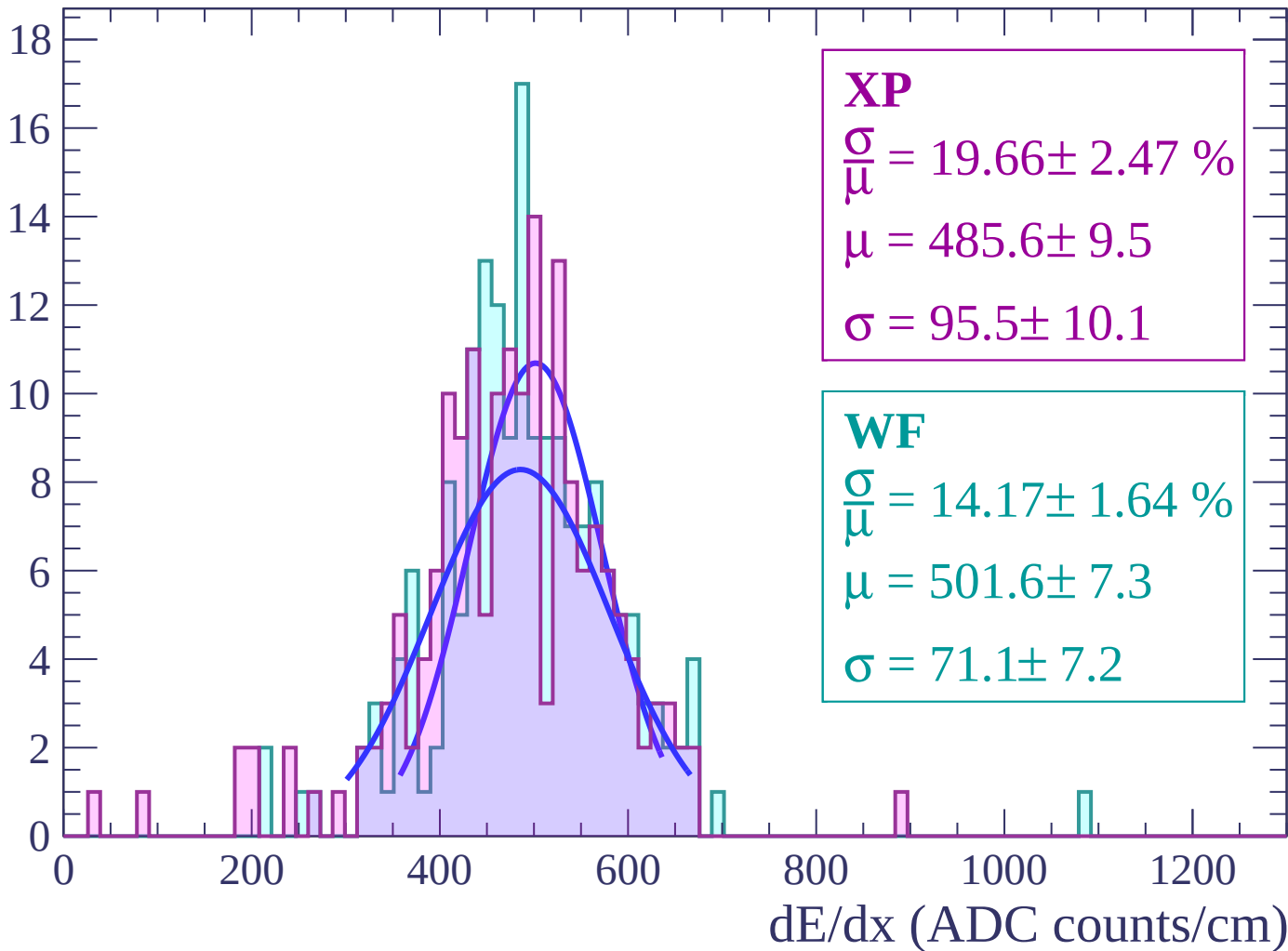
Energy loss | $-40 < \theta < -38$

Count



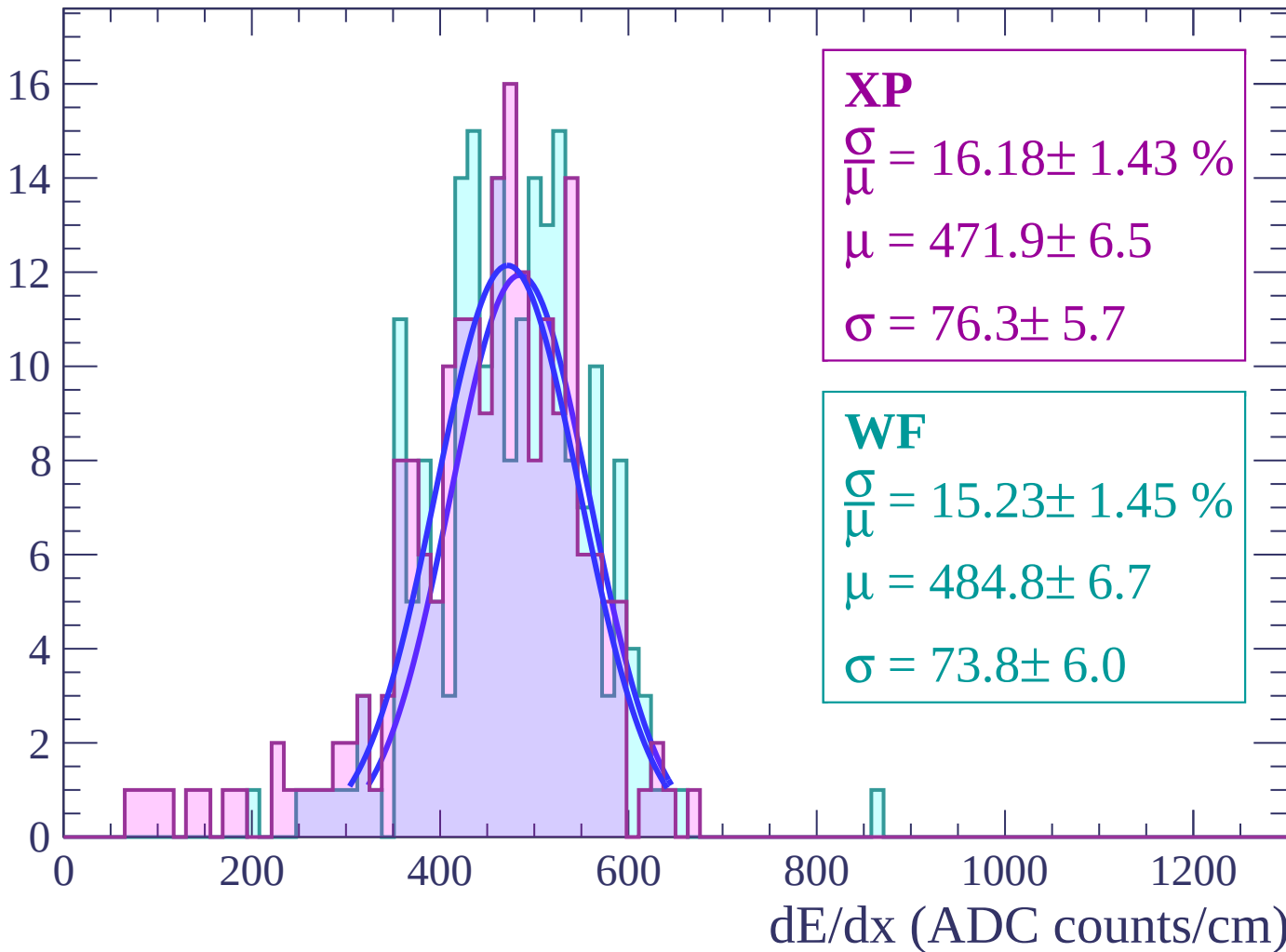
Energy loss | $-38 < \theta < -36$

Count



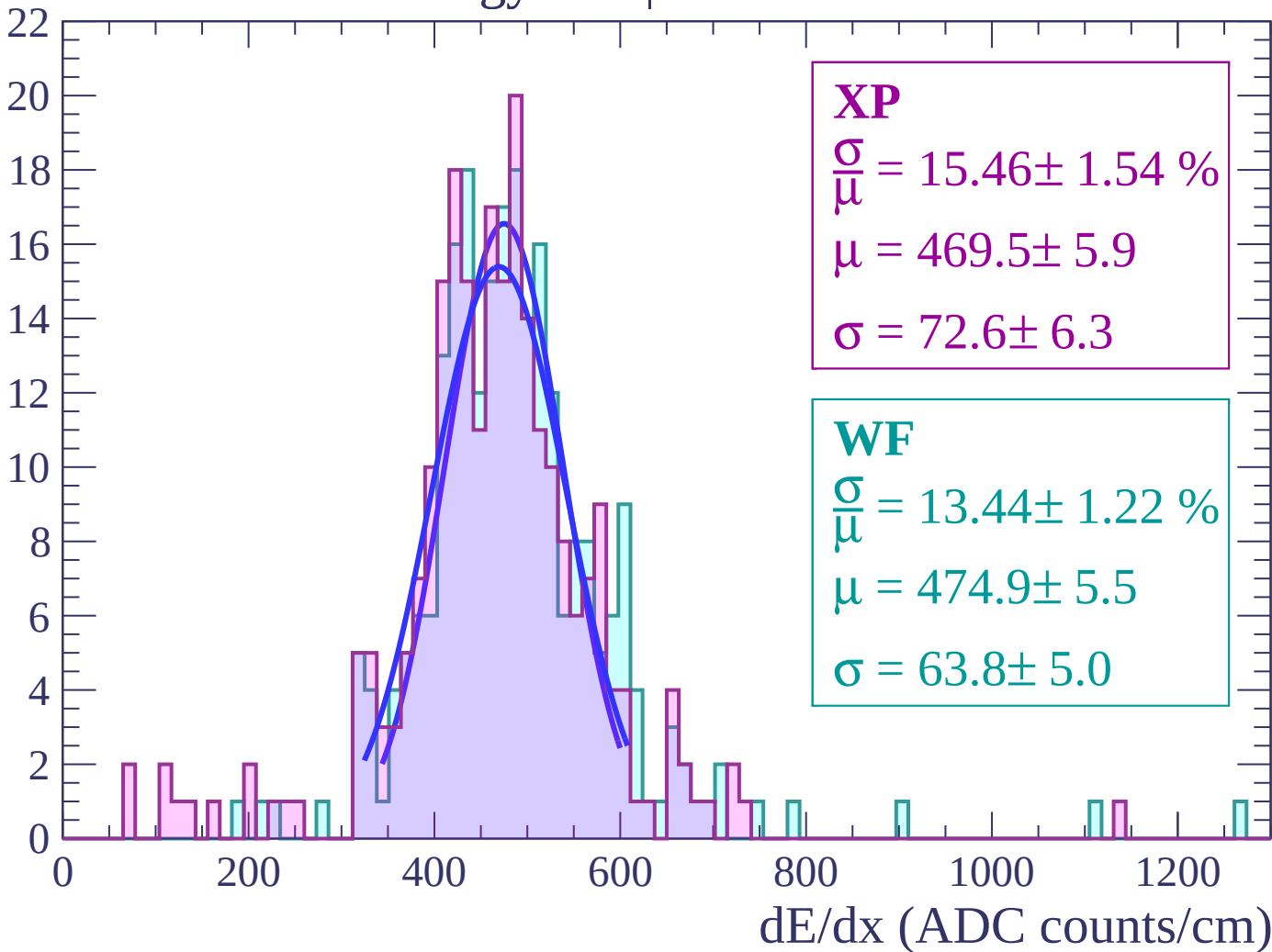
Energy loss | $-36 < \theta < -34$

Count



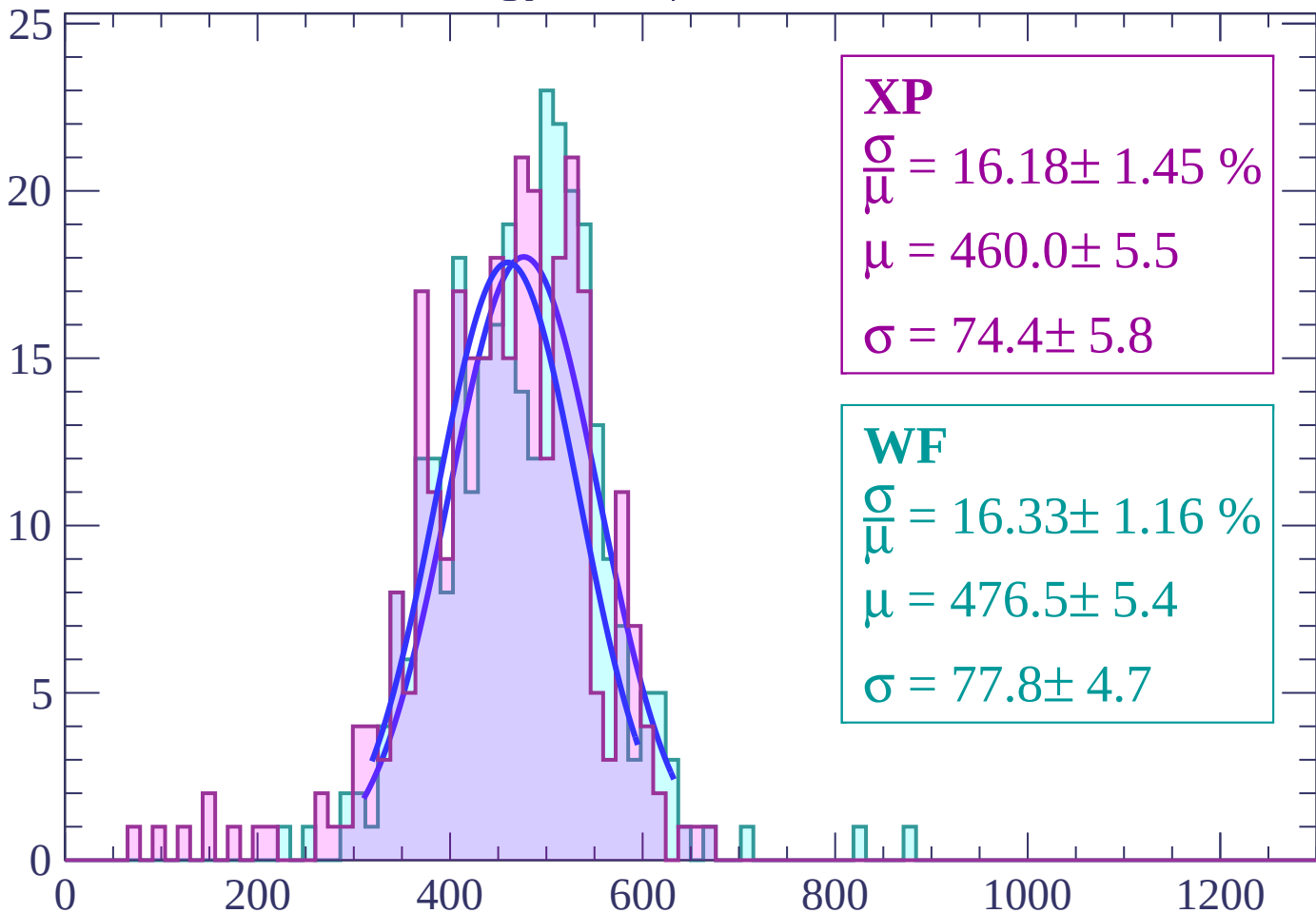
Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count



XP

$$\frac{\sigma}{\mu} = 16.18 \pm 1.45 \%$$

$$\mu = 460.0 \pm 5.5$$

$$\sigma = 74.4 \pm 5.8$$

WF

$$\frac{\sigma}{\mu} = 16.33 \pm 1.16 \%$$

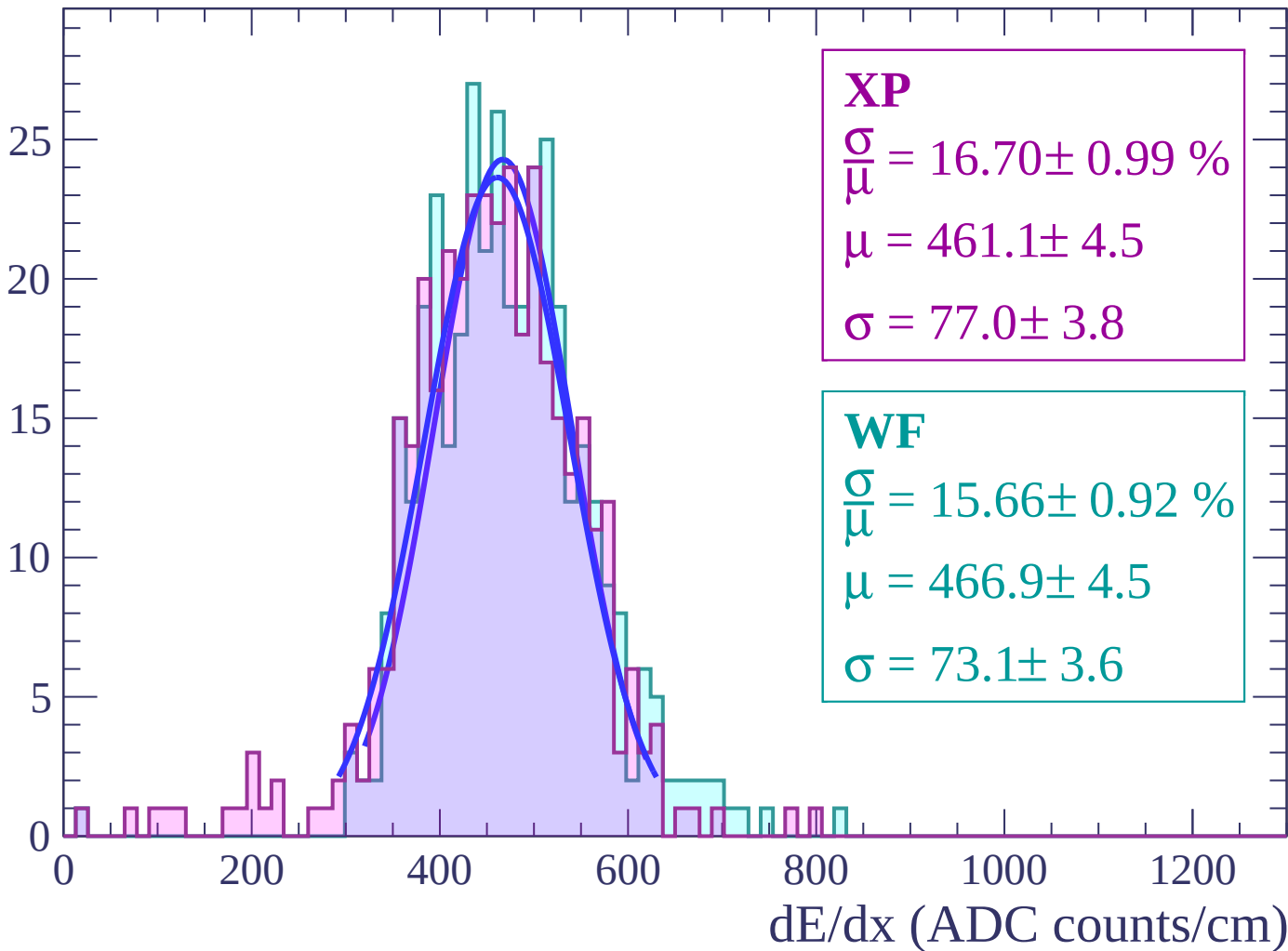
$$\mu = 476.5 \pm 5.4$$

$$\sigma = 77.8 \pm 4.7$$

dE/dx (ADC counts/cm)

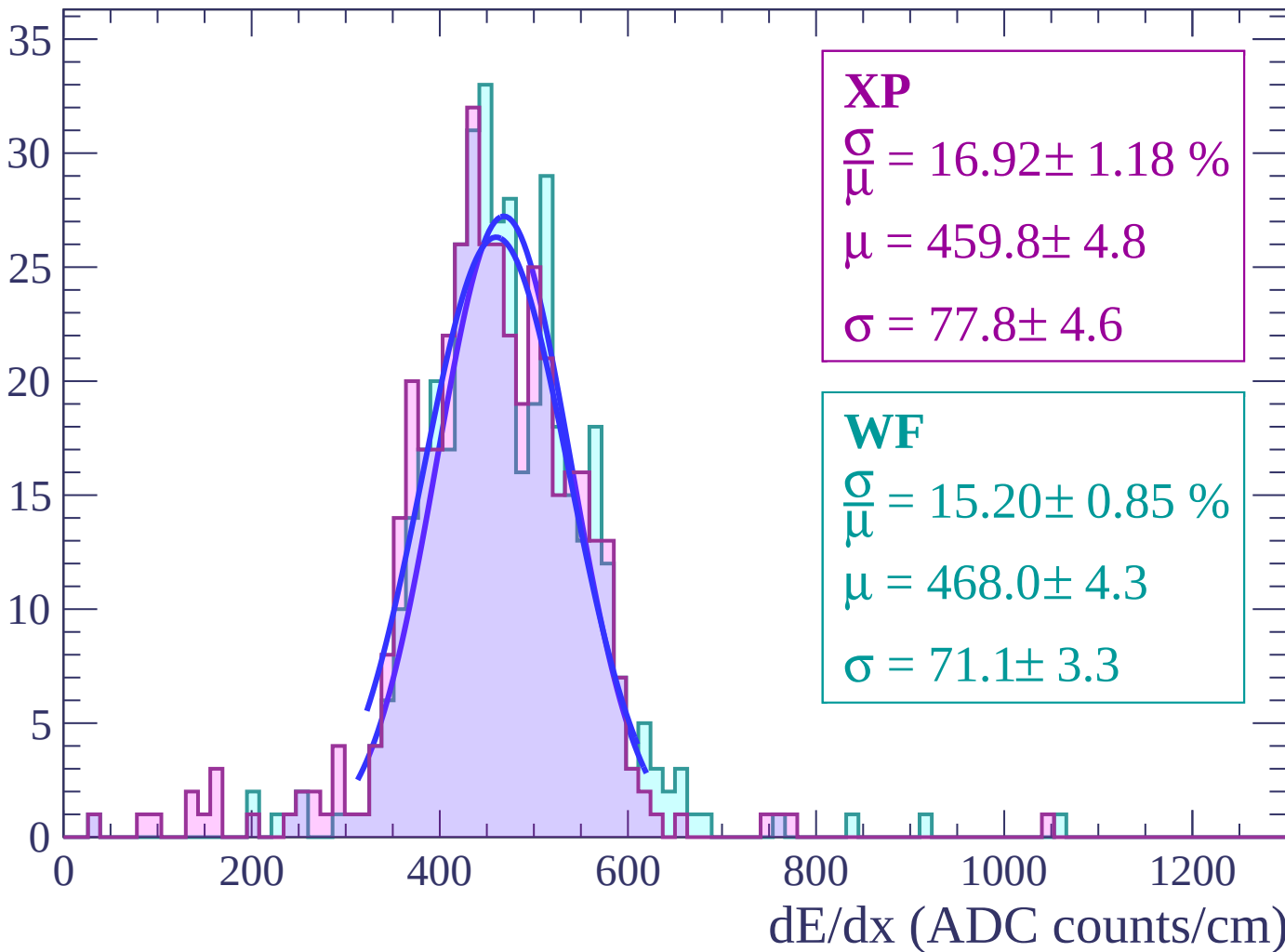
Energy loss | $-30 < \theta < -28$

Count



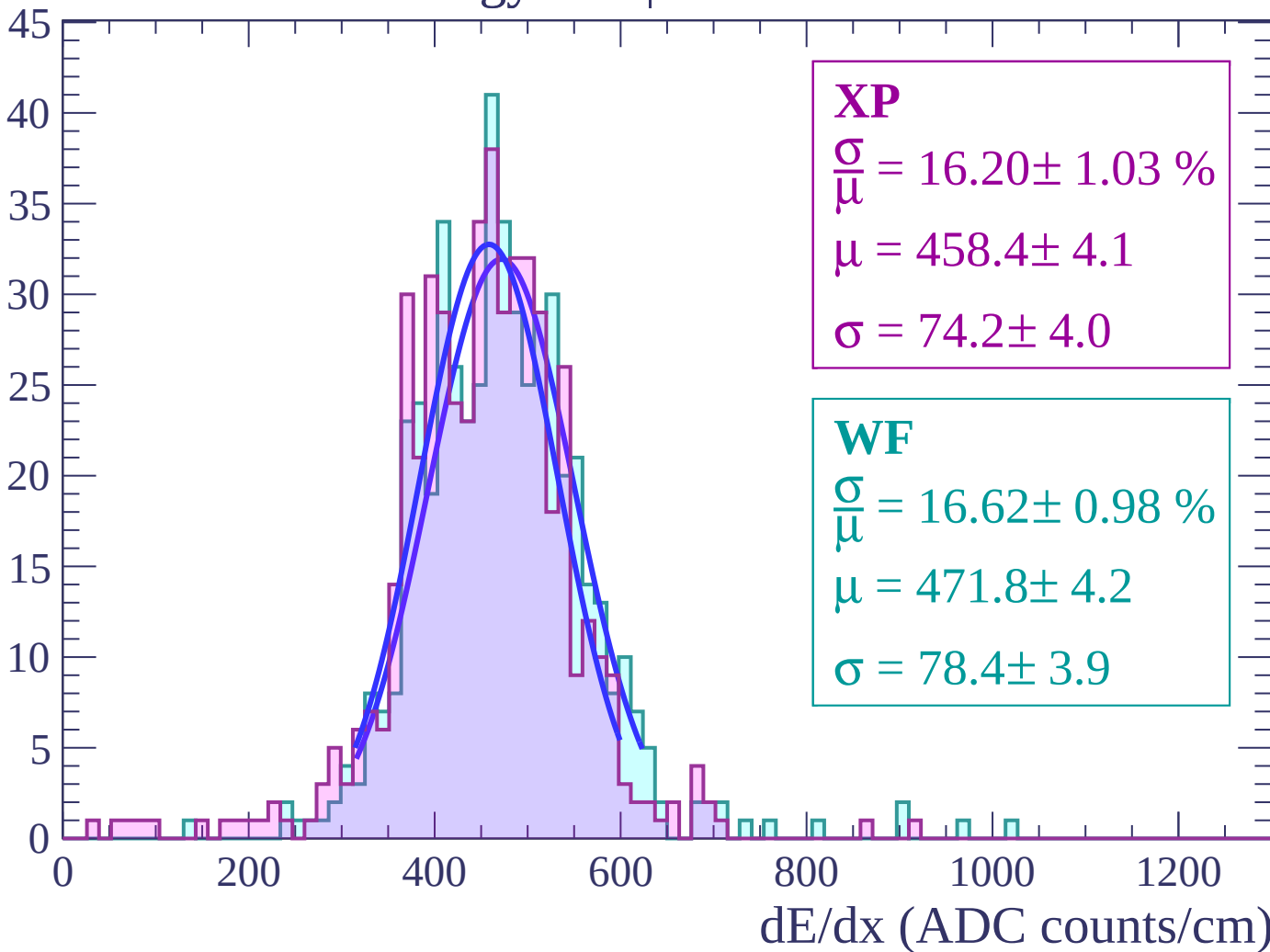
Energy loss | $-28 < \theta < -26$

Count



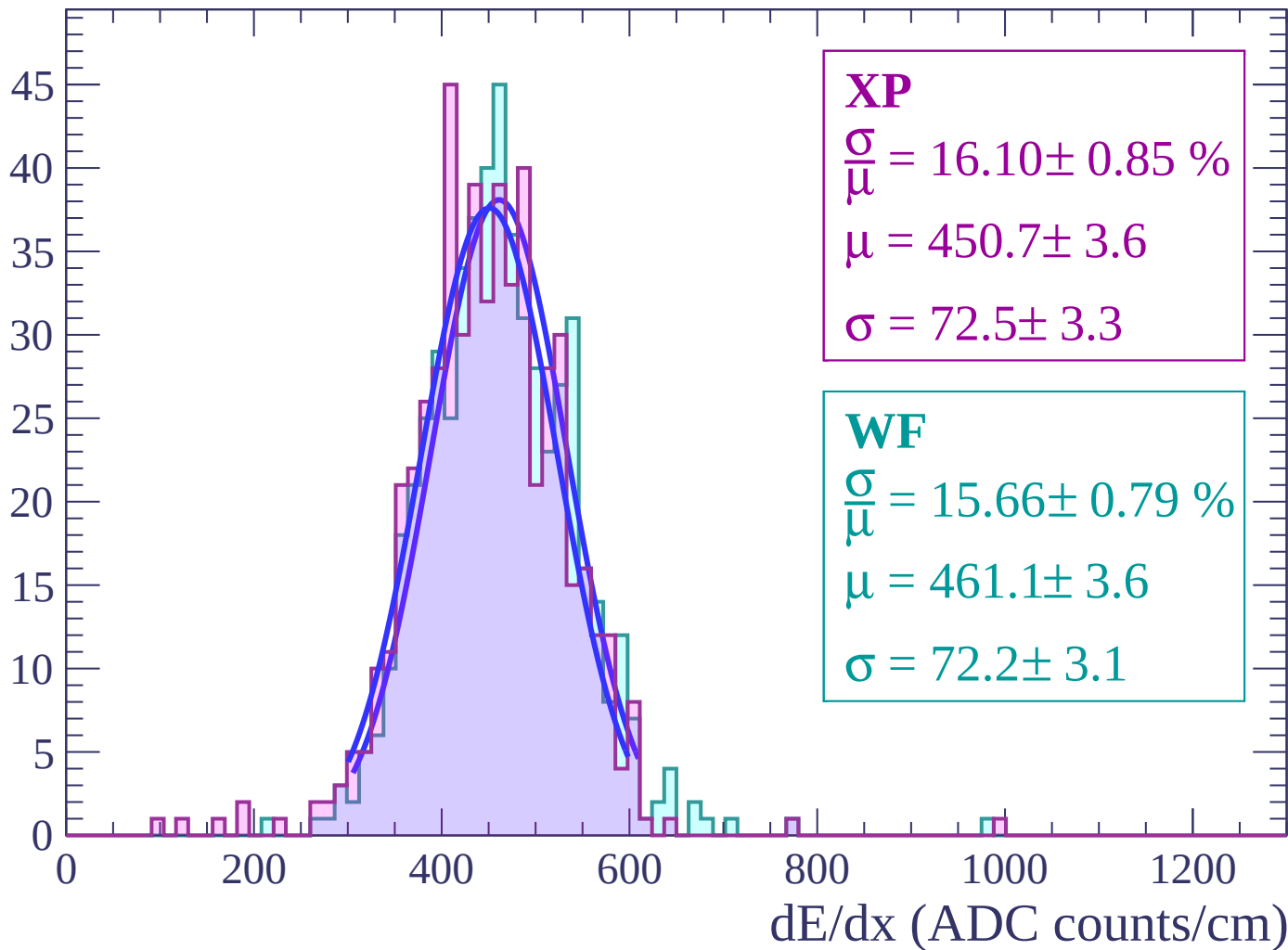
Energy loss | $-26 < \theta < -24$

Count



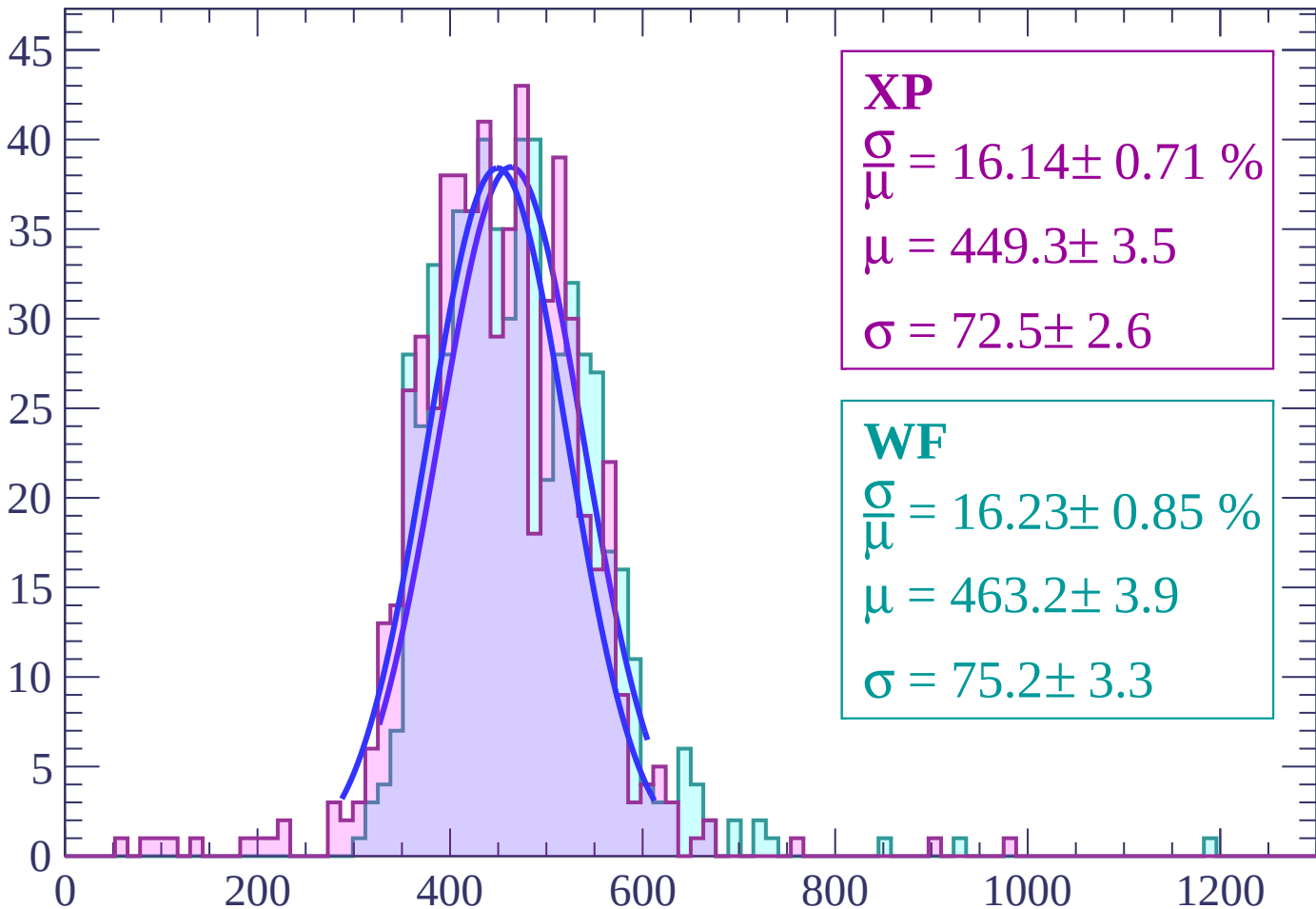
Energy loss | $-24 < \theta < -22$

Count



Energy loss | $-22 < \theta < -20$

Count



XP

$$\frac{\sigma}{\mu} = 16.14 \pm 0.71 \%$$

$$\mu = 449.3 \pm 3.5$$

$$\sigma = 72.5 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 16.23 \pm 0.85 \%$$

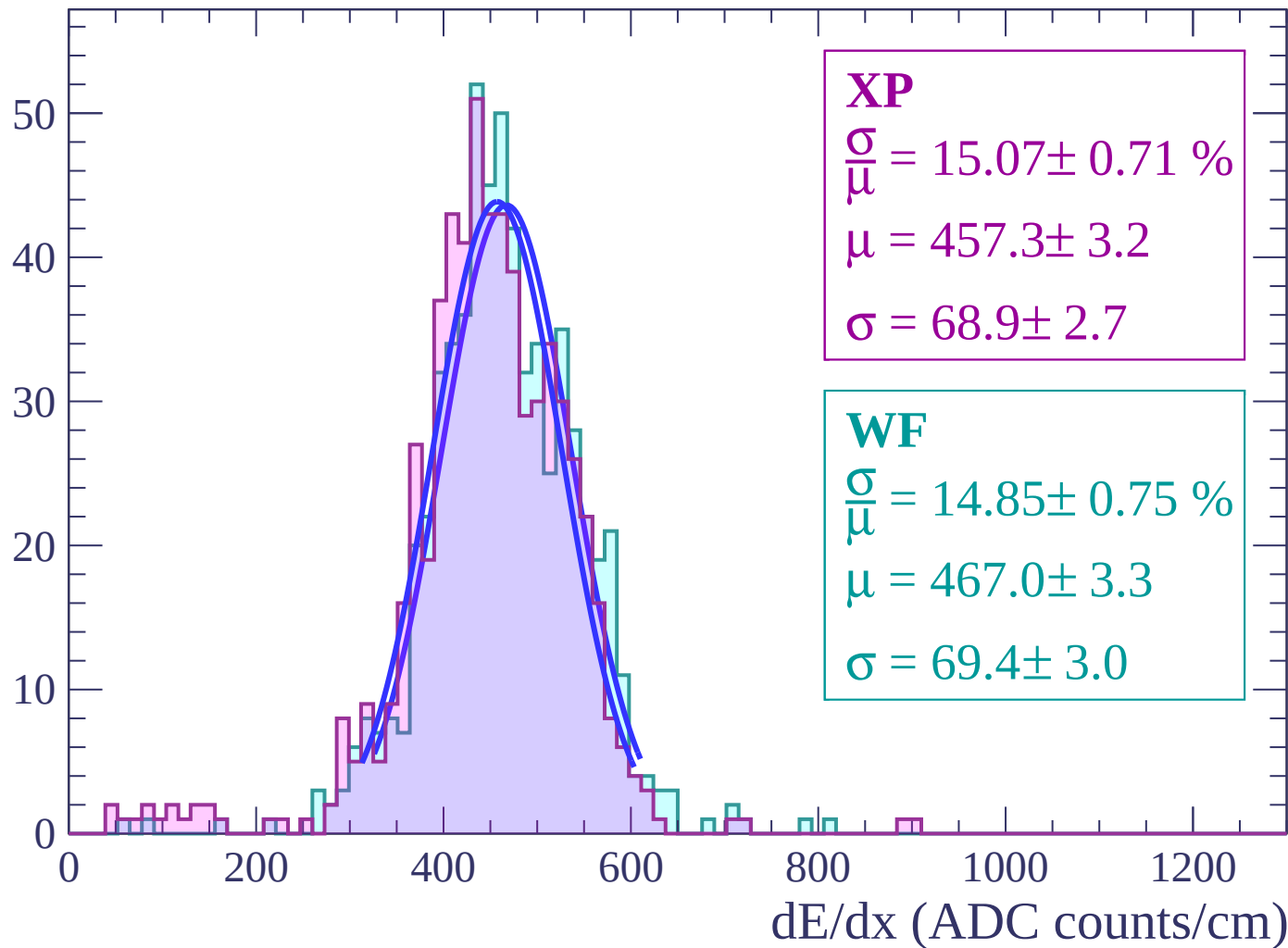
$$\mu = 463.2 \pm 3.9$$

$$\sigma = 75.2 \pm 3.3$$

dE/dx (ADC counts/cm)

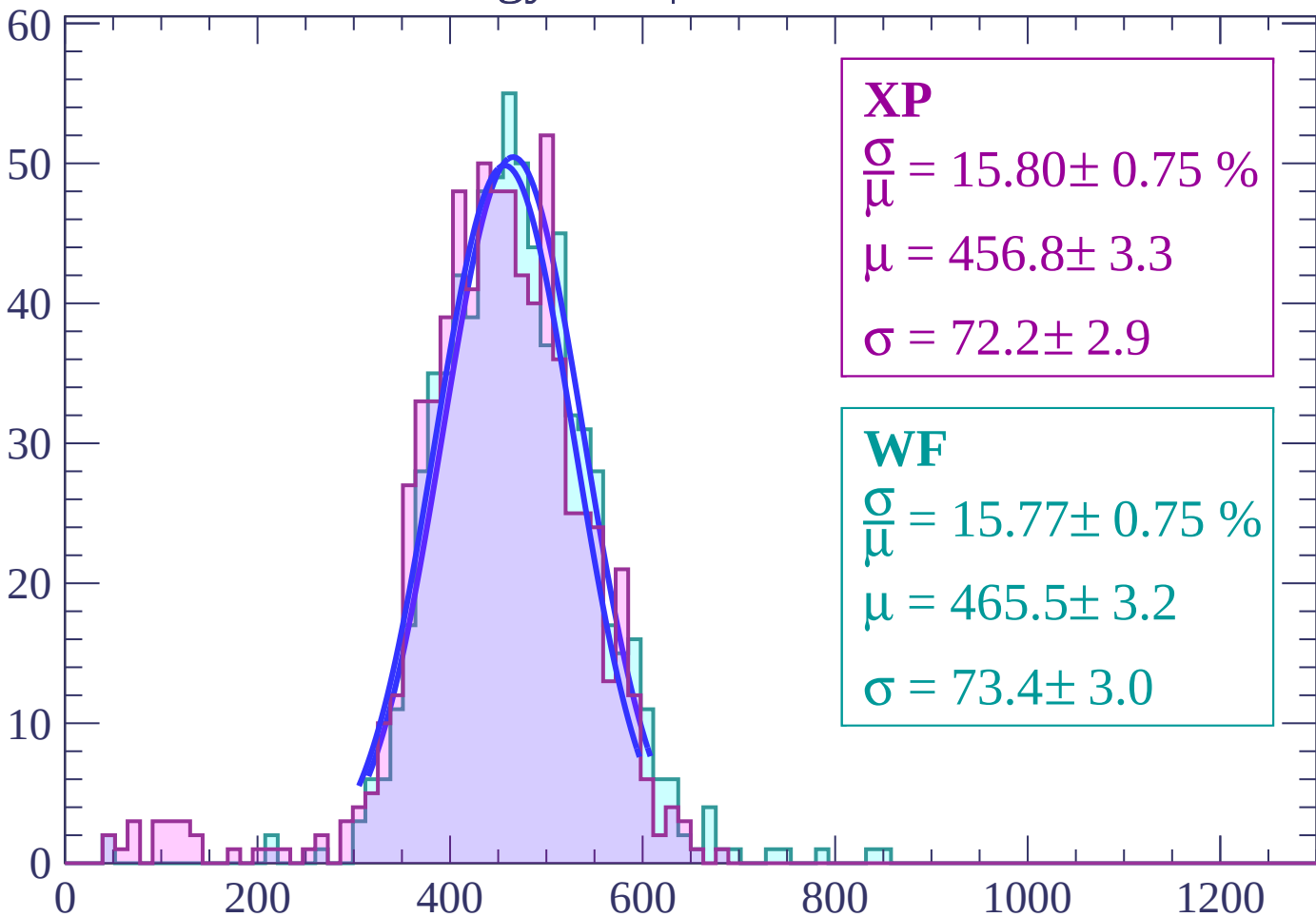
Energy loss | $-20 < \theta < -18$

Count



Energy loss | $-18 < \theta < -16$

Count



XP

$$\frac{\sigma}{\mu} = 15.80 \pm 0.75 \%$$

$$\mu = 456.8 \pm 3.3$$

$$\sigma = 72.2 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 15.77 \pm 0.75 \%$$

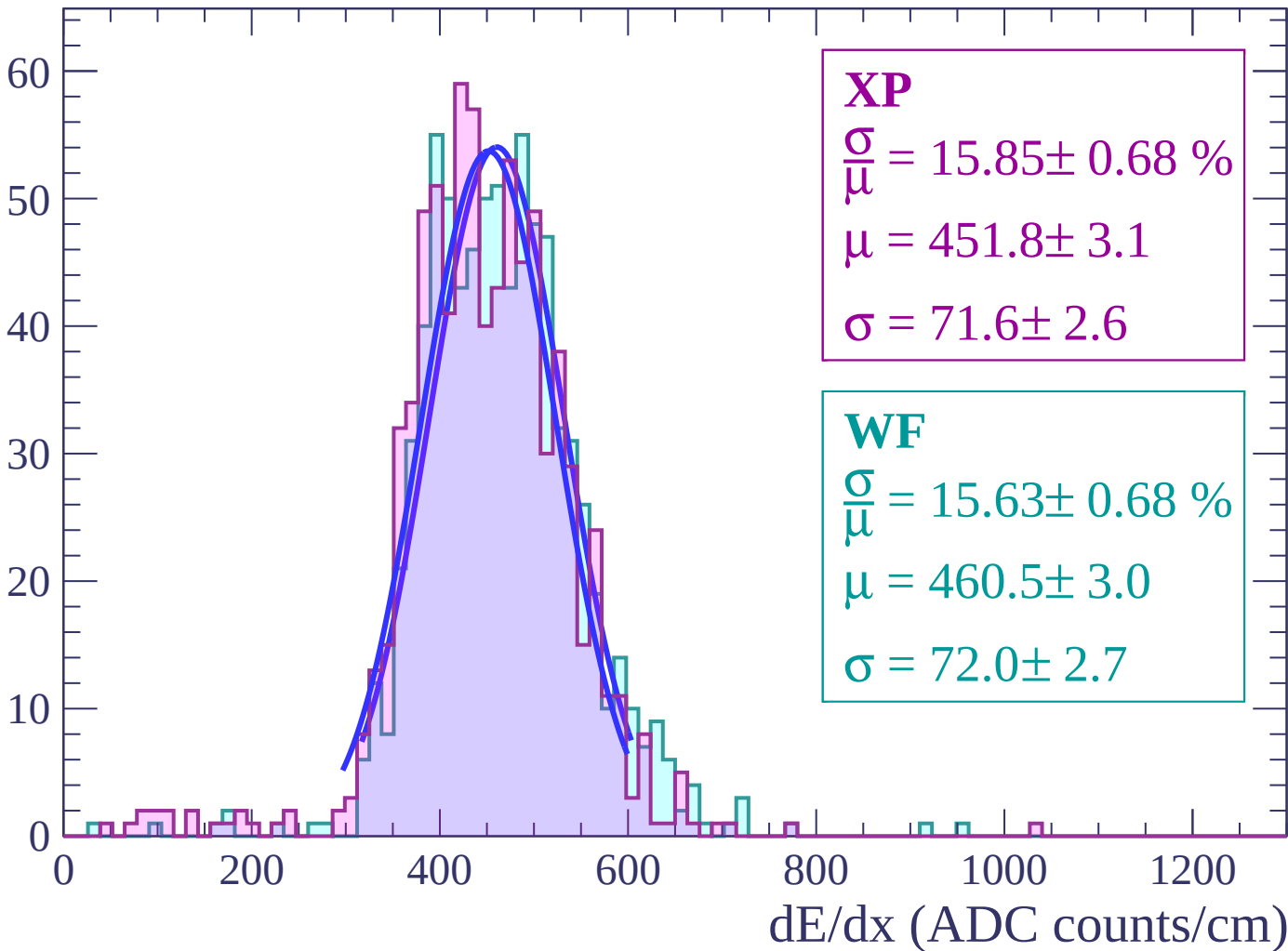
$$\mu = 465.5 \pm 3.2$$

$$\sigma = 73.4 \pm 3.0$$

dE/dx (ADC counts/cm)

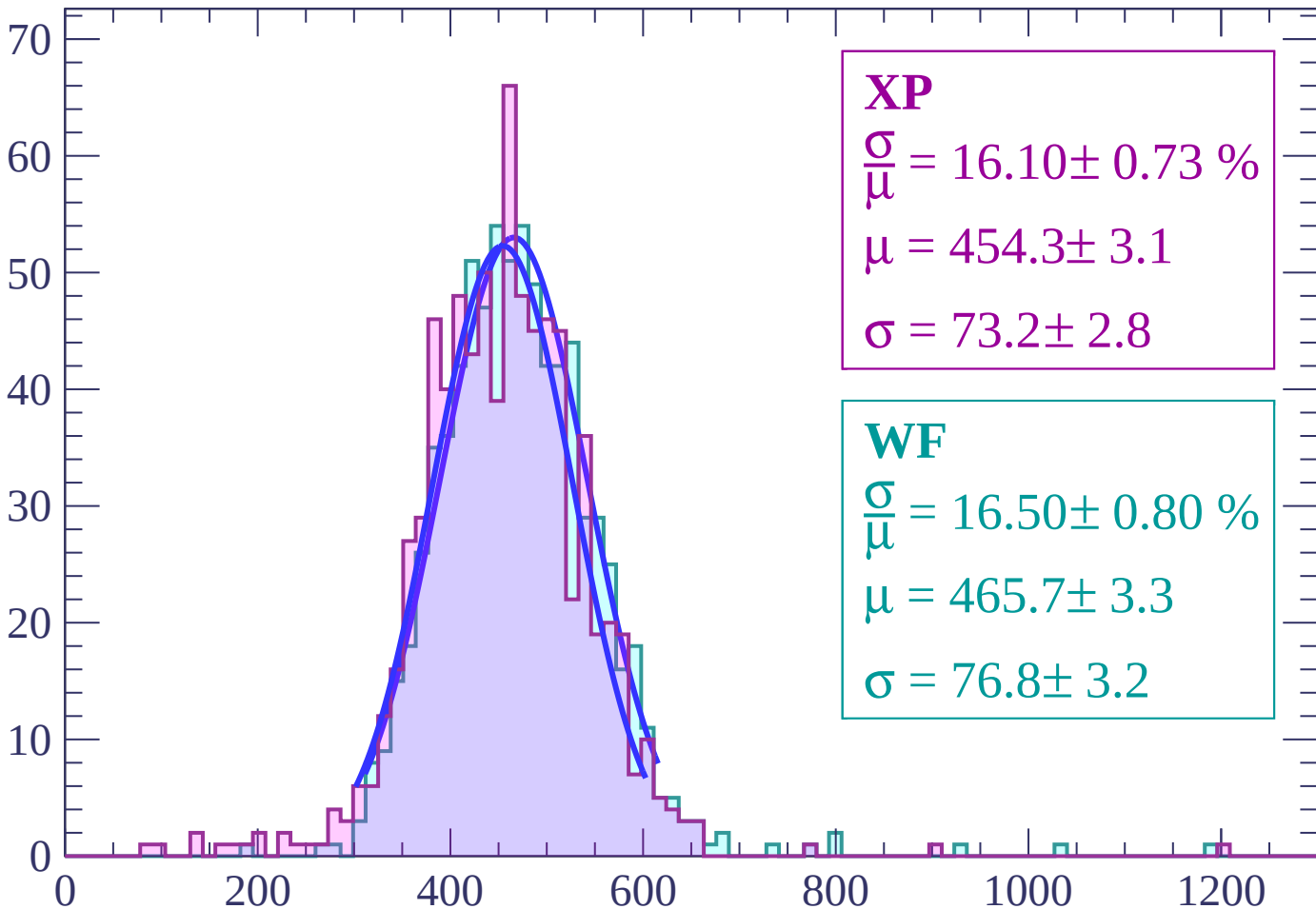
Energy loss | $-16 < \theta < -14$

Count



Energy loss | $-14 < \theta < -12$

Count



XP

$$\frac{\sigma}{\mu} = 16.10 \pm 0.73 \%$$

$$\mu = 454.3 \pm 3.1$$

$$\sigma = 73.2 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 16.50 \pm 0.80 \%$$

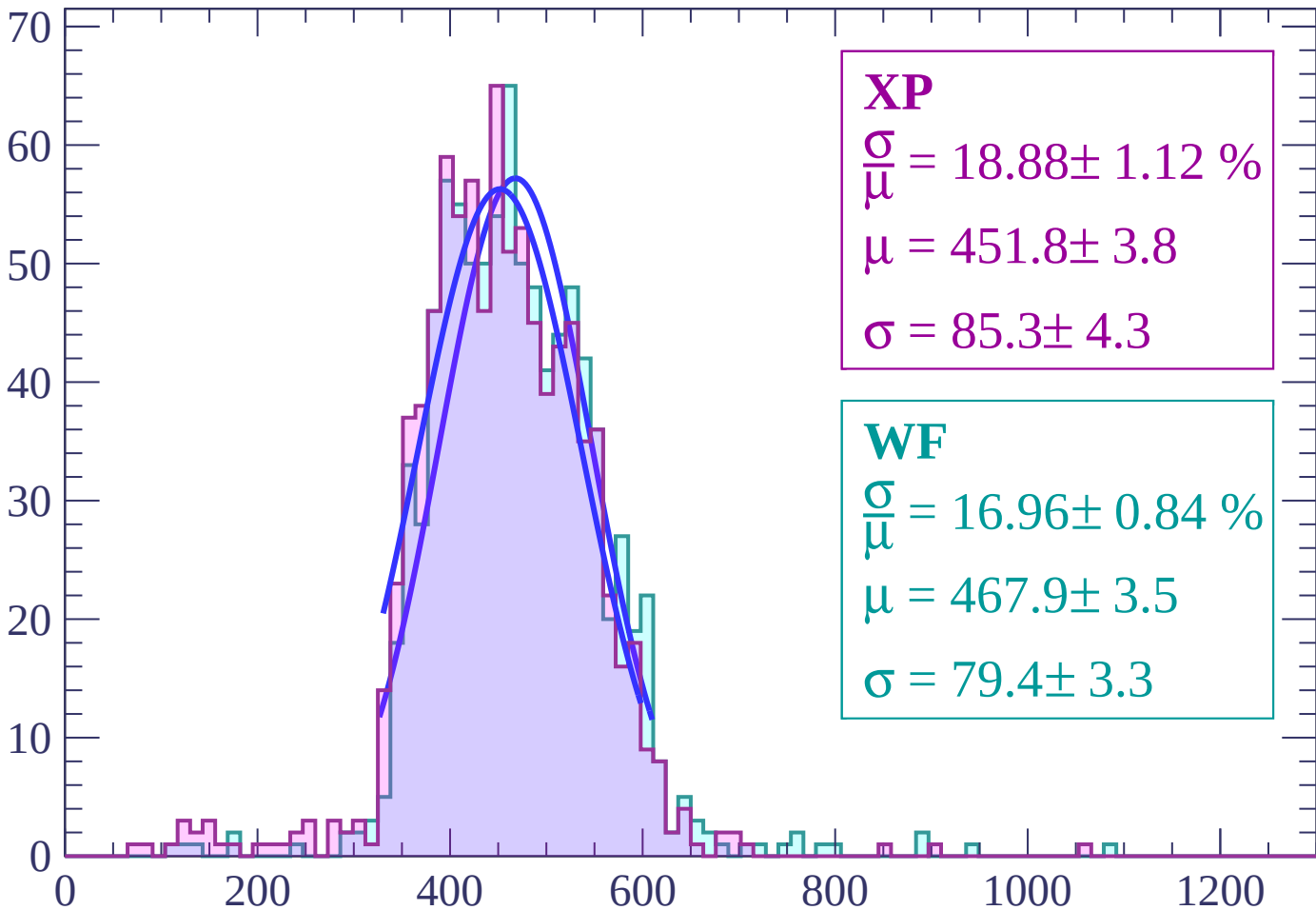
$$\mu = 465.7 \pm 3.3$$

$$\sigma = 76.8 \pm 3.2$$

dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

Count



XP

$$\frac{\sigma}{\mu} = 18.88 \pm 1.12 \%$$

$$\mu = 451.8 \pm 3.8$$

$$\sigma = 85.3 \pm 4.3$$

WF

$$\frac{\sigma}{\mu} = 16.96 \pm 0.84 \%$$

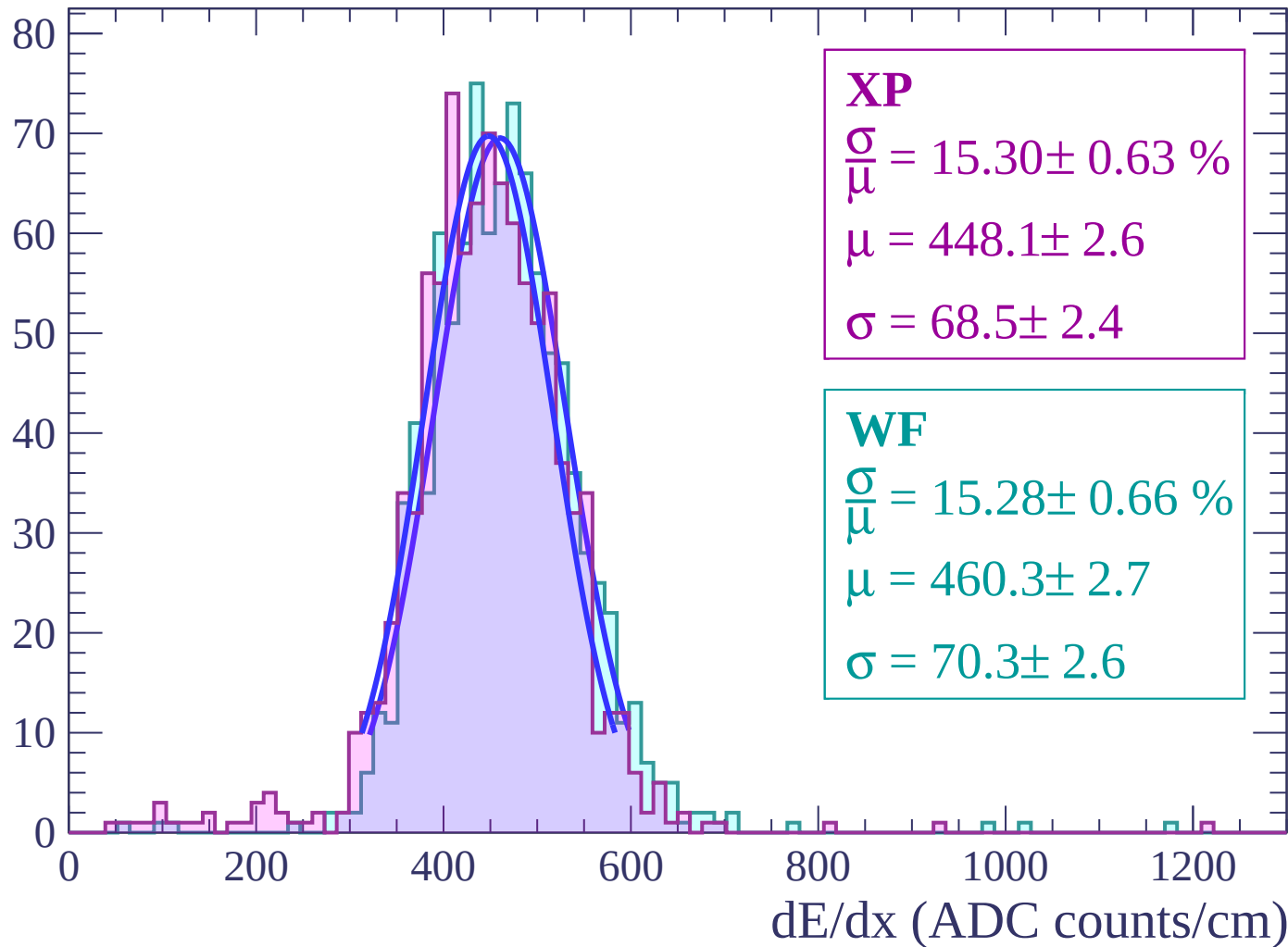
$$\mu = 467.9 \pm 3.5$$

$$\sigma = 79.4 \pm 3.3$$

dE/dx (ADC counts/cm)

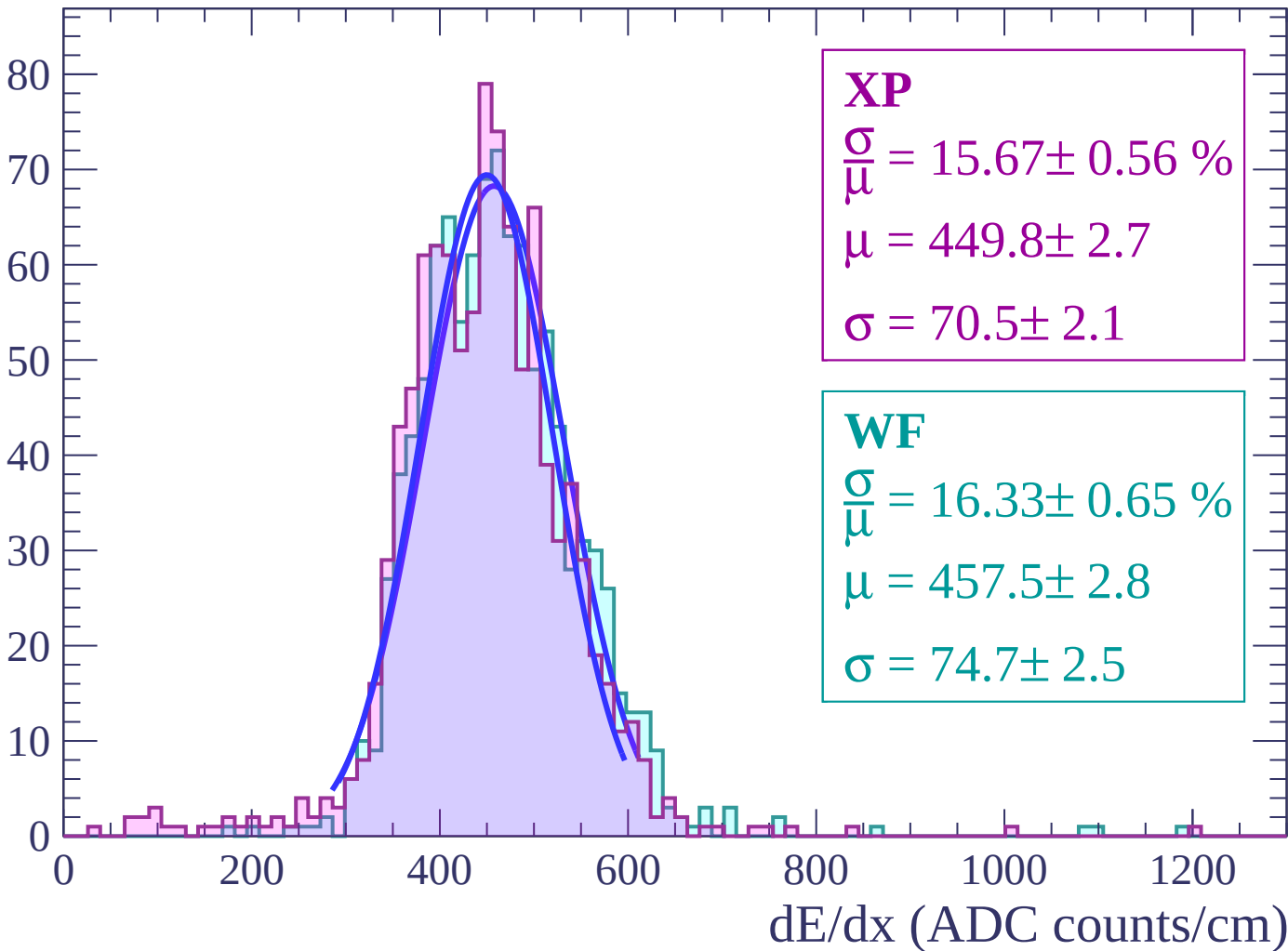
Energy loss | $-10 < \theta < -8$

Count



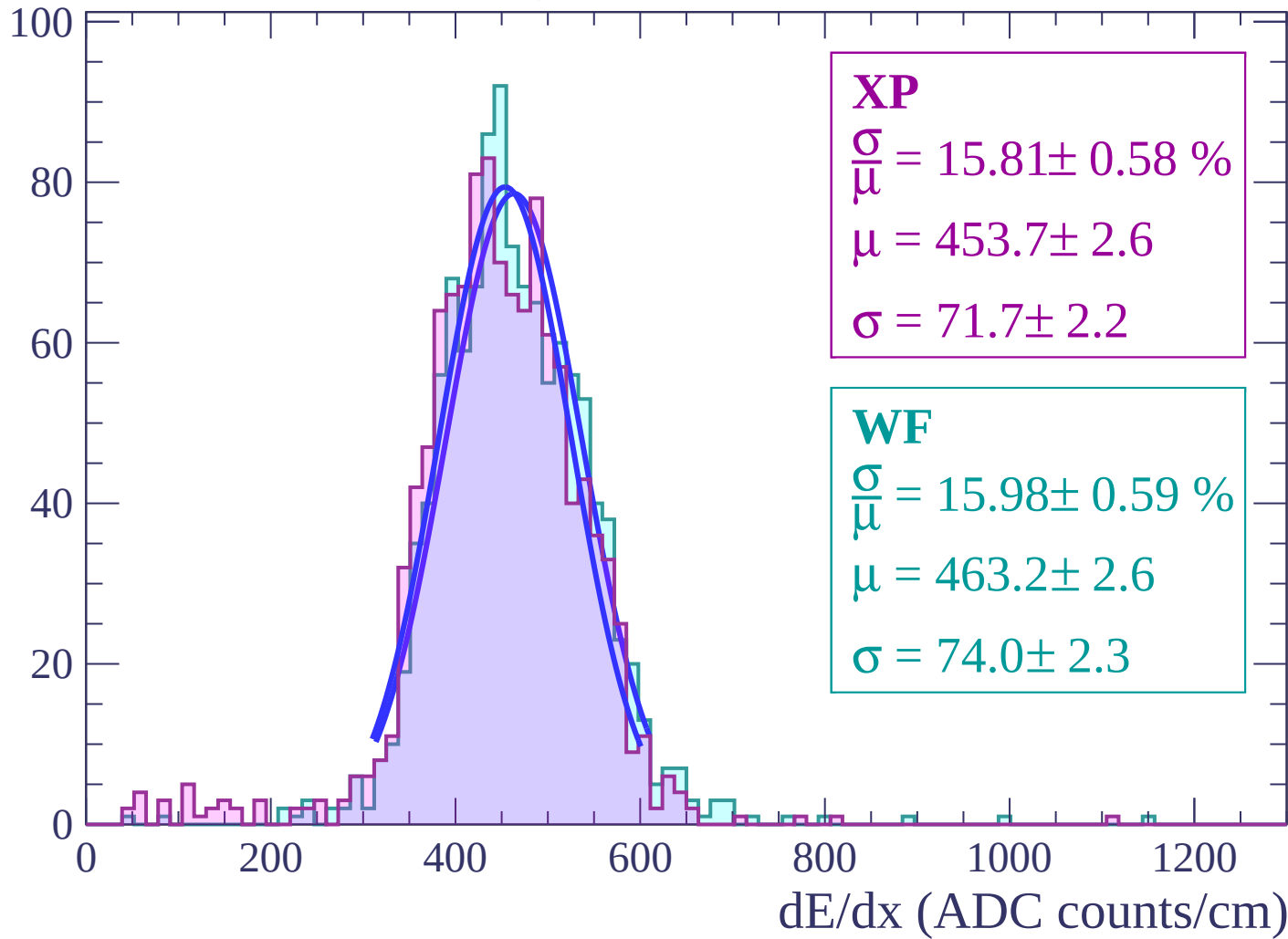
Energy loss | $-8 < \theta < -6$

Count



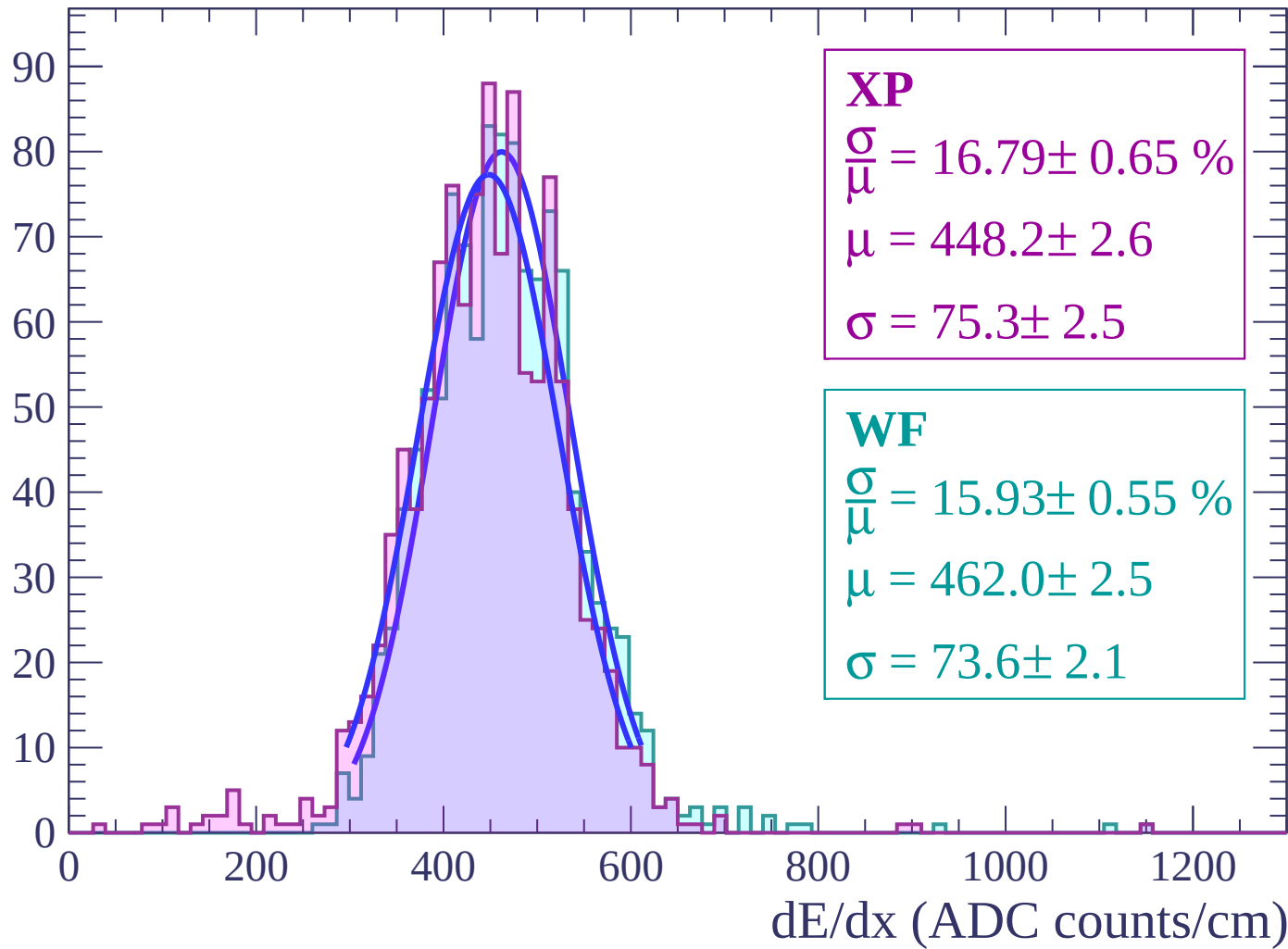
Energy loss | $-6 < \theta < -4$

Count



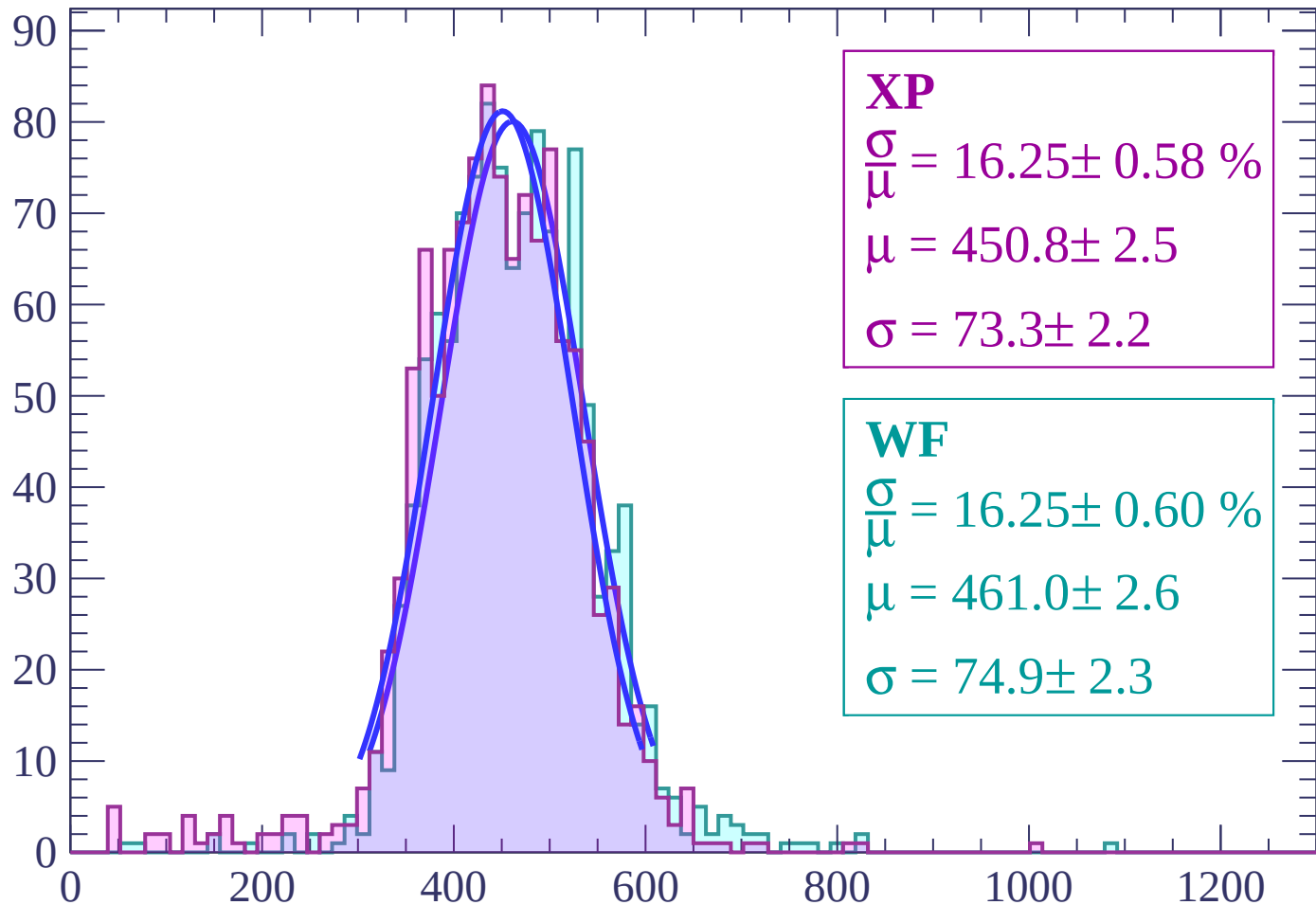
Energy loss | $-4 < \theta < -2$

Count



Energy loss | $-2 < \theta < 0$

Count



XP

$$\frac{\sigma}{\mu} = 16.25 \pm 0.58 \%$$

$$\mu = 450.8 \pm 2.5$$

$$\sigma = 73.3 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 16.25 \pm 0.60 \%$$

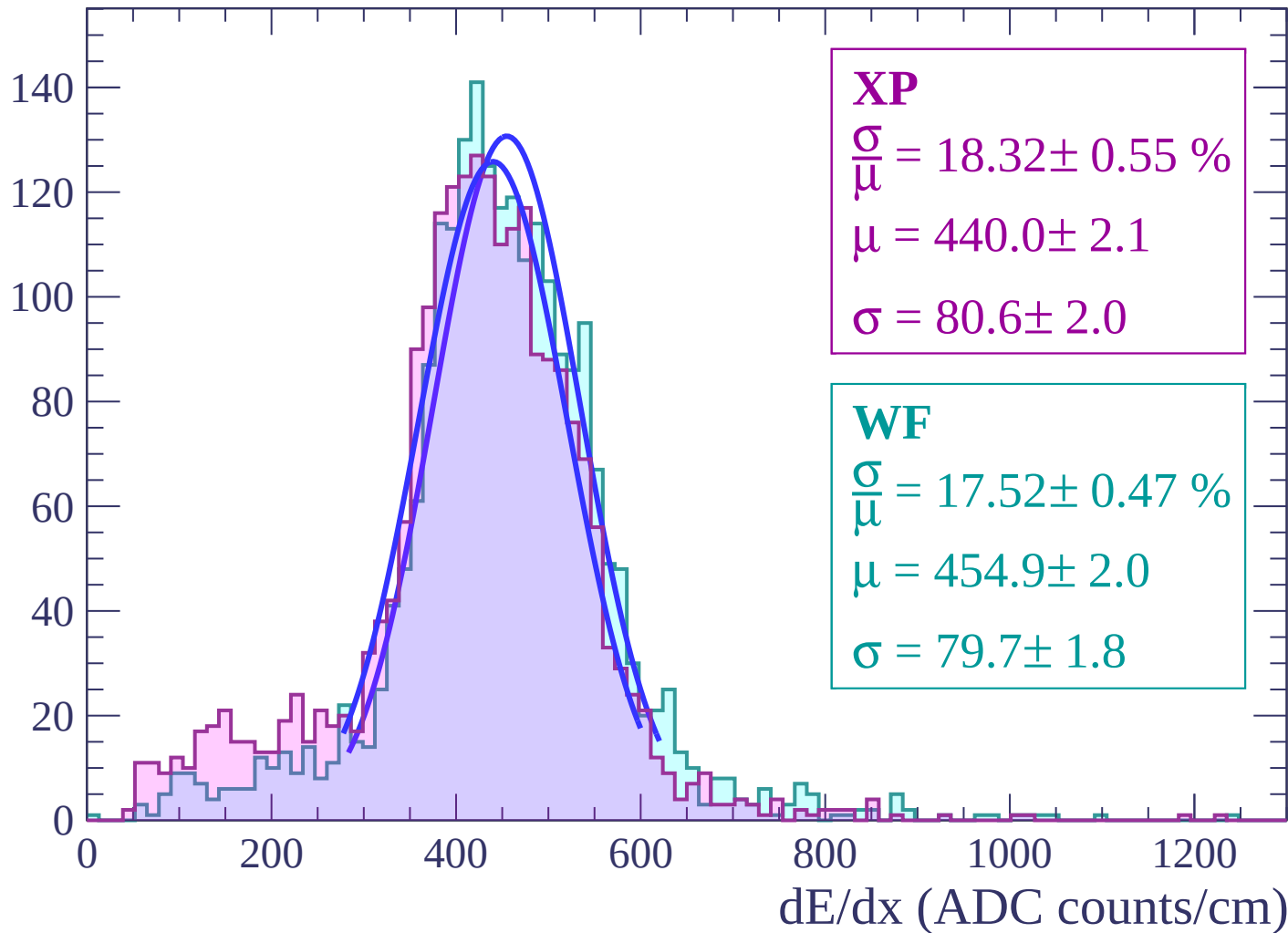
$$\mu = 461.0 \pm 2.6$$

$$\sigma = 74.9 \pm 2.3$$

dE/dx (ADC counts/cm)

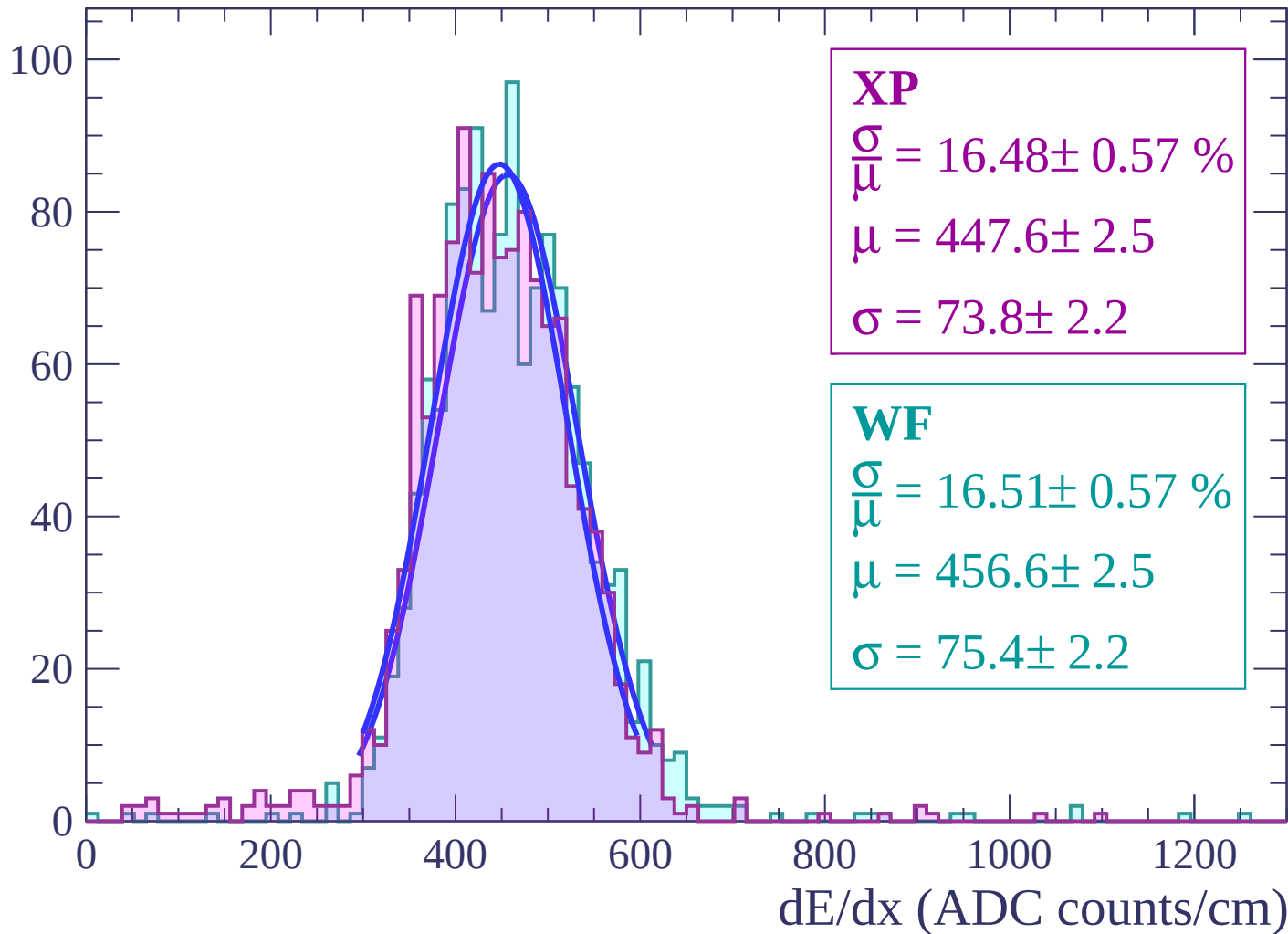
Energy loss | $0 < \theta < 2$

Count



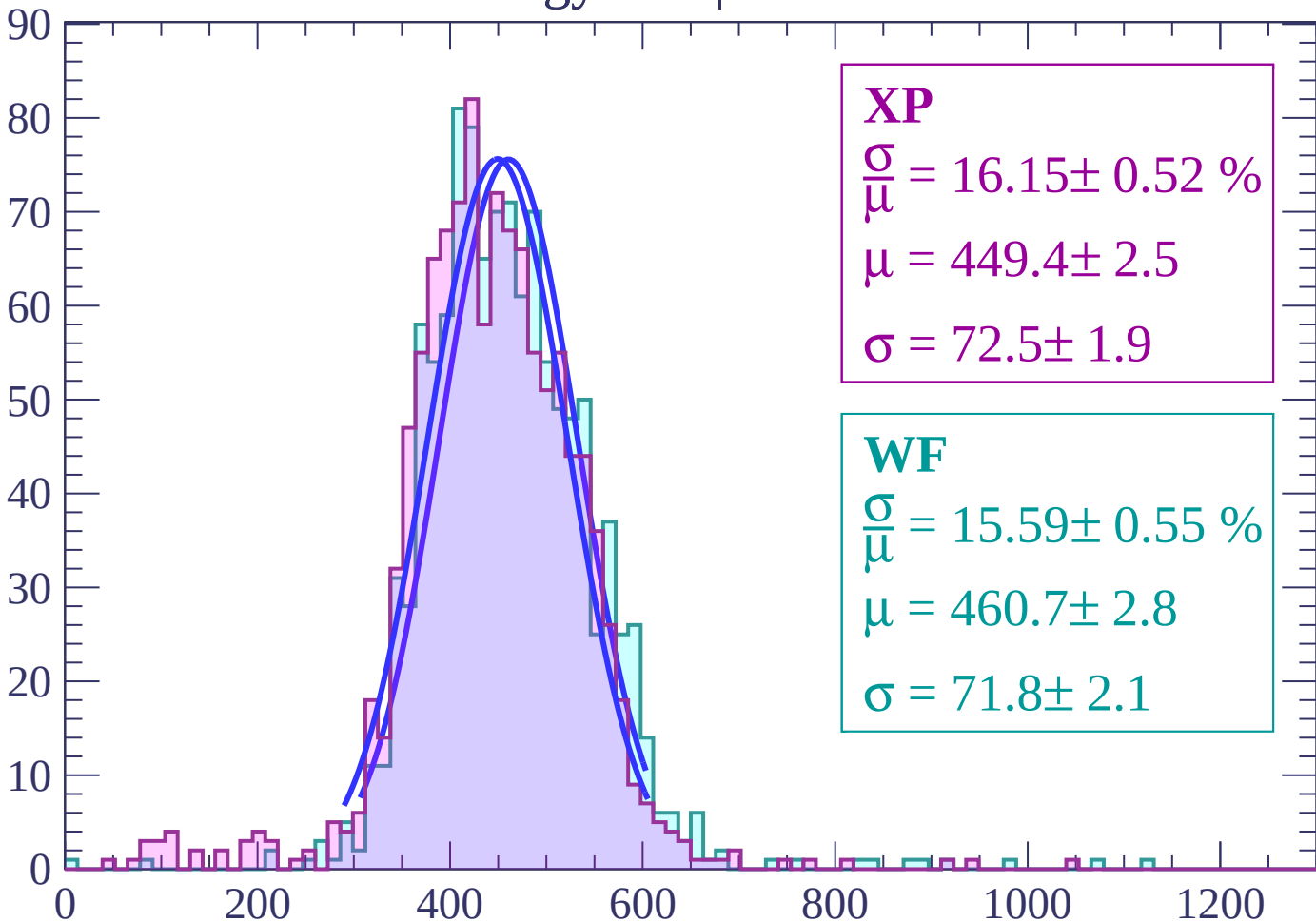
Energy loss | $2 < \theta < 4$

Count



Energy loss | $4 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 16.15 \pm 0.52 \%$$

$$\mu = 449.4 \pm 2.5$$

$$\sigma = 72.5 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 15.59 \pm 0.55 \%$$

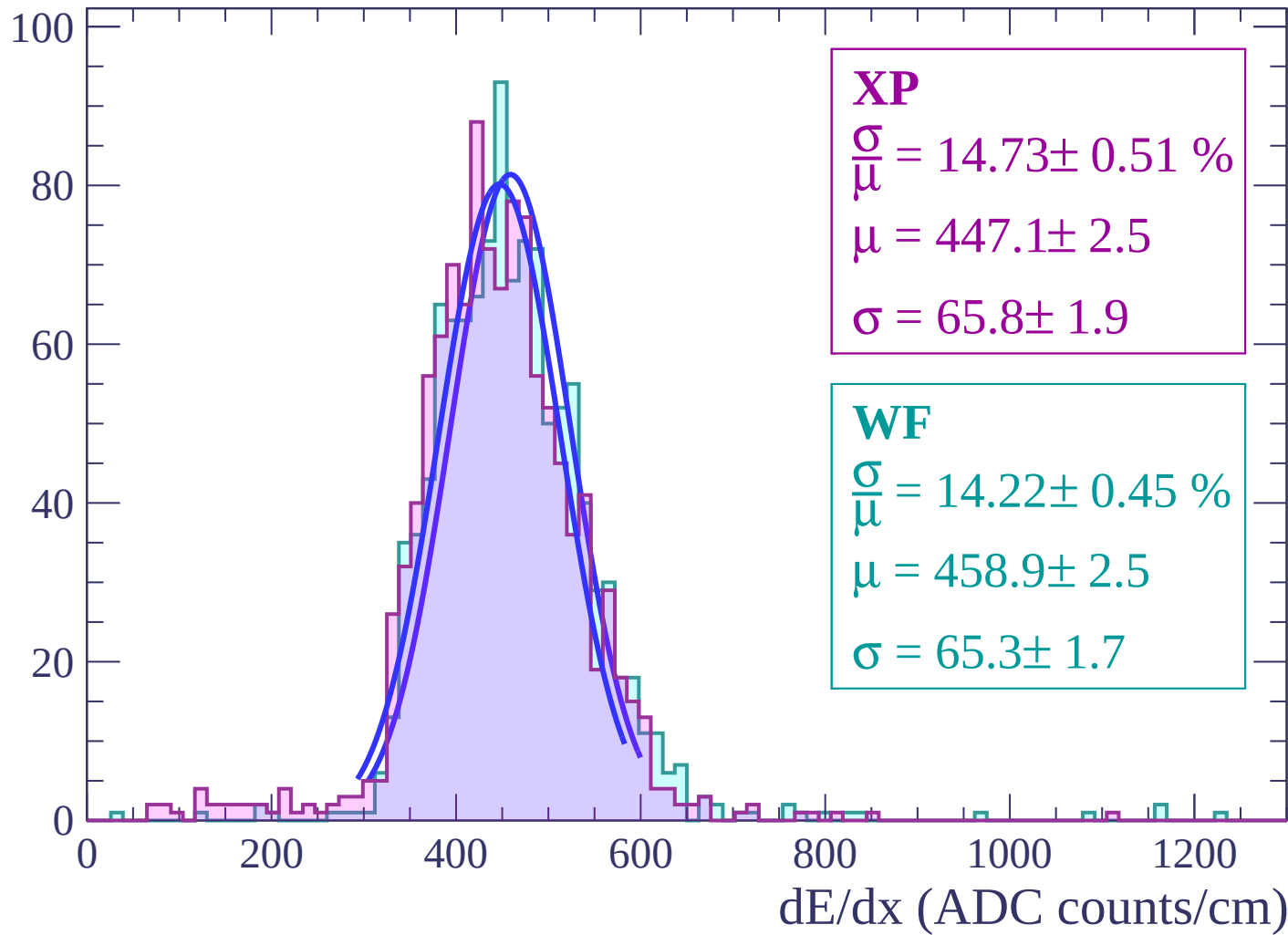
$$\mu = 460.7 \pm 2.8$$

$$\sigma = 71.8 \pm 2.1$$

dE/dx (ADC counts/cm)

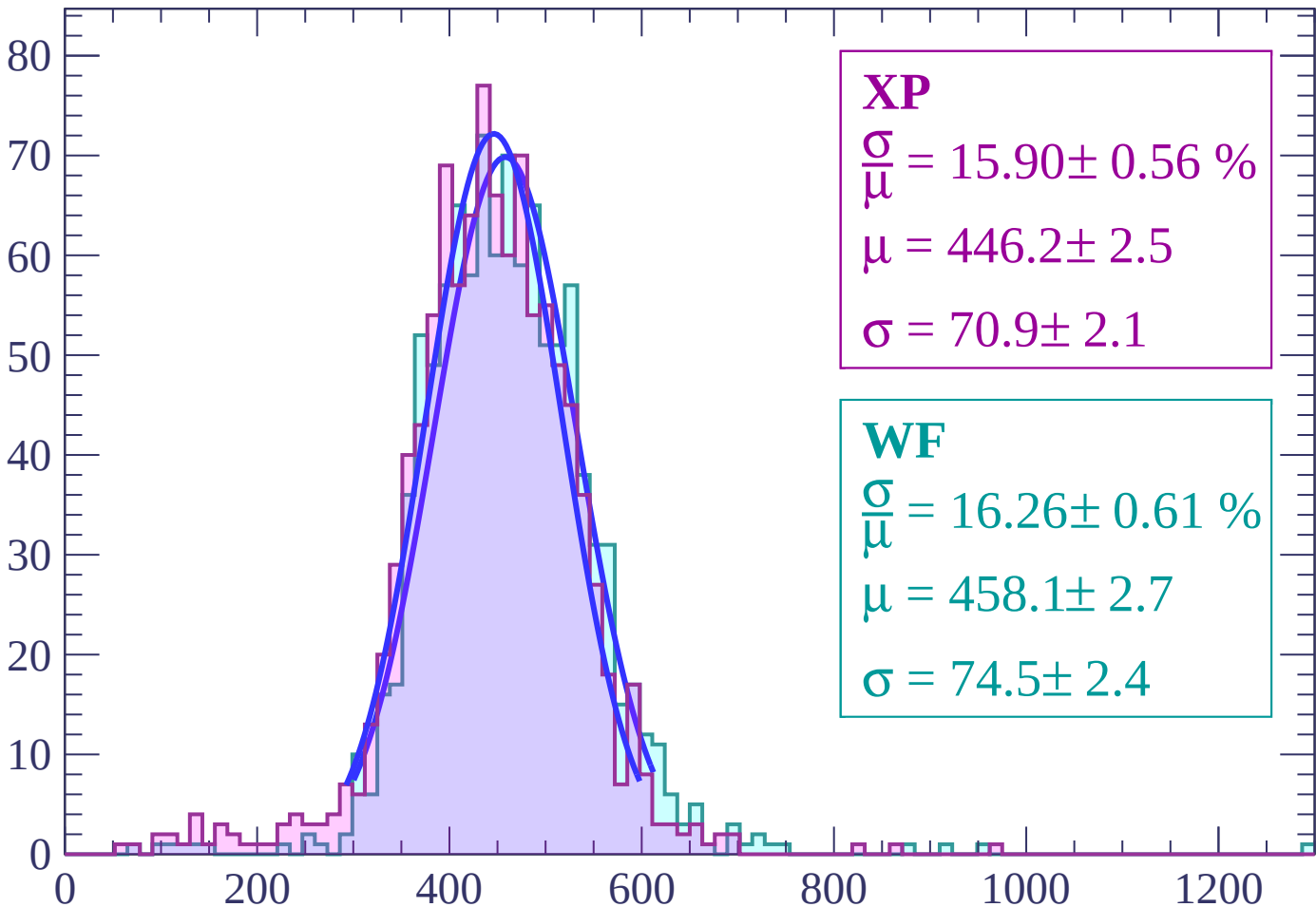
Energy loss | $6 < \theta < 8$

Count



Energy loss | $8 < \theta < 10$

Count



XP

$$\frac{\sigma}{\mu} = 15.90 \pm 0.56 \%$$

$$\mu = 446.2 \pm 2.5$$

$$\sigma = 70.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 16.26 \pm 0.61 \%$$

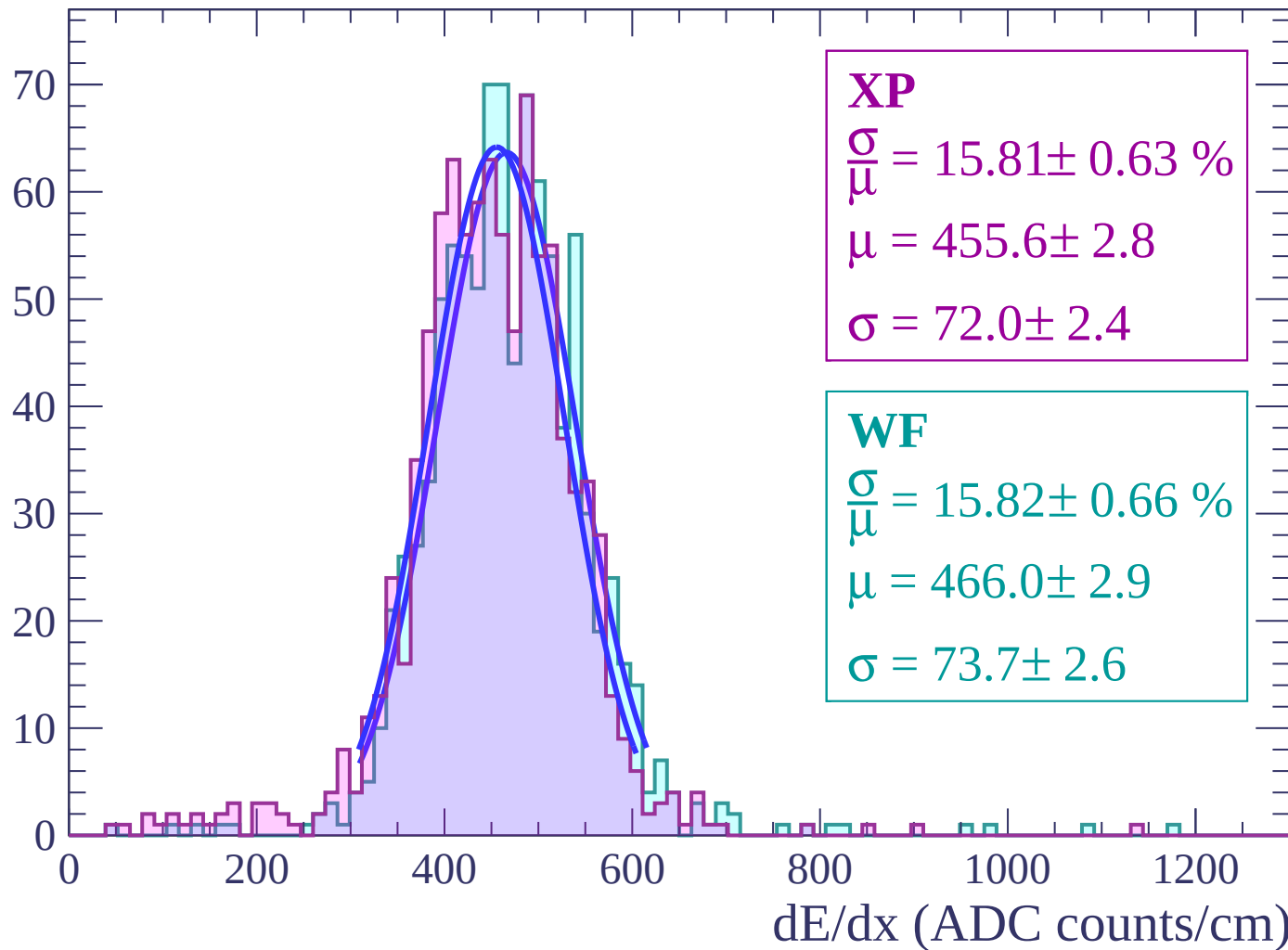
$$\mu = 458.1 \pm 2.7$$

$$\sigma = 74.5 \pm 2.4$$

dE/dx (ADC counts/cm)

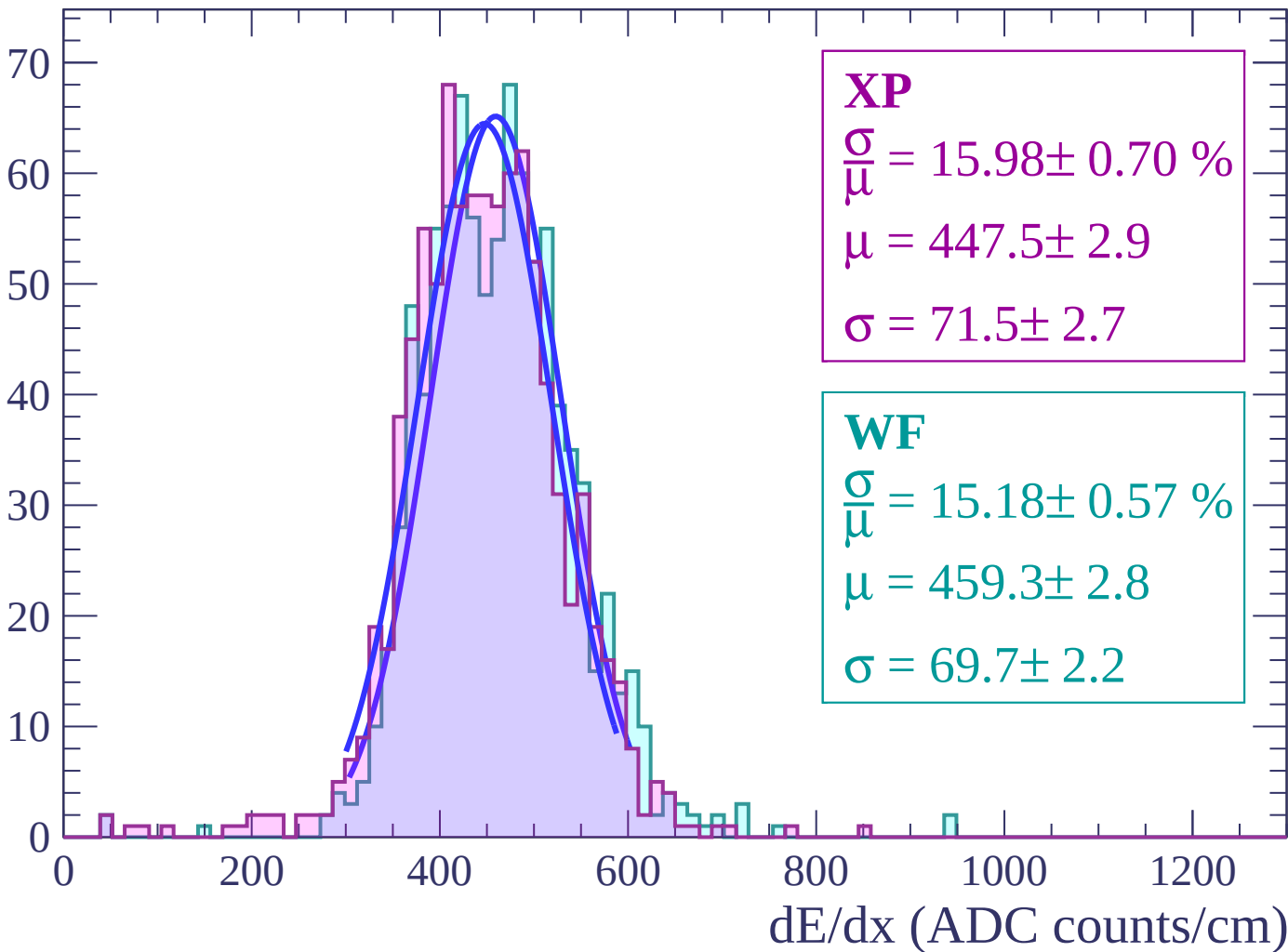
Energy loss | $10 < \theta < 12$

Count



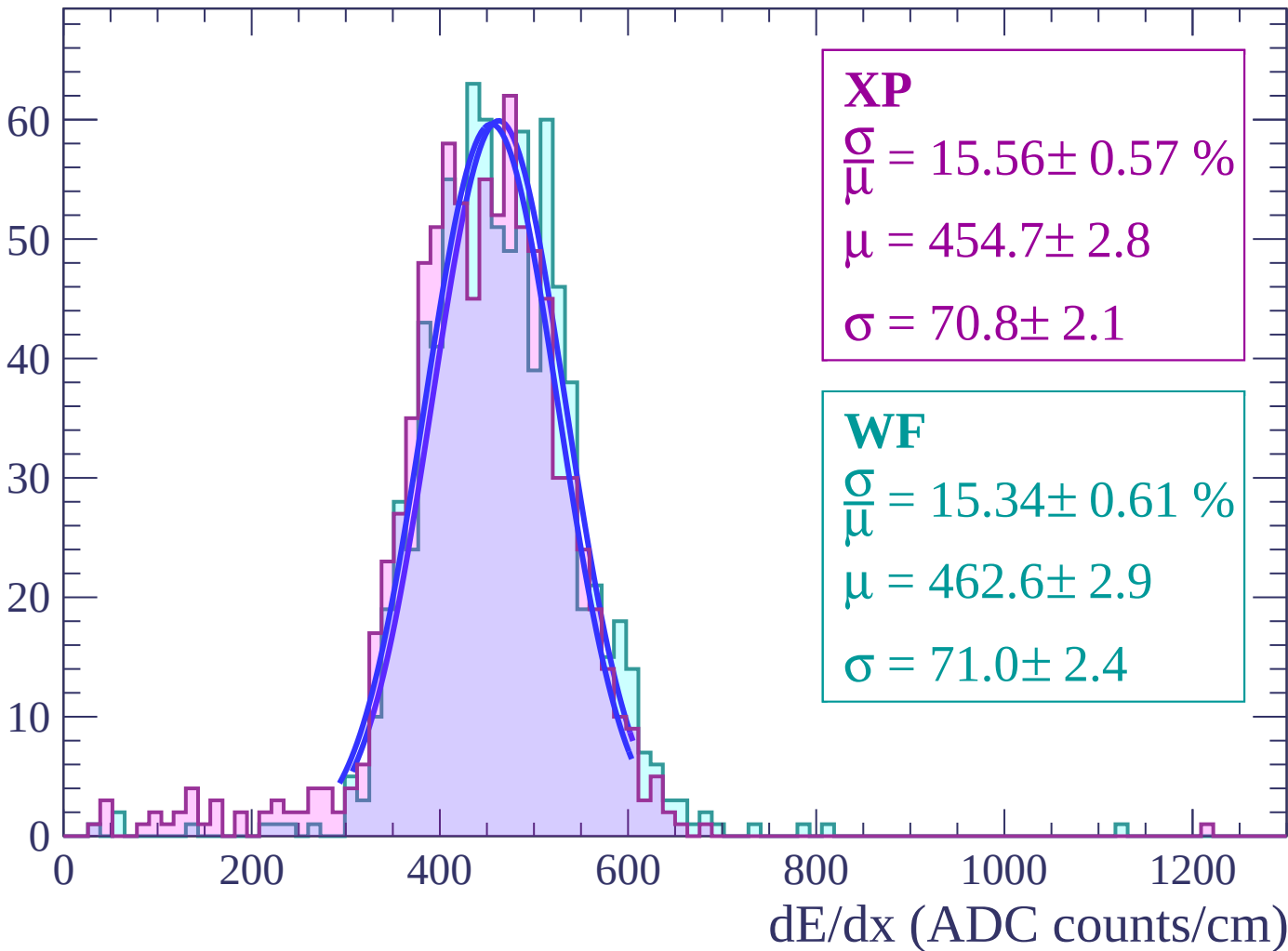
Energy loss | $12 < \theta < 14$

Count



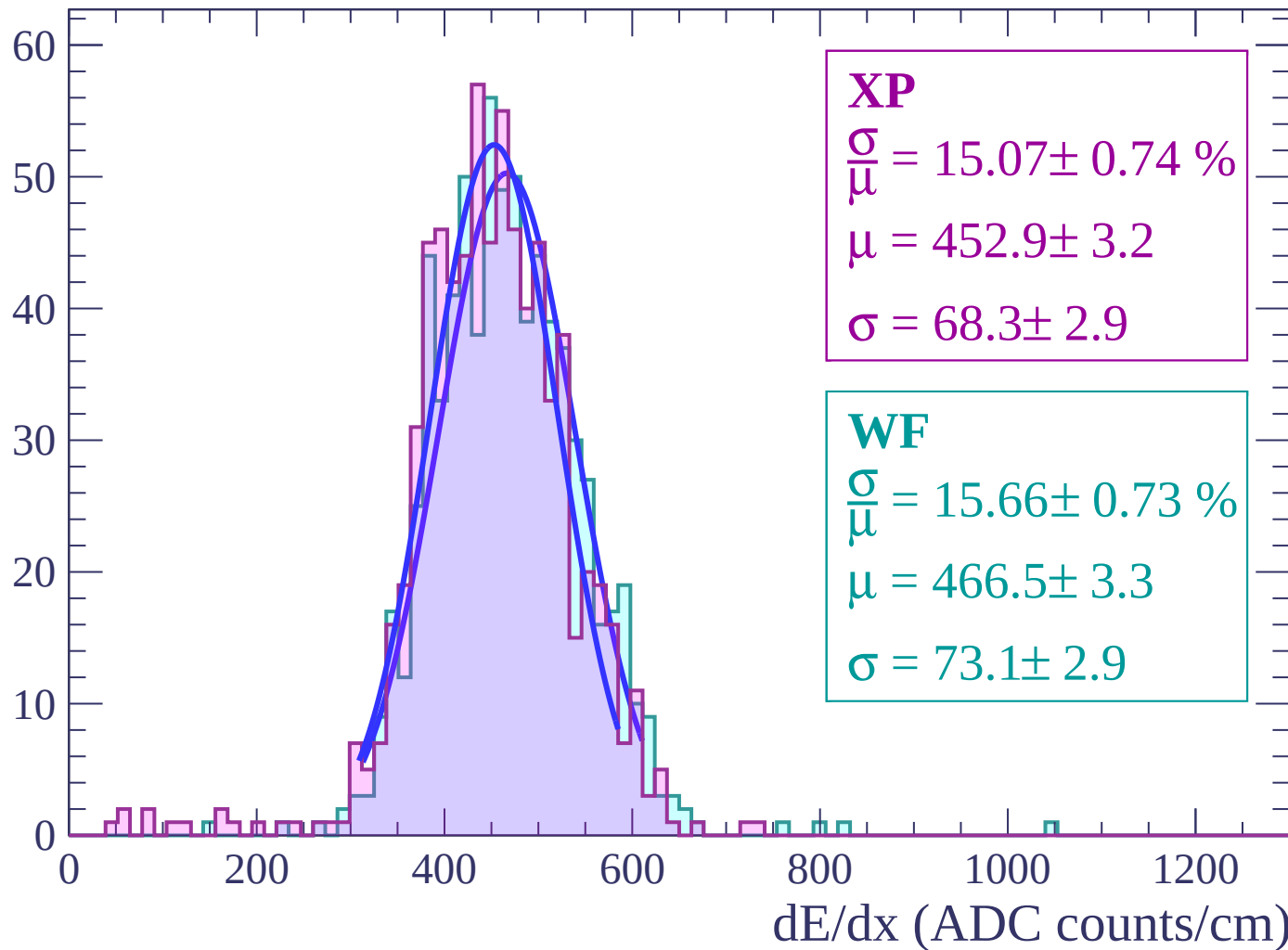
Energy loss | $14 < \theta < 16$

Count



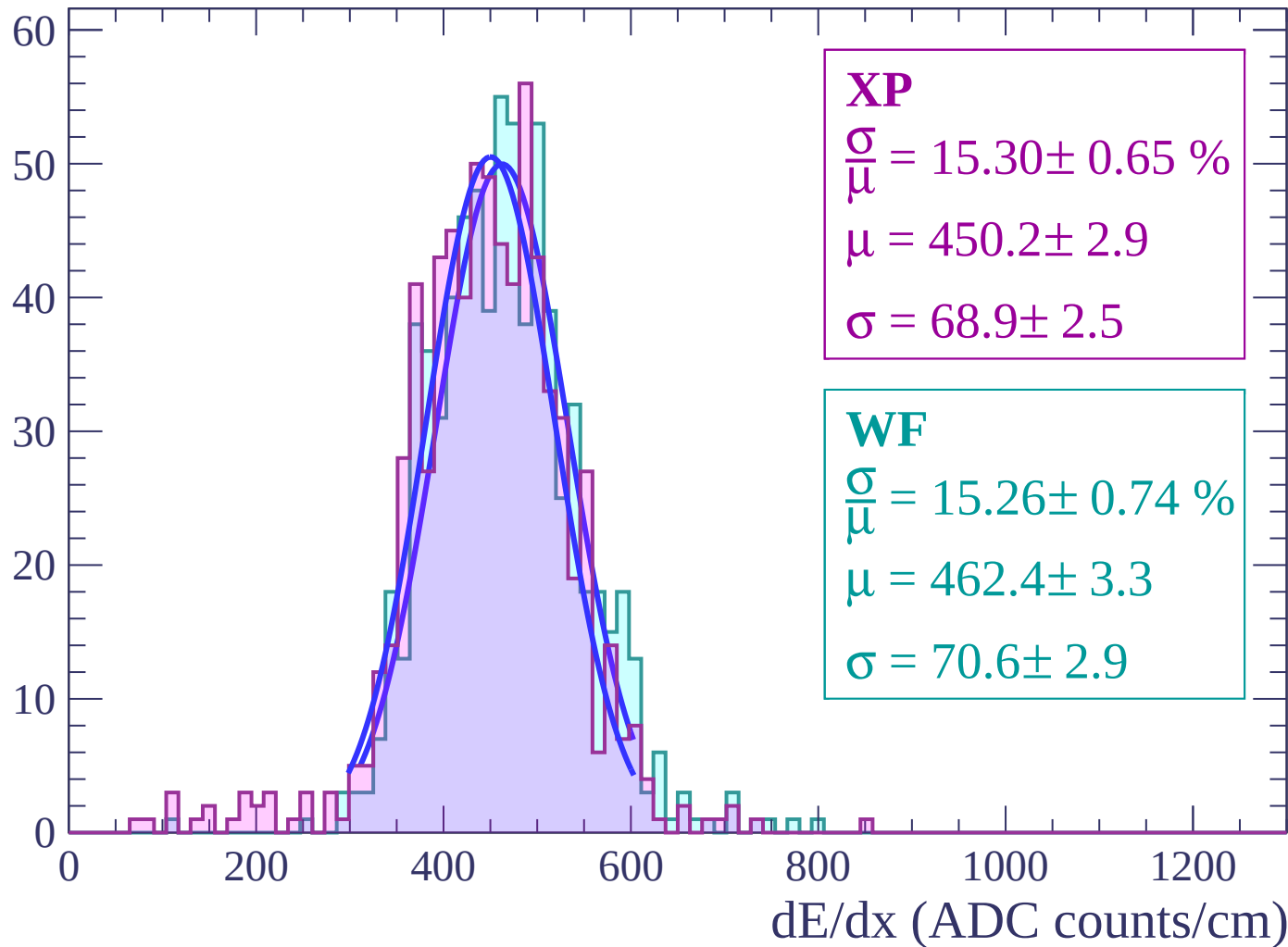
Energy loss | $16 < \theta < 18$

Count



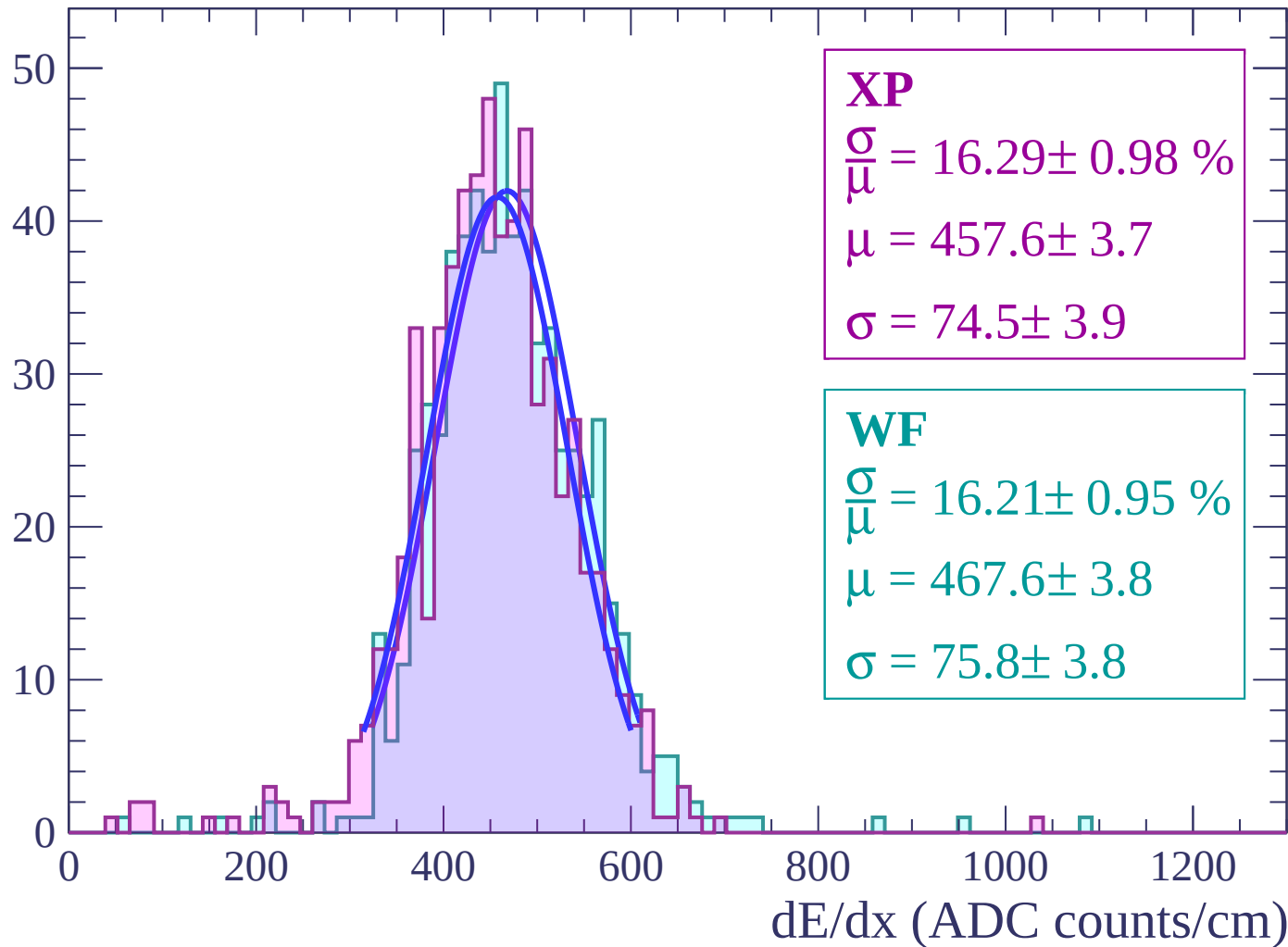
Energy loss | $18 < \theta < 20$

Count



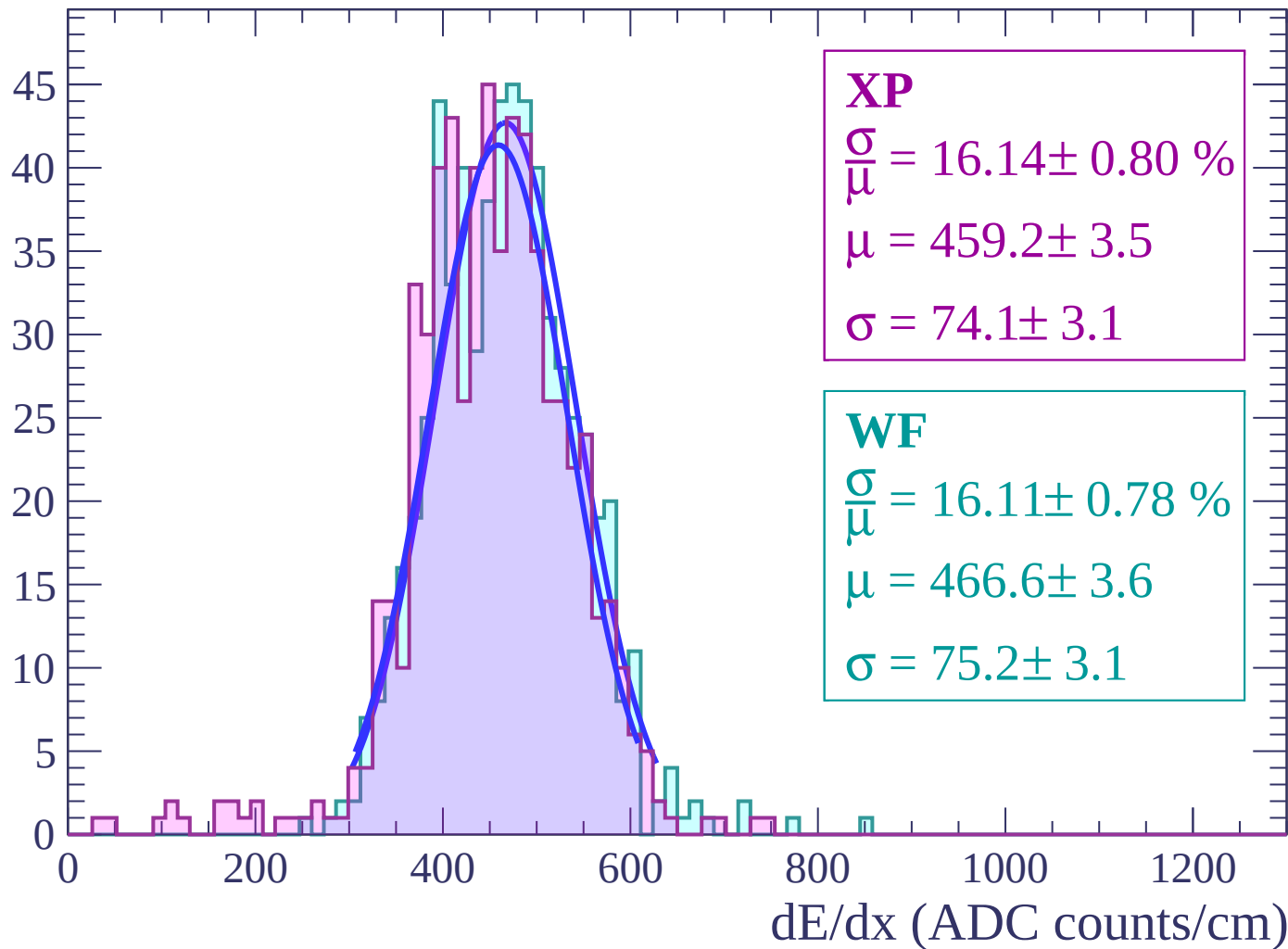
Energy loss | $20 < \theta < 22$

Count



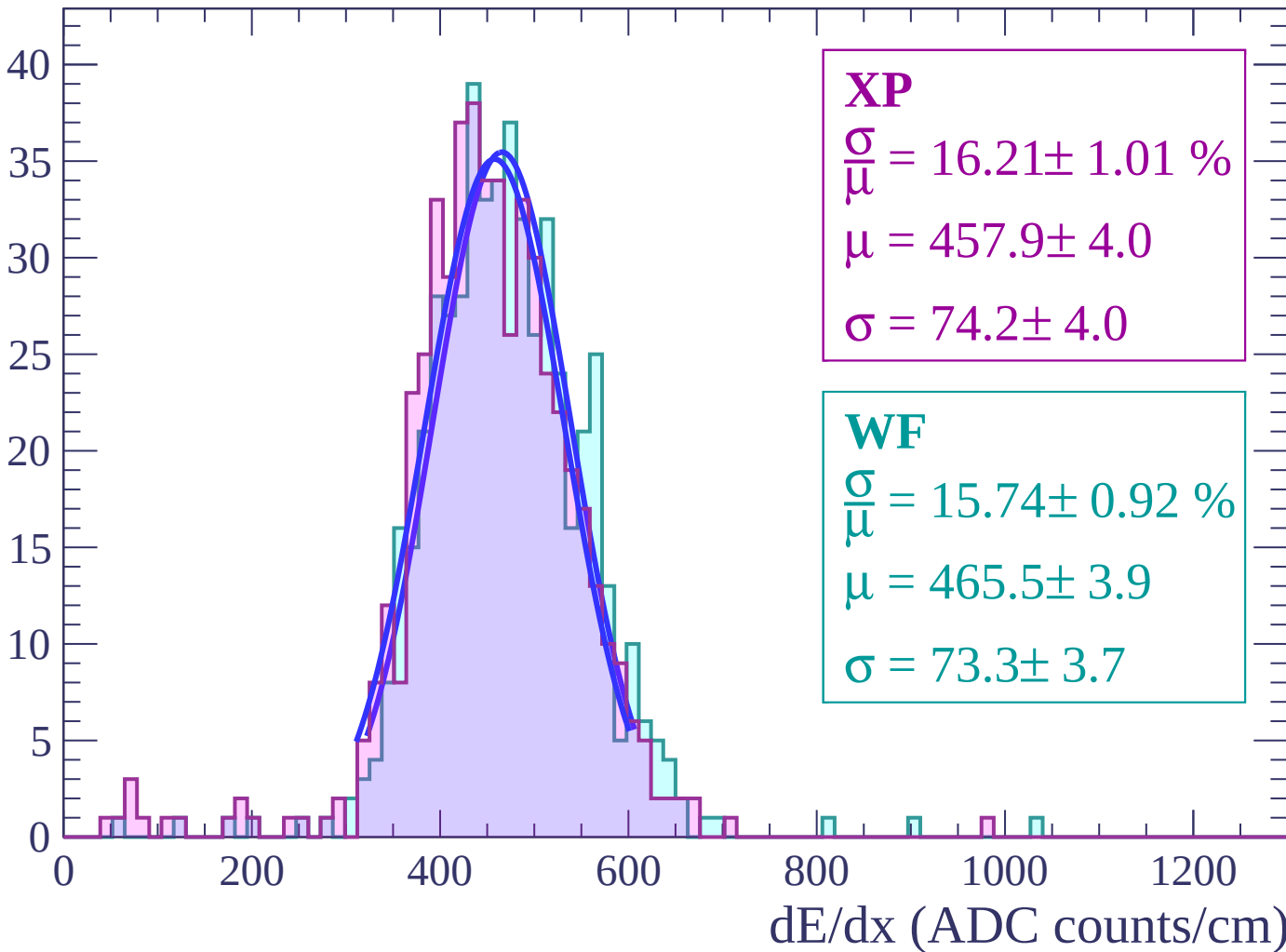
Energy loss | $22 < \theta < 24$

Count



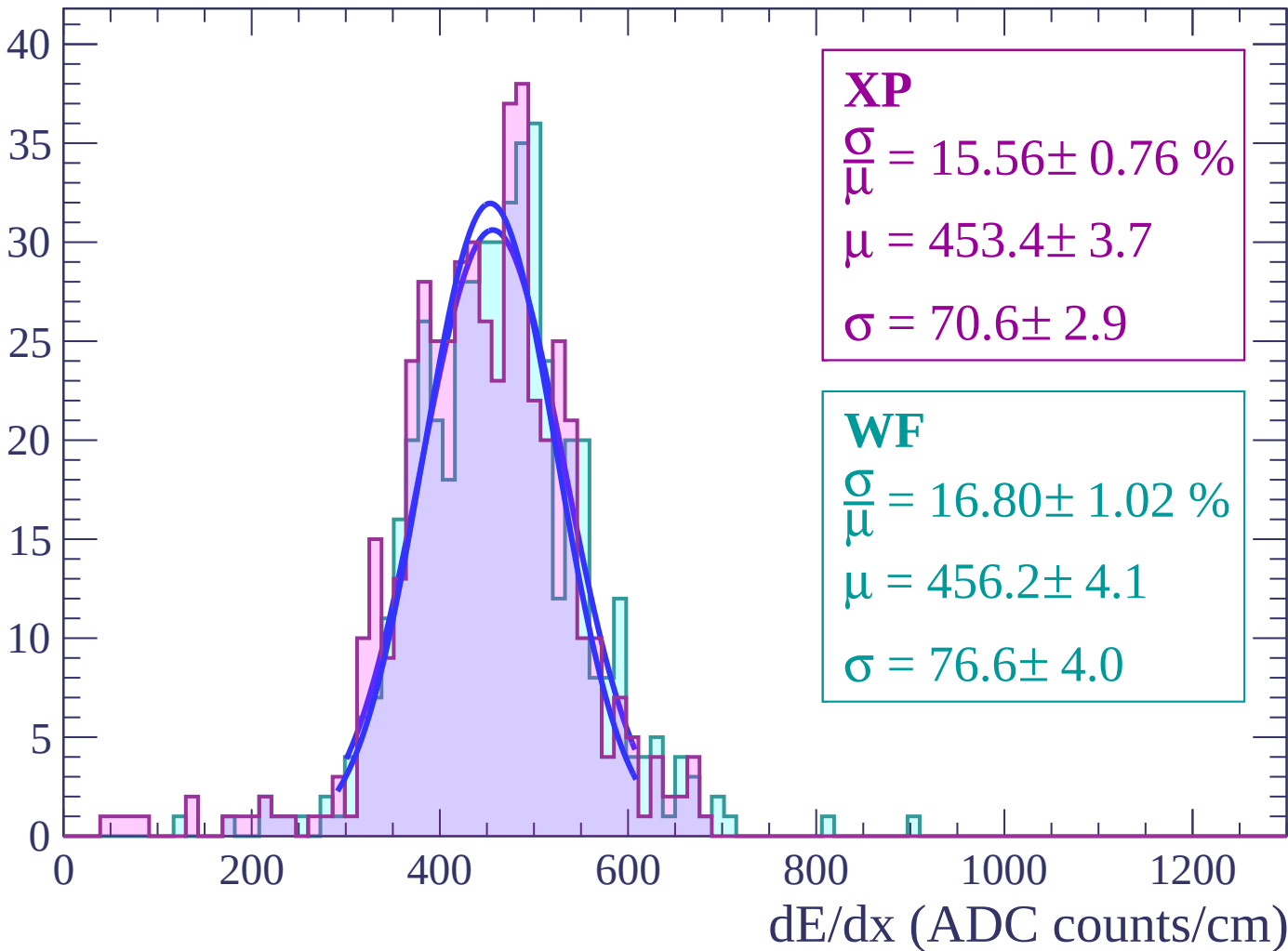
Energy loss | $24 < \theta < 26$

Count



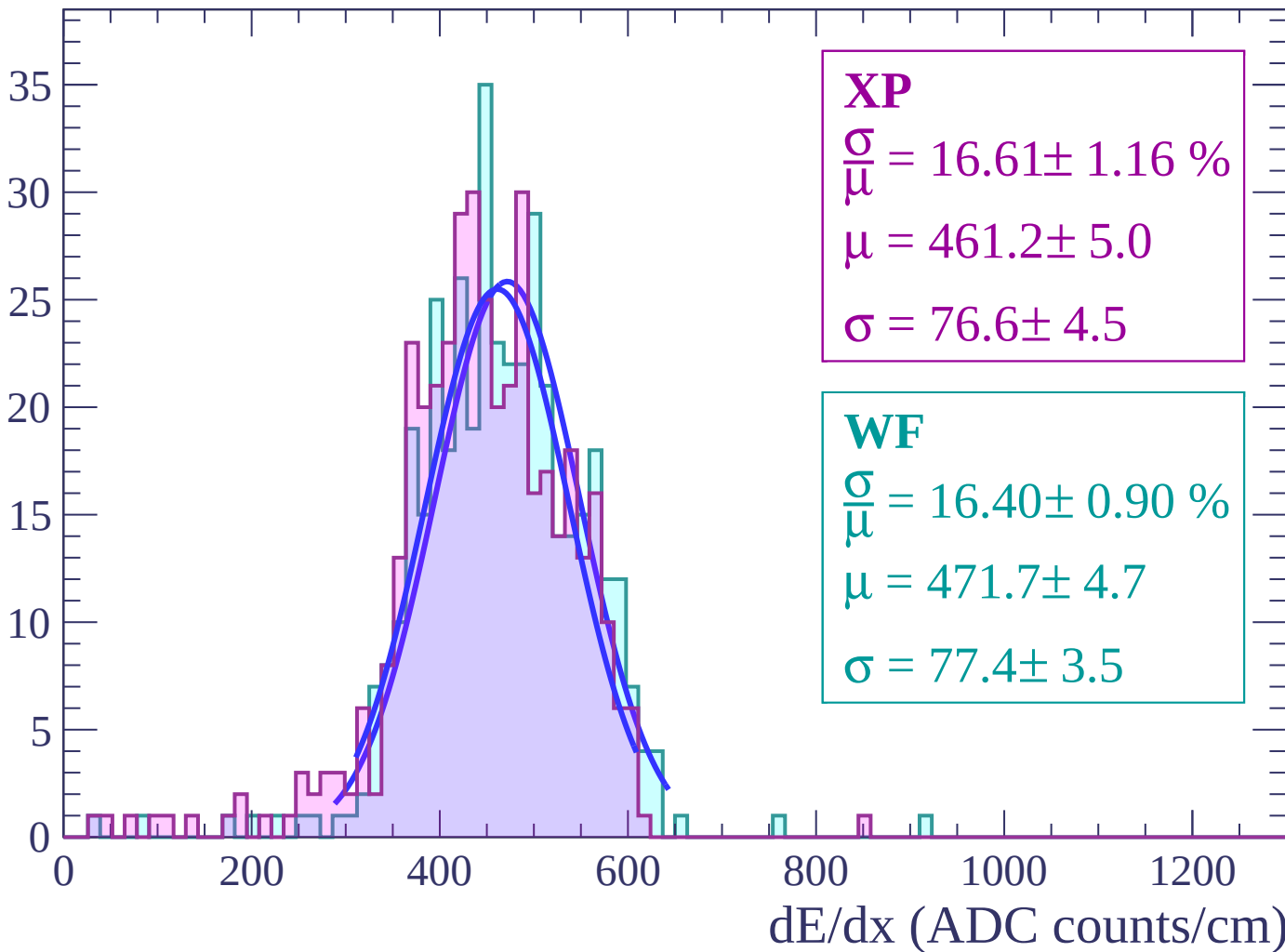
Energy loss | $26 < \theta < 28$

Count



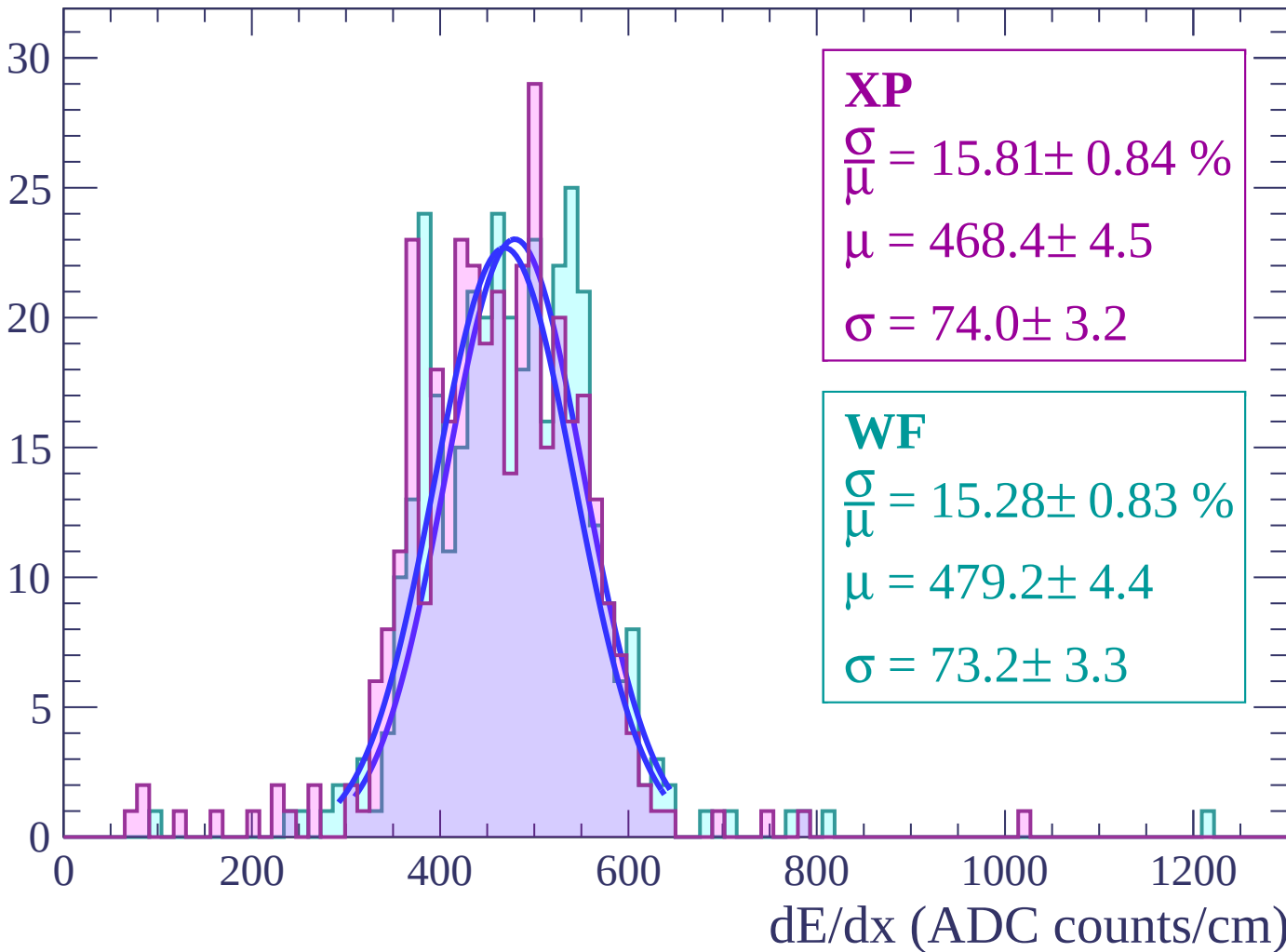
Energy loss | $28 < \theta < 30$

Count



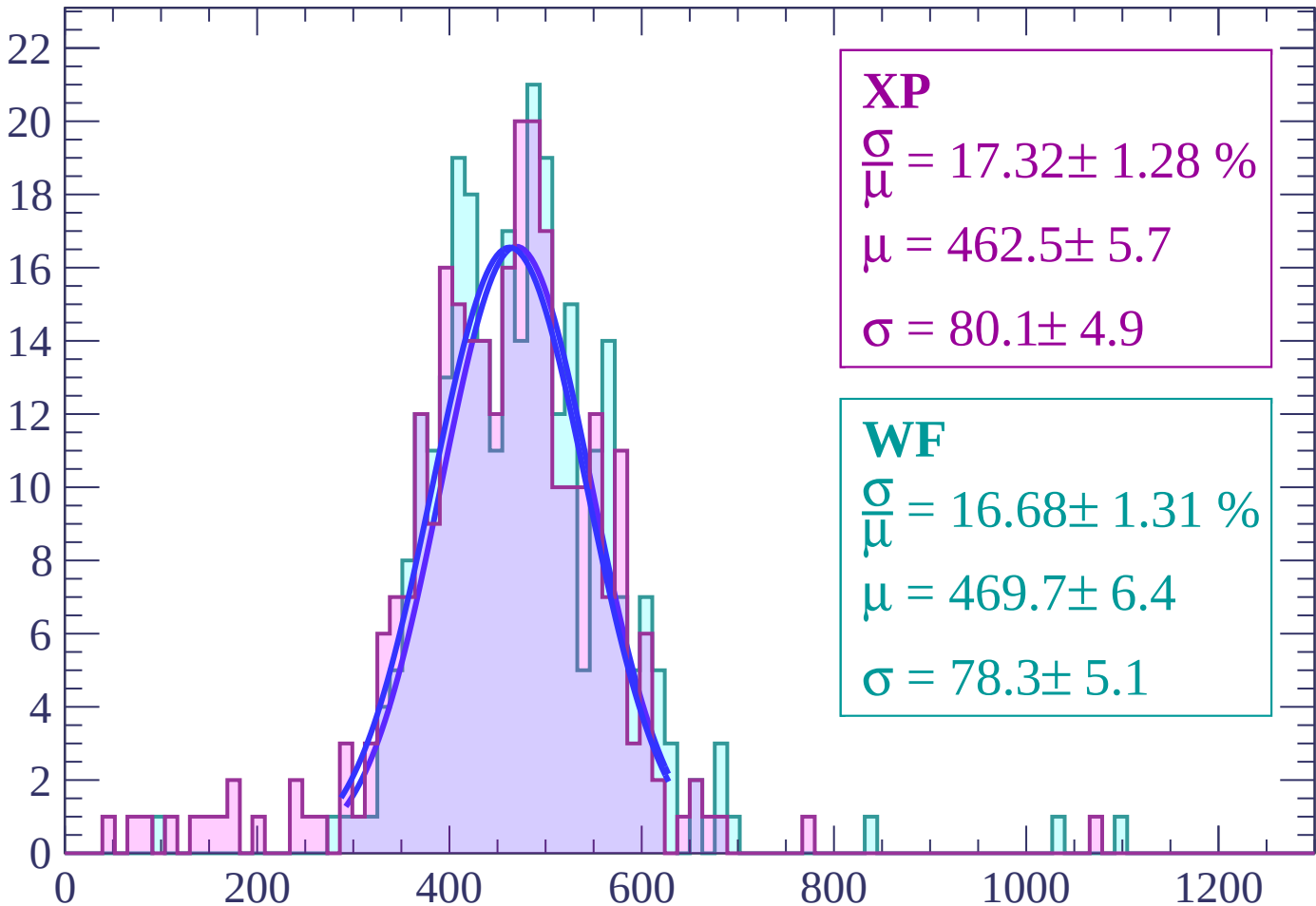
Energy loss | $30 < \theta < 32$

Count



Energy loss | $32 < \theta < 34$

Count



XP

$$\frac{\sigma}{\mu} = 17.32 \pm 1.28 \%$$

$$\mu = 462.5 \pm 5.7$$

$$\sigma = 80.1 \pm 4.9$$

WF

$$\frac{\sigma}{\mu} = 16.68 \pm 1.31 \%$$

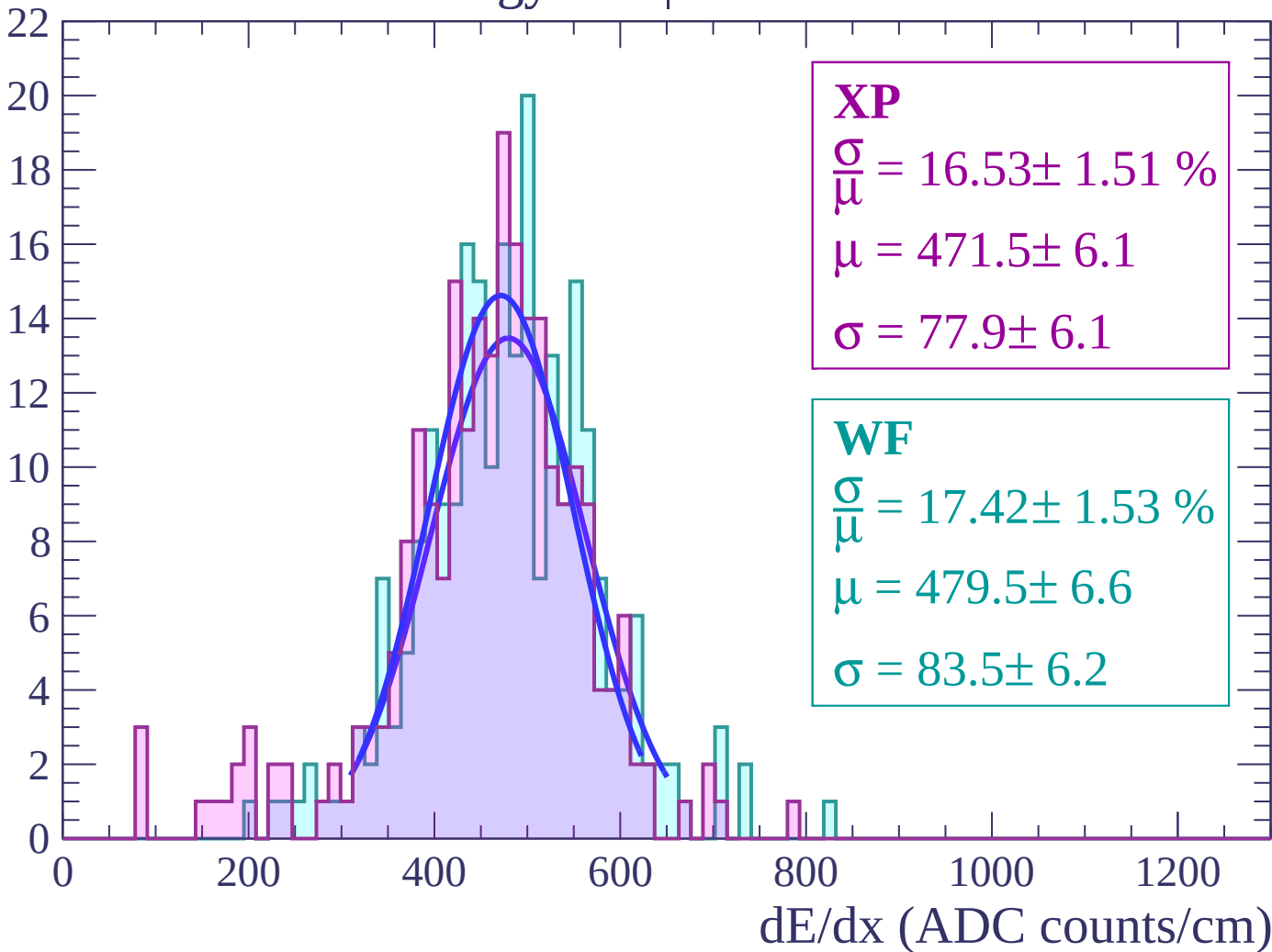
$$\mu = 469.7 \pm 6.4$$

$$\sigma = 78.3 \pm 5.1$$

dE/dx (ADC counts/cm)

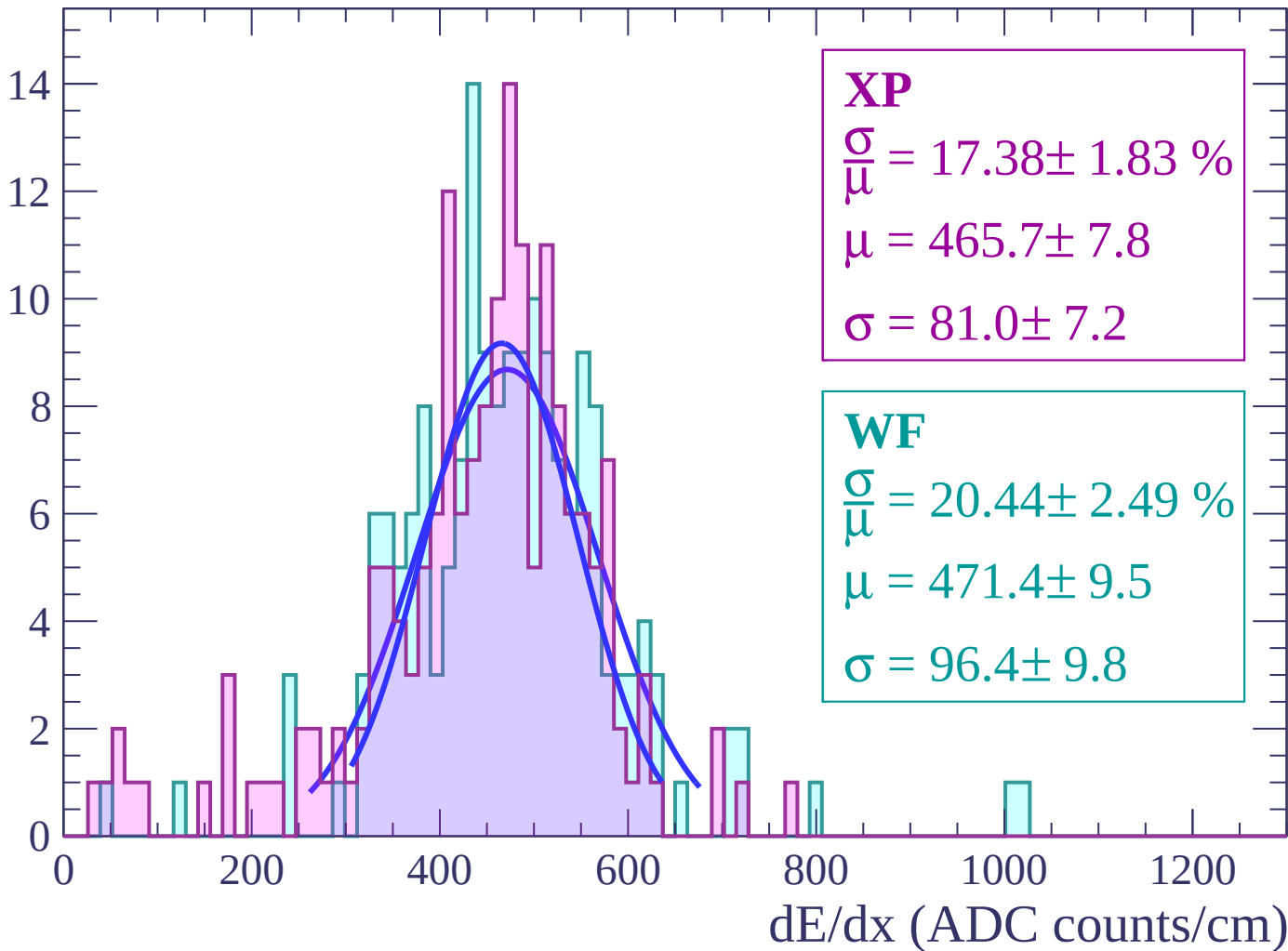
Energy loss | $34 < \theta < 36$

Count



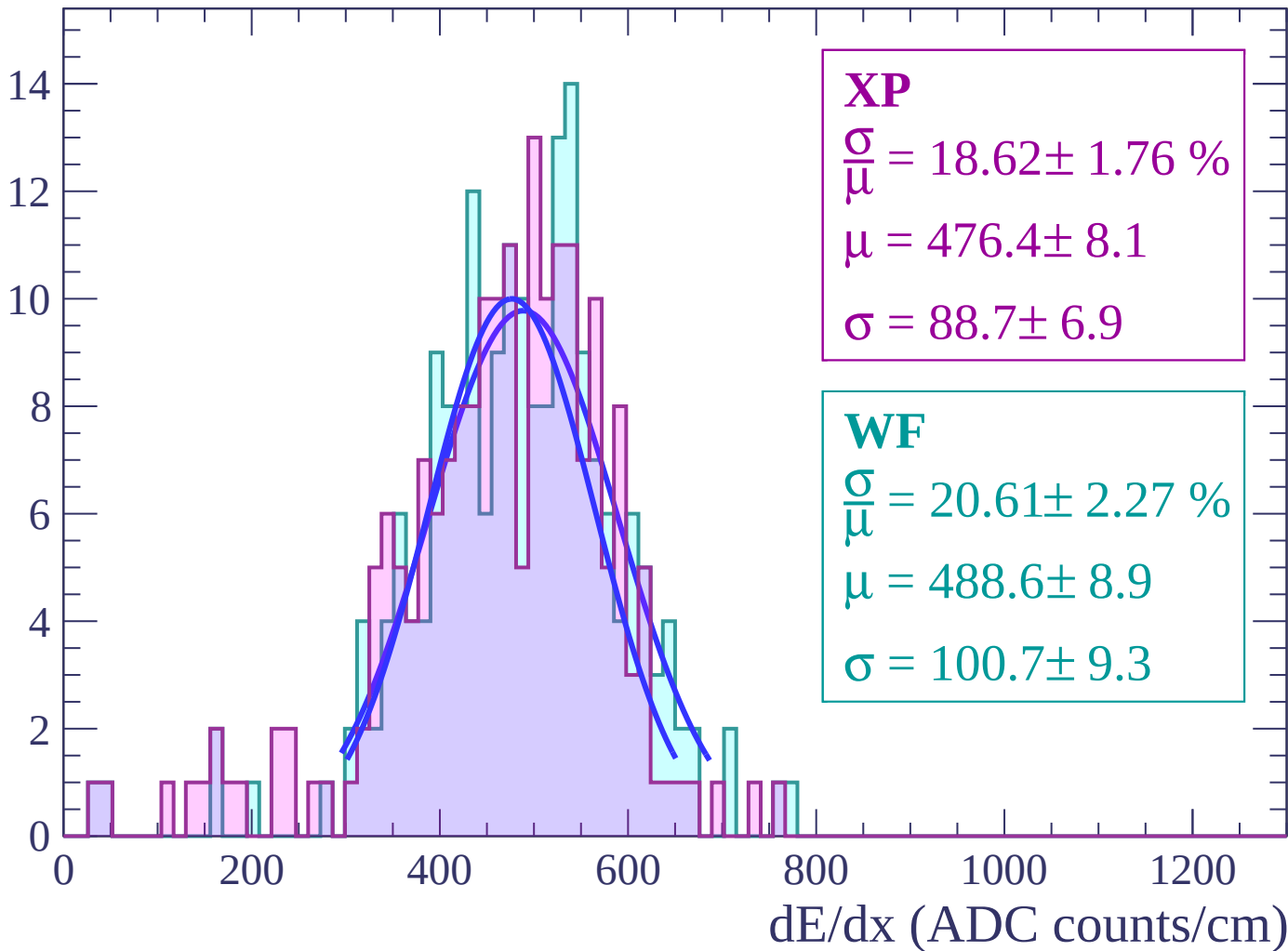
Energy loss | $36 < \theta < 38$

Count



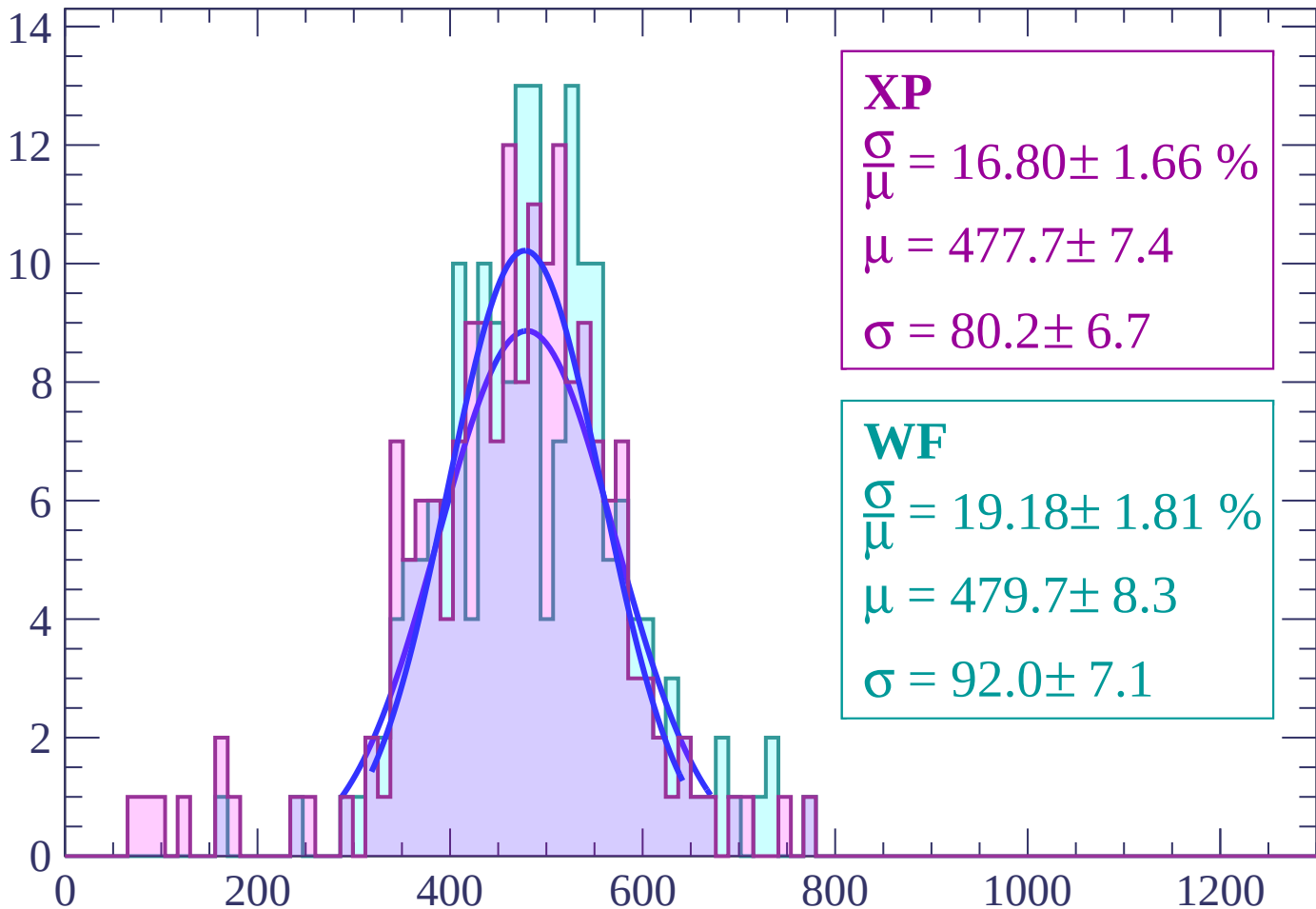
Energy loss | $38 < \theta < 40$

Count



Energy loss | $40 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 16.80 \pm 1.66 \%$$

$$\mu = 477.7 \pm 7.4$$

$$\sigma = 80.2 \pm 6.7$$

WF

$$\frac{\sigma}{\mu} = 19.18 \pm 1.81 \%$$

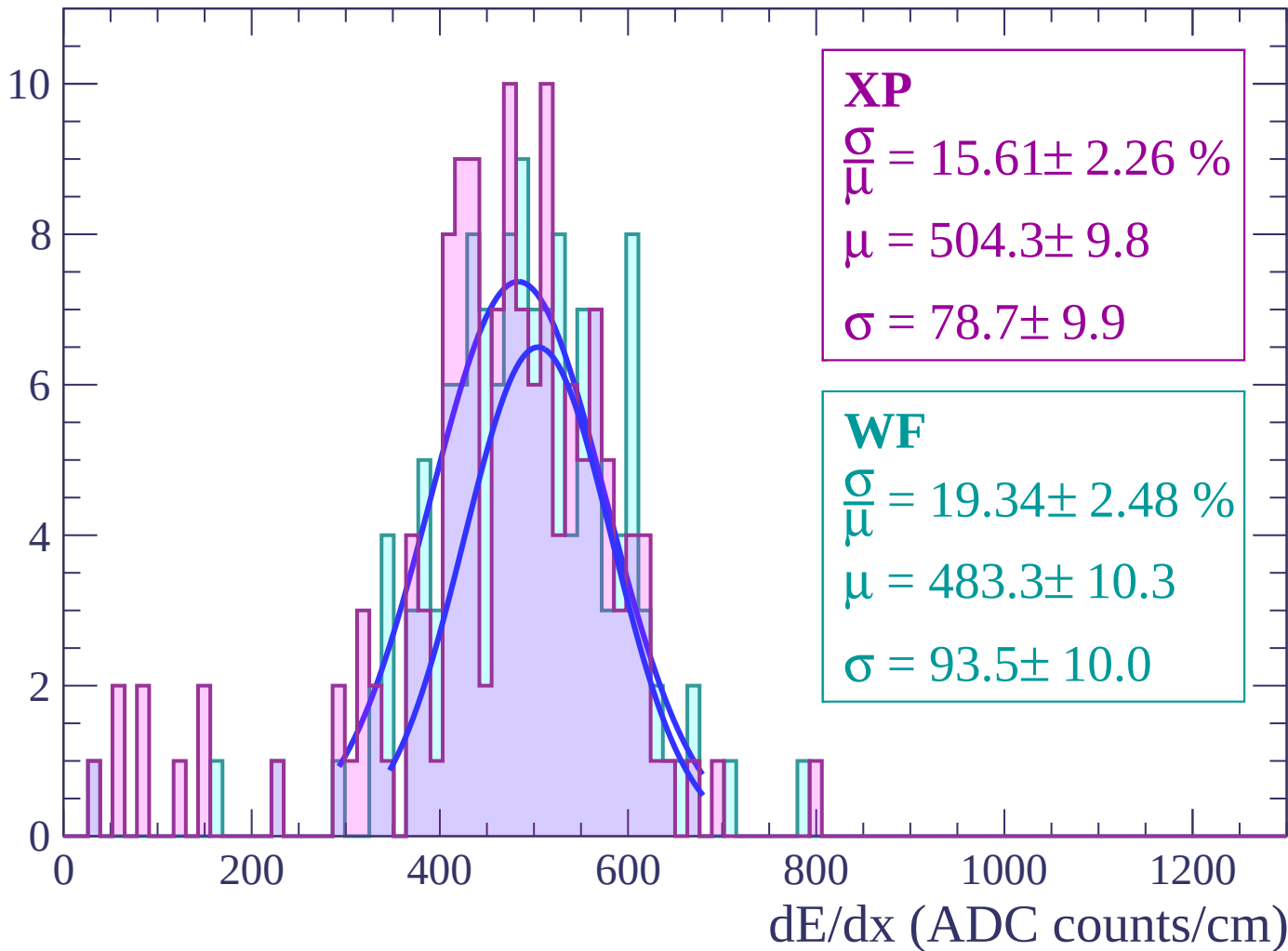
$$\mu = 479.7 \pm 8.3$$

$$\sigma = 92.0 \pm 7.1$$

dE/dx (ADC counts/cm)

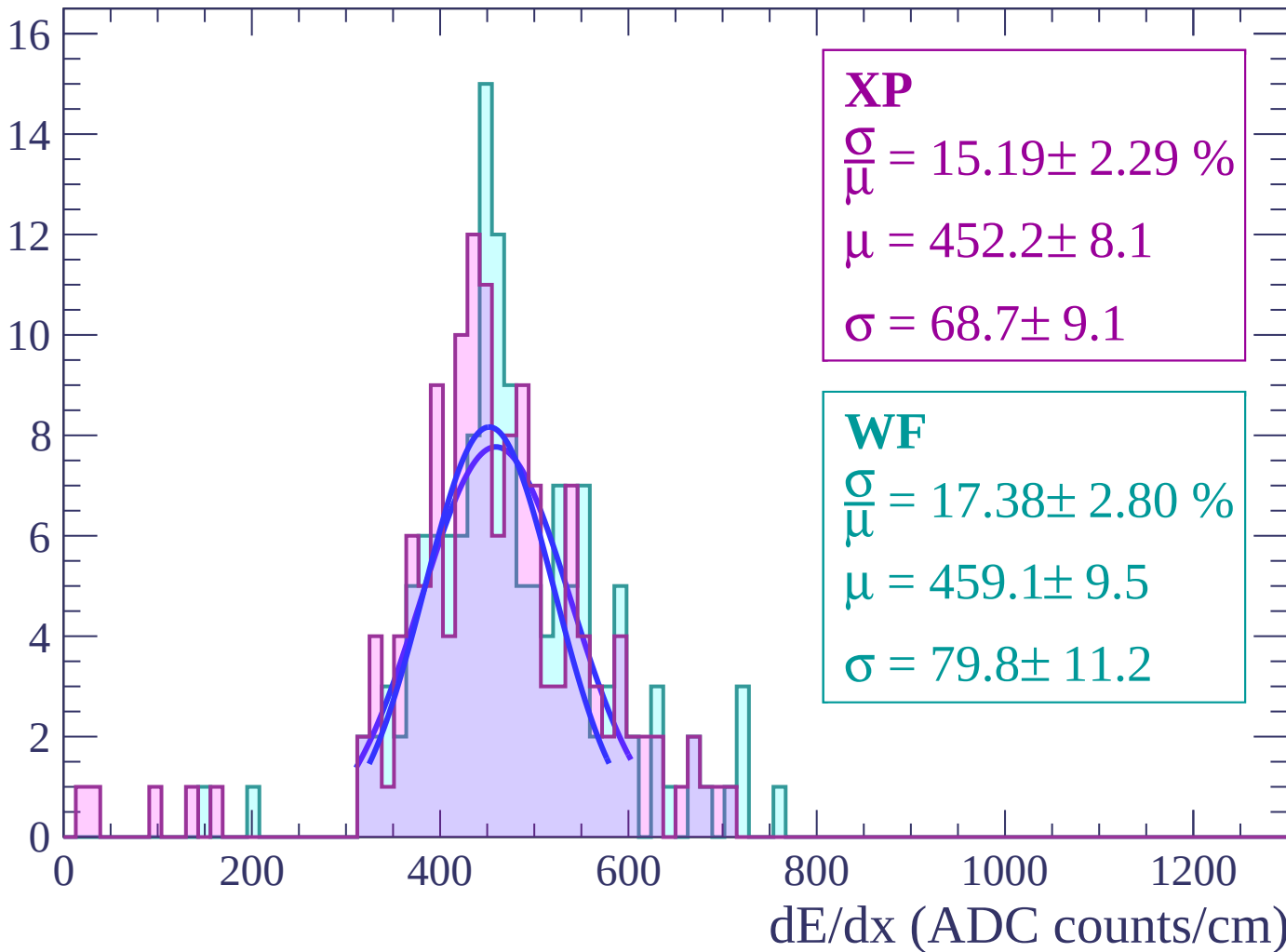
Energy loss | $42 < \theta < 44$

Count



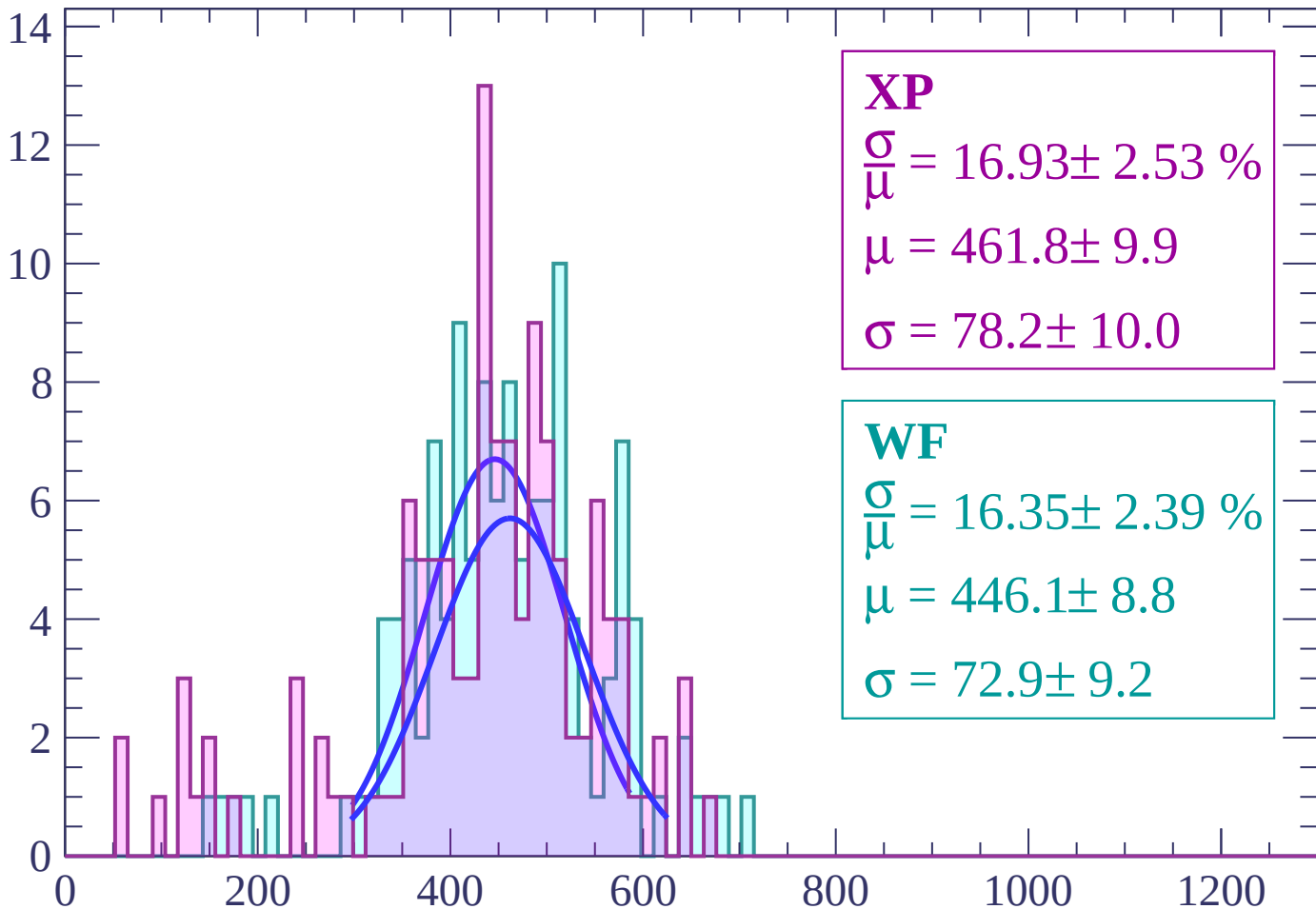
Energy loss | $44 < \theta < 46$

Count



Energy loss | $46 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 16.93 \pm 2.53 \%$$

$$\mu = 461.8 \pm 9.9$$

$$\sigma = 78.2 \pm 10.0$$

WF

$$\frac{\sigma}{\mu} = 16.35 \pm 2.39 \%$$

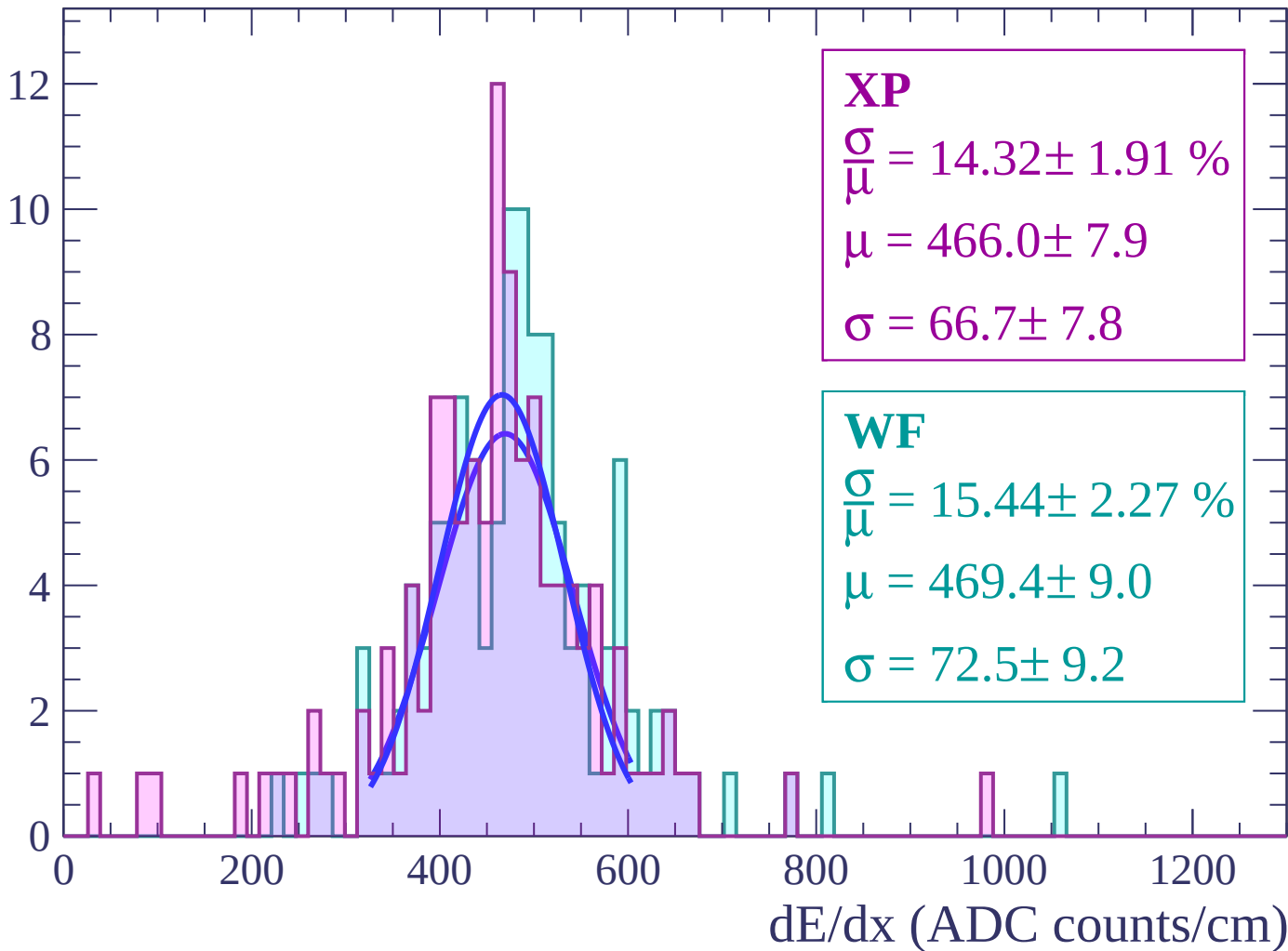
$$\mu = 446.1 \pm 8.8$$

$$\sigma = 72.9 \pm 9.2$$

dE/dx (ADC counts/cm)

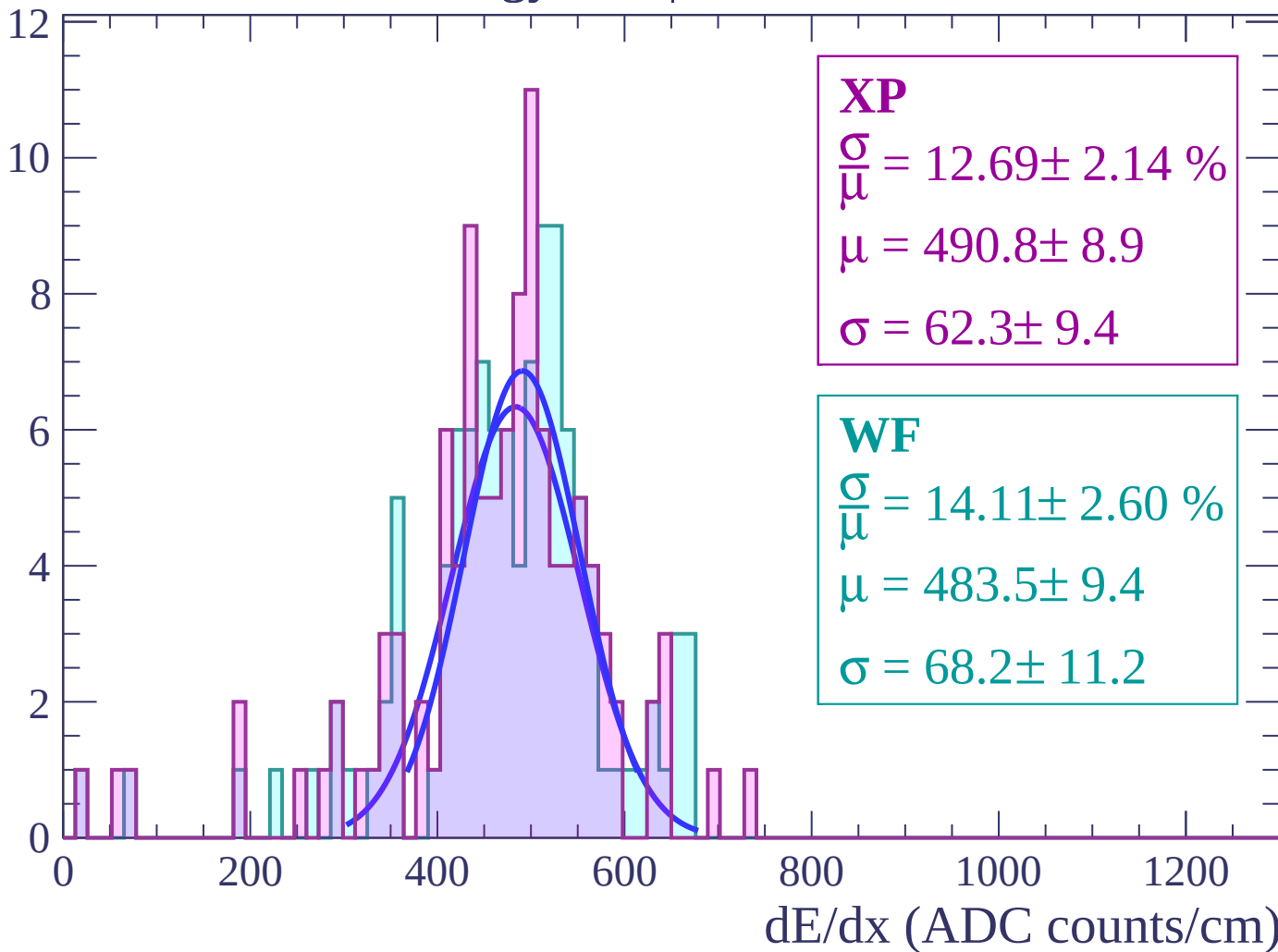
Energy loss | $48 < \theta < 50$

Count



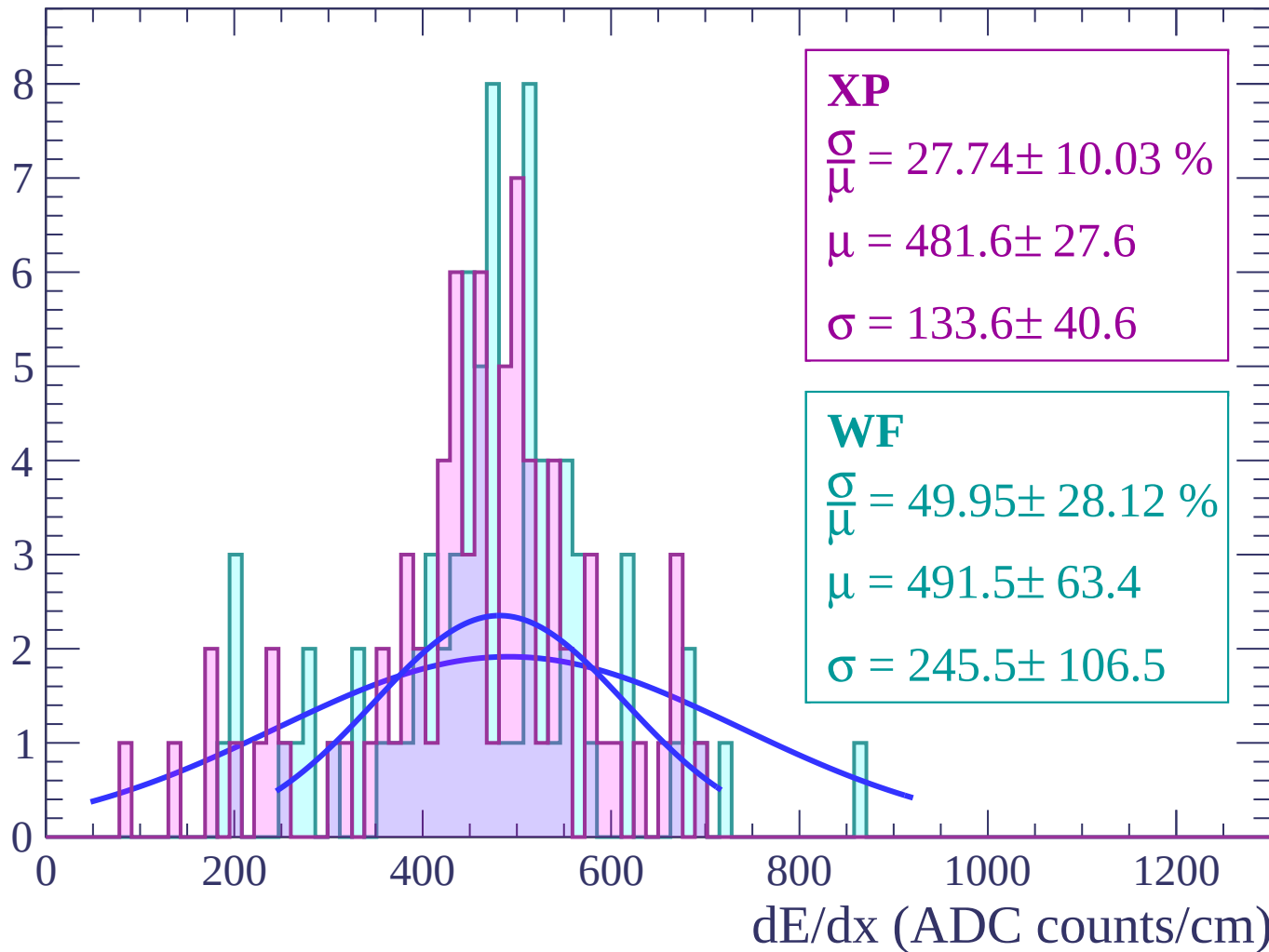
Energy loss | $50 < \theta < 52$

Count



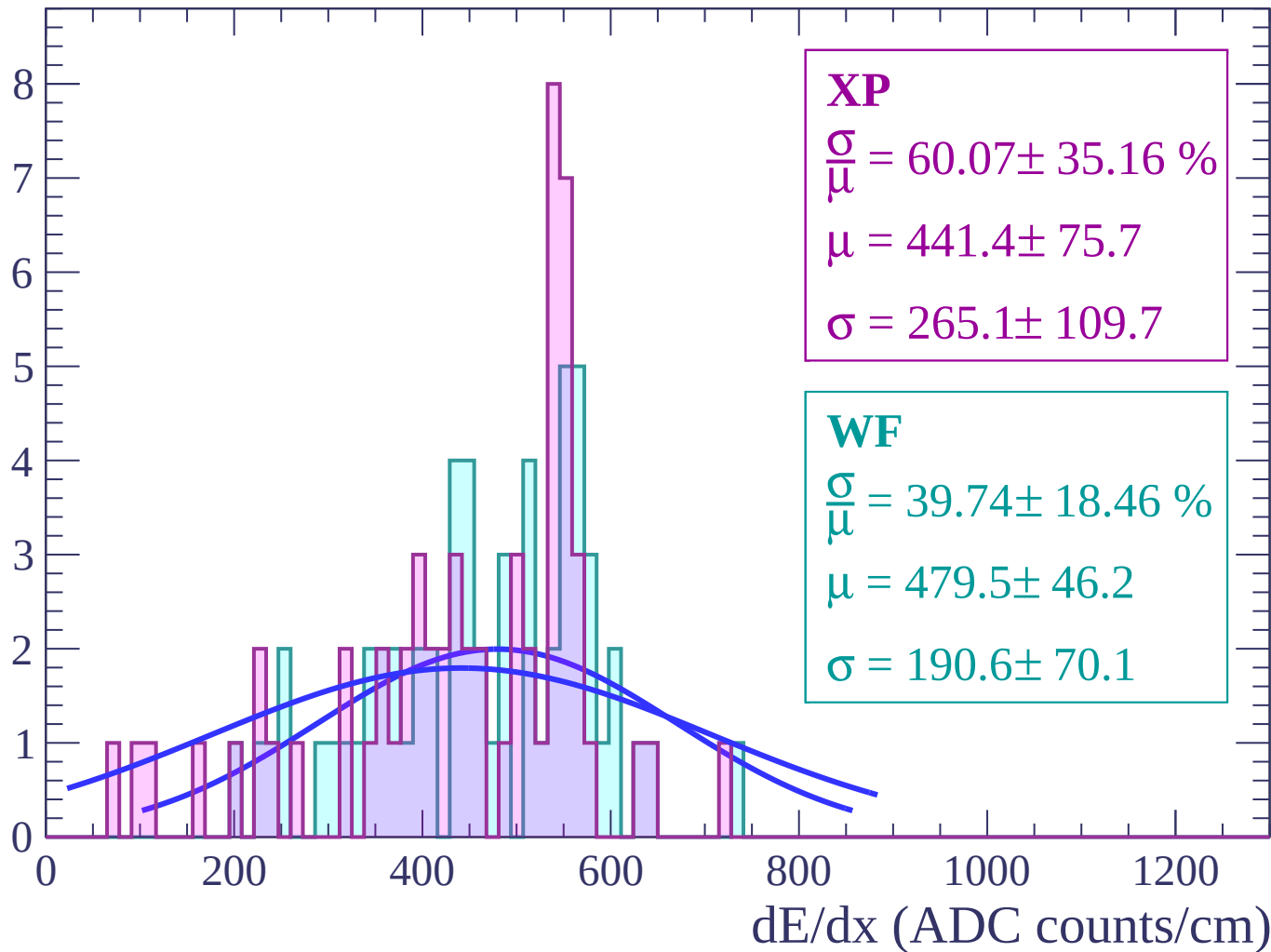
Energy loss | $52 < \theta < 54$

Count



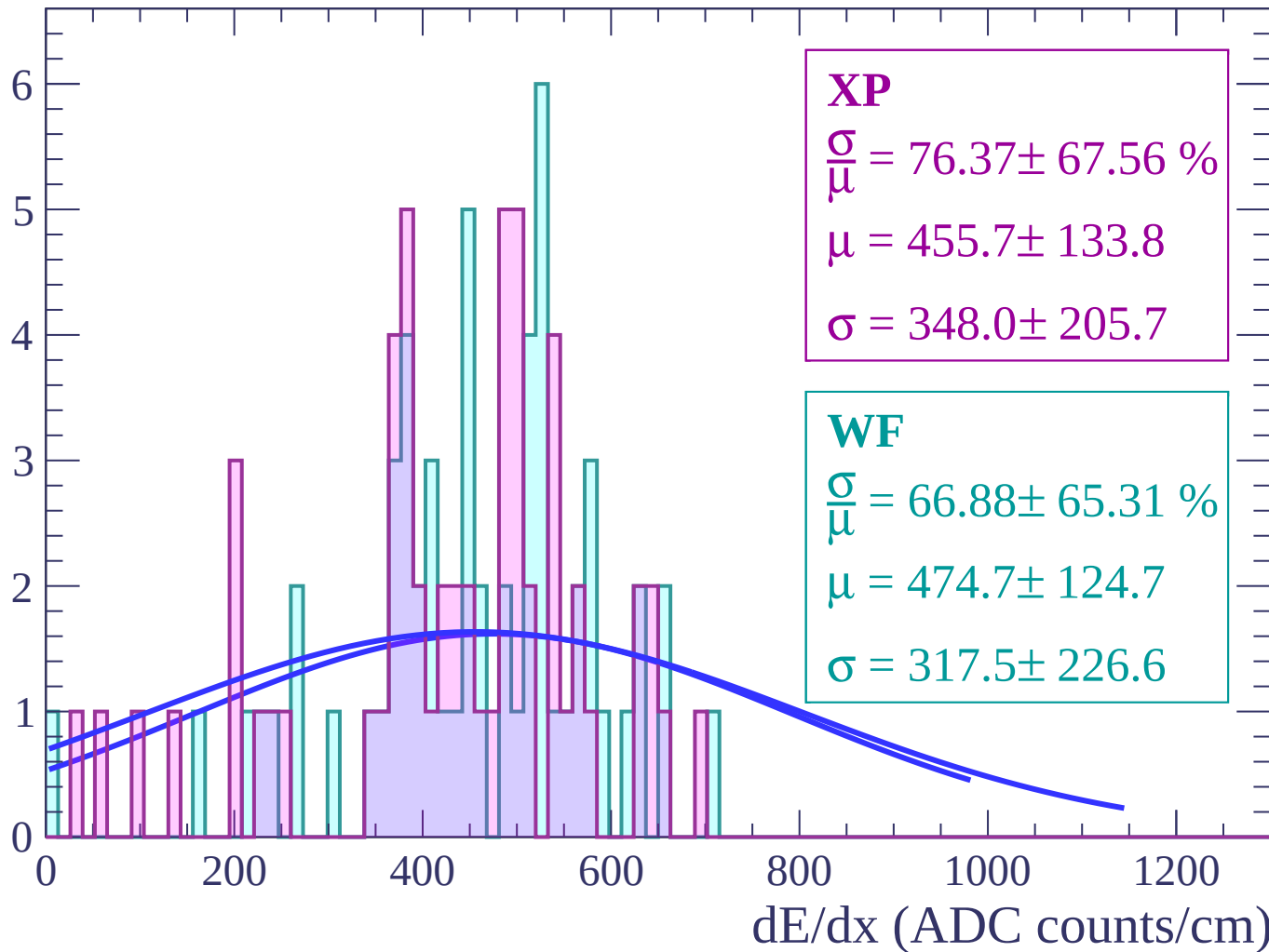
Energy loss | $54 < \theta < 56$

Count



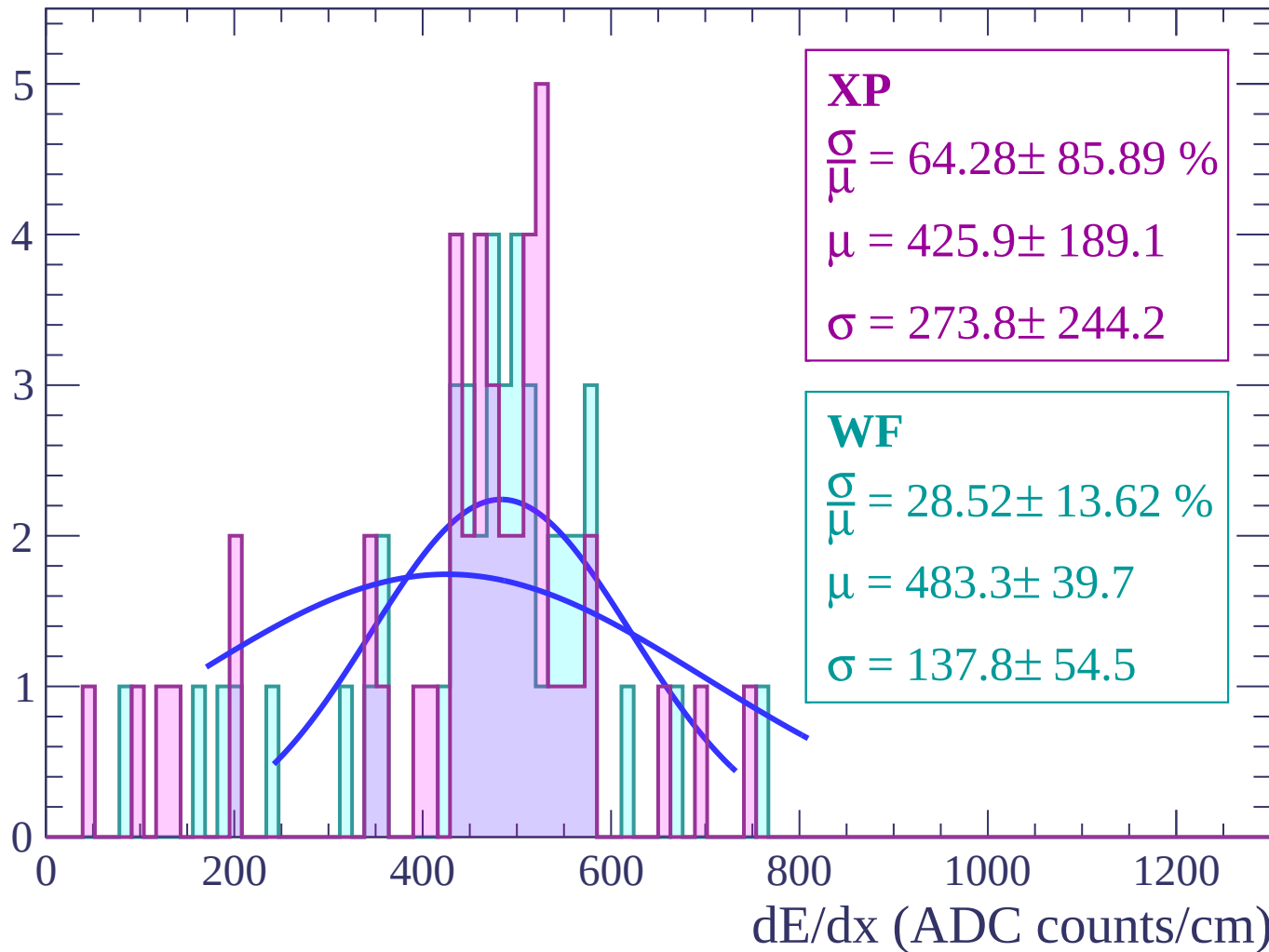
Energy loss | $56 < \theta < 58$

Count



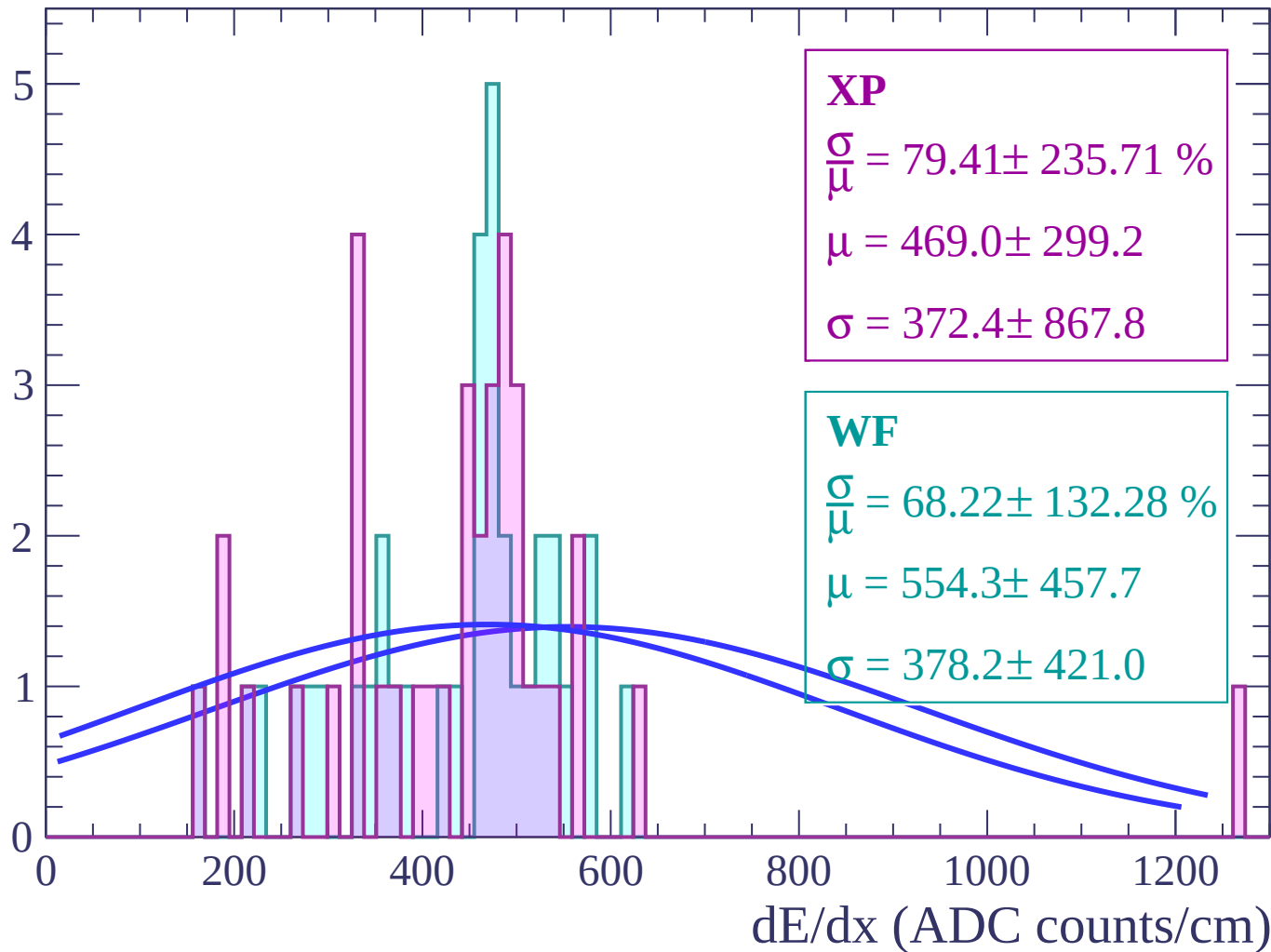
Energy loss | $58 < \theta < 60$

Count



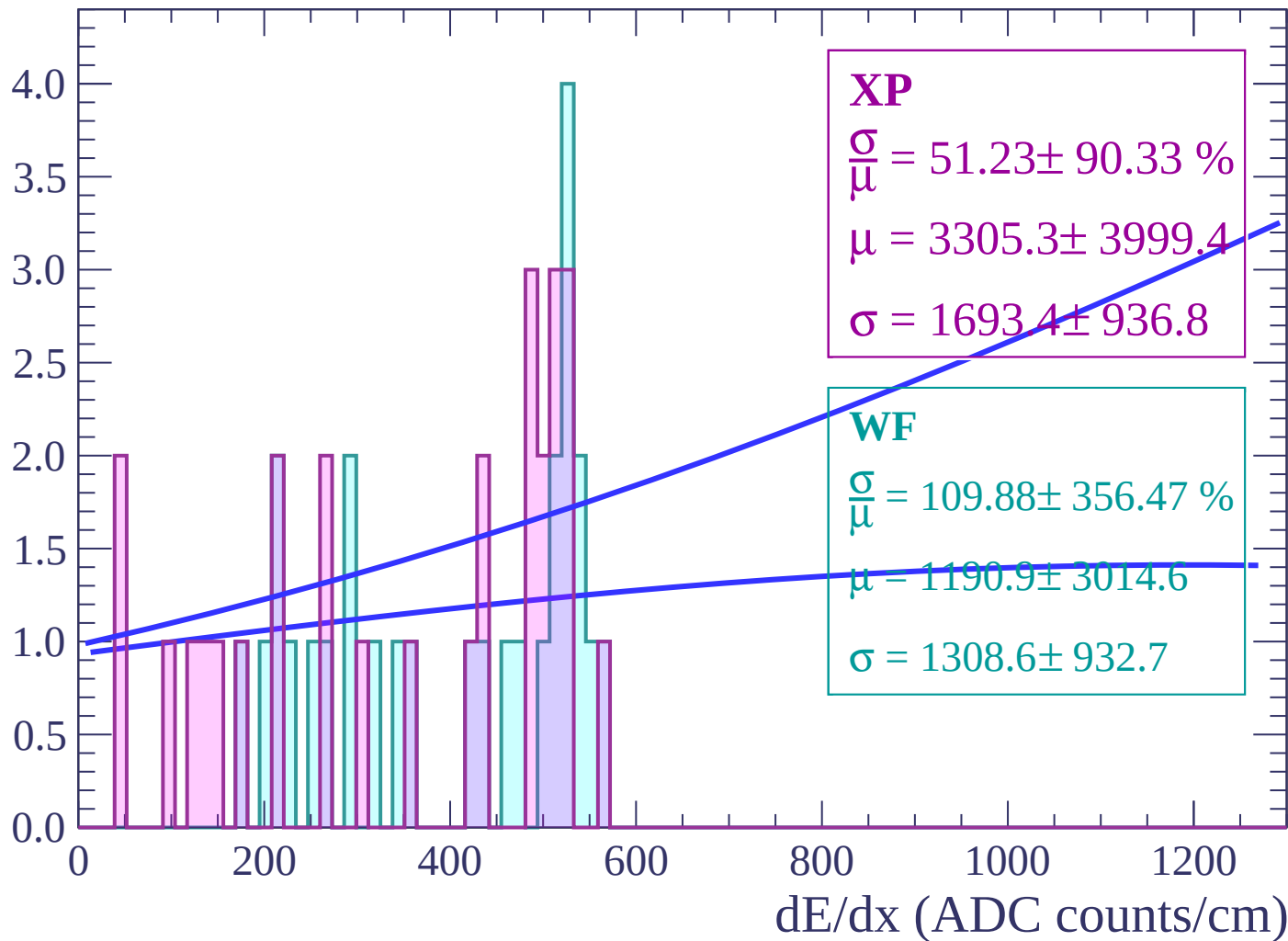
Energy loss | $60 < \theta < 62$

Count



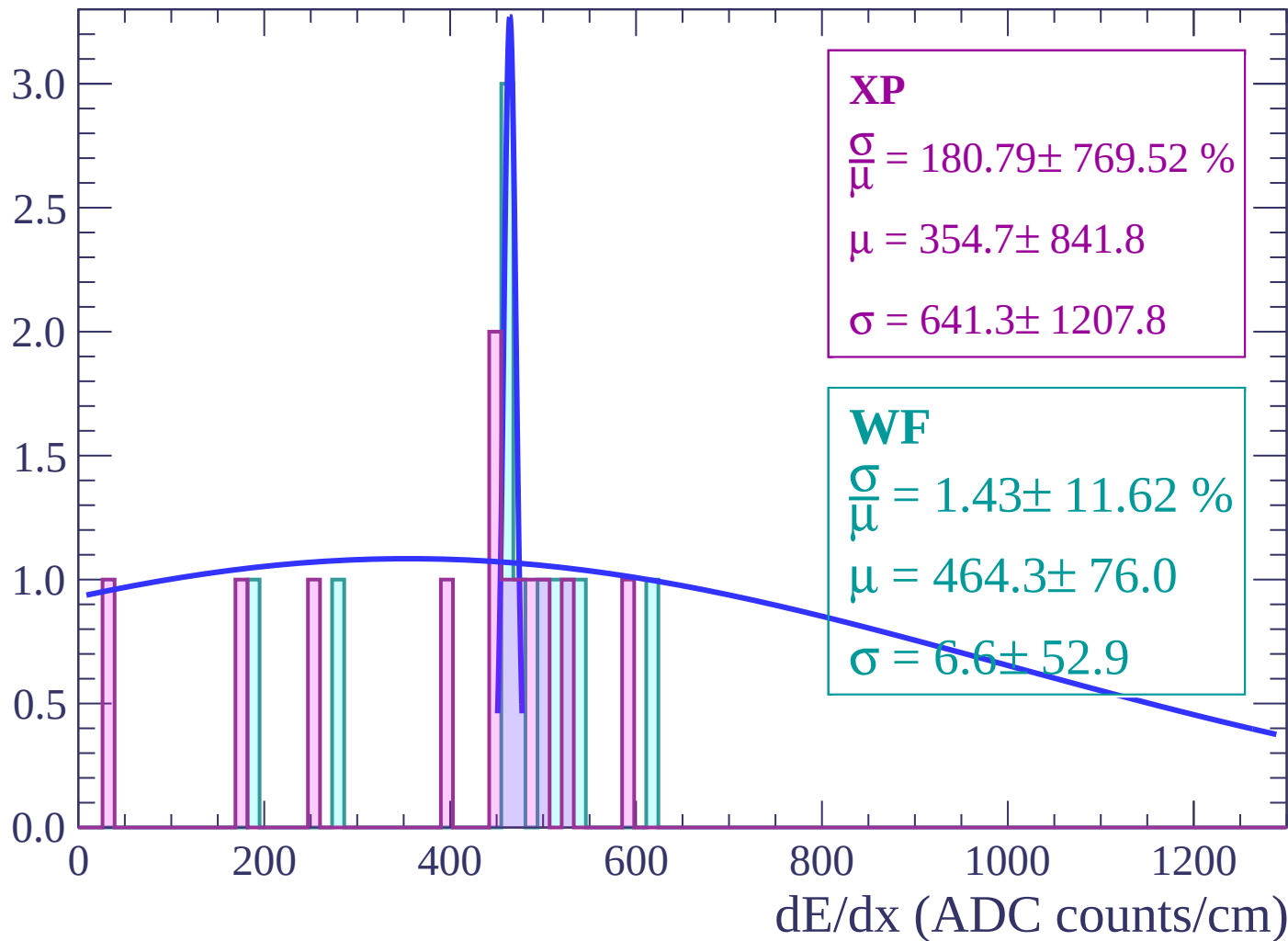
Energy loss | $62 < \theta < 64$

Count



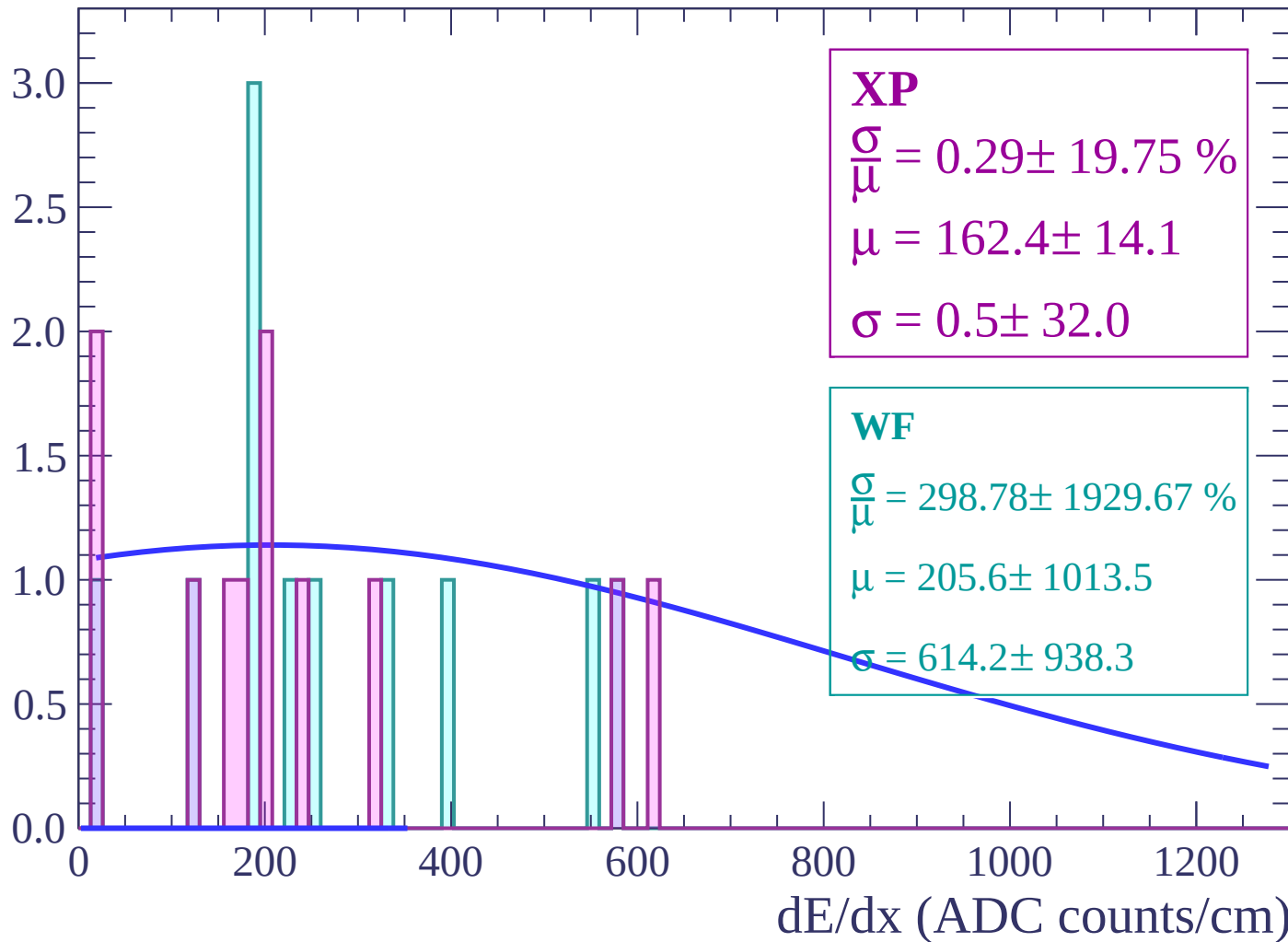
Energy loss | $64 < \theta < 66$

Count



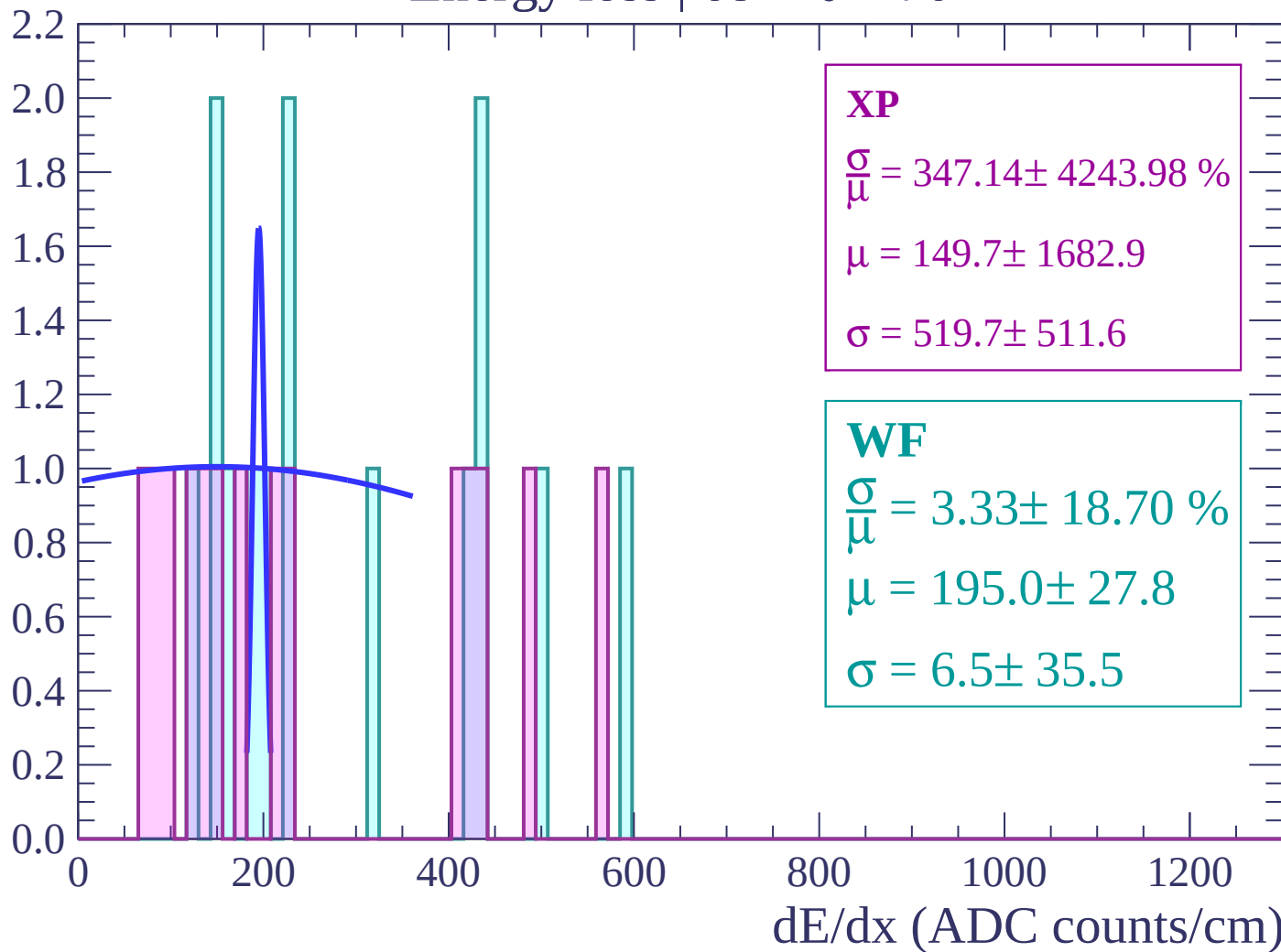
Energy loss | $66 < \theta < 68$

Count



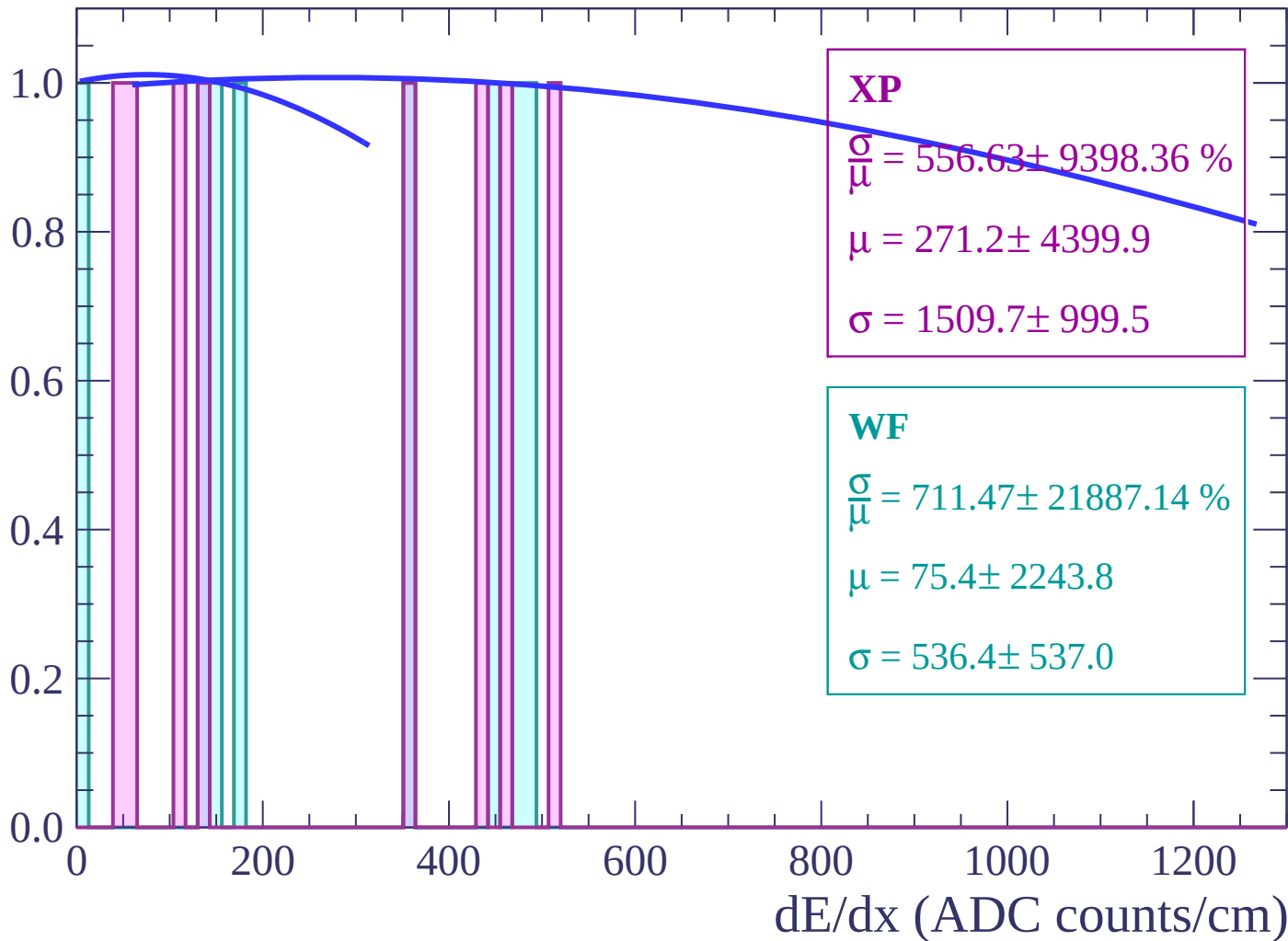
Energy loss | $68 < \theta < 70$

Count



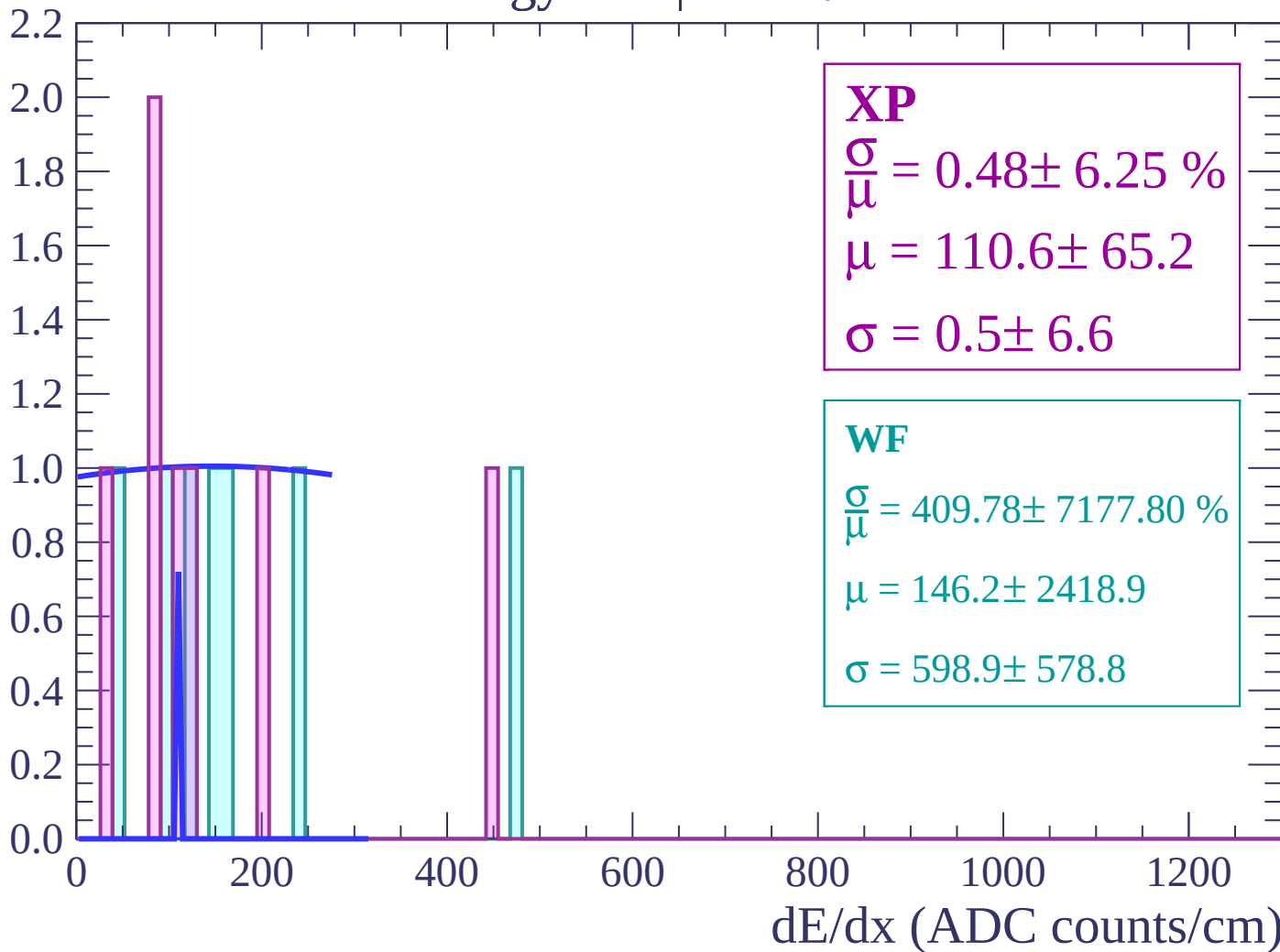
Energy loss | $70 < \theta < 72$

Count



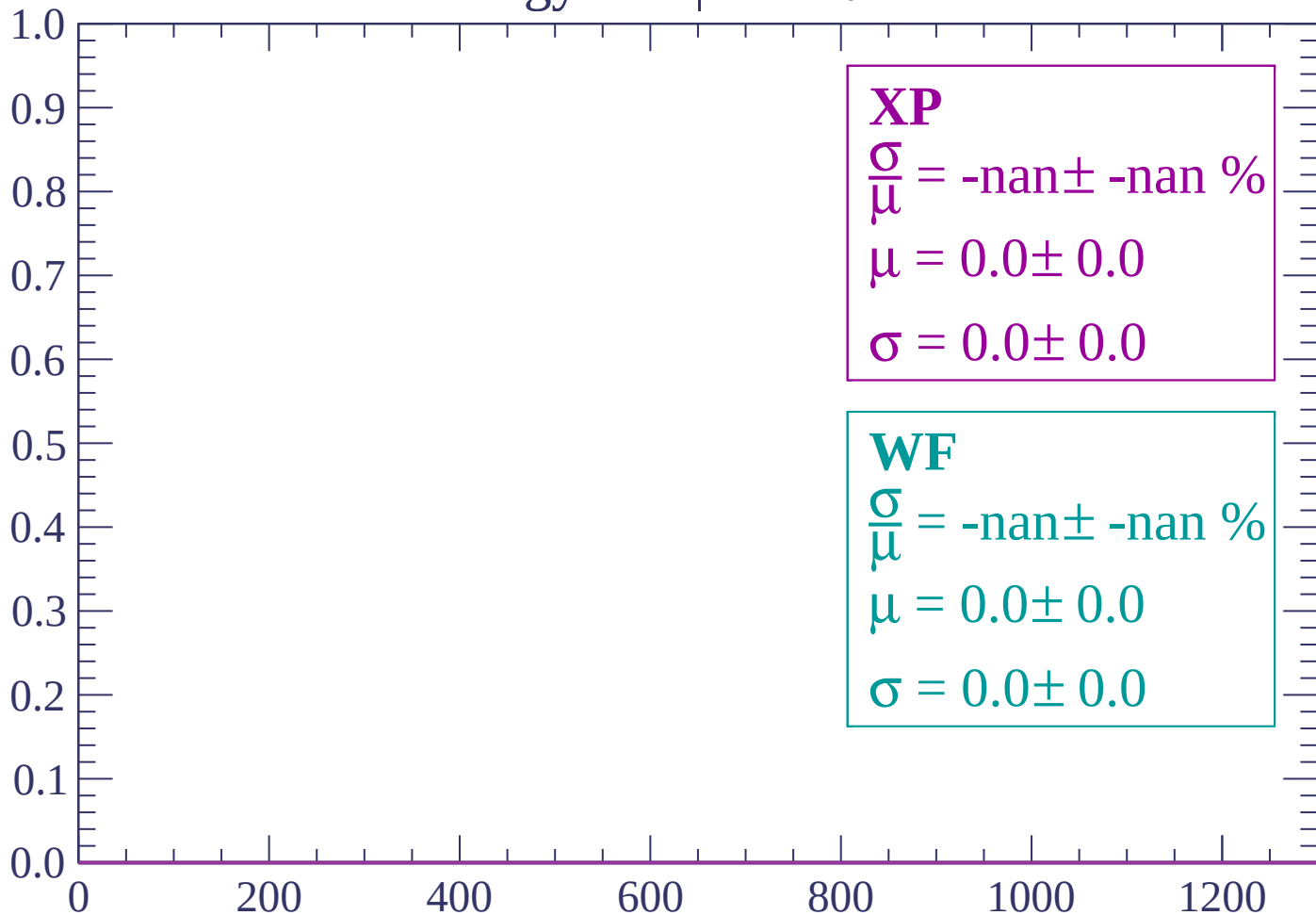
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

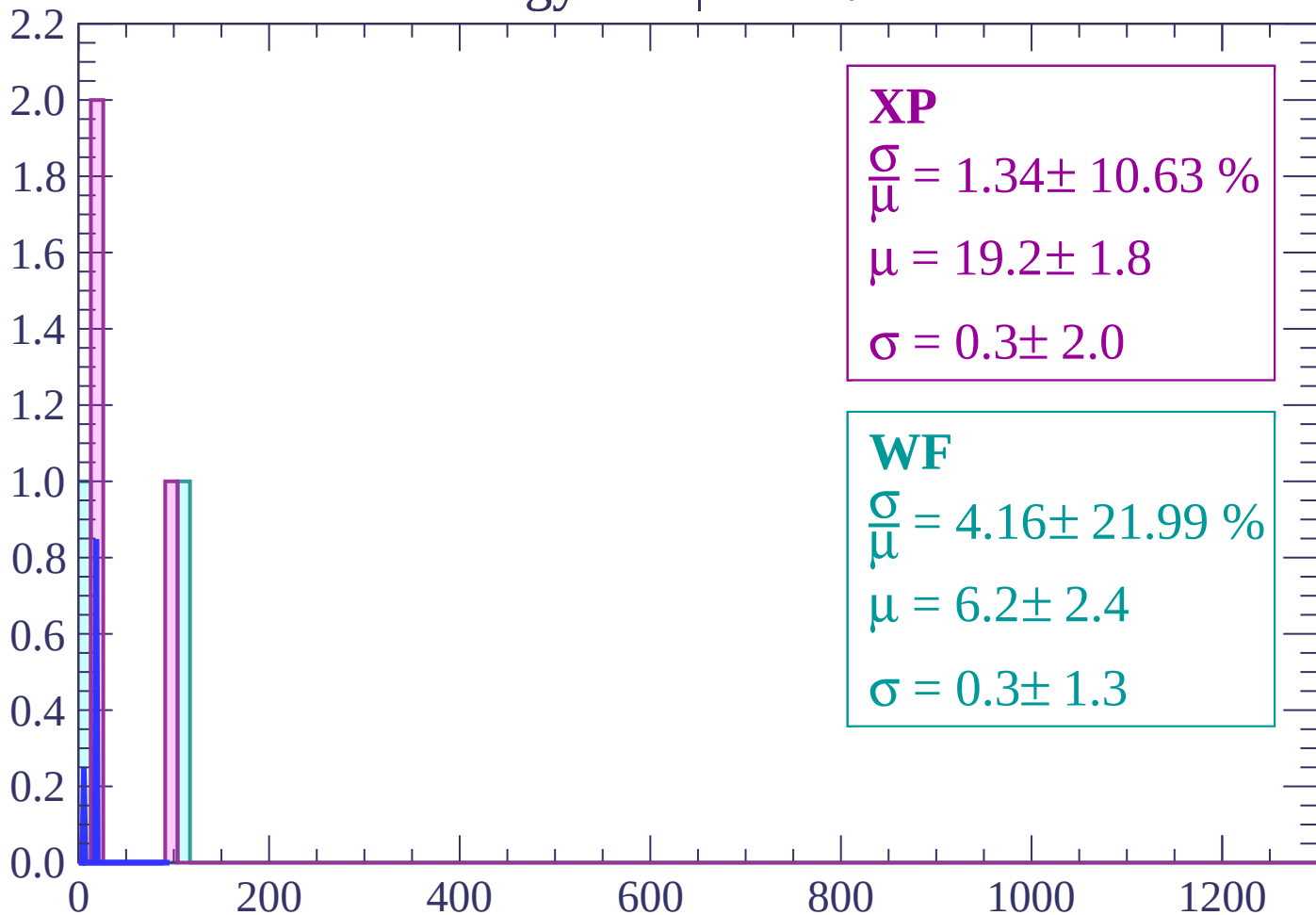
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

Count



XP

$$\frac{\sigma}{\mu} = 1.34 \pm 10.63 \%$$

$$\mu = 19.2 \pm 1.8$$

$$\sigma = 0.3 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

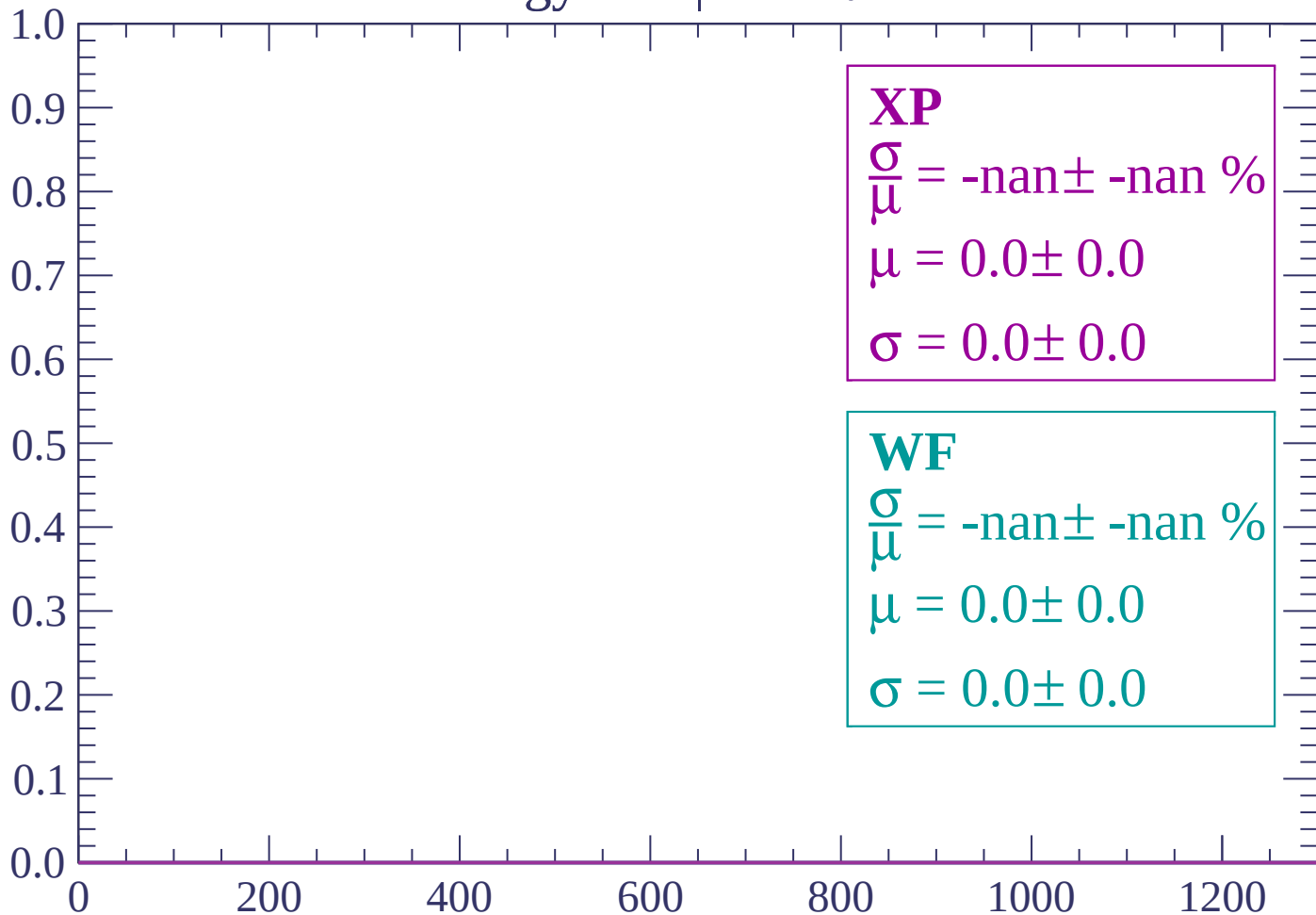
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

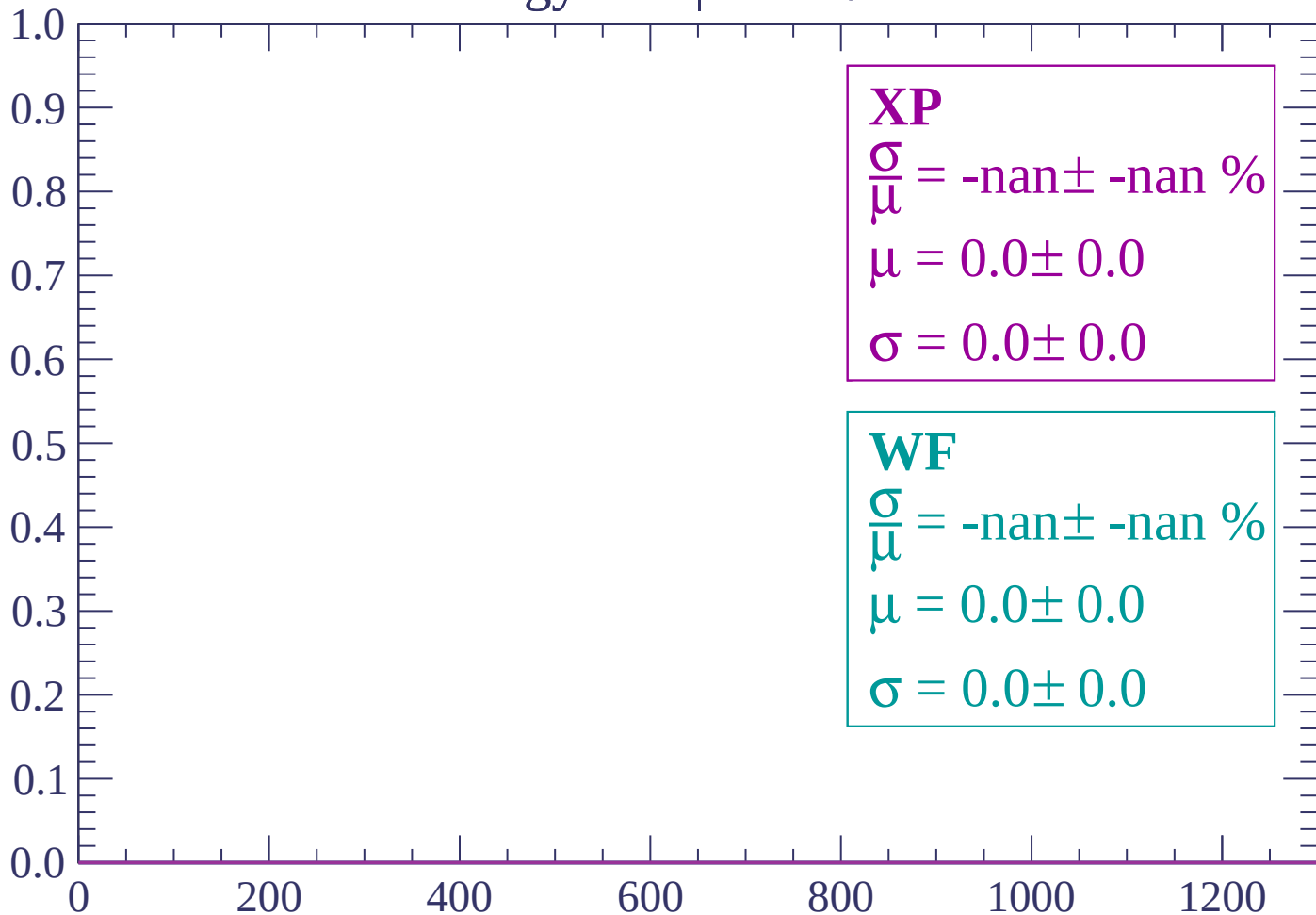
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

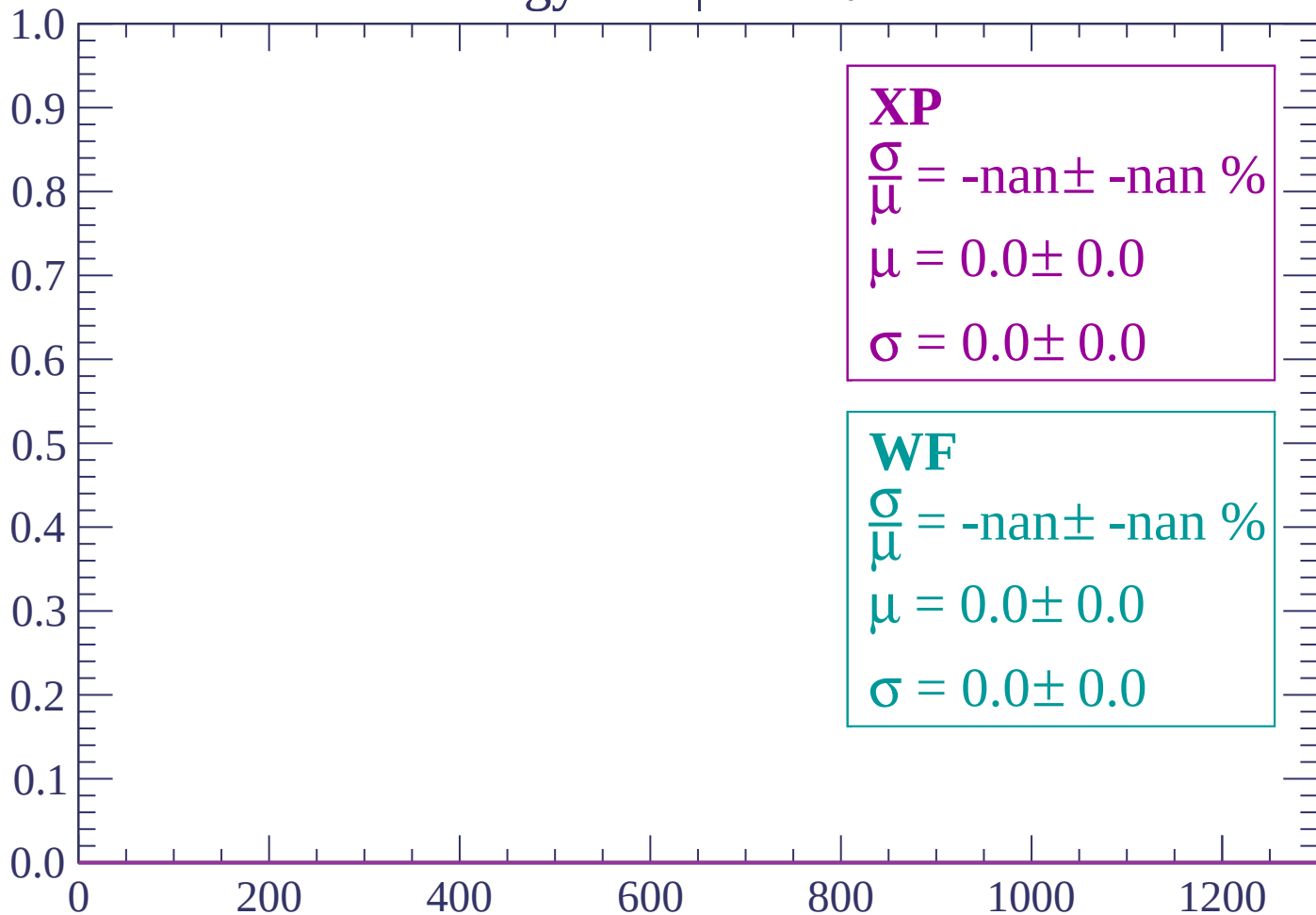
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

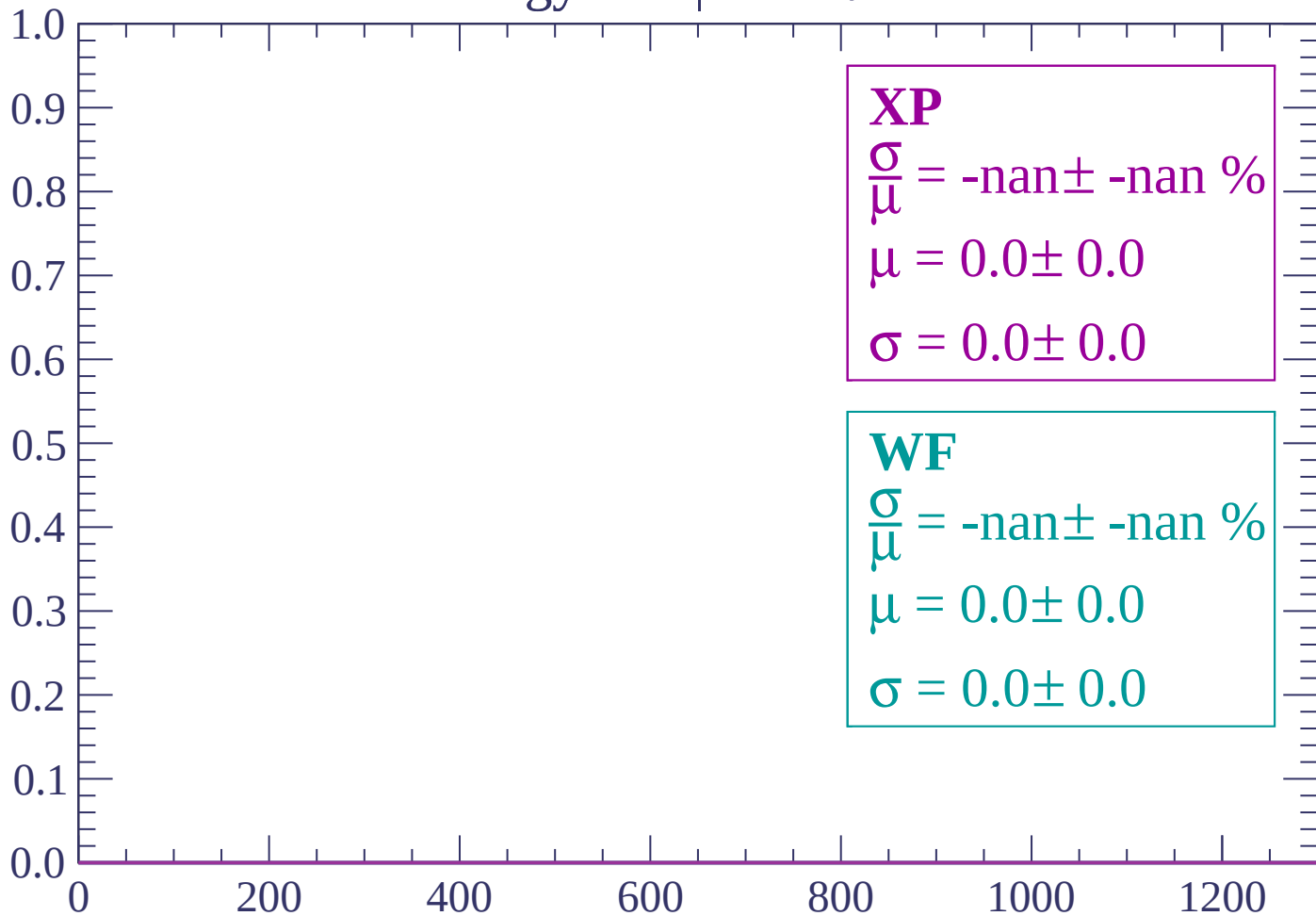
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

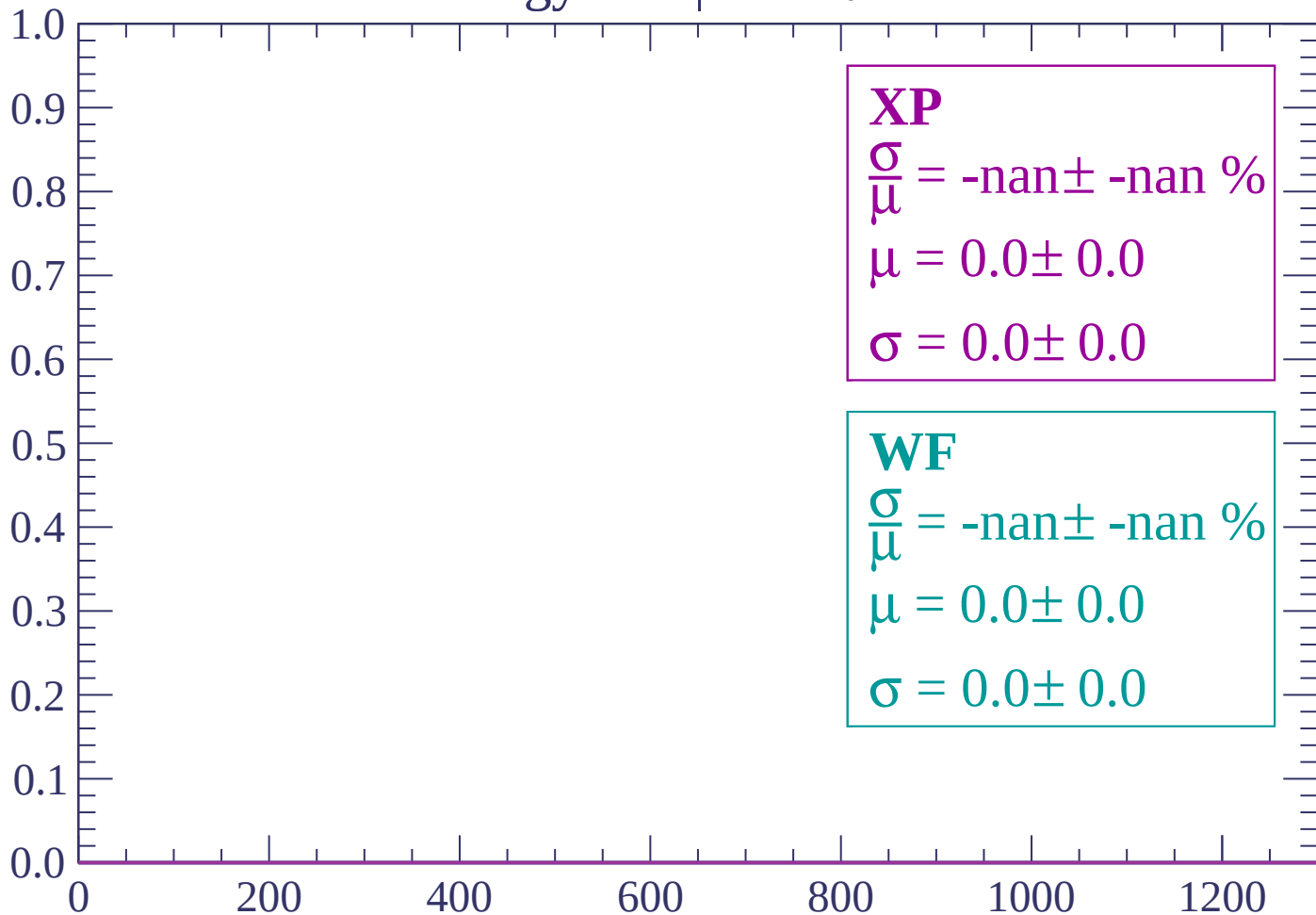
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

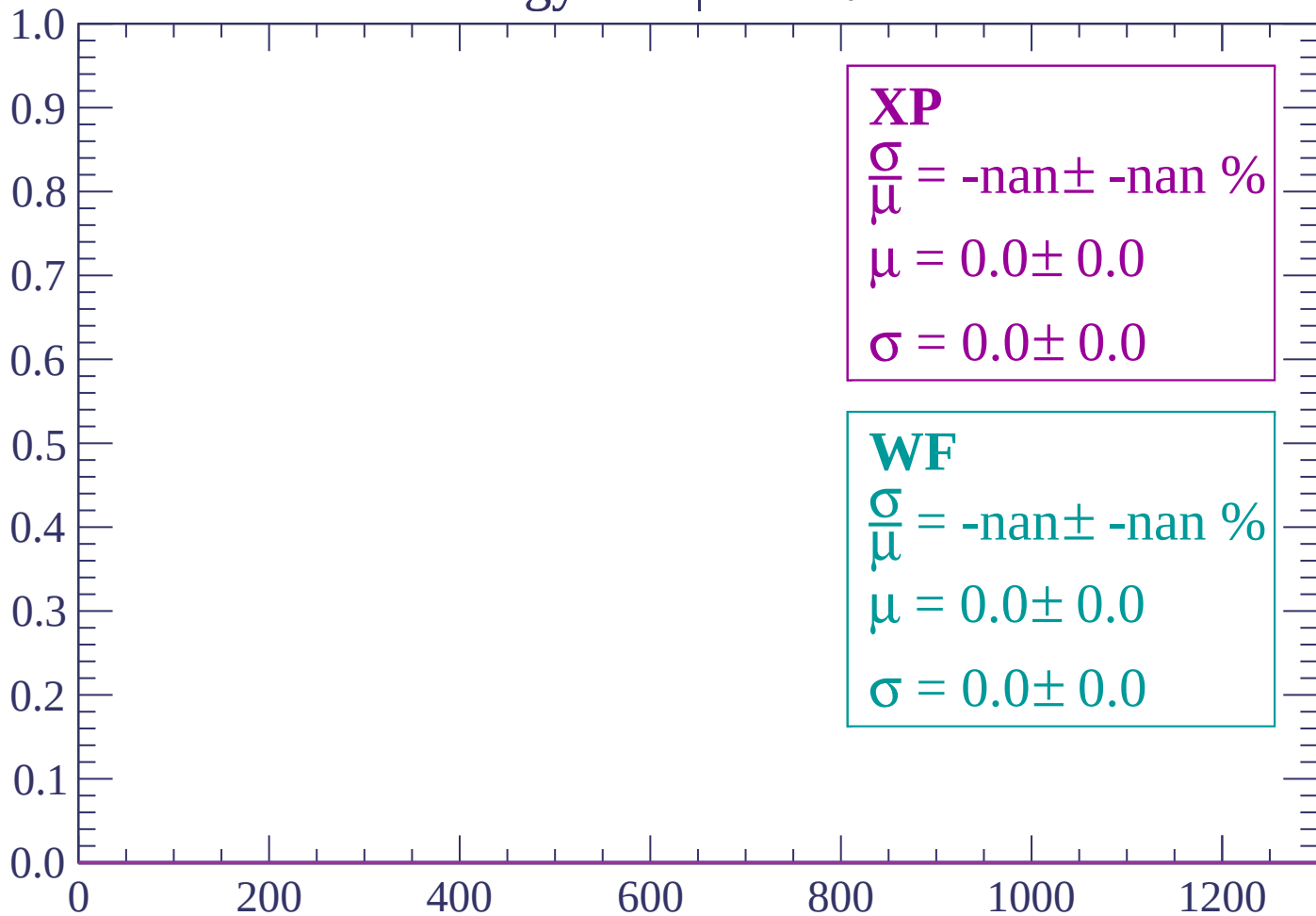
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

